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Quantum Electrodynamics

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Contents

[2]In §§ 49, 51, 53–55 we use ordinary units.

Introduction

§ 1. Uncertainty Relations in the Relativistic Domain

The quantum theory presented in Vol. III of this course is non-relativistic in its very substance, and it fails for phenomena in which motion occurs at speeds no longer small next to the speed of light. One might at first imagine that a relativistic theory could be reached by generalizing the formalism of non-relativistic quantum mechanics in a more or less straightforward manner. A closer look shows, however, that a logically complete relativistic theory cannot be built without invoking fresh physical principles.

We begin by recalling several physical notions on which non-relativistic quantum mechanics rests (Vol. III, § 1). A fundamental part was played there, as we saw, by the concept of measurement — the process in which a quantum system interacts with a "classical object" (an "apparatus") and thereby comes to possess definite values of particular dynamical variables (coordinates, velocities, and so forth). We saw, too, that quantum mechanics places severe limits on the extent to which different dynamical variables of an electron[3] can exist together. Thus the uncertainties Δq and Δp with which coordinate and momentum can coexist obey the relation $\Delta q \Delta p \sim \hbar$;[4] the more precisely either of these quantities is measured, the less precisely the other can be known at the same moment.

What matters, however, is that any single dynamical variable of the electron, on its own, admitted measurement of unlimited precision, and within a time interval that could be taken as short as desired. On this circumstance the whole of non-relativistic quantum mechanics depends in a fundamental way. It alone makes it possible to introduce the wave function, the central object of the theory's formalism. The physical content of the wave function $\psi(q)$ is, after all, that its squared modulus gives the probability that a measurement performed at a given instant will yield one or another value of the electron's coordinate. Evidently the very notion of such a probability presupposes that a coordinate measurement of unlimited precision and speed can be realized in principle; failing that, the notion would be empty of content and would forfeit its physical meaning.

The existence of a limiting speed (the speed of light c) leads to new limitations, of a fundamental kind, on the possibilities of measuring various physical quantities (*L. D. Landau, R. Peierls*, 1930).

In Vol. III, § 44 the relation

$$(v' - v)\,\Delta p\,\Delta t \sim \hbar \tag{1.1}$$

was obtained; it relates the uncertainty Δp of a momentum measurement on the electron to the length Δt of the measuring process itself, v and v' denoting the electron's velocity before and after that process. It follows that momentum can be measured accurately within a short time (small Δp together with small Δt) only if the measurement itself

[3] As in Vol. III, § 1, we say "electron" for brevity, understanding by it any quantum system.

[4] Ordinary units are used in this section.

alters the velocity strongly enough. In the non-relativistic theory this expresses the fact that a momentum measurement repeated after a short interval will not reproduce its result, but it places no restriction of principle on how accurate a single measurement of the momentum may be, the difference $v' - v$ being capable of unlimited increase.

The presence of a limiting speed, however, changes the state of affairs radically. The difference $v' - v$, like the velocities themselves, can now not exceed c (more precisely, $2c$). Replacing $v' - v$ in (1.1) by c, we obtain the relation

$$\Delta p \, \Delta t \sim \hbar / c, \tag{1.2}$$

which determines the best accuracy of momentum measurement attainable in principle for a given duration Δt of the measurement. Thus in the relativistic theory an arbitrarily accurate and arbitrarily rapid measurement of the momentum turns out to be impossible in principle. An exact measurement of the momentum ($\Delta p \to 0$) is possible only in the limit of an infinitely long duration of the measurement.

One has reason to expect that the measurability of the electron's coordinate as such is likewise affected. Within the mathematical formalism of the theory this appears as a conflict between an exact coordinate measurement and the statement that a free particle's energy is positive. As we shall see, the full set of eigenfunctions of a free particle's relativistic wave equation contains, besides the solutions whose time dependence is the "proper" one, solutions of "negative frequency" as well. These will in general appear also in the expansion of a wave packet describing an electron confined to a small region of space.

As will be shown, the "negative-frequency" wave functions are tied to the existence of antiparticles — the positrons. Their presence in the expansion of the wave packet reflects the creation of electron–positron pairs, unavoidable in the general case, during a measurement of the electron's coordinates. Since the emergence of these new particles cannot be controlled by the measuring process itself, coordinate measurement for the electron is drained of meaning.

In the rest frame of the electron the minimum error in a measurement of its coordinates is

$$\Delta q \sim \hbar / mc. \tag{1.3}$$

This value (the only one that dimensional considerations would permit) answers to a momentum uncertainty $\Delta p \sim mc$, which corresponds in turn to the minimum threshold energy needed to create a pair.

In a reference frame where the electron moves with energy ε, (1.3) is replaced by

$$\Delta q \sim c\hbar / \varepsilon. \tag{1.4}$$

In the extreme ultra-relativistic limit, in particular, energy and momentum are related by $\varepsilon \approx cp$, so that

$$\Delta q \sim \hbar / p, \tag{1.5}$$

the error Δq then coinciding with the particle's de Broglie wavelength.[5]

[5]What is meant here are measurements any outcome of which permits an inference about the state of

For photons the ultra-relativistic case always holds, so that expression (1.5) is valid. This means that it makes sense to speak of the coordinates of a photon only in those cases where the characteristic dimensions are large compared with the wavelength. But this is nothing other than the "classical" limiting case, corresponding to geometrical optics, in which one may speak of the propagation of light along definite trajectories — rays. In the quantum case, however, when the wavelength cannot be regarded as small, the concept of the coordinates of a photon becomes vacuous. We shall see later (see § 4) that in the mathematical formalism of the theory the non-measurability of the photon's coordinates already shows itself in the impossibility of forming from its wave function a quantity that could play the part of a probability density satisfying the requirements which relativistic invariance demands.

On the basis of all that has been said, it is natural to suppose that the future theory will renounce altogether the consideration of the time development of the processes of interaction of particles. It will show that in these processes no exactly definable characteristics exist (even within the limits of the usual quantum-mechanical accuracy), so that the description of a process in time will turn out to be just as illusory as classical trajectories proved to be in non-relativistic quantum mechanics. The only observable quantities will be the characteristics (momenta, polarizations) of free particles — the initial particles that enter into the interaction and the final particles that have arisen as a result of the process (*L. D. Landau, R. Peierls*, 1930).

The characteristic formulation of a problem in relativistic quantum theory consists in the determination of the probability amplitudes of transitions connecting given initial and final states (that is, at $t \to \mp\infty$) of a system of particles. The totality of the transition amplitudes between all possible states constitutes the *scattering matrix*, or *S-matrix*. This matrix will be the carrier of all the information about the processes of interaction of particles that possesses an observable physical meaning (*W. Heisenberg*, 1938).

At the present time a complete, logically closed relativistic quantum theory does not yet exist. We shall see that the existing theory introduces new physical aspects into the manner of describing the states of particles, which acquires certain features of field theory (see § 10). The theory is constructed, however, to a considerable degree on the model of, and with the help of the concepts of, ordinary quantum mechanics. This way of building the theory has led to success in the domain of quantum electrodynamics. The absence of complete logical closure shows itself in this theory in the existence of divergent expressions when its mathematical apparatus is applied directly; but for the removal of these divergences there exist entirely unambiguous procedures. Nevertheless these procedures retain to a considerable degree the character of semi-empirical recipes, and our confidence in the correctness of the results obtained in this way rests, in the end, on their excellent agreement with experiment, and not on the internal consistency and logical coherence of the fundamental principles of the theory.

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the electron; we thus set aside coordinate measurements by collision, in which over the time of observation the result occurs with a probability short of unity. In such a case a deflection of the particle does allow the electron's position to be inferred, but from the absence of a deflection nothing at all can be concluded.

CHAPTER I

Photon

§ 2. Quantization of the Free Electromagnetic Field

In order to treat the electromagnetic field as a quantum object, it is convenient to start from a classical description in which the field is characterized by a discrete, albeit infinite, set of variables; such a description will allow one to apply the standard formalism of quantum mechanics directly. A description by means of potentials specified at every point of space is, in essence, a description in terms of a continuous set of variables.

Let $\mathbf{A}(\mathbf{r}, t)$ be the vector potential of the free electromagnetic field, satisfying the "transversality condition"

$$\nabla \cdot \mathbf{A} = 0. \tag{2.1}$$

The scalar potential then vanishes, $\Phi = 0$, and the fields $\mathbf{E}$ and $\mathbf{H}$ are

$$\mathbf{E} = -\dot{\mathbf{A}}, \qquad \mathbf{H} = \nabla \times \mathbf{A}. \tag{2.2}$$

Maxwell's equations reduce to the wave equation for $\mathbf{A}$:

$$\Delta \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0. \tag{2.3}$$

As is well known (see Vol. II, § 52), in classical electrodynamics the passage to a description in terms of a discrete set of variables is effected by considering the field inside a large but finite volume V.[1] Let us recall, omitting the details of the calculation, how this is done.

A field in a finite volume can be expanded in travelling plane waves, so that its potential takes the form of a series

$$\mathbf{A} = \sum_{\mathbf{k}} (\mathbf{a}_{\mathbf{k}} e^{i\mathbf{k}\mathbf{r}} + \mathbf{a}^*_{\mathbf{k}} e^{-i\mathbf{k}\mathbf{r}}), \tag{2.4}$$

where the coefficients $\mathbf{a}_{\mathbf{k}}$ depend on time according to

$$\mathbf{a}_{\mathbf{k}} \sim e^{-i\omega t}, \quad \omega = |\mathbf{k}|. \tag{2.5}$$

By virtue of condition (2.1) the complex vectors $\mathbf{a}_{\mathbf{k}}$ are orthogonal to the corresponding wave vectors: $\mathbf{a}_{\mathbf{k}}\mathbf{k} = 0$.

The summation in (2.4) runs over an infinite discrete set of values of the wave vector (its three components k_x, k_y, k_z). The transition to an integration over a continuous distribution is made with the aid of the expression

$$\frac{d^3 k}{(2\pi)^3}$$

[1] To avoid burdening formulae with superfluous factors, we set $V = 1$.

for the number of allowed values of $\mathbf{k}$ falling within a volume element $d^3k = dk_x\, dk_y\, dk_z$ of $\mathbf{k}$-space.

Once the vectors $\mathbf{a_k}$ are given, the field inside the volume in question is completely determined. One may therefore view these quantities as a discrete set of classical "field variables". In order to find the passage to a quantum theory, however, a further transformation of these quantities is needed, one that brings the field equations into a form analogous to the canonical (Hamiltonian) equations of classical mechanics. The canonical variables of the field are introduced by means of

$$\mathbf{Q_k} = \frac{1}{\sqrt{4\pi}}(\mathbf{a_k} + \mathbf{a_k^*}), \qquad \mathbf{P_k} = -\frac{i\omega}{\sqrt{4\pi}}(\mathbf{a_k} - \mathbf{a_k^*}) = \dot{\mathbf{Q}}_\mathbf{k} \tag{2.6}$$

(they are evidently real). The vector potential is expressed through the canonical variables as

$$\mathbf{A} = \sqrt{4\pi}\sum_\mathbf{k}\left(\mathbf{Q_k}\cos\mathbf{kr} - \frac{1}{\omega}\mathbf{P_k}\sin\mathbf{kr}\right). \tag{2.7}$$

To find the Hamiltonian H one must evaluate the total field energy

$$\frac{1}{8\pi}\int(\mathbf{E}^2 + \mathbf{H}^2)\, d^3x,$$

expressed in terms of the quantities $\mathbf{Q_k}$, $\mathbf{P_k}$. Substituting the expansion (2.7) for $\mathbf{A}$, computing $\mathbf{E}$ and $\mathbf{H}$ by means of (2.2), and integrating, one obtains

$$H = \frac{1}{2}\sum_\mathbf{k}(\mathbf{P_k^2} + \omega^2\mathbf{Q_k^2}).$$

Each of the vectors $\mathbf{P_k}$ and $\mathbf{Q_k}$ is perpendicular to the wave vector $\mathbf{k}$ and therefore has two independent components. The direction of these vectors fixes the polarization direction of the corresponding wave. Denoting the two components of $\mathbf{Q_k}$, $\mathbf{P_k}$ (in the plane perpendicular to $\mathbf{k}$) by $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ ($\alpha = 1, 2$), we rewrite the Hamiltonian as

$$H = \sum_{\mathbf{k}\alpha}\frac{1}{2}(P_{\mathbf{k}\alpha}^2 + \omega^2 Q_{\mathbf{k}\alpha}^2). \tag{2.8}$$

The Hamiltonian has thus broken up into a sum of independent terms, each involving just one pair $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$. Every term answers to a travelling wave of a given wave vector and polarization and reproduces the structure of a single harmonic-oscillator Hamiltonian. One accordingly speaks of this decomposition as the expansion of the field in *oscillators*.

We now turn to the quantization of the free electromagnetic field. The method of classical description just set out makes the route to quantum theory obvious. The canonical variables — the generalized coordinates $Q_{\mathbf{k}\alpha}$ and generalized momenta $P_{\mathbf{k}\alpha}$ — are now to be treated as operators obeying the commutation rule

$$\hat{P}_{\mathbf{k}\alpha}\hat{Q}_{\mathbf{k}\alpha} - \hat{Q}_{\mathbf{k}\alpha}\hat{P}_{\mathbf{k}\alpha} = -i \tag{2.9}$$

(operators with different $\mathbf{k}\alpha$ all commute with one another). Together with them the potential $\mathbf{A}$ and, by (2.2), the field strengths $\mathbf{E}$ and $\mathbf{H}$ also become (Hermitian) operators.

A consistent determination of the field Hamiltonian requires evaluation of the integral

$$\hat{H} = \frac{1}{8\pi}\int (\hat{\mathbf{E}}^2 + \hat{\mathbf{H}}^2)\, d^3x, \tag{2.10}$$

in which $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ are expressed through the operators $\hat{P}_{\mathbf{k}\alpha}$, $\hat{Q}_{\mathbf{k}\alpha}$. In practice, however, the non-commutativity of the latter does not make itself felt here, since the products $Q_{\mathbf{k}\alpha}P_{\mathbf{k}\alpha}$ enter with the factor $\cos\mathbf{kr}\cdot\sin\mathbf{kr}$, which vanishes upon integration over the full volume. Consequently the Hamiltonian takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha}\frac{1}{2}(\hat{P}^2_{\mathbf{k}\alpha} + \omega^2\hat{Q}^2_{\mathbf{k}\alpha}), \tag{2.11}$$

exactly reproducing the classical Hamiltonian, as one would naturally expect.

Finding the eigenvalues of this Hamiltonian requires no special computation, since the problem reduces to the familiar one of the energy levels of linear oscillators (see Vol. III, § 23). We can therefore write down at once the energy levels of the field:

$$E = \sum_{\mathbf{k}\alpha}\left(N_{\mathbf{k}\alpha} + \frac{1}{2}\right)\omega, \tag{2.12}$$

where $N_{\mathbf{k}\alpha}$ are non-negative integers.

Discussion of this formula will be taken up in the next section; here we write out the matrix elements of the quantities $Q_{\mathbf{k}\alpha}$, which can be obtained directly from the known formulae for the matrix elements of the oscillator coordinate (see Vol. III, § 23). The non-vanishing matrix elements are

$$\langle N_{\mathbf{k}\alpha}|Q_{\mathbf{k}\alpha}|N_{\mathbf{k}\alpha}-1\rangle = \sqrt{\frac{N_{\mathbf{k}\alpha}}{2\omega}}. \tag{2.13}$$

The matrix elements of $P_{\mathbf{k}\alpha} = \dot{Q}_{\mathbf{k}\alpha}$ differ from those of $Q_{\mathbf{k}\alpha}$ only by a factor $\pm i\omega$.

For subsequent calculations it will, however, be more convenient to work not with $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$ themselves but with their linear combinations $\omega Q_{\mathbf{k}\alpha} \pm iP_{\mathbf{k}\alpha}$, which have matrix elements only for the transitions $N_{\mathbf{k}\alpha} \to N_{\mathbf{k}\alpha} \pm 1$. Accordingly we introduce the operators

$$\hat{c}_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}}(\omega\hat{Q}_{\mathbf{k}\alpha} + i\hat{P}_{\mathbf{k}\alpha}), \qquad \hat{c}^+_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}}(\omega\hat{Q}_{\mathbf{k}\alpha} - i\hat{P}_{\mathbf{k}\alpha}) \tag{2.14}$$

(the classical quantities $c_{\mathbf{k}\alpha}$, $c^*_{\mathbf{k}\alpha}$ coincide, up to a factor $\sqrt{2\pi/\omega}$, with the coefficients $a_{\mathbf{k}\alpha}$, $a^*_{\mathbf{k}\alpha}$ in the expansion (2.4)).

The matrix elements of these operators are

$$\langle N_{\mathbf{k}\alpha}-1|\hat{c}_{\mathbf{k}\alpha}|N_{\mathbf{k}\alpha}\rangle = \langle N_{\mathbf{k}\alpha}|\hat{c}^+_{\mathbf{k}\alpha}|N_{\mathbf{k}\alpha}-1\rangle = \sqrt{N_{\mathbf{k}\alpha}}. \tag{2.15}$$

The commutation rule between $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$ follows from the definition (2.14) and the rule (2.9):

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}^+_{\mathbf{k}\alpha} - \hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha} = 1. \tag{2.16}$$

Returning to the vector potential, we adopt an expansion of the same type as (2.4), but with the coefficients now promoted to operators:

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha}\mathbf{A}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\mathbf{A}^{*}_{\mathbf{k}\alpha}), \tag{2.17}$$

where

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,e^{i\mathbf{kr}}. \tag{2.18}$$

Here $\mathbf{e}^{(\alpha)}$ denotes a unit vector specifying the polarization direction of the oscillator; the vectors $\mathbf{e}^{(\alpha)}$ are perpendicular to the wave vector $\mathbf{k}$, and for each $\mathbf{k}$ there are two independent polarizations.

Analogously, for the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ we write

$$\hat{\mathbf{E}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha}\mathbf{E}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\mathbf{E}^{*}_{\mathbf{k}\alpha}), \quad \hat{\mathbf{H}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha}\mathbf{H}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\mathbf{H}^{*}_{\mathbf{k}\alpha}), \tag{2.19}$$

with

$$\mathbf{E}_{\mathbf{k}\alpha} = i\omega\mathbf{A}_{\mathbf{k}\alpha}, \qquad \mathbf{H}_{\mathbf{k}\alpha} = \mathbf{n}\times\mathbf{E}_{\mathbf{k}\alpha} \qquad (\mathbf{n} = \mathbf{k}/\omega). \tag{2.20}$$

The vectors $\mathbf{A}_{\mathbf{k}\alpha}$ are mutually orthogonal in the sense that

$$\int \mathbf{A}_{\mathbf{k}\alpha}\mathbf{A}^{*}_{\mathbf{k}'\alpha'}\,d^3x = \frac{2\pi}{\omega}\,\delta_{\alpha\alpha'}\,\delta_{\mathbf{k}\mathbf{k}'}. \tag{2.21}$$

To see this, note that when $\mathbf{A}_{\mathbf{k}\alpha}$ and $\mathbf{A}^{*}_{\mathbf{k}'\alpha'}$ carry different wave vectors their product involves $e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}}$, which integrates to zero over the volume; when only the polarizations differ, the orthogonality $\mathbf{e}^{(\alpha)}\mathbf{e}^{(\alpha')*} = 0$ of the two independent polarization directions provides the required vanishing. Analogous relations hold for the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$. Their normalization is conveniently written as

$$\frac{1}{4\pi}\int (\mathbf{E}_{\mathbf{k}\alpha}\mathbf{E}^{*}_{\mathbf{k}'\alpha'} + \mathbf{H}_{\mathbf{k}\alpha}\mathbf{H}^{*}_{\mathbf{k}'\alpha'})\,d^3x = \omega\,\delta_{\mathbf{k}\mathbf{k}'}\,\delta_{\alpha\alpha'}. \tag{2.22}$$

Substituting the operators (2.19) into (2.10) and integrating with the help of (2.22), we get the field Hamiltonian expressed through the operators $\hat{c}$, $\hat{c}^{+}$:

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2}\omega(\hat{c}_{\mathbf{k}\alpha}\hat{c}^{+}_{\mathbf{k}\alpha} + \hat{c}^{+}_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}). \tag{2.23}$$

In the representation under discussion (with the matrix elements of $\hat{c}$, $\hat{c}^{+}$ given by (2.15)) this operator is diagonal, and its eigenvalues coincide, naturally, with (2.12).

In classical theory the field momentum is defined by the integral

$$\mathbf{P} = \frac{1}{4\pi}\int \mathbf{E}\times\mathbf{H}\,d^3x.$$

On passing to the quantum theory by replacing $\mathbf{E}$ and $\mathbf{H}$ with the operators (2.19), one readily finds

$$\hat{\mathbf{P}} = \sum_{\mathbf{k}\alpha} \frac{1}{2}(\hat{P}^2_{\mathbf{k}\alpha} + \omega^2\hat{Q}^2_{\mathbf{k}\alpha})\mathbf{n} \tag{2.24}$$

— which mirrors the familiar classical link between energy and momentum of plane waves. This operator has eigenvalues

$$\mathbf{P} = \sum_{\mathbf{k}\alpha} \mathbf{k}\left(N_{\mathbf{k}\alpha} + \frac{1}{2}\right). \tag{2.25}$$

The operator representation realized by the matrix elements (2.15) is called the "occupation-number representation": it amounts to characterizing the state of the system (the field) through the quantum numbers $N_{\mathbf{k}\alpha}$ (*occupation numbers*). Within this representation the field operators (2.19) (and hence the Hamiltonian (2.11)) act on the system's wave function, written as a function of the numbers $N_{\mathbf{k}\alpha}$; we denote it $\Phi(N_{\mathbf{k}\alpha}, t)$. The field operators (2.19) do not depend on time explicitly. This corresponds to the Schrödinger representation of operators, familiar from non-relativistic quantum mechanics. What does depend on time is the state of the system, $\Phi(N_{\mathbf{k}\alpha}, t)$, its time dependence being governed by the Schrödinger equation

$$i\frac{\partial \Phi}{\partial t} = \hat{H}\Phi.$$

This description of the field is relativistically invariant in essence, since it rests on the invariant Maxwell equations. Yet the invariance is not exhibited explicitly — above all because the spatial coordinates and the time enter the description in a highly asymmetric way.

Within a relativistic framework it is desirable to give the description a more overtly invariant appearance. For this purpose one uses the so-called Heisenberg representation, wherein the explicit dependence on time is shifted onto the operators (see Vol. III, § 13). Time and spatial coordinates then appear symmetrically in the field-operator expressions, while the system state Φ depends only on the occupation numbers.

For the operator $\hat{\mathbf{A}}$ the passage to the Heisenberg representation amounts to replacing, in each term of the sum in (2.17), (2.18), the factor $e^{i\mathbf{kr}}$ by $e^{i(\mathbf{kr}-\omega t)}$; that is, one is to understand by $\mathbf{A}_{\mathbf{k}\alpha}$ the time-dependent functions

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,e^{-i(\omega t-\mathbf{kr})}. \tag{2.26}$$

This is easily verified by noting that the matrix element of a Heisenberg operator for the transition $i \to f$ must contain the factor $\exp\{-i(E_i - E_f)t\}$, where E_i and E_f are the energies of the initial and final states (see Vol. III, § 13). For a transition in which $N_{\mathbf{k}}$ decreases or increases by one, this factor reduces to $e^{-i\omega t}$ or $e^{i\omega t}$, respectively. The indicated replacement satisfies this requirement.

Henceforth (in treating both the electromagnetic field and the fields of particles) we shall always understand the Heisenberg representation of operators.

§ 3. Photons

Let us now turn to a discussion of the quantization formulae derived above.

To begin with, formula (2.12) for the energy of the field exposes the following difficulty. At the lowest energy level, all oscillator quantum numbers $N_{\mathbf{k}\alpha}$ vanish (this state is known as the *electromagnetic field vacuum*). Yet even in this state every oscillator retains a nonzero "zero-point energy" $\omega/2$. Summing over the entire infinite set of oscillators then produces an infinite result. One is thus confronted with one of the "divergences" stemming from the absence of full logical self-consistency in the existing theory.

As long as one is concerned only with the eigenvalues of the field energy, this difficulty can be eliminated by simply subtracting away the zero-point oscillation energy, i.e. by writing for the energy and momentum of the field[2]

$$E = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha}\omega, \qquad \mathbf{P} = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha}\mathbf{k}. \tag{3.1}$$

These formulae make it possible to introduce the concept fundamental to all of quantum electrodynamics: that of *light quanta*, or *photons*[3]. Namely, the free electromagnetic field can be regarded as a collection of particles, each carrying energy $\omega\ (= \hbar\omega)$ and momentum $\mathbf{k}\ (= \mathbf{n}\hbar\omega/c)$. The relationship between a photon's energy and momentum is precisely what relativistic mechanics requires for particles of zero rest mass that propagate at the speed of light. The occupation numbers $N_{\mathbf{k}\alpha}$ acquire the meaning of the number of photons with given momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. The polarization property of a photon is analogous to the notion of spin for other particles (the specific features of the photon in this regard will be examined below, in § 6).

It is easy to see that the mathematical formalism developed in the preceding section is fully consistent with the picture of the electromagnetic field as an assembly of photons; it is nothing other than the apparatus of so-called second quantization as applied to a system of photons[4]. In this method (see Vol. III, § 64) the occupation numbers serve as the independent variables, and the operators act on functions of those numbers. The central role is played by the "annihilation" and "creation" operators for particles, which respectively decrease or increase an occupation number by one. Exactly such operators are $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$: the operator $\hat{c}_{\mathbf{k}\alpha}$ destroys a photon in the state $\mathbf{k}\alpha$, while $\hat{c}^+_{\mathbf{k}\alpha}$ creates one in that state.

The commutation rule (2.16) corresponds to particles obeying Bose statistics. Photons are therefore bosons, as one might have anticipated: the number of photons occupying any given state can be arbitrary (we shall return to the significance of this fact in § 5).

The plane waves $\mathbf{A}_{\mathbf{k}\alpha}$ (2.26), which appear in the operator $\hat{\mathbf{A}}$ (2.17) as coefficients multiplying the photon annihilation operators, can be interpreted as the wave functions of photons possessing definite momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. This interpretation corresponds to expanding the ψ-operator as a series in the stationary-state wave functions of the particle within the non-relativistic second-quantization framework (unlike the

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[2]This subtraction can be carried out in a formally consistent manner by agreeing to interpret the operator products in (2.10) as "normal-ordered", i.e. arranged so that the operators $\hat{c}^+$ always stand to the left of the operators $\hat{c}$. Formula (2.23) then takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha}\left(\omega\, \hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}\right).$$

[3]The photon concept was first put forward by *Einstein* (A. Einstein, 1905).

[4]The method of second quantization as applied to radiation theory was developed by *Dirac* (P. A. M. Dirac, 1927).

non-relativistic case, however, the expansion (2.17) involves both annihilation and creation operators; the significance of this distinction will become clear later, see § 12).

The wave function (2.26) is normalized by the condition

$$\int \frac{1}{4\pi}\left(|\mathbf{E}_{\mathbf{k}\alpha}|^2 + |\mathbf{H}_{\mathbf{k}\alpha}|^2\right) d^3x = \omega \tag{3.2}$$

This amounts to normalizing to one photon in a volume $V = 1$. Indeed, the integral on the left-hand side of the equation represents the quantum-mechanical expectation value of the photon's energy in a state described by the given wave function[5]. On the right-hand side of (3.2) stands the energy of a single photon.

The role of the "Schrödinger equation" for the photon is played by Maxwell's equations. In the present case (for the potential $\mathbf{A}(\mathbf{r}, t)$ satisfying condition (2.1)) this is the wave equation

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta\mathbf{A} = 0.$$

In the general case of arbitrary stationary states, the photon "wave functions" are complex solutions of this equation whose time dependence enters through the factor $e^{-i\omega t}$.

Speaking of the photon wave function, let us emphasize once more that it cannot by any means be treated as a probability amplitude for the spatial localization of a photon — in contrast to the basic meaning of the wave function in non-relativistic quantum mechanics. This is linked to the fact that (as noted in § 1) the notion of photon coordinates has no physical meaning at all. We shall revisit the mathematical side of this issue at the end of the next section.

The Fourier-transform components of $\mathbf{A}(\mathbf{r}, t)$ with respect to the coordinates constitute the photon wave function in the momentum representation; we denote it $\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k})e^{-i\omega t}$. Thus, for a state with a definite momentum $\mathbf{k}$ and polarization $\mathbf{e}^{(\alpha)}$ the momentum-representation wave function is given simply by the coefficient of the exponential factor in (2.26):

$$\mathbf{A}_{\mathbf{k}\alpha}(\mathbf{k}', \alpha') = \sqrt{4\pi}\,\frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}}\,\delta_{\mathbf{k}'\mathbf{k}}\delta_{\alpha\alpha'}. \tag{3.3}$$

In keeping with the measurability of the momentum of a free particle, the momentum-representation wave function carries deeper physical content than its coordinate-space counterpart: it provides a way to compute the probabilities $w_{\mathbf{k}\alpha}$ of the various values of momentum and polarization for a photon in a given state. According to the general rules of quantum mechanics, $w_{\mathbf{k}\alpha}$ is given by the squared modulus of the expansion coefficients of $\mathbf{A}(\mathbf{k}')$ in the wave functions of states with definite $\mathbf{k}$ and $\mathbf{e}^{(\alpha)}$:

$$w_{\mathbf{k}\alpha} \propto \left|\sum_{\mathbf{k}'\alpha'} \mathbf{A}^*_{\mathbf{k}\alpha}(\mathbf{k}', \alpha')\mathbf{A}(\mathbf{k}')\right|^2$$

[5] Note that the coefficient $1/(4\pi)$ in the integral (3.2) is twice the usual coefficient $1/(8\pi)$ in (2.10). This difference is ultimately connected with the complex character of the vectors $\mathbf{E}_{\mathbf{k}\alpha}$, $\mathbf{H}_{\mathbf{k}\alpha}$, as opposed to the Hermitian field operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$.

(the proportionality coefficient depends on the choice of normalization for the functions). Substituting (3.3), we obtain

$$w_{\mathbf{k}\alpha} \propto |\mathbf{e}^{(\alpha)}\mathbf{A}(\mathbf{k})|^2. \tag{3.4}$$

Summing over the two polarizations, we find the probability that a photon has momentum $\mathbf{k}$:

$$w_{\mathbf{k}} \propto |\mathbf{A}(\mathbf{k})|^2. \tag{3.5}$$

§ 4. Gauge invariance

As is well known, the choice of field potentials in classical electrodynamics is not unique: the components of the 4-potential A_μ may be subjected to an arbitrary *gauge* (or *gradient*) transformation of the form

$$A_\mu \to A_\mu + \partial_\mu \chi, \tag{4.1}$$

where χ is an arbitrary function of the coordinates and time (see Vol. II, § 18). In the case of a plane wave, restricting oneself to transformations that preserve the functional form of the potential (its proportionality to $\exp(-ik_\mu x^\mu)$), the non-uniqueness amounts to the freedom of adding to the wave amplitude an arbitrary 4-vector proportional to k^μ.

The ambiguity of the potential persists, of course, in the quantum theory as well — as applied to the field operators or to the photon wave functions. Without prejudging the choice of potentials, one should write in place of (2.17) an analogous expansion for the operator 4-potential

$$\hat{A}^\mu = \sum_{\mathbf{k}\alpha} \left(\hat{c}_{\mathbf{k}\alpha} A^\mu_{\mathbf{k}\alpha} + \hat{c}^+_{\mathbf{k}\alpha} A^{\mu *}_{\mathbf{k}\alpha} \right), \tag{4.2}$$

where the wave functions $A^\mu_{\mathbf{k}\alpha}$ are 4-vectors of the form

$$A^\mu_{\mathbf{k}} = \sqrt{4\pi}\, \frac{e^\mu}{\sqrt{2\omega}}\, e^{-ik_\nu x^\nu}, \qquad e_\mu e^{\mu *} = -1,$$

or, in a compact notation that suppresses the four-dimensional vector indices:

$$A_{\mathbf{k}} = \sqrt{4\pi}\, \frac{e}{\sqrt{2\omega}}\, e^{-ikx}, \qquad ee^* = -1. \tag{4.3}$$

Here the 4-momentum is $k_\mu = (\omega, \mathbf{k})$ (so that $kx = \omega t - \mathbf{kr}$), and e is the unit polarization 4-vector[6].

If one restricts oneself to gauge transformations that do not alter how the function (4.3) depends on the coordinates and time, they amount to the replacement

$$e_\mu \to e_\mu + \chi k_\mu, \tag{4.4}$$

[6]Expression (4.3) does not have an entirely relativistically covariant (4-vector) form, which is related to the non-invariant character of our normalization to a finite volume $V = 1$. This is, however, of no fundamental significance and is fully compensated by the convenience of such a normalization scheme. We shall see later that it ensures simple and automatic derivation of real physical quantities in the proper invariant form.

where $\chi = \chi(k^\mu)$ is an arbitrary function. The transversality of the polarization means that one can always choose a gauge in which the 4-vector e takes the form

$$e^\mu = (0, \mathbf{e}), \qquad \mathbf{ek} = 0 \tag{4.5}$$

(we shall call such a gauge *three-dimensionally transverse*). In an invariant four-dimensional notation this requirement is written as the condition of *four-dimensional transversality*

$$ek = 0. \tag{4.6}$$

Note that this condition (as well as the normalization condition $ee^* = -1$) is not violated by the transformation (4.4), by virtue of $k^2 = 0$. On the other hand, the vanishing of the 4-momentum squared of a particle signifies that its mass is zero. A direct link between gauge invariance and the vanishing of the photon mass is thereby revealed (further aspects of this connection will be discussed in § 14).

No measurable physical quantity should change under a gauge transformation of the wave functions of the photons taking part in a given process. This requirement of *gauge invariance* plays an even greater role in quantum electrodynamics than in the classical theory. As we shall see from numerous examples, it serves here, alongside the requirement of relativistic invariance, as a powerful heuristic principle.

In turn, the gauge invariance of the theory is closely bound up with the law of conservation of electric charge; we shall dwell on this aspect in § 43.

We already remarked in the preceding section that the coordinate-space wave function of a photon cannot be interpreted as a probability amplitude for its spatial localization. From a mathematical standpoint this circumstance manifests itself in the impossibility of constructing, out of the wave function, a quantity whose formal properties alone would qualify it to serve as a probability density. Such a quantity would have to be an essentially positive bilinear combination of the wave function A_μ and its complex conjugate. Moreover, it would need to satisfy definite requirements of relativistic covariance — namely, to be the time component of a 4-vector (the point being that the continuity equation expressing conservation of particle number takes the four-dimensional form of a vanishing 4-divergence of the current 4-vector; the time component of this current is, in the present context, the probability density for the localization of the particle; see Vol. II, § 29). On the other hand, gauge invariance demands that A_μ enter the current only through the antisymmetric tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -i(k_\mu A_\nu - k_\nu A_\mu)$. The current 4-vector would therefore have to be assembled bilinearly from $F_{\mu\nu}$ and $F^*_{\mu\nu}$ (together with components of the 4-vector k_μ). But no such 4-vector can be constructed at all: every expression satisfying the stated conditions (for instance, $k^\lambda F^*_{\mu\nu} F_\lambda{}^\nu$) vanishes by the transversality condition ($k^\lambda F_{\nu\lambda} = 0$), quite apart from the fact that it would not be essentially positive (since it contains odd powers of the components k_μ).

§ 5. The electromagnetic field in quantum theory

Treating the field as a collection of photons is the sole description that fully captures the physical meaning of the electromagnetic field within quantum theory. It replaces the classical characterization through field strengths. The field strengths themselves enter the mathematical apparatus of the photon picture as operators of second quantization.

It is well known that a quantum system behaves classically when the quantum numbers characterizing its stationary states become large. For the free electromagnetic field (within a given volume) this requires that the oscillator quantum numbers — that is, the photon occupation numbers $N_{\mathbf{k}\alpha}$ — be large. The fact that photons obey Bose statistics is of profound importance in this context. In the mathematical formalism, the link between Bose statistics and classical field behavior is encoded in the commutation rules for $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}^+_{\mathbf{k}\alpha}$. When $N_{\mathbf{k}\alpha}$ is large, so that the matrix elements of these operators are also large, the unity appearing on the right-hand side of the commutation relation (2.16) may be dropped, giving

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}^+_{\mathbf{k}\alpha} \simeq \hat{c}^+_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha},$$

i.e. the operators go over into mutually commuting classical quantities $c_{\mathbf{k}\alpha}$, $c^*_{\mathbf{k}\alpha}$ that determine the classical field strengths.

The quasi-classicality condition for the field requires, however, a further refinement. The point is that if all the numbers $N_{\mathbf{k}\alpha}$ are large, then upon summing over every state $\mathbf{k}\alpha$ the field energy is in any case infinite, rendering the condition meaningless.

A physically sensible formulation of the quasi-classicality criterion rests on examining values of the field averaged over some small time interval Δt. If one writes the classical electric field $\bar{\mathbf{E}}$ (or the magnetic field $\bar{\mathbf{H}}$) as a Fourier integral over time, then upon time-averaging over an interval Δt only those Fourier components whose frequencies satisfy $\omega\Delta t \lesssim 1$ contribute appreciably to the mean value of $\bar{\mathbf{E}}$; in the opposite limit the oscillating factor $e^{-i\omega t}$ nearly averages to zero. Consequently, in establishing the quasi-classicality condition for the time-averaged field, one need consider only those quantum oscillators whose frequencies satisfy $\omega \lesssim 1/\Delta t$. It suffices to require that the quantum numbers of these oscillators be large. The number of oscillators with frequencies between zero and $\omega \sim 1/\Delta t$ (referred to the volume $V = 1$) is of order[7]

$$\left(\frac{\omega}{c}\right)^3 \sim \frac{1}{(c\Delta t)^3}. \tag{5.1}$$

The total energy of the field per unit volume is of order $\bar{\mathbf{E}}^2$. Dividing this quantity by the number of oscillators and by a characteristic energy of one photon ($\sim \hbar\omega$), one obtains an estimate for the photon occupation numbers

$$N_{\mathbf{k}} \sim \frac{\bar{\mathbf{E}}^2 c^3}{\hbar\omega^4}.$$

Demanding that this number be large, we arrive at the inequality

$$|\mathbf{E}| \gg \frac{\sqrt{\hbar c}}{(c\Delta t)^2}. \tag{5.2}$$

This is the sought-after condition permitting a classical treatment of the time-averaged (over intervals Δt) field. We see that the field must be sufficiently strong — all the stronger, the shorter the averaging interval Δt. For time-varying fields this interval must not, of course, exceed the characteristic periods over which the field changes appreciably.

[7] In this section we use ordinary units.

Consequently, sufficiently weak time-varying fields can under no circumstances be quasi-classical. Only for static (time-independent) fields may one set $\Delta t \to \infty$, so that the right-hand side of the inequality (5.2) vanishes. This means that a static field is always classical.

It has already been noted that the classical expressions for the electromagnetic field in the form of a superposition of plane waves must be treated as operator-valued in the quantum theory. The physical meaning of these operators is, however, rather limited. Indeed, a physically meaningful field operator ought to yield vanishing field values in the photon-vacuum state. Yet the expectation value of the squared-field operator $\hat{\mathbf{E}}^2$ in the ground state, which coincides up to a factor with the zero-point energy of the field, turns out to be infinite (by "expectation value" we mean the quantum-mechanical average, i.e. the corresponding diagonal matrix element of the operator). This divergence cannot be circumvented even by any formal subtraction procedure (as can be done for the field energy), since in the present case we would need to carry out the subtraction through some sensible modification of the operators $\hat{\mathbf{E}}$, $\hat{\mathbf{H}}$ themselves (rather than their squares), which proves to be impossible.

§ 6. Angular momentum and parity of the photon

As with any particle, the photon can possess a definite angular momentum. In order to elucidate the properties of this quantity for the photon, let us begin by recalling how the angular momentum of a particle is connected with its wave function in the formalism of quantum mechanics.

The total angular momentum $\mathbf{j}$ of a particle consists of the orbital part $\mathbf{l}$ and the intrinsic part — the spin $\mathbf{s}$. A particle of spin s is described by a symmetric spinor of rank $2s$, i.e. a set of $2s + 1$ components that transform into one another under coordinate-system rotations according to a specific law. The orbital angular momentum is instead linked to how the wave functions depend on coordinates: states with orbital angular momentum l have wave-function components that are expressed (linearly) in terms of spherical harmonics of order l.

The ability to distinguish consistently between spin and orbital angular momentum therefore demands that the "spin" and "coordinate" properties of the wave functions be independent: the coordinate dependence of each spinor component (at a given instant of time) should not be constrained by any supplementary conditions. Passing to the momentum representation, coordinate dependence becomes dependence on $\mathbf{k}$. In the three-dimensionally transverse gauge the photon wave function is the vector $\mathbf{A}(\mathbf{k})$. Because a vector is equivalent to a rank-2 spinor, one could formally ascribe spin 1 to the photon. This vector wave function, however, must satisfy the transversality requirement $\mathbf{k}\mathbf{A}(\mathbf{k}) = 0$, an extra constraint on the way $\mathbf{A}(\mathbf{k})$ depends on momentum. It follows that the dependence on $\mathbf{k}$ can no longer be chosen freely for every vector component at once, making it impossible to disentangle orbital angular momentum and spin.

We note also that for the photon one cannot define spin as the angular momentum of the particle at rest, since for a photon traveling at the speed of light no rest frame exists at all.

Thus, for a photon one can speak only of its total angular momentum. It is clear from the outset that the total angular momentum can assume only integer values. This is already evident from the fact that among the quantities characterizing the photon there are no spinors of odd rank.

As for any particle, the state of a photon is further characterized by its parity, associated with the behavior of the wave function under spatial inversion (see Vol. III, § 30). In the momentum representation, a reversal of the sign of the coordinates corresponds to reversing the sign of every component of $\mathbf{k}$. The action of the inversion operator $\hat{P}$ on a scalar function $\varphi(\mathbf{k})$ amounts simply to this sign change: $\hat{P}\varphi(\mathbf{k}) = \varphi(-\mathbf{k})$. When acting on the vector function $\mathbf{A}(\mathbf{k})$, one must additionally take into account that reversing the axes also reverses the sign of every component of the vector; hence[8]

$$\hat{P}\mathbf{A}(\mathbf{k}) = -\mathbf{A}(-\mathbf{k}). \tag{6.1}$$

Although the decomposition of the photon's angular momentum into orbital and spin parts lacks physical meaning, it is nevertheless convenient to introduce "spin" s and "orbital" angular momentum l in a formal sense as auxiliary concepts describing the transformation properties of the wave function with respect to rotations: the value $s = 1$ reflects the vectorial character of the wave function, while l is the order of the spherical harmonics entering it. Here we have in mind the wave functions of states with definite values of the photon's angular momentum, which for a free particle take the form of spherical waves. The number l determines, in particular, the parity of the photon state, equal to

$$P = (-1)^{l+1} \tag{6.2}$$

In the same formal sense one can write the angular-momentum operator $\hat{\mathbf{j}}$ as the sum $\hat{\mathbf{s}} + \hat{\mathbf{l}}$. As is well known, the angular-momentum operator is associated with infinitesimally small rotations of the coordinate system; here, with the action of such a rotation on a vector field. Within the decomposition $\hat{\mathbf{s}} + \hat{\mathbf{l}}$, the operator $\hat{\mathbf{s}}$ acts on the vector index and intermixes the various components of the vector. The operator $\hat{\mathbf{l}}$ acts on these same components viewed as functions of momentum (or of coordinates). Let us count the number of states (at a given energy) that are possible for a given value j of the photon's angular momentum (setting aside the trivial $(2j + 1)$-fold degeneracy with respect to the direction of the angular momentum).

For independent $\mathbf{l}$ and $\mathbf{s}$ this counting is carried out simply by enumerating the ways in which, according to the rules of the vector model, the angular momenta $\hat{\mathbf{l}}$ and $\mathbf{s}$ can be composed so as to yield the required value of j. For a particle with spin $s = 1$ we would find in this way (for a given nonzero value of j) three states with the following values of l and parity:

$$l = j,\ P = (-1)^{l+1} = (-1)^{j+1}; \quad l = j \pm 1,\ P = (-1)^{l+1} = (-1)^{j}.$$

[8]We adopt the convention of defining the parity of a state by the action of the inversion operator on the polar vector $\mathbf{A}$ (or, equivalently, the corresponding electric vector $\mathbf{E} = i\omega\mathbf{A}$). This differs in sign from the action on the axial vector $\mathbf{H} = i[\mathbf{kA}]$, since inversion does not reverse the direction of such a vector:

$$\hat{P}\mathbf{H}(\mathbf{k}) = \mathbf{H}(-\mathbf{k}).$$

If $j = 0$, there is only a single state (with $l = 1$) having parity $P = +1$.

This counting, however, has not accounted for the transversality condition on $\mathbf{A}$; all three components were regarded as independent. From the result obtained one must therefore subtract the number of states associated with a longitudinal vector. A longitudinal vector can be expressed as $\mathbf{k}\varphi(\mathbf{k})$, and it is apparent that, owing to its transformation behavior (with respect to rotations), its three components reduce to a single scalar φ [9]. The extra state that is excluded by transversality would therefore belong to a particle whose wave function is a scalar (a spinor of rank zero), that is, a particle with "spin 0" [10]. For this state the angular momentum j equals the order of the spherical harmonics appearing in φ. Its parity, viewed as a photon state, follows from how the inversion operator acts on the vector function $\mathbf{k}\varphi$:

$$\hat{P}(\mathbf{k}\varphi) = -(-\mathbf{k})\varphi(-\mathbf{k}) = (-1)^j \mathbf{k}\varphi(\mathbf{k}),$$

i.e. it equals $(-1)^j$. Thus, from the above count of states with parity $(-1)^j$ (two for $j \neq 0$ and one for $j = 0$) one must subtract one.

We arrive finally at the result that for nonzero photon angular momentum j there exist one even-parity and one odd-parity state. For $j = 0$ we obtain no states whatsoever. This means that the photon can never have zero angular momentum, so that j takes only the values $1, 2, 3, \ldots$ The impossibility of $j = 0$ is, moreover, obvious: the wave function of a state with vanishing angular momentum would have to be spherically symmetric, which is manifestly impossible for a transverse wave.

There is a standard nomenclature for the various photon states. When a photon carries angular momentum j with parity $(-1)^j$ it is called an *electric* 2^j*-pole* (or Ej) photon; when its parity is $(-1)^{j+1}$ it is called a *magnetic* 2^j*-pole* (or Mj) photon. In particular, the electric dipole photon has odd parity and $j = 1$; the electric quadrupole photon has even parity and $j = 2$; the magnetic dipole photon has even parity and $j = 1$ [11].

§ 7. Spherical waves of photons

Having established the admissible values of the photon's angular momentum, our task is now to find the wave functions that correspond to them [12].

Let us first consider a formal problem: to identify the vector functions that serve as eigenfunctions of the operators $\hat{\mathbf{j}}^2$ and $\hat{j}_z$; in doing so we do not stipulate in advance which particular functions enter the photon wave functions of interest, nor do we impose the transversality constraint.

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[9] Indeed, when one speaks of the transformation character of a quantity under rotation, one means a transformation at a given point, i.e. at a given $\mathbf{k}$. Under such a transformation $\mathbf{k}\varphi(\mathbf{k})$ does not change at all, i.e. it behaves as a scalar.

[10] Let us emphasize once more that what is meant here is not the state of any real particle. The counting carried out here is purely formal and amounts, mathematically, to sorting the complete collection of quantities that mix under rotations into irreducible representations of the rotation group.

[11] This nomenclature mirrors the usage in the classical theory of radiation: as will be shown later (§§ 46–47), the production of electric- and magnetic-type photons is controlled by the respective electric and magnetic multipole moments of the system of charges.

[12] This problem was first studied by Heitler (W. Heitler, 1936). The form of the solution presented here is due to V. B. Berestetskii (1947).

The functions will be sought in the momentum representation, where the coordinate operator takes the form $\hat{\mathbf{r}} = i\partial/\partial\mathbf{k}$ (see III, (15.12)). For the orbital angular momentum operator one then obtains

$$\hat{\mathbf{l}} = [\hat{\mathbf{r}}\mathbf{k}] = -i\left[\mathbf{k}\frac{\partial}{\partial\mathbf{k}}\right],$$

i.e. it differs from the angular momentum operator in coordinate space only by the replacement of $\mathbf{r}$ with $\mathbf{k}$. Accordingly, the formal solution of the problem is identical in both representations.

Let us denote the desired eigenfunctions by $\mathbf{Y}_{jm}$ and call them *spherical vectors*. They must satisfy the equations

$$\hat{\mathbf{j}}^2 = j(j+1)\mathbf{Y}_{jm}, \qquad \hat{j}_z\mathbf{Y}_{jm} = m\mathbf{Y}_{jm} \tag{7.1}$$

(the z axis being a fixed direction in space). We shall demonstrate that any function of the form $\mathbf{a}Y_{jm}$ possesses this property, where $\mathbf{a}$ is a vector constructed from the unit vector $\mathbf{n} = \mathbf{k}/\omega$ and Y_{jm} are the ordinary (scalar) spherical harmonics. The latter will everywhere be defined in accordance with III, § 28:

$$Y_{lm}(\mathbf{n}) = (-1)^{\frac{m+|m|}{2}} i^l \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}\, P_l^{|m|}(\cos\theta)\, e^{im\varphi} \tag{7.2}$$

(θ, φ being the spherical angles specifying the direction of $\mathbf{n}$) [13].

To show this, recall the commutation rule III, (29.4):

$$\{\hat{l}_i, a_k\}_- = ie_{ikl}a_l.$$

The right-hand side can be written as $(-\hat{s}_i a_k)$, where $\hat{\mathbf{s}}$ is the spin-1 operator (its action on a vector function is precisely $\hat{s}_i a_k = -ie_{ikl}a_l$; see III, § 57, problem 2). Consequently,

$$\hat{l}_i a_k - a_k \hat{l}_i = -\hat{s}_i a_k.$$

Making use of this relation, we find

$$\hat{j}_i a_k = (\hat{l}_i + \hat{s}_i)a_k = a_k \hat{l}_i.$$

It follows that

$$\hat{\mathbf{j}}^2(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}^2 Y_{jm}, \qquad \hat{j}_z(\mathbf{a}Y_{jm}) = \mathbf{a}\hat{l}_z Y_{jm}.$$

Since the spherical harmonic Y_{jm} is an eigenfunction of $\hat{l}^2$ and $\hat{l}_z$ with eigenvalues $j(j+1)$ and m respectively, we arrive at equations (7.1).

[13] For future reference, note the value of these functions at $\theta = 0$ ($\mathbf{n}$ along the z axis):

$$Y_{lm}(\mathbf{n}_z) = i^l\sqrt{\frac{2l+1}{4\pi}}\,\delta_{m0}. \tag{7.2a}$$

Three essentially distinct types of spherical vectors are obtained by choosing the vector **a** to be one of the following [14]:

$$\frac{\nabla_{\mathbf{n}}}{\sqrt{j(j+1)}}, \qquad \frac{[\mathbf{n}\nabla_{\mathbf{n}}]}{\sqrt{j(j+1)}}, \qquad \mathbf{n}. \tag{7.3}$$

The spherical vectors are thus defined as follows:

$$\begin{aligned} \mathbf{Y}^{(\mathrm{e})}_{jm} &= \frac{1}{\sqrt{j(j+1)}}\,\nabla_{\mathbf{n}} Y_{jm}, & P &= (-1)^j; \\ \mathbf{Y}^{(\mathrm{m})}_{jm} &= [\mathbf{n}\mathbf{Y}^{(\mathrm{e})}_{jm}], & P &= (-1)^{j+1}; \\ \mathbf{Y}^{(\mathrm{l})}_{jm} &= \mathbf{n} Y_{jm}, & P &= (-1)^j. \end{aligned} \tag{7.4}$$

Next to each vector we have indicated its parity P. The spherical vectors of the three types are mutually orthogonal; $\mathbf{Y}^{(\mathrm{l})}_{jm}$ is longitudinal, while $\mathbf{Y}^{(\mathrm{e})}_{jm}$ and $\mathbf{Y}^{(\mathrm{m})}_{jm}$ are transverse with respect to **n**.

The spherical vectors can be expressed through scalar spherical harmonics. In this expansion, $\mathbf{Y}^{(\mathrm{m})}_{jm}$ involves harmonics of a single order $l = j$ only, whereas $\mathbf{Y}^{(\mathrm{e})}_{jm}$ and $\mathbf{Y}^{(\mathrm{l})}_{jm}$ involve harmonics of orders $l = j \pm 1$. This is self-evident: one need only compare the parities listed in (7.4) with the parity $(-1)^{l+1}$ of a vector field, expressed via the order of the spherical harmonics it contains.

The spherical vectors within each type are mutually orthogonal and normalized according to

$$\int \mathbf{Y}_{jm}\mathbf{Y}^*_{j'm'}\,do = \delta_{jj'}\delta_{mm'}. \tag{7.5}$$

For the vectors $\mathbf{Y}^{(\mathrm{l})}_{jm}$ this follows immediately from the normalization of the spherical harmonics Y_{jm}. For $\mathbf{Y}^{(\mathrm{e})}_{jm}$ the normalization integral becomes

$$\frac{1}{j(j+1)}\int \nabla_{\mathbf{n}} Y_{jm}\,\nabla_{\mathbf{n}} Y^*_{j'm'}\,do = -\frac{1}{j(j+1)}\int Y^*_{j'm'}\,\Delta_{\mathbf{n}} Y_{jm}\,do,$$

and since $\Delta_{\mathbf{n}} Y_{jm} = -j(j+1)Y_{jm}$, we recover (7.5). The normalization of the vectors $\mathbf{Y}^{(\mathrm{m})}_{jm}$ reduces to an identical integral.

We remark that the spherical vectors (7.4) could also have been obtained without the direct verification of equations (7.1) carried out above — simply on the basis of general transformation-property arguments. Such reasoning led us in the preceding section to conclude that a vector function of the form $\mathbf{n}\varphi$ corresponds to a value of j equal to

[14] The operator $\nabla_{\mathbf{n}} \equiv |\mathbf{k}|\nabla_{\mathbf{k}}$ acts on functions that depend only on the direction of **n**. In spherical coordinates it has just two components:

$$\nabla_{\mathbf{n}} = \left(\frac{\partial}{\partial\theta}, \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi}\right).$$

The operator denoted below by $\Delta_{\mathbf{n}}$ is the angular part of the Laplacian:

$$\Delta_{\mathbf{n}} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}.$$

the order of the spherical harmonics appearing in φ; setting $\varphi = Y_{jm}$ further fixes the projection m. In this way one immediately arrives at the spherical vectors $\mathbf{Y}^{(1)}_{jm}$. The transformation-property arguments of § 6, however, remain valid if the factor $\mathbf{n}$ in the product $\mathbf{n}\varphi$ is replaced by $\boldsymbol{\nabla}_{\mathbf{n}}$ or by $[\mathbf{n}\boldsymbol{\nabla}_{\mathbf{n}}]$, yielding in this manner the spherical vectors of the other two types.

Let us return to the photon wave functions. For an electric-type photon (Ej) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^j$. Among the spherical vectors, $\mathbf{Y}^{(e)}_{jm}$ and $\mathbf{Y}^{(1)}_{jm}$ possess this parity; of the two, however, only the former satisfies the transversality condition. For a magnetic-type photon (Mj) the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^{j+1}$; only the spherical vector $\mathbf{Y}^{(m)}_{jm}$ has this parity. The wave functions of a photon with definite angular momentum j, projection m, and energy ω are therefore

$$\mathbf{A}_{\omega jm}(\mathbf{k}) = \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}| - \omega)\mathbf{Y}_{jm}(\mathbf{n}), \tag{7.6}$$

where one should write $\mathbf{Y}^{(e)}_{jm}$ or $\mathbf{Y}^{(m)}_{jm}$ for $\mathbf{Y}_{jm}$ according as the photon is of electric or magnetic type; the specified energy is incorporated through the factor $\delta(|\mathbf{k}| - \omega)$. The functions (7.6) are normalized by the condition

$$\frac{1}{(2\pi)^4}\int \omega\omega'\mathbf{A}^*_{\omega' j'm'}(\mathbf{k})\mathbf{A}_{\omega jm}(\mathbf{k})\,d^3k = \omega\delta(\omega' - \omega)\delta_{jj'}\delta_{mm'}. \tag{7.7}$$

For the coordinate-representation wave functions, condition (7.7) is equivalent to [15]

$$\frac{1}{4\pi}\int \left\{\mathbf{E}^*_{\omega' j'm'}(\mathbf{r})\mathbf{E}_{\omega jm}(\mathbf{r}) + \mathbf{H}^*_{\omega' j'm'}(\mathbf{r})\mathbf{H}_{\omega jm}(\mathbf{r})\right\} d^3x = \omega\delta(\omega' - \omega)\delta_{jj'}\delta_{mm'}. \tag{7.8}$$

Indeed, when written in terms of potentials, the integral on the left-hand side takes the form

$$\frac{1}{2\pi}\int \mathbf{A}^*_{\omega' j'm'}(\mathbf{r})\mathbf{A}_{\omega jm}(\mathbf{r})\,\omega'\omega\,d^3x.$$

One must substitute here the Fourier expansions

$$\begin{aligned}\mathbf{A}_{\omega jm}(\mathbf{r}) &= \int \mathbf{A}_{\omega jm}(\mathbf{k})\,e^{i\mathbf{kr}}\,\frac{d^3k}{(2\pi)^3},\\ \mathbf{A}^*_{\omega' j'm'}(\mathbf{r}) &= \int \mathbf{A}^*_{\omega' j'm'}(\mathbf{k}')\,e^{-i\mathbf{k}'\mathbf{r}}\,\frac{d^3k'}{(2\pi)^3}.\end{aligned} \tag{7.9}$$

After that, integration over d^3x produces the delta function $(2\pi)^3\delta(\mathbf{k}' - \mathbf{k})$, which is removed by the integration over d^3k', bringing the integral to the form (7.7).

Up to this point we have assumed the transverse gauge for the potentials, in which the scalar potential $\Phi = 0$. In various applications, however, other gauge choices for a spherical wave may prove more convenient. An admissible gauge transformation of the potentials in the momentum representation consists of the substitution

$$\mathbf{A} \to \mathbf{A} + \mathbf{n}f(\mathbf{k}), \qquad \Phi \to \Phi + f(\mathbf{k}),$$

[15] This condition is of the same type as (2.22). The appearance of the factor $\delta(\omega' - \omega)$ on the right-hand side is due to the fact that the field (a spherical wave) is being considered in all of infinite space rather than in a finite volume $V = 1$.

with $f(\mathbf{k})$ an arbitrary function. We choose it here so that the new potentials are still expressed through the same spherical harmonics and continue to have definite parity. For an electric-type photon these requirements restrict the potentials to the following form:

$$\begin{aligned} \mathbf{A}^{(e)}_{\omega jm}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}|-\omega)\big(\mathbf{Y}^{(e)}_{jm} + C\mathbf{n}Y_{jm}\big), \\ \Phi^{(e)}_{\omega jm}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}}\,\delta(|\mathbf{k}|-\omega)\, CY_{jm}, \end{aligned} \tag{7.10}$$

where C is an arbitrary constant. For a magnetic-type photon, on the other hand, such an addition to $\mathbf{A}^{(m)}(\mathbf{k})$ would destroy its definite parity, so under the same conditions the choice (7.6) is uniquely fixed.

The probability that a photon of given angular momentum and parity is detected traveling in the direction $\mathbf{n}$ within the solid-angle element do equals, by (3.5) and (7.6),

$$w(\mathbf{n})do = \left|\mathbf{Y}^{(e)}_{jm}\right|^2 do. \tag{7.11}$$

We have written the expression for an E-type photon. Since, however, $\left|\mathbf{Y}^{(m)}_{jm}\right|^2 = \left|\mathbf{Y}^{(e)}_{jm}\right|^2$, the probability distributions $w(\mathbf{n})$ for photons of both types are identical.

The squared modulus $\left|\mathbf{Y}^{(e)}_{jm}\right|^2$ is independent of the azimuthal angle φ (the factors $e^{\pm im\varphi}$ in the spherical harmonics cancel). The probability distribution $w(\mathbf{n})$ is therefore axially symmetric about the z axis. Furthermore, since each spherical vector has definite parity, the squared moduli are even under inversion, i.e. under the replacement $\theta \to \pi - \theta$; this means that the function $w(\theta)$, when expanded in Legendre polynomials, contains only polynomials of even order. The determination of the expansion coefficients reduces to computing integrals of products of three spherical harmonics followed by summation over components. Both steps are carried out with the aid of formulas derived in III, §§ 107–108, and lead to the following result:

$$\begin{aligned} w(\theta) = (-1)^{m+1}\frac{(2j+1)^2}{4\pi}\sum_{n=0}^{\infty}(4n+1)\begin{pmatrix} j & j & 2n \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} j & j & 2n \\ m & -m & 0 \end{pmatrix}\times \\ \times\begin{Bmatrix} j & j & 1 \\ j & j & 2n \end{Bmatrix} P_{2n}(\cos\theta). \end{aligned} \tag{7.12}$$

Finally, we present expressions for the components of the spherical vectors as expansions in spherical harmonics. We employ the "spherical components" of a vector $\mathbf{f}$, defined in accordance with III, § 107; the components f_λ of the vector $\mathbf{f}$ are

$$f_0 = if_z, \quad f_{+1} = -\frac{i}{\sqrt{2}}(f_x + if_y), \quad f_{-1} = \frac{i}{\sqrt{2}}(f_x - if_y). \tag{7.13}$$

Introducing "circular basis vectors":

$$\mathbf{e}^{(0)} = i\mathbf{e}^{(z)}, \quad \mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}\big(\mathbf{e}^{(x)} + i\mathbf{e}^{(y)}\big), \quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}\big(\mathbf{e}^{(x)} - i\mathbf{e}^{(y)}\big) \tag{7.14}$$

($\mathbf{e}^{(x,y,z)}$ being the unit vectors along the x, y, z axes), we have

$$\mathbf{f} = \sum_\lambda (-1)^{1-\lambda} f_{-\lambda}\mathbf{e}^{(\lambda)}, \qquad f_\lambda = (-1)^{1-\lambda}\mathbf{f}\mathbf{e}^{(-\lambda)*} = \mathbf{f}\mathbf{e}^{(\lambda)}. \tag{7.15}$$

The spherical components of the spherical vectors are expressed through $3j$-symbols and spherical harmonics by the following formulas:

$$\begin{aligned}
(-1)^{j+m+\lambda+1}\left(\mathbf{Y}^{(e)}_{jm}\right)_\lambda &= -\sqrt{j}\begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m\end{pmatrix} Y_{j+1,m+\lambda} \\
&\quad + \sqrt{j+1}\begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m\end{pmatrix} Y_{j-1,m+\lambda}, \\
(-1)^{j+m+\lambda+1}\left(\mathbf{Y}^{(m)}_{jm}\right)_\lambda &= -\sqrt{2j+1}\begin{pmatrix} j & 1 & j \\ m+\lambda & -\lambda & -m\end{pmatrix} Y_{j,m+\lambda}, \\
(-1)^{j+m+\lambda+1}\left(\mathbf{Y}^{(l)}_{jm}\right)_\lambda &= \sqrt{j+1}\begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m\end{pmatrix} Y_{j+1,m+\lambda} \\
&\quad + \sqrt{j}\begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m\end{pmatrix} Y_{j-1,m+\lambda}.
\end{aligned} \tag{7.16}$$

These formulas are derived as follows. Each of the three spherical vectors has the form $\mathbf{Y}_{jm} = \mathbf{a}Y_{jm}$, where $\mathbf{a}$ is one of the three vectors (7.3). Therefore

$$\mathbf{Y}_{jm} = \sum_{lm'} \langle lm'|\mathbf{a}|jm\rangle Y_{lm'},$$

so that one must determine the matrix elements of the vectors $\mathbf{a}$ between eigenstates of the orbital angular momentum. From formula (107.6) of III it follows that

$$\langle lm'|a_\lambda|jm\rangle = i(-1)^{j_{\max}-m'}\begin{pmatrix} l & 1 & j \\ -m' & \lambda & m\end{pmatrix}\langle l\|a\|j\rangle,$$

where $j_{\max}$ is the larger of the numbers l and j. It therefore suffices to know the nonvanishing reduced matrix elements $\langle l\|a\|j\rangle$. The relevant formulas are:

$$\begin{aligned}
\langle l-1\|n\|l\rangle &= \langle l\|n\|l-1\rangle^* = i\sqrt{l}, \\
\langle l\|\nabla_\mathbf{n}\|l-1\rangle &= i(l-1)\sqrt{l}, \\
\langle l-1\|\nabla_\mathbf{n}\|l\rangle &= i(l+1)\sqrt{l}, \\
\langle l\|[\mathbf{n}\nabla_\mathbf{n}]\|l\rangle &= i\sqrt{l(l+1)(2l+1)}.
\end{aligned} \tag{7.17}$$

§ 8. Polarization of the photon

The polarization vector $\mathbf{e}$ plays for the photon the role of the "spin part" of the wave function (with the reservations expressed in § 6 regarding the notion of photon spin).

The various cases arising for the polarization of a photon are in no way distinct from the possible types of polarization of a classical electromagnetic wave (see II, § 48).

An arbitrary polarization $\mathbf{e}$ can be represented as a superposition of two mutually orthogonal polarizations $\mathbf{e}^{(1)}$ and $\mathbf{e}^{(2)}$ ($\mathbf{e}^{(1)}\mathbf{e}^{(2)*} = 0$), chosen in some definite manner. In the decomposition

$$\mathbf{e} = e_1\mathbf{e}^{(1)} + e_2\mathbf{e}^{(2)} \tag{8.1}$$

the squared moduli of the coefficients e_1 and e_2 give the probability that the photon has polarization $\mathbf{e}^{(1)}$ or $\mathbf{e}^{(2)}$.

As these two basis polarizations one may choose two mutually perpendicular linear polarizations. Alternatively, any polarization can be decomposed into two circular ones with opposite senses of rotation. We shall denote the right and left circular polarization vectors by $\mathbf{e}^{(+1)}$ and $\mathbf{e}^{(-1)}$ respectively; in a coordinate system $\xi\eta\zeta$ with the ζ axis along the photon's direction $\mathbf{n} = \mathbf{k}/\omega$,

$$\mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}\left(\mathbf{e}^{(\xi)} + i\mathbf{e}^{(\eta)}\right), \qquad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}\left(\mathbf{e}^{(\xi)} - i\mathbf{e}^{(\eta)}\right). \tag{8.2}$$

The existence of two distinct polarizations of the photon (at a given momentum) means, in other words, that every eigenvalue of the momentum is doubly degenerate. This circumstance is intimately connected with the fact that the photon mass equals zero.

For a freely moving particle of nonvanishing mass a rest frame always exists. It is clear that the intrinsic symmetry properties of the particle as such manifest themselves precisely in this frame. The symmetry to be considered is with respect to all possible rotations about the center, that is, with respect to the full spherical symmetry group. The quantity characterizing the symmetry properties of the particle under this group is its spin s, which determines the multiplicity of the degeneracy (the number $2s + 1$ of distinct wave functions that transform into one another). In particular, a particle whose wave function is a vector (three components) has spin 1.

For a particle of zero mass, however, no rest frame exists — in every reference frame it travels at the speed of light. For such a particle there is always a preferred spatial direction — the direction of the momentum vector $\mathbf{k}$ (the ζ axis). In this situation the full three-dimensional rotation symmetry is evidently absent, and one may speak only of axial symmetry about the distinguished axis.

Under axial symmetry only the *helicity* of the particle is conserved — the projection of angular momentum onto the ζ axis; we denote it by λ [16]. If one further demands symmetry under reflections in planes passing through the ζ axis, then states differing in the sign of λ are mutually degenerate; for $\lambda \neq 0$ there is therefore a twofold degeneracy [17]. The state of a photon with a definite momentum belongs to precisely this type of twofold-degenerate state. It is described by a "spin" wave function that is a vector $\mathbf{e}$ in the $\xi\eta$ plane; the two components of this vector transform into each other under all rotations about the ζ axis and under reflections in planes containing that axis.

The various polarization cases of a photon are in a definite correspondence with the possible values of its helicity. This correspondence can be established using formulas III,

[16]In contrast to the projection m of the angular momentum onto a fixed direction (the z axis) in space, discussed in the preceding section.

[17]We note that the electronic terms of a diatomic molecule are classified in exactly the same way (see III, § 78).

(57.9), which relate the components of a vector wave function to the components of the equivalent rank-2 spinor[18]. The projections $\lambda = +1$ and -1 correspond to vectors $\mathbf{e}$ that have only the component $e_\xi - ie_\eta$ or $e_\xi + ie_\eta$ nonzero, i.e. $\mathbf{e} = \mathbf{e}^{(+1)}$ or $\mathbf{e} = \mathbf{e}^{(-1)}$ respectively. In other words, $\lambda = +1$ and -1 correspond to the right and left circular polarizations of the photon (in § 16 the same result will be derived by directly computing the eigenfunctions of the spin-projection operator). Thus the projection of the photon's angular momentum onto the direction of its motion can take only the two values (± 1); the value 0 is not possible.

A photon state with definite momentum and polarization is a pure state (in the sense explained in III, § 14); it is described by a wave function and constitutes a complete quantum-mechanical specification of the particle (photon) state. There also exist "mixed" photon states corresponding to a less complete description, carried out not by a wave function but only by a density matrix.

Let us examine a photon state that is mixed in its polarization but possesses a definite momentum $\mathbf{k}$. Such a state (called a state of *partial polarization*) admits a "coordinate" wave function.

The polarization density matrix of a photon is a second-rank tensor $\rho_{\alpha\beta}$ defined in the plane orthogonal to $\mathbf{n}$ (the $\xi\eta$ plane; the indices α, β assume only two values). This tensor is Hermitian:

$$\rho_{\alpha\beta} = \rho^*_{\beta\alpha}, \tag{8.3}$$

and normalized by the condition

$$\rho_{\alpha\alpha} = \rho_{11} + \rho_{22} = 1. \tag{8.4}$$

By virtue of (8.3) the diagonal components ρ_{11} and ρ_{22} are real, and one determines the other through condition (8.4). The component ρ_{12} is complex, while $\rho_{21} = \rho^*_{12}$. Altogether, then, the density matrix is characterized by three real parameters.

Once the polarization density matrix is known, one can find the probability that the photon has any given definite polarization $\mathbf{e}$. This probability is obtained by "projecting" the tensor $\rho_{\alpha\beta}$ onto the direction of the vector $\mathbf{e}$, i.e. it equals the quantity

$$\rho_{\alpha\beta} e^*_\alpha e_\beta. \tag{8.5}$$

Thus the components ρ_{11} and ρ_{22} are the probabilities of linear polarizations along the ξ and η axes. Projection onto the vectors (8.2) yields the probabilities of the two circular polarizations:

$$\tfrac{1}{2}\left[1 \pm i(\rho_{12} - \rho_{21})\right]. \tag{8.6}$$

In both form and substance the properties of the tensor $\rho_{\alpha\beta}$ coincide with those of the tensor $J_{\alpha\beta}$ that describes partially polarized light in the classical theory (see II, § 50). Let us recall some of these properties here.

For a pure state of definite polarization $\mathbf{e}$, the tensor $\rho_{\alpha\beta}$ reduces to products of the components of the vector $\mathbf{e}$:

$$\rho_{\alpha\beta} = e_\alpha e^*_\beta. \tag{8.7}$$

[18] Recall that the wave-function components, viewed as probability amplitudes for the various values of the angular-momentum projection of the particle (which is what is meant here), correspond to the contravariant components of the spinor.

In this case the determinant $|\rho_{\alpha\beta}| = 0$. In the opposite case of an unpolarized photon, all polarization directions are equally probable, i.e.

$$\rho_{\alpha\beta} = \delta_{\alpha\beta}/2, \tag{8.8}$$

and then $|\rho_{\alpha\beta}| = 1/4$.

For the general case of partial polarization it is convenient to introduce three real Stokes parameters ξ_1, ξ_2, ξ_3 [19]; the density matrix is then expressible as

$$\rho_{\alpha\beta} = \frac{1}{2}\begin{pmatrix} 1+\xi_3 & \xi_1 - i\xi_2 \\ \xi_1 + i\xi_2 & 1-\xi_3 \end{pmatrix}. \tag{8.9}$$

All three parameters range between -1 and $+1$. In the unpolarized state $\xi_1 = \xi_2 = \xi_3 = 0$; for a completely polarized photon $\xi_1^2 + \xi_2^2 + \xi_3^2 = 1$.

The parameter ξ_3 characterizes linear polarization along the ξ or η axis; the probability of linear polarization along these axes is $(1+\xi_3)/2$ and $(1-\xi_3)/2$ respectively. The values $\xi_3 = +1$ or -1 therefore correspond to complete polarization in these directions.

The parameter ξ_1 characterizes linear polarization along directions making an angle $\varphi = \pi/4$ or $\varphi = -\pi/4$ with the ξ axis. The probability of linear polarization in these directions is $(1+\xi_1)/2$ or $(1-\xi_1)/2$; this is easily verified by projecting the tensor $\rho_{\alpha\beta}$ onto the directions $\mathbf{e} = (1, \pm 1)/\sqrt{2}$.

Lastly, the parameter ξ_2 represents the degree of circular polarization; by (8.6), the probability of right or left circular polarization is $(1+\xi_2)/2$ or $(1-\xi_2)/2$ respectively. Because these two polarizations answer to helicities $\lambda = \pm 1$, it is evident that in general ξ_2 coincides with the mean helicity of the photon. Note further that for a pure state with polarization $\mathbf{e}$,

$$\xi_2 = i[\mathbf{e}\mathbf{e}^*]\mathbf{n}. \tag{8.10}$$

Recall (see II, § 50) that the quantities ξ_2 and $\sqrt{\xi_1^2 + \xi_3^2}$ are invariant under Lorentz transformations.

In what follows we shall also need to know how the Stokes parameters transform under time reversal. One readily verifies that they are invariant under this operation. Since the result clearly does not depend on whether the polarization state is pure or mixed, it is enough to check invariance for a pure state. In quantum mechanics, time reversal replaces the wave function with its complex conjugate (see III, § 18). For a plane-polarized wave the corresponding substitution is [20]

$$\mathbf{k} \to -\mathbf{k}, \qquad \mathbf{e} \to -\mathbf{e}^*. \tag{8.11}$$

Under this transformation the symmetric part of the density matrix, $(1/2)(e_\alpha e_\beta^* + e_\beta e_\alpha^*)$, and hence the parameters ξ_1 and ξ_3 remain unchanged. The invariance of ξ_2 under the

[19] The notation for the parameters should not be confused with that for the ξ axis!

[20] The extra sign reversal of $\mathbf{e}$ is due to the fact that time reversal changes the sign of the electromagnetic vector potential. The scalar potential is unaffected; consequently, for the 4-vector e time reversal takes the form

$$(e_0, \mathbf{e}) \to (e_0^*, -\mathbf{e}^*). \tag{8.11a}$$

same transformation is apparent from (8.10); it is also obvious already from the physical meaning of ξ_2 as the mean helicity. Indeed, helicity is the projection of the angular momentum **j** onto the direction **n**, that is, the product **jn**; time reversal reverses the sign of both these vectors.

For the calculations that follow, the photon density matrix must be cast in four-dimensional form, that is, expressed as a certain 4-tensor $\rho_{\mu\nu}$. When the photon is polarized and described by a 4-vector e_μ, a natural definition of this tensor is

$$\rho_{\mu\nu} = e_\mu e_\nu^*. \tag{8.12}$$

In the three-dimensionally transverse gauge, $e = (0, \mathbf{e})$, if one of the spatial coordinate axes is chosen along **n**, the nonvanishing components of this 4-tensor coincide with (8.7).

For an unpolarized photon the three-dimensionally transverse gauge yields a tensor $\rho_{\mu\nu}$ with components

$$\rho_{ik} = \tfrac{1}{2}(\delta_{ik} - n_i n_k), \qquad \rho_{0i} = \rho_{i0} = \rho_{00} = 0 \tag{8.13}$$

(if one of the axes coincides with the direction of **n**, we return to (8.8)). Using the tensor $\rho_{\mu\nu}$ directly in this three-dimensional form is, however, inconvenient. We may instead perform a gauge transformation; for the density matrix this is a transformation of the type

$$\rho_{\mu\nu} \to \rho_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \tag{8.14}$$

where χ_μ are arbitrary functions. Setting

$$\chi_0 = -1/(4\omega), \qquad \chi_i = k_i/(4\omega^2)$$

we obtain in place of (8.13) the simple four-dimensional expression

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \tag{8.15}$$

The four-dimensional representation of the density matrix of a partially polarized photon is readily obtained by first rewriting the two-dimensional tensor (8.9) in three-dimensional form. In the three-dimensionally transverse gauge: $e^{(1)} = (0, \mathbf{e}^{(1)})$, $e^{(2)} = (0, \mathbf{e}^{(2)})$. The four-dimensional photon density matrix is therefore

$$\begin{aligned}\rho_{\mu\nu} = (1/2)(e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + (\xi_1/2)(e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ - (i\xi_2/2)(e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + (\xi_3/2)(e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}).\end{aligned} \tag{8.17}$$

The convenience of any particular choice of the 4-vectors $e^{(1)}$, $e^{(2)}$ depends on the specific conditions of the problem at hand.

It should be borne in mind that the conditions (8.16) do not fix the choice of $e^{(1)}$ and $e^{(2)}$ uniquely. If any 4-vector e_μ satisfies these conditions, so does any 4-vector of the form $e_\mu + \chi k_\mu$ (since $k^2 = 0$). This ambiguity is tied to the gauge freedom of the density matrix.

The first term in (8.17) corresponds to the unpolarized state. It could therefore be replaced, by (8.15), with $-g_{\mu\nu}/2$. Such a replacement is again equivalent to a gauge transformation. When working with 4-tensors of the form (8.17), decomposed in two independent 4-vectors, it is useful to adopt the following formal device. Writing (8.17) as

$$\rho_{\mu\nu} = \sum_{a,b=1}^{2} \rho^{(ab)} e_\mu^{(a)} e_\nu^{(b)},$$

we represent the coefficients $\rho^{(ab)}$ as a 2×2 matrix

$$\rho = \begin{pmatrix} \rho^{(11)} & \rho^{(12)} \\ \rho^{(21)} & \rho^{(22)} \end{pmatrix}.$$

Any Hermitian 2×2 matrix can be expanded over four linearly independent 2×2 matrices — the Pauli matrices σ_x, σ_y, σ_z and the identity matrix 1:

$$\rho = (1/2)(1 + \boldsymbol{\xi}\boldsymbol{\sigma}), \qquad \boldsymbol{\xi} = (\xi_1, \xi_2, \xi_3), \tag{8.18}$$

as is easily verified by direct comparison with (8.17) using the well-known explicit forms of the Pauli matrices (18.5) (grouping the three quantities ξ_1, ξ_2, ξ_3 into a "vector" $\boldsymbol{\xi}$ is of course purely formal and serves only to shorten the notation).

The requisite generalization is achieved by replacing these 3-vectors by spacelike real 4-vectors $e^{(1)}$, $e^{(2)}$, orthogonal to each other and to the photon 4-momentum k:

$$e^{(1)2} = e^{(2)2} = -1, \qquad e^{(1)}e^{(2)} = 0, \qquad e^{(1)}k = e^{(2)}k = 0. \tag{8.16}$$

Problem

Write the photon density matrix in the representation whose "coordinate axes" are the circular basis vectors (8.2).

Solution. The components of the tensor $\rho'_{\alpha\beta}$ in the new axes ($\alpha, \beta = \pm 1$) are obtained by projecting the tensor (8.9) onto the basis vectors (8.2):

$$\rho'_{11} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(+1)}, \qquad \rho'_{1,-1} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(-1)}, \qquad \dots,$$

$$\rho' = \frac{1}{2}\begin{pmatrix} 1 + \xi_2 & -\xi_3 + i\xi_1 \\ -\xi_3 - i\xi_1 & 1 - \xi_2 \end{pmatrix}.$$

§ 9. The two-photon system

Arguments analogous to those developed in § 6 make it possible to count the number of allowed states in the more involved case of a system of two photons as well (*L. Landau*, 1948).

Let us work in the center-of-inertia frame of the two photons, where $\mathbf{k}_1 = -\mathbf{k}_2 \equiv \mathbf{k}$[21]. In the momentum representation the wave function of the pair can be cast as a three-dimensional rank-2 tensor $A_{ik}(\mathbf{n})$ built bilinearly from the components of the individual

[21]A center-of-inertia frame can be found in every case except when the two photons propagate in the same direction. For such a pair the total momentum $\mathbf{k}_1 + \mathbf{k}_2$ and total energy $\omega_1 + \omega_2$ are connected by exactly the same relation as for one photon, so no frame with $\mathbf{k}_1 + \mathbf{k}_2 = 0$ exists.

photon vector wave functions; each of its two indices refers to one photon ($\mathbf{n}$ denotes the unit vector in the direction of $\mathbf{k}$). The transversality requirement for each photon is encoded in the orthogonality of A_{ik} to $\mathbf{n}$:

$$A_{il}n_l = 0, \qquad A_{lk}n_l = 0. \tag{9.1}$$

Interchanging the two photons amounts to transposing the indices of the tensor A_{ik} together with a simultaneous reversal of the sign of $\mathbf{n}$. Because photons obey Bose statistics,

$$A_{ik}(-\mathbf{n}) = A_{ki}(\mathbf{n}). \tag{9.2}$$

The tensor A_{ik} is, in general, not symmetric in its indices. We decompose it into its symmetric part (s_{ik}) and antisymmetric part (a_{ik}): $A_{ik} = s_{ik} + a_{ik}$. Each of these parts must individually satisfy the relation (9.2) (as well as the orthogonality conditions (9.1)). This gives

$$s_{ik}(-\mathbf{n}) = s_{ik}(\mathbf{n}), \tag{9.3}$$

$$a_{ik}(-\mathbf{n}) = -a_{ik}(\mathbf{n}). \tag{9.4}$$

Spatial inversion does not by itself reverse the sign of the components of a second-rank tensor, but it does reverse the sign of $\mathbf{n}$. From (9.3) it is therefore apparent that the wave function s_{ik} is symmetric under inversion, i.e. it corresponds to even-parity states of the photon system; the wave function a_{ik}, on the other hand, corresponds to odd-parity states.

An antisymmetric second-rank tensor is equivalent (dual) to an axial vector $\mathbf{a}$ whose components are expressed via the tensor components as $a_i = \frac{1}{2}e_{ikl}a_{kl}$, where e_{ikl} is the antisymmetric unit tensor (see II, § 6). The orthogonality of the tensor a_{kl} to the vector $\mathbf{n}$ implies that the vectors $\mathbf{a}$ and $\mathbf{n}$ are parallel[22]. One may therefore write $\mathbf{a} = \mathbf{n}\varphi(\mathbf{n})$, where φ is a scalar; by (9.4) we must have $\mathbf{a}(-\mathbf{n}) = -\mathbf{a}(\mathbf{n})$, and hence

$$\varphi(-\mathbf{n}) = \varphi(\mathbf{n}).$$

This relation means that the scalar φ can be built linearly only from spherical harmonics of even order L (including zero).

We see that the antisymmetric tensor a_{ik}, by its transformation properties (under rotations), is equivalent to a single scalar (cf. the footnote on p. 34). Assigning to the latter a “spin” of 0, we find that the angular momentum of the state is $J = L$. Thus the tensor a_{ik} corresponds to odd-parity states of the photon system with even total angular momentum J.

Let us turn to the symmetric tensor s_{ik}. Since it is even under reversal of the sign of $\mathbf{n}$, it describes even-parity states of the photon system. It follows likewise that all components of s_{ik} are expressed through spherical harmonics of even order L (including $L = 0$). An arbitrary symmetric second-rank tensor can be decomposed, as is well known, into a scalar (s_{ii}) and a traceless symmetric tensor (s'_{ik}) with $s'_{ii} = 0$.

[22]We have: $a_{ik} = e_{ikl}a_l$, and the orthogonality condition yields

$$a_{ik}n_k = e_{ikl}a_l n_k = [\mathbf{na}]_i = 0.$$

The scalar s_{ii} is assigned "spin" 0, so the angular momentum of the corresponding states is $J = L$, which is even. The tensor s'_{ik} corresponds to "spin" 2 (see III, § 57). Composing this "spin" with the even "orbital angular momentum" L according to the addition rule for angular momenta, we find that for a given even $J \neq 0$ there are three states (with $L = J \pm 2, J$), while for odd $J \neq 1$ there are two states (with $L = J \pm 1$). The exceptions are $J = 0$ with a single state ($L = 2$) and $J = 1$ with a single state ($L = 2$).

In this counting, however, we have not yet accounted for the orthogonality of the tensor s_{ik} to the vector $\mathbf{n}$. One must therefore subtract from the number of states obtained those corresponding to a symmetric second-rank tensor that is "parallel" to the vector $\mathbf{n}$. Such a tensor (denote it s''_{ik}) can be written as

$$s''_{ik} = n_i b_k + n_k b_i,$$

where $\mathbf{b}$ is some vector. By (9.3) this vector must satisfy $\mathbf{b}(-\mathbf{n}) = -\mathbf{b}(\mathbf{n})$. The tensor s''_{ik} responsible for the "extra" states is thus equivalent to an odd-parity vector. This vector must consequently be expressed through spherical harmonics of odd orders L only. Noting further that a vector corresponds to "spin" 1, we find that for each even $J \neq 0$ there are two states (with $L = J \pm 1$), and for each odd J a single state (with $L = J$); the special case $J = 0$ has one state (with $L = 1$).

Collecting all the results obtained, we arrive at the following table giving the number of possible even- and odd-parity states of a system of two photons (with vanishing total momentum) for the various values of the total angular momentum J:

J	Even	Odd
0	1	1
1	–	–
$2k$	2	1
$2k+1$	1	–

(9.5)

(k being a positive integer other than zero). We observe that for odd J no odd-parity states exist, while the value $J = 1$ is altogether excluded[23].

The wave function of the two-photon system A_{ik} determines the correlation between their polarizations. The probability that both photons simultaneously possess definite polarizations $\mathbf{e}_1$ and $\mathbf{e}_2$ is proportional to

$$A_{ik} e^*_{1i} e^*_{2k}.$$

In other words, if the polarization $\mathbf{e}_1$ of one photon is specified, then the polarization of the second photon, $\mathbf{e}_2$, is proportional to

$$e_{2k} \propto A_{ik} e^*_{1i}. \tag{9.6}$$

In odd-parity states of the system, A_{ik} coincides with the antisymmetric tensor a_{ik}. In that case

$$\mathbf{e}_2 \mathbf{e}^*_1 \propto a_{ik} e^*_{1i} e^*_{1k} = 0,$$

[23]For an alternative derivation of these results, see problem 1 to § 69.

i.e. the polarizations of the two photons are mutually orthogonal. For linear polarization this signifies that their directions are perpendicular, while for circular polarization it means opposite senses of rotation.

The even-parity state with $J = 0$ is described by a symmetric tensor reducing to a scalar,

$$s_{ik} = \text{const} \cdot (\delta_{ik} - n_i n_k).$$

From (9.6) we then obtain $\mathbf{e}_1 = \mathbf{e}_2^*$. For linear polarization this means that their directions are parallel, whereas for circular polarization it means opposite senses of rotation. The latter fact is self-evident: when $J = 0$ the sum of the angular-momentum projections of the two photons onto any common direction $\mathbf{k}$ must in all cases vanish (the projections onto the opposite directions $\mathbf{k}_1$ and $\mathbf{k}_2$, i.e. the helicities, are accordingly equal).

CHAPTER II

Bosons

§ 10. Wave equation for particles with spin 0

Chapter I demonstrated the construction of a quantum description for the free electromagnetic field, proceeding from the known classical-limit properties of the field and relying on the apparatus of ordinary quantum mechanics. The scheme of field description as a photon system obtained in this way retains many features that carry over to the relativistic quantum-theoretical treatment of particles as well.

The electromagnetic field is a system with an infinite number of degrees of freedom. For it there is no conservation law for the number of particles (photons), and among its possible states are those with any number of particles [1]. However, the same property must, generally speaking, also hold in relativistic theory for systems of any particles. The conservation of particle number in non-relativistic theory is tied to the conservation of mass: the sum of the masses (rest masses) of the particles does not change during their interaction; conservation of the total mass in a system of electrons, say, implies the constancy of their number as well. In relativistic mechanics, however, no law of mass conservation exists; only the total energy of the system (which includes the rest energies of the particles) must be conserved. For this reason the particle number need no longer be conserved, and consequently every relativistic particle theory must be a theory of systems possessing an infinite number of degrees of freedom. In other words, such a particle theory acquires the character of a field theory.

The adequate mathematical apparatus for treating systems whose particle number varies is that of second quantization (see III, §§ 64, 65). In the quantum treatment of the electromagnetic field the 4-potential $\hat{A}$ serves as the second-quantization operator. It is built from the (coordinate-space) wave functions of the individual particles (photons) together with their creation and annihilation operators. A completely analogous role in the description of a system of particles belongs to the operator of the quantized wave function. Its construction requires, first and foremost, knowledge of the wave-function form for a single free particle and of the equation governing this function.

It should be stressed that the notion of free-particle fields is auxiliary in character. Real particles interact, and the task of the theory lies in studying these interactions. But every interaction reduces to a collision, before and after which the system can be regarded as a collection of free particles. In § 1 it was noted that these are the only directly measurable objects. We therefore use the fields of free particles as a means for describing initial and final states.

We begin the relativistic description of free particles with the case of spin-0 particles. The mathematical simplicity of this case makes it possible to bring out most clearly the basic ideas and characteristic features of such a description.

[1] In practice, of course, the photon number changes only through various interaction processes.

The state of a free particle (without spin) can be completely specified by giving its momentum $\mathbf{p}$ alone. The energy ε of the particle [2] satisfies $\varepsilon^2 = \mathbf{p}^2 + m^2$ (where m is the particle mass), or in four-dimensional form:

$$p^2 = m^2. \tag{10.1}$$

It is well known that the conservation of momentum and energy is tied to the homogeneity of space and time, that is, to symmetry with respect to any parallel translation of the 4-coordinate system. Within the quantum description, this symmetry requirement implies that under such a coordinate transformation the wave function of a particle possessing a definite 4-momentum may only acquire a phase factor (whose modulus equals unity). This requirement is met solely by an exponential with an exponent that depends linearly on the 4-coordinates. Put differently, the wave function of a free particle in a state of definite 4-momentum $p^\mu = (\varepsilon, \mathbf{p})$ must have the form of a plane wave:

$$\text{const} \cdot e^{-ipx}, \qquad px = \varepsilon t - \mathbf{pr} \tag{10.2}$$

(the choice of sign in the exponent is in itself a convention within relativistic theory; it is made so as to agree with the non-relativistic case).

The wave equation must possess the functions (10.2) as particular solutions for every 4-vector p obeying the condition (10.1). It must be linear—this being a manifestation of the superposition principle: every linear combination of functions (10.2) likewise describes a possible particle state and hence must also satisfy the equation. Lastly, the equation should have the lowest attainable order; raising the order would bring in extraneous solutions.

Spin is the angular momentum that a particle possesses in the frame where it is at rest. When the particle's spin equals s, the wave function in the rest frame takes the form of a three-dimensional spinor of rank $2s$. For describing the particle in an arbitrary reference frame one must cast the wave function in terms of four-dimensional objects.

A particle of spin 0 is described in its rest frame by a three-dimensional scalar. This scalar may, however, have a different four-dimensional "ancestry": it can be a four-scalar ψ, but it can equally well be the temporal component of a timelike 4-vector ψ_μ whose only component that does not vanish in the rest frame is ψ_0 [3].

For a free particle the only operator that can enter the wave equation is the 4-momentum operator $\hat{p}$. Its components are the differentiation operators with respect to the coordinates and time:

$$\hat{p}^\mu = i\partial^\mu = \left(i\frac{\partial}{\partial t}, -i\boldsymbol{\nabla} \right). \tag{10.3}$$

The wave equation must represent a differential relation between the quantities ψ and ψ_μ, effected by means of the operator $\hat{p}$. This relation must, of course, be expressed through relativistically invariant equations. Such equations are

$$m\psi_\mu = \hat{p}_\mu \psi, \qquad \hat{p}^\mu \psi_\mu = m\psi, \tag{10.4}$$

[2] We denote the energy of an individual particle by ε, as distinct from the energy E of the system of particles.

[3] Or, by the same token, the temporal component of a higher-rank 4-tensor; that possibility, however, would give rise to equations of higher order.

where m is a dimensional constant characterizing the particle [4].

Substituting ψ_μ from the first equation into the second, we obtain

$$(\hat{p}^2 - m^2)\psi = 0 \tag{10.5}$$

(*O. Klein*, *V. A. Fock*, 1926; *W. Gordon*, 1927). Written out explicitly, this equation reads

$$-\partial_\mu \partial^\mu \psi \equiv \left(-\frac{\partial^2}{\partial^2 t} + \Delta \right) \psi = m^2 \psi. \tag{10.6}$$

If one inserts a plane wave (10.2) for ψ, the result is $p^2 = m^2$, showing that m is indeed the particle mass. It is worth noting that the structure of equation (10.5) could, of course, have been anticipated, since $\hat{p}^2$ is the sole scalar operator one can form from $\hat{p}$ (and for this reason every component of the wave function of a particle of arbitrary spin obeys the same equation—something we shall encounter many times below).

Thus a spin-0 particle is described essentially by a single (four-dimensional) scalar ψ obeying the second-order equation (10.5). In the first-order equations (10.4) the role of the wave function is played by the aggregate of the quantities ψ and ψ_μ, the 4-vector ψ_μ reducing to the 4-gradient of the scalar ψ. In the rest frame the particle's wave function does not depend on the (spatial) coordinates, so that the spatial components of the 4-vector ψ_μ vanish, as they should.

For carrying out second quantization it proves useful to write the particle's energy and momentum as spatial integrals of certain bilinear (in ψ and ψ^*) combinations playing the role of spatial densities of these quantities. This amounts to finding the energy–momentum tensor $T_{\mu\nu}$ that corresponds to equation (10.5). With the aid of this tensor, the energy–momentum conservation law takes the form

$$\partial_\mu T^\mu_\nu = 0. \tag{10.7}$$

Following the general rules of field theory (see II, § 32), let us write down the variational principle whose consequence would be equation (10.5). Such a principle must consist in requiring the minimality of the "action integral"

$$S = \int L \, d^4x \tag{10.8}$$

with respect to a certain real 4-scalar L—the Lagrangian density of the field [5]. From the scalar ψ (and the operator ∂^μ) one can construct a real scalar bilinear expression of the form

$$L = \partial_\mu \psi^* \cdot \partial^\mu \psi - m^2 \psi^* \psi, \tag{10.9}$$

where m is a dimensional constant. Treating ψ and ψ^* as independent variables describing the field ("generalized coordinates" of the field q), it is easy to verify that the Lagrange

[4] The constants m are introduced in (10.4) in such a way that ψ_μ and ψ have the same dimensionality. It would be pointless to introduce different constants m_1 and m_2 in these two equations, since they could always be made equal by redefining ψ or ψ_μ.

[5] The corresponding second-quantized operator $\hat{L}$ is called the field *Lagrangian*. For brevity of terminology we shall use this term both for the "quantized" and the "unquantized" Lagrangian density.

equations

$$\frac{\partial}{\partial x^{\mu}}\frac{\partial L}{\partial q_{,\mu}}=\frac{\partial L}{\partial q} \qquad (10.10)$$

($q_{,\mu} \equiv \partial_\mu q$) do indeed coincide with the equations (10.5) for ψ and ψ^*, with m being the particle mass. Note also that the expression (10.9) is written with an overall sign such that the square of the time derivative, $|\partial\psi/\partial t|^2$, enters L with a plus sign; in the opposite case the action could not have a minimum (cf. II, § 27). The choice of the overall numerical coefficient of L is a matter of convention (and is reflected only in the normalization coefficient in ψ).

The energy–momentum tensor is now computed by the formula

$$T_\mu^\nu = \sum q_{,\mu}\frac{\partial L}{\partial q_{,\nu}} - L\delta_\mu^\nu \qquad (10.11)$$

(summation over all q). Substituting (10.9), we obtain

$$T_{\mu\nu} = \partial_\mu\psi^* \cdot \partial_\nu\psi + \partial_\nu\psi^* \cdot \partial_\mu\psi - Lg_{\mu\nu} \qquad (10.12)$$

(these quantities are, as they should be, real, which is ensured by the reality of L). In particular,

$$T_{00} = 2\frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial t} - L = \frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial t} + \nabla\psi^*\cdot\nabla\psi + m^2\psi^*\psi, \qquad (10.13)$$

$$T_{i0} = \frac{\partial\psi^*}{\partial t}\frac{\partial\psi}{\partial x^i} + \frac{\partial\psi^*}{\partial x^i}\frac{\partial\psi}{\partial t}. \qquad (10.14)$$

The 4-momentum of the field is given by the integral

$$P_\mu = \int T_{\mu 0}\, d^3x, \qquad (10.15)$$

i.e. T_{00} and T_{i0} play the role of the energy and momentum densities. We note that the quantity T_{00} is essentially positive.

Formula (10.13) can be used to normalize the wave function. A plane wave normalized to one particle in a volume $V = 1$ takes the form

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}}e^{-ipx}. \qquad (10.16)$$

For this function one has $T_{00} = \varepsilon$, whence the total energy within the volume $V = 1$ coincides with the energy of one particle.

The angular momentum, whose conservation follows from the isotropy of space, likewise admits a representation as a spatial integral; we shall, however, have no occasion to use such a representation later on.

Finally, besides the conservation laws linked directly with space–time symmetry, equations (10.4) admit one more conservation law. It is easy to verify that, by virtue of (10.4) (and the analogous equations for ψ^*), the following equation holds:

$$\partial_\mu j^\mu = 0, \qquad (10.17)$$

where

$$j_\mu = m(\psi^*\psi_\mu + \psi^*_\mu\psi) = i[\psi^*\partial_\mu\psi - (\partial_\mu\psi^*)\psi]. \tag{10.18}$$

It is apparent from this that j^μ plays the role of the 4-vector current density. Equation (10.17) is then the continuity equation expressing the conservation of the quantity

$$Q = \int j_0\, d^3x, \tag{10.19}$$

where

$$j_0 = j^0 = i\left(\psi^*\frac{\partial\psi}{\partial t} - \frac{\partial\psi^*}{\partial t}\psi\right). \tag{10.20}$$

Let us call attention to the fact that j_0 is not a positive-definite quantity. This circumstance alone shows that in the general case it certainly cannot be interpreted as the probability density of the spatial localization of the particle. The meaning of the conservation law expressed by equation (10.17) will be clarified in the next section.

§ 11. Particles and antiparticles

In keeping with the general procedure of second quantization, one must expand an arbitrary wave function over the eigenfunctions spanning the full set of possible free-particle states, for example over the plane waves ψ_p:

$$\psi = \sum_{\mathbf{p}} a_{\mathbf{p}}\psi_p, \qquad \psi^* = \sum_{\mathbf{p}} a^*_{\mathbf{p}}\psi^*_p.$$

After that, the coefficients $a_{\mathbf{p}}$, $a^*_{\mathbf{p}}$ would have to be promoted to operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+{}_{\mathbf{p}}$ for annihilating and creating particles in the corresponding states [6].

At this point, however, we immediately encounter the following fundamentally new circumstance (compared with the non-relativistic theory). In a plane wave that solves equation (10.5), the energy ε must satisfy (for a given momentum $\mathbf{p}$) only the condition $\varepsilon^2 = \mathbf{p}^2 + m^2$, so that it can take two values: $\pm\sqrt{\mathbf{p}^2 + m^2}$. Physically meaningful values of the free-particle energy can, however, be only positive ones. Yet simply discarding the negative values is inadmissible: the general solution of the wave equation is formed only by a superposition of all its independent particular solutions. This points to the need for a certain re- interpretation of the expansion coefficients of ψ and ψ^* upon second quantization.

Let us write this expansion in the form

$$\psi = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a^{(+)}_{\mathbf{p}}\, e^{i(\mathbf{pr}-\varepsilon t)} + \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a^{(-)}_{\mathbf{p}}\, e^{i(\mathbf{pr}+\varepsilon t)}, \tag{11.1}$$

where the first sum contains plane waves normalized according to (10.16) with positive "frequencies," and the second with negative ones; everywhere ε stands for the positive

[6]We label the ψ-functions by the 4-momentum index p; functions with "negative frequency" will henceforth be denoted ψ_{-p}. The operators $\hat{a}$, $\hat{a}^+$, on the other hand, carry the three-momentum index $\mathbf{p}$, which completely specifies the state of a real particle.

quantity: $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$. In performing second quantization, the coefficients $a_{\mathbf{p}}^{(+)}$ in the first sum are replaced in the usual way by particle annihilation operators $\hat{a}_{\mathbf{p}}$. In the second sum we observe that upon subsequent evaluation of matrix elements the time dependence of its terms will correspond not to annihilation but to creation of particles; the factor $e^{i\varepsilon t} = (e^{-i\varepsilon t})^*$ corresponds to one extra particle with energy ε in the final state (cf. the end of § 2). Accordingly, the coefficients $a_{\mathbf{p}}^{(-)}$ are replaced by creation operators $\hat{b}^+_{-\mathbf{p}}$ for certain other particles. Changing the summation variable in the second sum of (11.1) from $\mathbf{p}$ to $-\mathbf{p}$ (so that the exponential factor takes the form $e^{-i(\mathbf{pr}-\varepsilon t)}$), we arrive at the ψ-operators

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}} e^{-ipx} + \hat{b}^+_{\mathbf{p}} e^{ipx}), \quad \hat{\psi}^+ = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}^+_{\mathbf{p}} e^{ipx} + \hat{b}_{\mathbf{p}} e^{-ipx}). \qquad (11.2)$$

As a result, every operator $\hat{a}_{\mathbf{p}}$ or $\hat{b}_{\mathbf{p}}$ appears multiplied by a function carrying the "correct" time factor ($\sim e^{-i\varepsilon t}$), whereas each $\hat{a}^+_{\mathbf{p}}$ or $\hat{b}^+_{\mathbf{p}}$ is accompanied by the complex conjugate of that function. This permits the identification, following the general rules, of $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ as annihilation operators and of $\hat{a}^+_{\mathbf{p}}$ and $\hat{b}^+_{\mathbf{p}}$ as creation operators for particles possessing momenta $\mathbf{p}$ and energies ε.

We thus arrive at the picture of particles of two kinds, appearing together and on an equal footing. They are referred to as *particles* and *antiparticles* (the reason for this name will become clear below). One kind corresponds, within the second-quantization formalism, to the operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+_{\mathbf{p}}$, and the other to $\hat{b}_{\mathbf{p}}$, $\hat{b}^+_{\mathbf{p}}$. Both species of particles, whose operators enter the same ψ-operator, thereby possess identical masses.

These conclusions can also be reached by starting directly from the demands of relativistic invariance.

From a mathematical viewpoint, Lorentz transformations are rotations of the four-dimensional coordinate system that change the orientation of the time axis (combined with purely spatial rotations, which leave the time axis unchanged, they constitute the transformation group known as the *Lorentz group*[7]). All these transformations share the property that they do not carry the t axis out of the corresponding half of the light cone, which expresses the physical principle of the existence of a limiting signal-propagation speed.

From a purely mathematical standpoint, however, a simultaneous reversal of the sign of all four coordinates is likewise a rotation (*four-dimensional inversion*): the determinant of this transformation equals +1, just as for every other rotational transformation. Under it the time axis passes from one half of the light cone to the other. Although this circumstance signifies the physical unrealizability of such a transformation (as a change of reference frame), mathematically the distinction reduces merely to the fact that (because of the pseudo-Euclidean character of the metric) such a rotation cannot be carried out without admitting a complex transformation of the coordinates in the process.

It is natural to expect that this distinction should be immaterial when one speaks of four-dimensional invariance. Then every expression that is invariant under Lorentz

[7]Note that the totality of all three-dimensional (spatial) rotations constitutes by itself a group, entering the Lorentz group as a subgroup. The totality of Lorentz transformations does not by itself form a group: the result of successive Lorentz transformations may reduce to a purely spatial rotation.

transformations should also be invariant under 4-inversion. The precise formulation of this requirement as applied to the scalar ψ-operator will be given in § 13. But let us immediately note that in any case it entails the simultaneous presence in the ψ-operators of terms with both signs in front of ε in the exponents, since the replacement $t \to -t$ reverses precisely this sign.

Returning to expressions (11.2), let us now determine the commutation relations among the operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+_{\mathbf{p}}$ (and $\hat{b}_{\mathbf{p}}$, $\hat{b}^+_{\mathbf{p}}$). In the photon case this was done (for $\hat{c}_{\mathbf{p}}$, $\hat{c}^+_{\mathbf{p}}$) by drawing on the oscillator analogy, that is, ultimately from the classical-limit behavior of the electromagnetic field. Here no comparable analogy exists. In order to decide whether the operators obey Bose or Fermi commutation rules, one can appeal only to the structure of the Hamiltonian built from these operators.

The Hamiltonian is obtained (see III, § 64) by substituting $\hat{\psi}$ and $\hat{\psi}^+$ for ψ and ψ^* in the integral $\int T_{00}\, d^3x$ [8]. In this way we find

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}} + \hat{b}_{\mathbf{p}}\hat{b}^+_{\mathbf{p}}). \tag{11.3}$$

It is easy to see that a sensible result for the eigenvalues of this Hamiltonian is obtained only if the operators satisfy Bose commutation rules:

$$\{\hat{a}_{\mathbf{p}}, \quad \hat{a}^+_{\mathbf{p}}\}_- = \{\hat{b}_{\mathbf{p}}, \quad \hat{b}^+_{\mathbf{p}}\}_- \tag{11.4}$$

(all other pairs of operators commute; in particular, all particle operators $\hat{a}_{\mathbf{p}}$, $\hat{a}^+_{\mathbf{p}}$ commute with all antiparticle operators $\hat{b}_{\mathbf{p}}$, $\hat{b}^+_{\mathbf{p}}$). Indeed, in that case

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}} + \hat{b}^+_{\mathbf{p}}\hat{b}_{\mathbf{p}} + 1).$$

The eigenvalues of the products $\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}}$ and $\hat{b}^+_{\mathbf{p}}\hat{b}_{\mathbf{p}}$ are non-negative integers $N_{\mathbf{p}}$ and $\bar{N}_{\mathbf{p}}$—the numbers of particles and antiparticles. The infinite additive constant $\sum \varepsilon$ ("vacuum energy") can once again simply be dropped:

$$E = \sum_{\mathbf{p}} \varepsilon(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}}) \tag{11.5}$$

(cf. formula (3.1) and the remark following it). This expression is manifestly positive and accords with the picture of two species of actually existing particles. In an analogous fashion, for the total momentum of the particle system we obtain

$$\mathbf{P} = \sum_{\mathbf{p}} \mathbf{p}(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}}). \tag{11.6}$$

Had we adopted, in place of (11.4), the Fermi commutation relations (anticommutators instead of commutators), we would have obtained

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon(\hat{a}^+_{\mathbf{p}}\hat{a}_{\mathbf{p}} - \hat{b}^+_{\mathbf{p}}\hat{b}_{\mathbf{p}} + 1)$$

[8] In non-relativistic theory one customarily places the conjugate operator $\hat{\psi}^+$ to the left of ψ. Here, however, the ordering is immaterial, since interchanging $\hat{\psi}^+$ and $\hat{\psi}$ would merely interchange the equivalent operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$. It is, however, necessary, once a particular ordering has been chosen, always to adhere to the same convention.

and, instead of formula (11.5), the physically meaningless expression $\sum \varepsilon(N_{\mathbf{p}} - \bar{N}_{\mathbf{p}})$. This expression is not positive- definite and therefore cannot represent the energy of a system of free particles.

Thus spin-0 particles are bosons.

Next, consider the integral Q (10.19). Replacing the functions ψ and ψ^* in j^0 by the operators $\hat{\psi}$ and $\hat{\psi}^+$ and carrying out the integration, we obtain

$$\hat{Q} = \sum_{\mathbf{p}} (\hat{a}^+_{\mathbf{p}} \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}^+_{\mathbf{p}}) = \sum_{\mathbf{p}} (\hat{a}^+_{\mathbf{p}} \hat{a}_{\mathbf{p}} - \hat{b}^+_{\mathbf{p}} \hat{b}_{\mathbf{p}} - 1). \tag{11.7}$$

The eigenvalues of this operator (apart from the inessential additive constant $\sum 1$) are

$$Q = \sum_{\mathbf{p}} (N_{\mathbf{p}} - \bar{N}_{\mathbf{p}}), \tag{11.8}$$

i.e. they equal the difference between the total numbers of particles and antiparticles.

As long as we deal with free particles, abstracting from all interaction among them, the content of the conservation law for Q (as well as the conservation of total energy and momentum (11.5, 11.6)) remains, of course, largely formal: what is actually conserved is not merely this sum, but each of the individual numbers $N_{\mathbf{p}}$, $\bar{N}_{\mathbf{p}}$ separately. Whether the quantity Q will remain conserved as a result of the interaction depends on the nature of that interaction. If Q is conserved (i.e. if the operator $\hat{Q}$ commutes with the interaction Hamiltonian), then expression (11.8) shows what restriction this imposes on the possible changes in particle number: only pairs "particle + antiparticle" can appear or disappear.

If a particle carries an electric charge, its antiparticle must bear a charge of the opposite sign: were both to carry identical charges, the creation or annihilation of a pair would violate the strict law of nature—conservation of the total electric charge. We shall see below (§ 32) how this opposition of charges arises automatically within the theory (in the interaction of particles with the electromagnetic field).

The quantity Q is sometimes called the *charge* of the field of the given particles. For electrically charged particles, Q determines, in particular, the total electric charge of the system (in units of the elementary charge e). Let us stress, however, that particles and antiparticles may well be electrically neutral.

We see in this way how the character of the relativistic energy–momentum relation (the two-valuedness of the root of the equation $\varepsilon^2 = \mathbf{p}^2 + m^2$), combined with the requirements of relativistic invariance, leads in quantum theory to the emergence of a new classification principle for particles—the possibility of existence of pairs of distinct particles ("particle + antiparticle"), standing in the correspondence described above. This remarkable prediction was first made (for spin-1/2 particles) by *Dirac* in 1930, even before the actual discovery of the first antiparticle—the positron[9].

§ 12. Strictly neutral particles

When carrying out the second quantization of the ψ-function (11.1), the coefficients $a^{(+)}_{\mathbf{p}}$ and $a^{(-)}_{\mathbf{p}}$ were treated as operators pertaining to different particles. This is, however,

[9]The notion of antiparticles was extended to bosons by *Weisskopf* and *Pauli* (*V. Weisskopf, W. Pauli*, 1934).

not obligatory: as a special case, the annihilation and creation operators appearing in $\hat{\psi}$ may refer to one and the same type of particle (as was the case for photons—cf. (2.17)). Denoting these operators $\hat{c}_{\mathbf{p}}$ and $\hat{c}^{+}_{\mathbf{p}}$ in this case, we write the ψ-operator as

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} (\hat{c}_{\mathbf{p}} e^{-ipx} + \hat{c}^{+}_{\mathbf{p}} e^{ipx}). \tag{12.1}$$

The field described by such an operator corresponds to a system of identical particles that can be said to "coincide with their own antiparticles."

The operator (12.1) is Hermitian ($\hat{\psi}^{+} = \hat{\psi}$); in this sense such a field has half as many "degrees of freedom" as a complex field, for which the operators $\hat{\psi}$ and $\hat{\psi}^{+}$ do not coincide.

In connection with this, the Lagrangian of the field, when expressed through the Hermitian operator $\hat{\psi}$, must contain an extra factor of $1/2$ (compared with (10.9)) [10]

$$\hat{L} = (1/2)(\partial_\mu \hat{\psi} \cdot \partial^\mu \hat{\psi} - m^2 \hat{\psi}^2). \tag{12.2}$$

The corresponding energy–momentum tensor is

$$\hat{T}_{\mu\nu} = \partial_\mu \hat{\psi} \cdot \partial_\nu \hat{\psi} - \hat{L} g_{\mu\nu}, \tag{12.3}$$

so that the energy-density operator reads

$$\hat{T}_{00} = \left(\frac{\partial \hat{\psi}}{\partial t}\right)^2 - \hat{L} = \frac{1}{2}\left[\left(\frac{\partial \hat{\psi}}{\partial t}\right)^2 + (\boldsymbol{\nabla}\hat{\psi})^2 + m^2 \hat{\psi}^2\right]. \tag{12.4}$$

Inserting (12.1) into the integral $\int \hat{T}_{00}\, d^3x$, we obtain the field Hamiltonian

$$\hat{H} = \frac{1}{2} \sum_{\mathbf{p}} \varepsilon (\hat{c}^{+}_{\mathbf{p}} \hat{c}_{\mathbf{p}} + \hat{c}_{\mathbf{p}} \hat{c}^{+}_{\mathbf{p}}). \tag{12.5}$$

This again reveals the necessity of Bose quantization:

$$\{\hat{c}_{\mathbf{p}},\ \hat{c}^{+}_{\mathbf{p}}\}_{-} = 1, \tag{12.6}$$

and the energy eigenvalues (once more dropping the additive constant) are

$$E = \sum_{\mathbf{p}} \varepsilon_{\mathbf{p}} N_{\mathbf{p}}. \tag{12.7}$$

Fermi quantization, on the other hand, would yield a meaningless result—a value of E independent of $N_{\mathbf{p}}$.

The "charge" Q of the field under consideration vanishes. This is already clear from the fact that the charge Q must change sign when particles are replaced by antiparticles,

[10] This is analogous to the extra factor $1/2$ in the operator (2.10) for the electromagnetic field energy density (expressed via the Hermitian operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$), as compared to the photon energy density (3.2) expressed through the photon's complex wave function; cf. the remark on p. 26.

and here the two coincide. Correspondingly, no 4-vector current density exists either. Indeed, the expression

$$\hat{j}_\mu = i[\hat{\psi}^+\partial_\mu\hat{\psi} - (\partial_\mu\hat{\psi}^+)\hat{\psi}\,] \tag{12.8}$$

for the operator of the conserved 4-current $\hat{j}$ vanishes when $\hat{\psi} = \hat{\psi}^+$ (and the vector $\hat{\psi}\partial_\mu\hat{\psi}$ is not by itself conserved). This in turn signifies the absence of any special conservation law that would restrict the possible changes of the particle number. It is evident that such particles are, in any case, electrically neutral.

Particles of this kind are called *strictly neutral*, as distinct from electrically neutral particles that possess an antiparticle. While the latter can annihilate (converting into photons) only in pairs, strictly neutral particles can annihilate singly.

The structure of the ψ-operator (12.1) is the same as that of the operators (2.17)–(2.20) of the electromagnetic field. In this sense one can say that photons themselves are strictly neutral particles. In the case of the electromagnetic field, the Hermiticity of the operators was linked to the reality of the field strengths as measurable (in the classical limit) physical quantities. For the ψ-operators of particles, however, no such link exists, since they do not correspond to any directly measurable quantities at all.

The absence of a conserved 4-vector current is a general property of strictly neutral particles and is not connected with zero spin (it holds for photons as well). Physically it expresses the absence of the corresponding prohibitions on changes in particle number. From a formal standpoint there is a direct connection between the absence of a conserved current and the reality of the field—the Hermiticity of the operator $\hat{\psi}$.

The Lagrangian of a complex field

$$\hat{L} = \partial_\mu\hat{\psi}^+ \cdot \partial^\mu\hat{\psi} - m^2\hat{\psi}^+\hat{\psi} \tag{12.9}$$

is invariant under multiplication of the ψ-operator by an arbitrary phase factor, i.e. under transformations of the form

$$\hat{\psi} \to e^{i\alpha}\hat{\psi}, \qquad \hat{\psi}^+ \to e^{-i\alpha}\hat{\psi}^+ \tag{12.10}$$

(these are termed *gauge* transformations). In particular, the Lagrangian remains unchanged under an infinitesimally small gauge transformation

$$\hat{\psi} \to \hat{\psi} + i\delta\alpha\cdot\hat{\psi}, \qquad \hat{\psi}^+ \to \hat{\psi}^+ - i\delta\alpha\cdot\hat{\psi}^+. \tag{12.11}$$

Under an infinitesimally small variation of the "generalized coordinates" q, the Lagrangian undergoes the change

$$\delta\hat{L} = \sum\left(\frac{\partial\hat{L}}{\partial q}\delta q + \frac{\partial\hat{L}}{\partial q_{,\mu}}\delta q_{,\mu}\right) =$$

$$= \sum\left(\frac{\partial\hat{L}}{\partial q} - \frac{\partial}{\partial x^\mu}\frac{\partial\hat{L}}{\partial q_{,\mu}}\right)\delta q + \sum\frac{\partial}{\partial x^\mu}\left(\frac{\partial\hat{L}}{\partial q_{,\mu}}\delta q\right)$$

(summation over all q). The first term vanishes by virtue of the "equations of motion" (Lagrange's equations). Taking $\hat{\psi}$ and $\hat{\psi}^+$ as the "coordinates" q and setting

$$\delta\hat{\psi} = i\delta\alpha\cdot\hat{\psi}, \quad \delta\hat{\psi}^+ = -i\delta\alpha\cdot\hat{\psi}^+,$$

we get

$$\delta\hat{L} = i\delta\alpha \frac{\partial}{\partial x^\mu}\left(\hat{\psi}\frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}} - \hat{\psi}^+ \frac{\partial \hat{L}}{\partial \hat{\psi}^+_{,\mu}}\right).$$

It is apparent that the condition of invariance of the Lagrangian ($\delta\hat{L} = 0$) is equivalent to the continuity equation ($\partial_\mu \hat{j}^\mu$=0) for the 4-vector

$$\hat{j}^\mu = i\left(\hat{\psi}^+ \frac{\partial \hat{L}}{\partial \hat{\psi}^+_{,\mu}} - \hat{\psi}\frac{\partial \hat{L}}{\partial \hat{\psi}_{,\mu}}\right). \tag{12.12}$$

It is readily verified that for the Lagrangian (12.9) this formula yields the current (12.8).

We see, therefore, that in the mathematical framework of the theory the existence of a conserved current is tied to gauge invariance of the Lagrangian (*W. Pauli*, 1941). The Lagrangian (12.2) of a strictly neutral field, however, lacks this symmetry.

§ 13. The transformations C, P, T

Unlike four-dimensional inversion, a three-dimensional (spatial) inversion cannot be reduced to any rotation of the 4-coordinate system: the determinant of this transformation equals not $+1$ but -1. The symmetry properties of particles with respect to inversion (P-transformation) are therefore not dictated by the requirements of relativistic invariance [11].

Applied to a scalar wave function, the inversion operation amounts to the transformation

$$\hat{P}\psi(t, \mathbf{r}) = \pm\psi(t, -\mathbf{r}), \tag{13.1}$$

where the sign "+" or "−" on the right-hand side corresponds, respectively, to a true scalar or a pseudoscalar.

It is apparent from this that one must distinguish two aspects of the wave function's behavior under inversion. One of them is related to the dependence of the wave function on the coordinates. In nonrelativistic quantum mechanics only this aspect was considered—it leads to the notion of the parity of a state (which we shall henceforth call the *orbital parity*), characterizing the symmetry properties of the particle's motion. If a state possesses a definite orbital parity ($+1$ or -1), this means that

$$\psi(t, -\mathbf{r}) = \pm\psi(t, \mathbf{r}).$$

The second aspect is the behavior of the wave function at a given point (which one may conveniently take to be the coordinate origin) when the coordinate axes are inverted. This behavior gives rise to the concept of the *intrinsic parity of the particle*. For a spin-0 particle the two signs in definition (13.1) correspond to intrinsic-parity values of $+1$ and -1. The overall parity of a particle system equals the product of their individual intrinsic parities and the orbital parity associated with the relative motion.

The "intrinsic" symmetry properties of different particles manifest themselves, of course, only in the processes of their mutual transformations. The analogue of intrinsic

[11] The Lorentz group supplemented by spatial inversion is called the *extended Lorentz group* (as opposed to the original group, which does not contain P and which in this context is termed *proper*). The extended group includes all transformations that do not carry the t axis out of the corresponding half of the light cone.

parity in nonrelativistic quantum mechanics is the parity of a bound state of a composite system (for instance, a nucleus). From the standpoint of relativistic theory, which draws no fundamental distinction between composite and elementary particles, this kind of intrinsic parity is no different from the intrinsic parity of the particles that figure as elementary ones in nonrelativistic theory. In the nonrelativistic domain, where the latter behave as immutable objects, their intrinsic symmetry properties are unobservable, and consequently any consideration of them would be devoid of physical content.

Within the second-quantization formalism, the intrinsic parity is expressed through the behavior of the ψ-operators under inversion. For scalar and pseudoscalar fields the transformation laws read

$$P : \hat{\psi}(t, \mathbf{r}) \to \pm\hat{\psi}(t, -\mathbf{r}). \tag{13.2}$$

The very meaning of the action of inversion on the ψ-operator must, however, be formulated as a definite transformation of the particle annihilation and creation operators—one such that the change (13.2) results from it. It is easy to see that the required transformation is

$$P : \hat{a}_{\mathbf{p}} \to \pm\hat{a}_{-\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \pm\hat{b}_{-\mathbf{p}} \tag{13.3}$$

(and likewise for the conjugate operators). Indeed, carrying out this substitution in the operator

$$\hat{\psi}(t, \mathbf{r}) = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}} e^{-i\omega t + i\mathbf{pr}} + \hat{b}^{+}_{\mathbf{p}} e^{i\omega t - i\mathbf{pr}} \right) \tag{13.4}$$

and then relabeling the summation variable ($\mathbf{p} \to -\mathbf{p}$), we bring it to the form $\pm\hat{\psi}(t, -\mathbf{r})$. Thus, denoting by $\hat{\psi}^P(t, \mathbf{r})$ the operator in which transformation (13.3) has been performed, one may write

$$\hat{\psi}^P(t, \mathbf{r}) = \pm\hat{\psi}(t, -\mathbf{r}). \tag{13.5}$$

Note that transformation (13.3) has a perfectly natural form: inversion reverses the sign of the polar vector $\mathbf{p}$, so that particles with momentum $\mathbf{p}$ are replaced by particles with momentum $-\mathbf{p}$.

In (13.3) the operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ both transform with either the upper or the lower signs. In the second-quantization formalism this expresses the equality of the intrinsic parities of particle and antiparticle (with spin 0). This equality is in itself already obvious from the fact that particles and antiparticles (of spin 0) are described by the very same (scalar or pseudoscalar) wave functions.

In relativistic theory there also arises a symmetry with respect to a transformation that has no counterpart in nonrelativistic theory; it is called *charge conjugation* (C-transformation). If all operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ are mutually interchanged:

$$C : \hat{a}_{\mathbf{p}} \to \hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \hat{a}_{\mathbf{p}} \tag{13.6}$$

(i.e. particles and antiparticles are interchanged), then $\hat{\psi}$ goes over into the "charge-conjugate" operator $\hat{\psi}^C$, with

$$\hat{\psi}^C(t, \mathbf{r}) = \hat{\psi}^{+}(t, \mathbf{r}) \tag{13.7}$$

This equality expresses the symmetry with which the notions of particles and antiparticles enter the theory.

Let us note that the definition of the charge-conjugation transformation contains a certain inessential formal arbitrariness. The content of the transformation is unaltered if one introduces into definition (13.6) an arbitrary phase factor:

$$\hat{a}_{\mathbf{p}} \to e^{i\alpha}\hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to e^{-i\alpha}\hat{a}_{\mathbf{p}}.$$

Then one would have

$$\hat{\psi} \to e^{i\alpha}\hat{\psi}^{+}, \quad \hat{\psi}^{+} \to e^{-i\alpha}\hat{\psi},$$

and a twofold application of this transformation would still yield the identity ($\hat{\psi} \to \hat{\psi}$). All such definitions are, however, equivalent to one another. Since the properties of the ψ-operators are unaffected by multiplication by a phase factor (cf. the end of the preceding section), one may simply relabel $\hat{\psi}$ as $\hat{\psi}e^{i\alpha/2}$ and then return to the definition of charge conjugation in the form (13.6),(13.7).

Because charge conjugation replaces a particle by a non-identical antiparticle, it does not in the general case give rise to any new characteristic of a particle or a system of particles as such.

Systems containing equal numbers of particles and antiparticles constitute an exception. The operator $\hat{C}$ carries such a system into itself, so that eigenstates exist with eigenvalues $C = \pm 1$ (which follow from $\hat{C}^2 = 1$). When describing charge symmetry one may treat particle and antiparticle as two distinct "charge states" of a single particle, distinguished by the charge quantum number $Q = \pm 1$. The system's wave function then factorizes into an orbital part and a "charge" part, and must be symmetric with respect to the simultaneous interchange of all variables (spatial and charge) belonging to any pair of particles. The symmetry character of the "charge" factor fixes the charge parity of the system (see the Problem) [12].

The concept of charge parity, which arises naturally for "strictly neutral" systems, should apply also to strictly neutral "elementary" particles. In the second-quantization formalism this concept is described by the relation

$$\hat{\psi}^{C} = \pm\hat{\psi}; \tag{13.8}$$

the signs "+" and "−" correspond to charge-even and charge-odd particles.

In § 11 it was pointed out that relativistic invariance must also entail invariance under 4-inversion. For the operator of a scalar (in the sense of 4-rotations) field this means that under such a transformation one must have

$$\hat{\psi}(t, \mathbf{r}) = \hat{\psi}(-t, -\mathbf{r})$$

always with the same sign "+" on the right-hand side. In terms of the operators $\hat{a}_{\mathbf{p}}$, $\hat{b}_{\mathbf{p}}$, the conversion of $\hat{\psi}(t, \mathbf{r})$ into $\hat{\psi}(-t, -\mathbf{r})$ is accomplished by interchanging the coefficients of e^{-ipx} and e^{ipx} in (13.4), i.e. by the replacement

$$\hat{a}_{\mathbf{p}} \to \hat{b}^{+}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \hat{a}^{+}_{\mathbf{p}} \tag{13.9}$$

[12] In these arguments we have in mind particles of spin 0. The method of analysis described here generalizes directly to other cases—see, for example, the Problem in § 27.

Since it replaces the a-operators by b-operators, this transformation incorporates the mutual interchange of particles and antiparticles. We see that in relativistic theory there naturally arises a requirement of invariance under a transformation in which spatial inversion (P) and time reversal (T) are accompanied by charge conjugation (C); this statement is known as the *CPT theorem*[13].

In this connection it is, however, appropriate to stress that although the arguments presented here and in §§ 11, 12 appear to be a natural development of the concepts of ordinary quantum mechanics and classical relativity, the results obtained in this way go beyond that framework both in form (ψ-operators that simultaneously contain creation and annihilation operators for particles) and in substance (particles and antiparticles). These results cannot therefore be regarded as a matter of pure logical necessity. They embody new physical principles whose correctness can be judged only by experiment.

If we denote by $\hat{\psi}^{CPT}(t, \mathbf{r})$ the operator (13.4) in which the transformation (13.9) has been carried out, one may write

$$\hat{\psi}^{CPT}(t, \mathbf{r}) = \hat{\psi}(-t, -\mathbf{r}). \tag{13.10}$$

Having thus formulated 4-inversion as the transformation (13.9), we thereby establish for the ψ-operator also the formulation of the time-reversal transformation: together with the CP transformation (called the *combined inversion*) it must yield (13.9). Taking into account the definitions (13.3) and (13.6), we therefore find

$$T: \hat{a}_{\mathbf{p}} \to \pm\hat{a}^{+}_{-\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \to \pm\hat{b}^{+}_{-\mathbf{p}} \tag{13.11}$$

(the signs "±" match those in (13.3)). The meaning of this transformation is perfectly natural: time reversal not only converts motion with momentum $\mathbf{p}$ into motion with momentum $-\mathbf{p}$, but also interchanges the initial and final states in matrix elements; the annihilation operators for particles with momenta $\mathbf{p}$ are therefore replaced by creation operators for particles with momenta $-\mathbf{p}$. Performing the substitution (13.11) in (13.4) and relabeling the summation variable ($\mathbf{p} \to -\mathbf{p}$), we find that[14]

$$\hat{\psi}^{T}(t, \mathbf{r}) = \pm\hat{\psi}^{+}(-t, \mathbf{r}). \tag{13.12}$$

This result mirrors the standard time-reversal prescription of quantum mechanics: when a state has wave function $\psi(t, \mathbf{r})$, the "temporally reversed" state is represented by $\psi^*(-t, \mathbf{r})$; complex conjugation is needed in order to recover the "proper" character of the time dependence, which the sign flip of t would otherwise spoil (*E. P. Wigner*, 1932).

Since the transformation T (and with it CPT) interchange initial and final states, the concepts of eigenstates and eigenvalues have no meaning for them. They therefore do not lead to new characteristics of particles as such. The consequences to which they lead when applied to scattering processes will be discussed in §§ 69, 71.

Let us examine how the operator 4-vector of the current $\hat{j}^{\mu}$ (12.8) transforms under C, P, and T. Transformation (13.2) together with the replacement $(\partial_0, \partial_i) \to (\partial_0, -\partial_i)$

[13] It was formulated by *Lüders* (G. Lüders, 1954) and *Pauli* (W. Pauli, 1955).

[14] If one were to define the operation T independently of the other transformations, the same arbitrariness in the choice of a phase factor would arise as for the operation C. The requirement of CPT symmetry, however, leaves the phase-factor choice free in only one of the two transformations, C or T.

gives

$$P : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (\hat{j}^0, -\hat{\mathbf{j}})_{t,-\mathbf{r}}, \tag{13.13}$$

as it must for a true 4-vector. Transformation (13.7) would simply give

$$C : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (-\hat{j}^0, -\hat{\mathbf{j}})_{t,\mathbf{r}}, \tag{13.14}$$

were the operators $\hat{\psi}$ and $\hat{\psi}^+$ commutative. The noncommutativity of these operators arises, however, solely from the noncommutativity of $\hat{a}_{\mathbf{p}}$ and $\hat{a}^+_{\mathbf{p}}$ (or of $\hat{b}_{\mathbf{p}}$ and $\hat{b}^+_{\mathbf{p}}$) at the same $\mathbf{p}$; but by virtue of the commutation rules (11.4) the interchange of these operators merely introduces terms independent of the occupation numbers, i.e. of the field state. Discarding these terms as inessential (just as in (11.5),(11.6)), we recover rule (13.14), which has a natural meaning: by replacing particles with antiparticles, charge conjugation reverses the sign of every component of the 4-current.

Since the time-reversal operation is linked with the transposition of initial and final states, its application to a product of operators reverses the order of the factors. Thus,

$$\left(\hat{\psi}^+ \partial_\mu \hat{\psi}\right)^T = \left(\partial_\mu \hat{\psi}\right)^T \left(\hat{\psi}^+\right)^T .$$

In the present case, however, this is immaterial: by virtue of the commutativity of the ψ-operators (in the sense indicated above), reverting to the original order of the factors does not affect the result. Noting also that under time reversal $(\partial_0, \partial_i) \to (-\partial_0, \partial_i)$, we find the transformation rule for the current:

$$T : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (\hat{j}^0, -\hat{\mathbf{j}})_{-t,\mathbf{r}}. \tag{13.15}$$

The three-dimensional vector $\mathbf{j}$ changes sign in accordance with the classical meaning of this quantity.

Finally, under the CPT transformation we have

$$CPT : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \to (-\hat{j}^0, -\hat{\mathbf{j}})_{-t,-\mathbf{r}}, \tag{13.16}$$

in accord with the meaning of this operation as a 4-inversion. Let us stress in this connection that, since 4-inversion reduces to a rotation of the 4-coordinate system, with respect to it there is no distinction at all between two types (true and pseudo) of 4-tensors of any rank.

Up to now we have been considering free particles. But the quantum numbers of parity acquire real significance only when one examines interacting particles, where definite selection rules are associated with them, permitting or forbidding particular processes. Such significance, however, can attach only to conserved quantities—eigenvalues of operators that commute with the Hamiltonian of the interacting particles.

Relativistic invariance guarantees that the CPT-transformation operator commutes with the Hamiltonian in every case. As for the individual transformations C and P (together with T), experiment demonstrates that electromagnetic and strong interactions respect these symmetries, so that the associated parity quantum numbers are conserved within those interactions. In weak processes, however, these conservation laws break down [15].

[15] The suggestion that parity conservation might fail in weak interactions was first put forward by *Lee* and *Yang* (*T. D. Lee*, *C. N. Yang*, 1956). Earlier still, the broader idea that P- and T-invariance of physical laws is not mandatory was advanced by *Dirac* (1949).

Anticipating later developments, we point out that the operator governing the coupling of charged particles to the electromagnetic field is formed as the product of the operator 4-vectors $\hat{A}$ and $\hat{j}$. Because charge conjugation flips the sign of $\hat{j}$, electromagnetic-interaction invariance under this transformation requires that $\hat{A}$ likewise change sign. Put differently, photons are charge-odd particles.

The stated behavior of the operators $\hat{A}$ is in agreement with the properties of the 4-potential in classical theory. Indeed, from the transformations

$$C : (\hat{A}_0, \hat{\mathbf{A}}) \to (-\hat{A}_0, -\hat{\mathbf{A}})_{t,\mathbf{r}},$$

$$P : (\hat{A}_0, \hat{\mathbf{A}}) \to (\hat{A}_0, -\hat{\mathbf{A}})_{t,-\mathbf{r}},$$

$$CPT : (\hat{A}_0, \hat{\mathbf{A}}) \to (-\hat{A}_0, -\hat{\mathbf{A}})_{-t,-\mathbf{r}},$$

it follows that

$$T : (\hat{A}_0, \hat{\mathbf{A}}) \to (\hat{A}_0, -\hat{\mathbf{A}})_{-t,\mathbf{r}},$$

which corresponds to the classical rule for transforming the electromagnetic-field potentials under time reversal.

The requirement of CPT invariance does not impose any restrictions on the properties of particles in themselves. It does, however, lead to a definite connection between the properties of particles and antiparticles. This includes, first and foremost, the equality of their masses—a fact already clear from the connection, set out in § 11, between 4-inversion and the very origin of the particle–antiparticle concept.

Furthermore, CPT invariance implies that the proportionality coefficients relating the electric and magnetic moment vectors to the spin vector differ for particle and antiparticle only in sign. Indeed, the magnetic moment changes sign under C and T transformations and (being an axial vector) is P-invariant. Consequently, the CPT transformation, while converting a particle into its antiparticle, does not at the same time change the sign of the magnetic moment; the spin vector, however, does change sign. The same applies to the electric moment, which remains unchanged under time reversal and changes sign under C-transformation and (by the properties of a polar vector) under spatial inversion.

The requirements of P- or T-invariance (if they are obeyed), on the other hand, constrain the properties of each individual particle: they forbid the particle from possessing an electric dipole moment. Indeed, the only vector that can be constructed for an elementary particle at rest from its ψ-operators is the vector of its spin operator. This vector is P-even and T-odd; it can therefore determine only the magnetic, but not the electric moment. Let us emphasize that for this prohibition it suffices to require merely one of the two—P- or T-invariance.

Problem

Determine the charge parity and spatial parity of a system consisting of a spin-0 particle and its antiparticle, with orbital angular momentum l of the relative motion.

Solution. Permutation of the particle coordinates is equivalent to inversion (about the center of inertia) and therefore multiplies the orbital function by $(-1)^l$; permutation of the charge variables is equivalent to charge conjugation and multiplies the "charge" factor in the wave function by the sought quantity C. From the condition $C(-1)^l = 1$ we have

$$C = (-1)^l.$$

The spatial parity P of the system is the product of the orbital parity with the intrinsic parities of the two particles. Because particle and antiparticle share the same intrinsic parity, here P simply equals the orbital parity: $P = (-1)^l$.

§ 14. Wave equation for spin-1 particles

In its rest frame a particle of spin 1 is characterized by a wave function with three components—a three-dimensional vector (one therefore often speaks of a *vector* particle). Viewed four-dimensionally, these three components can be either the spatial part of a (spacelike) 4-vector ψ^μ, or the mixed components of an antisymmetric 4-tensor of rank two, $\psi^{\mu\nu}$, for which the time (ψ^0) and purely spatial (ψ^{ik}) components vanish in the rest frame [16].

The wave equation—a differential relation between the quantities ψ^μ and $\psi^{\mu\nu}$—is established by the relations that we write as

$$i\psi_{\mu\nu} = \hat{p}_\mu\psi_\nu - \hat{p}_\nu\psi_\mu, \tag{14.1}$$

$$im^2\psi_\mu = \hat{p}^\nu\psi_{\mu\nu}, \tag{14.2}$$

where $\hat{p} = i\partial$ (*A. Proca*, 1936). Applying the operator $\hat{p}^\mu$ to both sides of equation (14.2), we obtain (on account of the antisymmetry of $\psi_{\mu\nu}$)

$$\hat{p}^\mu\psi_\mu = 0. \tag{14.3}$$

From (14.1, 14.2) one can eliminate $\psi_{\mu\nu}$ by inserting the first equation into the second. With the aid of (14.3) this yields

$$(\hat{p}^2 - m^2)\psi_\mu = 0, \tag{14.4}$$

from which it is again apparent (cf. § 10) that m is the particle mass. A free spin-1 particle can therefore be described entirely by a single 4-vector ψ^μ whose components satisfy the second-order equation (14.4) together with the supplementary condition (14.3), which removes from ψ^μ the part belonging to spin 0.

In the rest frame, where ψ_μ is independent of the spatial coordinates, we find that $\hat{p}^0\psi_0 = 0$. Since at the same time $\hat{p}^0\psi_0 = m\psi_0$, it follows that $\psi_0 = 0$ in the rest frame, as required. Together with ψ_0 the quantities ψ_{ik} also vanish.

A spin-1 particle can possess different intrinsic parities, depending on whether ψ^μ is a true vector or a pseudovector. In the first case

$$\hat{P}\psi^\mu = \left(\psi^0, -\psi^i\right),$$

[16] Looking ahead, let us mention that the set consisting of the 4-vector ψ_μ and the 4-tensor $\psi^{\nu\mu}$ corresponds to the set of four-dimensional second-rank spinors $\xi^{\alpha\beta}$, $\eta_{\dot{\alpha}\dot{\beta}}$, $\zeta^{\alpha\dot{\beta}}$, where $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ are symmetric spinors that transform into each other under inversion (see § 19).

and in the second

$$\hat{P}\psi^{\mu} = \left(-\psi^{0}, \psi^{i}\right).$$

Equations (14.1),(14.2) can be derived from a variational principle with the Lagrangian

$$L = (1/2)\,\psi_{\mu\nu}\psi^{\mu\nu*} - (1/2)\psi^{\mu\nu*}(\partial_{\mu}\psi_{\nu} - \partial_{\nu}\psi_{\mu}) - \\ - (1/2)\psi^{\mu\nu}(\partial_{\mu}\psi^{*}_{\nu} - \partial_{\nu}\psi^{*}_{\mu}) + m^{2}\psi_{\mu}\psi^{\mu*}. \quad (14.5)$$

The independent generalized coordinates here are ψ_{μ}, ψ^{*}_{μ}, $\psi_{\mu\nu}$, $\psi^{*}_{\mu\nu}$ [17].

For obtaining the energy–momentum tensor, formula (10.11) is not entirely convenient in the present case, since it would lead to a non-symmetric tensor requiring additional symmetrization. One may instead use the formula

$$\frac{1}{2}\,T_{\mu\nu}\sqrt{-g} = -\frac{\partial}{\partial x^{\lambda}}\frac{\partial\sqrt{-g}\,L}{\partial g^{\mu\nu}_{,\lambda}} + \frac{\partial\sqrt{-g}\,L}{\partial g^{\mu\nu}}, \quad (14.6)$$

in which it is assumed that L is written in a form valid for arbitrary curvilinear coordinates (see II, § 94). If L involves only the components of the metric tensor $g_{\mu\nu}$ itself (but not their coordinate derivatives), the formula simplifies to

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}}\frac{\partial\sqrt{-g}\,L}{\partial g^{\mu\nu}} = 2\frac{\partial L}{\partial g^{\mu\nu}} - g_{\mu\nu}L$$

(recall that $d\ln g = -g_{\mu\nu}dg^{\mu\nu}$).

Since the differentiation in formula (14.6) is not performed with respect to ψ_{μ} or $\psi_{\mu\nu}$, when applying it there is no need to treat these quantities as independent; one may directly use relation (14.1) and rewrite the Lagrangian (14.5) as

$$L = (-1/2)\,\psi_{\mu\nu}\psi^{*}_{\lambda\rho}g^{\mu\lambda}g^{\nu\rho} + m^{2}\psi_{\mu}\psi^{*}_{\nu}g^{\mu\nu}. \quad (14.7)$$

Then

$$T_{\mu\nu} = -\psi_{\mu\lambda}\psi_{\nu}{}^{\lambda*} - \psi^{*}_{\mu\lambda}\psi_{\nu}{}^{\lambda} + m^{2}\left(\psi^{*}_{\mu}\psi_{\nu} + \psi^{*}_{\nu}\psi_{\mu}\right) + \\ + g_{\mu\nu}\left((1/2)\,\psi_{\lambda\rho}\psi^{\lambda\rho*} - m^{2}\psi^{*}_{\lambda}\psi^{\lambda}\right). \quad (14.8)$$

In particular, the energy density is given by a manifestly positive expression

$$T_{00} = (1/2)\,\psi_{ik}\psi^{*}_{ik} + \psi_{0i}\psi^{*}_{0i} + m^{2}\left(\psi_{0}\psi^{*}_{0} + \psi_{i}\psi^{*}_{i}\right). \quad (14.9)$$

The conserved 4-vector of the current density reads

$$j^{\mu} = i\left(\psi^{\mu\nu*}\psi_{\nu} - \psi^{\mu\nu}\psi^{*}_{\nu}\right). \quad (14.10)$$

[17] Had we carried out the variation with respect to ψ_{μ} alone (assuming from the outset that $\psi_{\mu\nu}$ is expressed through ψ_{μ} via (14.1)), equation (14.3) would have to be imposed as an auxiliary condition unrelated to the variational principle.

It can be obtained from formula (12.12) by differentiating the Lagrangian (14.5) with respect to the derivatives $\partial_\mu \psi_\nu$. In particular,

$$j^0 = i\left(\psi^{0k*}\psi_k - \psi^{0k}\psi_k^*\right) \tag{14.11}$$

and is not a positive-definite quantity.

A plane wave normalized to one particle in volume $V = 1$:

$$\psi_\mu = \frac{1}{\sqrt{2\varepsilon}} u_\mu e^{-ipx}, \quad u_\mu u^{\mu *} = -1, \tag{14.12}$$

where u_μ is a unit polarization 4-vector obeying (by virtue of (14.3)) the four-dimensional transversality condition

$$u_\mu p^\mu = 0. \tag{14.13}$$

Indeed, substituting the function (14.12) into (14.9) and (14.11) gives

$$T_{00} = -2\varepsilon^2 \psi_\mu \psi^{\mu *} = \varepsilon, \qquad j^0 = 1.$$

In contrast to the photon, a vector particle of nonzero mass possesses three independent polarization directions. For the corresponding amplitudes see (16.21).

For partially polarized vector particles the polarization density matrix is introduced in such a way that for a pure state it becomes simply the product

$$\rho_{\mu\nu} = u_\mu u_\nu^*$$

(analogously to expression (8.7) for photons). According to (14.12), (14.13) it satisfies the conditions

$$p^\mu \rho_{\mu\nu} = 0, \qquad \rho_\mu^\mu = -1. \tag{14.14}$$

For unpolarized particles the matrix $\rho_{\mu\nu}$ must have the form $a g_{\mu\nu} + b p_\mu p_\nu$. Fixing the coefficients a and b from (14.14), one finds

$$\rho_{\mu\nu} = -\frac{1}{3}\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2}\right). \tag{14.15}$$

Quantization of the vector-particle field proceeds in the same way as for the scalar case, and there is no need to repeat the whole argument anew. The ψ-operators of the vector field take the form

$$\begin{aligned}
\hat{\psi}_\mu &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{b}_{\mathbf{p}\alpha}^{+} u_\mu^{(\alpha)*} e^{ipx}\right), \\
\hat{\psi}_\mu^{+} &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\alpha}^{+} u_\mu^{(\alpha)*} e^{ipx} + \hat{b}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx}\right),
\end{aligned} \tag{14.16}$$

where the index α labels the three independent polarizations.

The positive definiteness of expression (14.9) for T_{00} and the indefiniteness of j^0 (14.11) lead, just as in the scalar case, to the requirement of Bose quantization.

There exists a close connection between the properties of the strictly neutral vector field and the electromagnetic field. A neutral vector field is described by a Hermitian ψ-operator:

$$\hat{\psi}_\mu = \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{c}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{c}^+_{\mathbf{p}\alpha} u_\mu^{(\alpha)*} e^{ipx} \right). \tag{14.17}$$

The Lagrangian of this field is

$$\hat{L} = \frac{1}{4} \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} - \frac{1}{2} \hat{\psi}^{\mu\nu} (\partial_\mu \hat{\psi}_\nu - \partial_\nu \hat{\psi}_\mu) + \frac{1}{2} m^2 \hat{\psi}_\mu \hat{\psi}^\mu. \tag{14.18}$$

The electromagnetic field corresponds to the case $m = 0$. The 4-vector ψ^μ then becomes the 4-potential A^μ, and the 4-tensor $\psi^{\mu\nu}$ becomes the field tensor $F^{\mu\nu}$, related to the potential through definition (14.1). Equation (14.2) turns into $\partial^\nu \psi_{\mu\nu} = 0$, which corresponds to the second pair of Maxwell's equations. Condition (14.3) no longer follows from it and thus ceases to be obligatory. In the absence of the supplementary condition there is no need to treat $\hat{\psi}_\mu$ and $\hat{\psi}_{\mu\nu}$ as independent "coordinates" in the Lagrangian, and (14.18) reduces to

$$\hat{L} = -(1/4) \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} \tag{14.19}$$

in agreement with the familiar classical expression for the electromagnetic-field Lagrangian. This Lagrangian, together with the tensor $\hat{\psi}_{\mu\nu}$, is invariant under an arbitrary gauge transformation of the "potentials" $\hat{\psi}_\mu$. The connection between this circumstance and the vanishing mass is clearly visible: the Lagrangian (14.18) lacks this property on account of the term $m^2 \hat{\psi}_\mu \hat{\psi}^\mu$.

§ 15. Wave equation for particles with higher integer spins

Since the wave equations (14.3, 14.4) follow directly from specifying the mass and spin of the particle, the practical use of the Lagrangian comes down not so much to deriving these equations as to constructing the expressions for the energy, momentum, and charge of the field.

For this purpose, as already noted, one may use expression (14.7) in place of (14.5), and the former can be further transformed. Using (14.1) and rewriting the Lagrangian (14.7) as

$$L = -\left(\partial_\mu \psi^*_\nu\right)\left(\partial^\mu \psi^\nu\right) + \left(\partial_\nu \psi^*_\mu\right)\left(\partial^\mu \psi^\nu\right) + m^2 \psi_\mu \psi^{\mu *} = \tag{15.1}$$

$$= -\left(\partial_\mu \psi^*_\nu\right)\left(\partial^\mu \psi^\nu\right) + m^2 \psi^*_\mu \psi^\mu + \partial_\nu \left(\psi^*_\mu \partial^\mu \psi^\nu\right) - \psi^*_\mu \partial^\mu \partial_\nu \psi^\nu.$$

By virtue of (14.3) the last term vanishes, while the penultimate one is a total derivative. Dropping it, we arrive at the Lagrangian

$$L' = -\left(\partial_\mu \psi^*_\nu\right)\left(\partial^\mu \psi^\nu\right) + m^2 \psi^*_\mu \psi^\mu.$$

It has the same structure as the Lagrangian (10.9) for a spin-0 particle, differing only in that the scalar ψ is replaced by the 4-vector ψ_μ and the overall sign is reversed. The sign change is due to the spacelike character of ψ_μ, so that $\psi_\mu\psi^{\mu *} < 0$, whereas for a scalar particle $\psi\psi^* > 0$.

If one constructs the energy–momentum 4-tensor and the current 4-vector with the help of Lagrangian (15.1), one obtains expressions of the same form as expressions (10.12) and (10.18) for the scalar field:

$$T_{\mu\nu} = -\partial_\mu\psi^{\lambda *} \cdot \partial_\nu\psi_\lambda - \partial_\nu\psi^{\lambda *} \cdot \partial_\mu\psi_\lambda - L' g_{\mu\nu}, \tag{15.2}$$

$$j_\mu = -i\left[\psi^*_\lambda\partial_\mu\psi^\lambda - \left(\partial_\mu\psi^*_\lambda\right)\psi^\lambda\right]. \tag{15.3}$$

Their difference from (14.8) and (14.10) likewise reduces to total derivatives. But the local values of these quantities lack (as was already stressed) any deep physical significance. Only the volume integrals P_μ (10.15) and Q (10.19) matter, and they coincide for either choice of $T_{\mu\nu}$ and j_μ.

This method of description extends straightforwardly to particles of arbitrary (integer) spin. The wave function of a particle with spin s is an irreducible 4-tensor of rank s, that is, a tensor symmetric in all of its indices and vanishing upon contraction over any pair of them:

$$\psi_{\ldots\mu\ldots\nu\ldots} = \psi_{\ldots\nu\ldots\mu\ldots}, \quad \psi_{\ldots\mu\ldots}{}^{\mu}{}_{\ldots} = 0. \tag{15.4}$$

This tensor must obey the supplementary four-dimensional transversality condition

$$\hat{p}^\mu\psi_{\ldots\mu\ldots} = 0, \tag{15.5}$$

and each of its components must satisfy the second-order equation

$$(\hat{p}^2 - m^2)\psi_{\ldots} = 0. \tag{15.6}$$

In the rest frame, condition (15.5) causes every component of the 4-tensor that carries a 0 among its indices to vanish. In other words, the wave function in the rest frame (i.e. in the nonrelativistic limit) reduces, as expected, to an irreducible 3-tensor of rank s with $2s + 1$ independent components.

The Lagrangian, the energy–momentum tensor, and the current vector for particles of spin s differ from (15.1)–(15.3) solely by the replacement of ψ_λ with $\psi_{\lambda\mu\ldots}$.

The normalized plane wave is

$$\psi^{\mu\nu\cdots} = \frac{1}{\sqrt{2\varepsilon}}\, u^{\mu\nu\cdots} e^{-ipx}, \quad u^*_{\mu\nu\ldots} u^{\mu\nu\cdots} = -1, \tag{15.7}$$

with the wave amplitude satisfying the conditions

$$u^{\cdots\mu\cdots} p_\mu = 0. \tag{15.8}$$

There are $2s + 1$ independent polarization states.

Field quantization proceeds as an obvious generalization of the spin-0 or spin-1 cases.

The scheme presented above is entirely sufficient for the stated purpose—describing the field of free particles. The situation changes, however, if one poses the problem of describing the interaction of particles with the electromagnetic field. Such an interaction would need to be incorporated into a Lagrangian from which all equations could be obtained without having to impose supplementary conditions separately. In practice, however, it turns out that this approach to describing the interaction works only for electrons—particles of spin $1/2$ (see § 32). For other spins the problem would therefore be of purely methodological interest.

Let us note that for all (integer and half-integer) spins $s > 1$ it proves impossible to formulate a variational principle using only a single function (tensorial or spinorial) whose rank matches the given spin. For this purpose it becomes necessary to introduce, as auxiliary quantities, tensorial or spinorial objects of lower rank as well. The Lagrangian is then chosen so that these auxiliary quantities automatically vanish by virtue of the free-particle field equations that follow from the variational principle [18].

§ 16. Helicity states of a particle[19]

Within relativistic theory, the orbital angular momentum $\mathbf{l}$ and the spin $\mathbf{s}$ of a particle in motion lack individual conservation laws—only their sum, the total angular momentum $\mathbf{j} = \mathbf{l} + \mathbf{s}$, is conserved. Consequently, the spin projection onto a given direction (the z axis) is also nonconserved, rendering it unsuitable for labeling the polarization (spin) states of a particle in motion.

What is conserved, however, is the projection of the spin on the direction of the momentum: since $\mathbf{l} = [\mathbf{rp}]$, the product $\mathbf{sn}$ coincides with the conserved product $\mathbf{jn}$ ($\mathbf{n} = \mathbf{p}/|\mathbf{p}|$). This quantity is called the *helicity* [20] (we have already considered it for the photon in § 8). Its eigenvalues will be denoted by λ ($\lambda = -s, \dots, +s$), and states of the particle with definite values of λ will be called helicity states.

Let $\psi_{\mathbf{p}\lambda}$ be the wave function (a plane wave) describing a state of the particle with definite $\mathbf{p}$ and λ, and let $u^{(\lambda)}(\mathbf{p})$ be its amplitude; for brevity we suppress the component indices of this function (for integer spin these are 4-tensor indices).

We saw in the preceding sections that a relativistic description of particles with nonzero (integer) spin requires introducing a wave function whose number of components exceeds $2s+1$. The number of independent components, however, remains equal to $2s+1$; the "extra" components are eliminated by imposing supplementary conditions, by virtue of which these components vanish in the rest frame (in the next chapter we shall see the same for half-integer s).

According to the transformation rules for angular momentum (see II, § 14), the helicity is invariant under Lorentz transformations that leave unchanged the direction of $\mathbf{p}$ onto which the angular momentum is projected. Therefore the number λ retains its meaning as a quantum number under such transformations, and to study the symmetry properties of helicity states one may use a reference frame in which $|\mathbf{p}| \ll m$ (in the limit, the rest frame). Then $\psi_{\mathbf{p}\lambda}$ reduces to a nonrelativistic $(2s+1)$-component wave function. Let us denote its amplitude by $w^{(\lambda)}(\mathbf{n})$, indicating as the argument the direction $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ along which the angular momentum is quantized. The amplitude $w^{(\lambda)}$ is an eigenfunction

[18] See *Fierz M., Pauli W.*//Proc. Roy. Soc. — 1939. — V. A 173. — P. 211. In that work the program indicated here was carried out for particles of spin $3/2$ and 2.

[19] The content of this section applies to particles of any (integer or half-integer) spin.

[20] In the Russian literature the term is *spiralnost'*.

of the operator $\mathbf{n}\hat{\mathbf{s}}$:

$$\mathbf{n}\hat{\mathbf{s}}\, w^{(\lambda)}(\mathbf{n}) = \lambda w^{(\lambda)}(\mathbf{n}). \tag{16.1}$$

In the spinor representation $w^{(\lambda)}$ is a contravariant symmetric spinor of rank $2s$; by the correspondence formulas (57.2) (see III) its components can also be labeled by the values of the spin projection σ on a fixed axis z [21].

In the momentum representation the wave functions of the states under consideration essentially coincide with the amplitudes $u^{(\lambda)}(\mathbf{p})$. Specifically:

$$\psi_{\mathbf{p}\lambda}(\mathbf{k}) = u^{(\lambda)}(\mathbf{k})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) = u^{(\lambda)}(\mathbf{p})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}), \tag{16.2}$$

where the momentum as an independent variable is denoted $\mathbf{k}$, in contrast to its eigenvalue $\mathbf{p}$, and $\boldsymbol{\nu} = \mathbf{k}/|\mathbf{k}|$, in contrast to $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ [22]. In the nonrelativistic limit

$$\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}) = w_{\sigma}^{(\lambda)}(\boldsymbol{\nu})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) = w^{(\lambda)}(\mathbf{n})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}). \tag{16.3}$$

In more detail this expression should be written as

$$\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}, \sigma) = w_{\sigma}^{(\lambda)}(\boldsymbol{\nu})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}),$$

where the discrete independent variable σ is also displayed explicitly.

The helicity operator $\hat{\mathbf{s}}\mathbf{n}$ is commutative with $\hat{j}_z$ and $\hat{\mathbf{j}}^2$. This follows from the fact that the angular-momentum operator corresponds to an infinitesimal coordinate rotation, while the dot product of any two vectors remains unchanged under arbitrary rotations. Consequently, stationary states exist in which the particle has simultaneously well-defined angular momentum j, its projection $j_z = m$, and helicity λ. We shall refer to such states as *spherical helicity states*.

We now find the wave functions of these states in the momentum representation. The construction proceeds by direct analogy with the expressions derived in III, § 103 for the wave functions of a symmetric top. Those results were obtained from the transformation formulas for wave functions under finite rotations (see III, § 58). Since that derivation relies exclusively on rotational-symmetry properties, it applies to momentum-representation functions just as it does to coordinate-space functions.

Along with the space-fixed coordinate system xyz (with respect to which the functions $\psi_{jm\lambda}$ are written), let us also introduce a "body-fixed" frame $\xi\eta\zeta$ with the ζ axis along the direction of $\boldsymbol{\nu}$. Without repeating the corresponding reasoning (cf. the derivation of formula (103.8) (see III)), we write

$$\psi_{jm\lambda}(\mathbf{k}) = \psi_{j\lambda}^{(0)} D_{\lambda m}^{(j)}(\boldsymbol{\nu}),$$

[21] The reasoning given here (together with the listing of allowed values of λ) pertains to particles whose mass differs from zero. For a particle of zero mass no rest frame exists, and the helicity admits only the two values $\lambda = \pm s$. This restriction traces back to the circumstance noted already in § 8: states of such a particle fall into representations of the axial-symmetry group, which permits at most a twofold degeneracy of levels (from the viewpoint of the wave equation this signifies that upon passing to the limit $m \to 0$ the system of equations for a spin-s particle separates into independent equations describing massless particles of spins s, $s - 1, \ldots$). In particular, for the photon $\lambda = \pm 1$, and the three-dimensional vectors $\mathbf{e}^{(\pm 1)}$ (8.2) assume the role of the corresponding amplitudes $w^{(\lambda)}$.

[22] Here the delta function $\delta^{(2)}$ is defined so that $\int \delta^{(2)}(\boldsymbol{\nu} - \mathbf{n})\, do_{\nu} = 1$. In (16.2) (and in the analogous case below, see (16.4)) the delta function ensuring the given value of the energy has been omitted.

where $\psi^{(0)}_{j\lambda}$ is the wave function in the "body-fixed" frame describing a state with a definite value of the ζ-projection of the angular momentum: $j_\zeta = \lambda$; in the momentum representation this function evidently coincides with the amplitude $u^{(\lambda)}$. The normalized (see below) wave function is

$$\psi_{jm\lambda}(\mathbf{k}) = \sqrt{\frac{2j+1}{4\pi}}\, D^{(j)}_{\lambda m}(\boldsymbol{\nu}) u^{(\lambda)}(\mathbf{k}). \tag{16.4}$$

Here, however, a question arises regarding the choice of phases, connected with the following ambiguity. The rotation of the system xyz relative to $\xi\eta\zeta$ is specified by three Euler angles α, β, γ; the direction of $\boldsymbol{\nu}$, on which alone the wave function of the particle can depend, is determined only by two spherical angles $\alpha \equiv \varphi, \beta \equiv \theta$. One must therefore adopt some convention for the angle γ. We shall set $\gamma = 0$, i.e. define $D^{(j)}_{\lambda m}(\boldsymbol{\nu})$ as

$$D^{(j)}_{\lambda m}(\boldsymbol{\nu}) = D^{(j)}_{\lambda m}(\varphi, \theta, 0) = e^{im\varphi} d^{(j)}_{\lambda m}(\theta). \tag{16.5}$$

By virtue of (58.21) (see III) the functions (16.5) satisfy the orthogonality and normalization conditions:

$$\int D^{(j_1)*}_{\lambda_1 m_1}(\boldsymbol{\nu}) D^{(j_2)}_{\lambda_2 m_2}(\boldsymbol{\nu}) \frac{do_\nu}{4\pi} = \frac{1}{2j+1}\, \delta_{j_1 j_2} \delta_{m_1 m_2} \tag{16.6}$$

($do_\nu = \sin\theta\, d\theta\, d\varphi$). Orthogonality in the index λ is guaranteed by the amplitude factor $u^{(\lambda)}$. The functions $\psi_{jm\lambda}$ are therefore mutually orthogonal, as required, across all three labels $jm\lambda$, and the prefactor adopted in (16.4) ensures their normalization according to

$$\int |\psi_{jm\lambda}|^2\, do_\nu = 1. \tag{16.7}$$

Here it is assumed that the amplitudes $u^{(\lambda)}$ are normalized to unity: $u^{(\lambda)} u^{(\lambda)*} = 1$.

Let us consider the behavior of helicity-state wave functions under coordinate inversion. The product of the polar vector $\boldsymbol{\nu}$ and the axial vector $\mathbf{j}$ is a pseudoscalar. It is therefore clear that upon inversion a state of helicity λ goes over into a state with $-\lambda$; one must, however, determine the phase factors in these transformations.

Under inversion $\boldsymbol{\nu} \to -\boldsymbol{\nu}$. The vector $\boldsymbol{\nu}$ is specified by two angles φ, θ, and the transformation $\boldsymbol{\nu} \to -\boldsymbol{\nu}$ is effected by the replacement $\varphi \to \varphi + \pi$, $\theta \to \pi - \theta$. This determines the axis ζ, but the position of the axes ξ and η, which also depends on the third Euler angle γ, remains unspecified; the transformation of θ and φ alone does not allow one to distinguish between an inversion and a rotation of the axis ζ in this sense. In terms of all three Euler angles the inversion is the transformation

$$\alpha \equiv \varphi \to \varphi + \pi, \quad \beta \equiv \theta \to \pi - \theta, \quad \gamma \to \pi - \gamma. \tag{16.8}$$

Therefore, if $D^{(j)}_{\lambda m}(\boldsymbol{\nu})$ is defined according to (16.5) (i.e. with $\gamma = 0$), and the substitution $\boldsymbol{\nu} \to -\boldsymbol{\nu}$ is understood as the result of an inversion, then

$$D^{(j)}_{\lambda m}(-\boldsymbol{\nu}) = D^{(j)}_{\lambda m}(\varphi + \pi, \pi - \theta, \pi). \tag{16.9}$$

Using formulas (58.9), (58.16), (58.18) (see III) we therefore find

$$D^{(j)}_{\lambda m}(-\boldsymbol{\nu}) = e^{i\lambda\pi} d^{(j)}_{\lambda m}(\pi - \theta) e^{im(\varphi+\pi)} =$$

$$= (-1)^{j-\lambda} e^{im\varphi} d^{(j)}_{-\lambda m}(\theta) = (-1)^{j-\lambda} D^{(j)}_{-\lambda m}(\boldsymbol{\nu}),$$

or

$$D^{(j)}_{\lambda m}(-\boldsymbol{\nu}) = (-1)^{-\lambda} D^{(j)}_{-\lambda m}(\boldsymbol{\nu}) \tag{16.10}$$

($j - \lambda$ is an integer).

An analogous formula for the spinor $w^{(\lambda)}$ can be obtained by noting that its components $w^{(\lambda)}_\sigma$ coincide, up to a factor, with the functions

$$w^{(\lambda)}_\sigma(\boldsymbol{\nu}) \propto D^{(s)}_{\lambda\sigma}(\boldsymbol{\nu})^*. \tag{16.11}$$

Indeed, applying the transformation formula (58.7) (see III) to the spin eigenfunctions and setting the ζ-projection of the spin to the definite value λ (i.e. replacing $\psi_{jm'}$ on the right-hand side of (58.7) (see III) by $\delta_{m'\lambda}$), we find that $D^{(s)}_{\lambda\sigma}(\boldsymbol{\nu})$ are spin wave functions corresponding to definite values of the z- and ζ-projections (σ and λ). The totality of these functions ($\sigma = -s, \ldots, +s$) forms (by the correspondence formulas (57.6) (see III)) a covariant spinor of rank $2s$. The components of the contravariant spinor (57.2) (see III), however—corresponding to the components $w^{(\lambda)}_\sigma$—transform as the complex conjugates of the components of a covariant spinor of the same rank. From (16.10),(16.11) we have

$$w^{(\lambda)}(-\boldsymbol{\nu}) = (-1)^{s-\lambda} w^{(-\lambda)}(\boldsymbol{\nu}) \tag{16.12}$$

($s - \lambda$ is an integer). The inversion operation applied to $w^{(\lambda)}$, however, consists not only in the replacement $\boldsymbol{\nu} \to -\boldsymbol{\nu}$, but also in multiplication by an overall phase factor (the "intrinsic parity" of the particle), which we denote by η:

$$\hat{P} w^{(\lambda)}(\boldsymbol{\nu}) = \eta w^{(\lambda)}(-\boldsymbol{\nu}) = \eta(-1)^{s-\lambda} w^{(-\lambda)}(\boldsymbol{\nu}). \tag{16.13}$$

For the relativistic amplitude $u^{(\lambda)}(\mathbf{k})$ this transformation takes the form

$$\hat{P} u^{(\lambda)}(\mathbf{k}) = \eta\beta u^{(\lambda)}(-\mathbf{k}) = \eta(-1)^{s-\lambda} u^{(-\lambda)}(\mathbf{k}), \tag{16.14}$$

where β is a certain matrix that acts as the identity on the components of $u^{(\lambda)}$ surviving in the limit $|\mathbf{p}| \to 0$. It is important that this matrix is independent of the quantum numbers of the state, so that the difference between (16.13) and (16.14) is in this sense inessential [24].

Applying (16.14) to (16.2), we obtain the transformation law for the wave functions of the states $|\mathbf{n}\lambda\rangle$:

$$\hat{P}\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}) = \eta(-1)^{s-\lambda}\psi_{-\mathbf{n}\,-\lambda}(\boldsymbol{\nu}). \tag{16.15}$$

For the spherical helicity states, making use of (16.10) and (16.12), we obtain the transformation law:

$$\hat{P}\psi_{jm\lambda}(\boldsymbol{\nu}) = \eta(-1)^{j-s}\psi_{jm\,-\lambda}(\boldsymbol{\nu}). \tag{16.16}$$

[24] Thus, for $s = 1$ the amplitudes $u^{(\lambda)}$ are 4-vectors (16.22); in this case β is the full identity matrix with respect to 4-vector indices: $\beta_{\mu\nu} = \delta_{\mu\nu}$. For $s = 1/2$, $u^{(\lambda)}$ is a bispinor; the phase factor is then $\eta = i$, and β is the Dirac matrix γ^0 (see (21.10)).

The states ψ_{jm0} transform, according to (16.16), into themselves, i.e. they possess a definite parity. If, however, $\lambda \neq 0$, then only superpositions of states with opposite helicities have a definite parity:

$$\psi^{(\pm)}_{jm|\lambda|} = \frac{1}{\sqrt{2}}\left(\psi_{jm\lambda} \pm \psi_{jm-\lambda}\right). \tag{16.17}$$

Under inversion they transform into themselves according to

$$\hat{P}\psi^{(\pm)}_{jm|\lambda|}(\boldsymbol{\nu}) = \pm\eta(-1)^{j-s}\psi^{(\pm)}_{jm|\lambda|}(\boldsymbol{\nu}). \tag{16.18}$$

Let us draw attention to the fact that in this section we have classified the states of a free particle with given angular momentum by operating solely with conserved quantities, without appealing to the concept of orbital angular momentum (which was used, for instance, in §§ 6, 7 for the classification of photon states).

As an illustration consider the case of spin 1. In the rest frame the amplitudes $u^{(\lambda)}$ (4-vectors) reduce to three-dimensional vectors $\mathbf{e}^{(\lambda)}$, which play here the role of the amplitudes $w^{(\lambda)}$. The action of the spin-1 operator on a vector function $\mathbf{e}$ is given by

$$(\hat{s}_i\mathbf{e})_k = -ie_{ikl}e_l \tag{16.19}$$

(see III, § 57, Problem 2). Equation (16.1) therefore assumes the form

$$i[\mathbf{n}\mathbf{e}^{(\lambda)}] = \lambda\mathbf{e}^{(\lambda)}. \tag{16.20}$$

Its solutions (in a coordinate frame $\xi\eta\zeta$ with the ζ axis along $\mathbf{n}$) coincide with the circular unit vectors (7.14) [26]:

$$\mathbf{e}^{(0)} = i(0,0,1), \qquad \mathbf{e}^{(\pm1)} = \mp\frac{i}{\sqrt{2}}(1, \pm i, 0). \tag{16.21}$$

In a frame where the particle has momentum $\mathbf{p}$, the helicity-state amplitudes are the 4-vectors

$$u^{(0)\mu} = \left(\frac{|\mathbf{p}|}{m}, \frac{\varepsilon}{m}\mathbf{e}^{(0)}\right), \quad u^{(\pm1)\mu} = (0, \mathbf{e}^{(\pm1)}). \tag{16.22}$$

If $\mathbf{e}$ is a polar vector, then $\eta = -1$. In that case the functions (16.17) (which for $s = 1$ are three-dimensional vectors) have the following parities:

$$\psi^{(+)}_{jm|\lambda|}: \quad P = (-1)^j,$$

$$\psi^{(-)}_{jm|\lambda|}: \quad P = (-1)^{j+1},$$

$$\psi_{jm0}: \quad P = (-1)^j.$$

Comparing with the definition of vector spherical harmonics (7.4), we see that these functions are identical (up to phase factors) with $\mathbf{Y}^{(\mathrm{e})}_{jm}$, $\mathbf{Y}^{(\mathrm{m})}_{jm}$, $\mathbf{Y}^{(\mathrm{l})}_{jm}$ respectively.

[26]The choice of phase factors is fixed by the requirement that the matrix elements of the spin operators computed with the eigenfunctions (16.21) conform to the general definitions in III, §§ 27, 107.

Determining the phase factors (say, by comparing the values at $\theta = 0$), we arrive at the equalities

$$\mathbf{Y}^{(\mathrm{e})}_{jm} = i^{j-1}\sqrt{\frac{2j+1}{8\pi}}\left(\mathbf{e}^{(1)}D^{(j)}_{1m}(\boldsymbol{\nu}) + \mathbf{e}^{(-1)}D^{(j)}_{-1m}(\boldsymbol{\nu})\right),$$

$$\mathbf{Y}^{(\mathrm{m})}_{jm} = i^{j-1}\sqrt{\frac{2j+1}{8\pi}}\left(\mathbf{e}^{(1)\prime}D^{(j)}_{1m}(\boldsymbol{\nu}) + \mathbf{e}^{(-1)\prime}D^{(j)}_{-1m}(\boldsymbol{\nu})\right), \tag{16.23}$$

$$\mathbf{Y}^{(\mathrm{l})}_{jm} = i^{j-1}\sqrt{\frac{2j+1}{4\pi}}\,\mathbf{e}^{(0)}D^{(j)}_{0m}$$

(j is an integer!); $\mathbf{e}^{(\lambda)\prime} = [\mathbf{n}\mathbf{e}^{(\lambda)}]$ are circular unit vectors in axes $\xi'\eta'\zeta$ that are rotated relative to $\xi\eta\zeta$ by 90° about the ζ axis.

The last of formulas (16.23) is equivalent to the expression (58.23) (see III) for $d^{(j)}_{0m}(\theta)$. From the first (or the second) formula one can obtain a simple expression for the functions $d^{(j)}_{\pm 1m}$. We have

$$i^{j-1}\sqrt{\frac{2j+1}{8\pi}}\,D^{(j)}_{\pm 1m} = \mathbf{Y}^{(\mathrm{e})}_{jm}\mathbf{e}^{(\pm 1)*} = \frac{1}{\sqrt{j(j+1)}}\,\mathbf{e}^{(\pm 1)*}\boldsymbol{\nabla}Y_{jm}.$$

We expand the scalar product on the right-hand side in the frame $\xi\eta\zeta$, where

$$\left(\frac{\partial}{\partial\xi}, \frac{\partial}{\partial\eta}\right) \to \left(\frac{\partial}{\partial\theta}, \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi}\right).$$

Recalling definition (7.2) of the function Y_{jm} and definition (16.5), we obtain as the final result

$$d^{(j)}_{\pm 1m}(\theta) = (-1)^{m+1}\sqrt{\frac{(j-m)!}{(j+m)!\,j(j+1)}}\left(\pm\frac{\partial}{\partial\theta} + \frac{m}{\sin\theta}\right)P^m_j(\cos\theta), \quad m \geqslant 0. \tag{16.24}$$

CHAPTER III

Fermions

§ 17. Four-dimensional spinors

In nonrelativistic theory a particle of arbitrary spin s is described by a $(2s+1)$-component quantity—a symmetric spinor of rank $2s$. From the mathematical standpoint these are quantities that furnish the irreducible representations of the spatial rotation group.

In relativistic theory this group appears only as a subgroup of the broader group of four-dimensional rotations—the Lorentz group. In connection with this there arises the need to construct a theory of four-dimensional spinors (4-spinors)—quantities that carry the irreducible representations of the Lorentz group; the exposition of this theory occupies §§ 17–19. In §§ 17, 18 only the proper Lorentz group is considered, which does not include spatial inversion; the latter will be treated in § 19.

The construction of the 4-spinor theory proceeds along the same lines as the three-dimensional spinor theory (*B. L. van der Waerden*, 1929; *G. E. Uhlenbeck*, *O. Laporte*, 1931).

A spinor ξ^α is a two-component quantity ($\alpha = 1, 2$); regarded as components of the wave function of a spin-$1/2$ particle, ξ^1 and ξ^2 correspond to eigenvalues of the z-projection of the spin equal to $+1/2$ and $-1/2$ respectively. Under any (proper) Lorentz transformation the two quantities ξ^1, ξ^2 transform through each other:

$$\xi^{1\prime} = \alpha\xi^1 + \beta\xi^2, \qquad \xi^{2\prime} = \gamma\xi^1 + \delta\xi^2. \tag{17.1}$$

The coefficients α, β, γ, δ are definite functions of the rotation angles of the 4-coordinate system, subject to the condition

$$\alpha\delta - \beta\gamma = 1, \tag{17.2}$$

i.e. the determinant of the binary transformation (17.1) equals 1, just as the determinants of coordinate transformations in the Lorentz group do.

By virtue of conditions (17.2) the bilinear form $\xi^1\Xi^2 - \xi^2\Xi^1$ (where ξ^α and Ξ^α are two spinors) is invariant under the transformation (17.1) (it corresponds to a spin-0 particle "composed" of two spin-$1/2$ particles). For a natural way of writing such invariant expressions, alongside the "contravariant" components of the spinor ξ^α one also introduces "covariant" components ξ_α. The passage from one set to the other is carried out with the aid of the "spinor metric" $g_{\alpha\beta}$ [1]:

$$\xi_\alpha = g_{\alpha\beta}\xi^\beta, \tag{17.3}$$

where

$$g = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{17.4}$$

[1] Spinor indices will be denoted by the first letters of the Greek alphabet: $\alpha, \beta, \gamma, \dots$.

so that

$$\xi_1 = \xi^2, \qquad \xi_2 = -\xi^1. \tag{17.5}$$

The invariant $\xi^1\Xi^2 - \xi^2\Xi^1$ is then written as a scalar product $\xi^\alpha\Xi_\alpha$. Note that $\xi^\alpha\Xi_\alpha = -\xi_\alpha\Xi^\alpha$.

Up to this point the properties listed above have formally coincided with those of three-dimensional spinors. A difference arises, however, when one turns to complex-conjugate spinors.

In nonrelativistic theory the sum

$$\psi^1\psi^{1*} + \psi^2\psi^{2*}, \tag{17.6}$$

which governs the probability density of finding particles at a given point in space, had to be a scalar, and for this purpose the components $\psi^{\alpha*}$ had to transform as covariant spinor components; put differently, the transformation (17.1) had to be unitary ($\alpha = \delta^*$, $\beta = -\gamma^*$). In the relativistic case, however, the particle density does not behave as a scalar; it forms the time component of a 4-vector. Owing to this, the requirement just stated ceases to hold, and no further restrictions on the transformation coefficients arise (apart from (17.2)). Four complex quantities $\alpha, \beta, \gamma, \delta$ constrained only by (17.2) amount to $8 - 2 = 6$ real parameters—in agreement with the number of angles specifying a 4-coordinate rotation (rotations in six coordinate planes).

Consequently, complex-conjugate binary transformations prove to be essentially different from one another, and in relativistic theory two varieties of spinors arise. To distinguish between them one adopts special notation: the indices of spinors whose transformation law is the complex conjugate of (17.1) are written as digits with dots placed above them (*dotted indices*). Thus, by definition,

$$\eta^{\dot\alpha} \sim \xi^{\alpha*}, \tag{17.7}$$

where the sign $\sim$ stands for "transforms as." In other words, the transformation formulas for a "dotted" spinor are:

$$\eta^{\dot1\prime} = \alpha^*\eta^{\dot1} + \beta^*\eta^{\dot2}, \qquad \eta^{\dot2\prime} = \gamma^*\eta^{\dot1} + \delta^*\eta^{\dot2}. \tag{17.8}$$

The operations of lowering and raising dotted indices are performed in the same way as for undotted ones:

$$\eta_{\dot1} = \eta^{\dot2}, \qquad \eta_{\dot2} = -\eta^{\dot1}. \tag{17.9}$$

With respect to spatial rotations the behavior of 4-spinors coincides with that of 3-spinors. For the latter, as we know, $\psi^*_\alpha \sim \psi^\alpha$. By virtue of definition (17.7) the 4-spinor $\eta_{\dot\alpha}$ therefore behaves under rotations as a contravariant 3-spinor ψ^α. The eigenvalues of the spin projection $1/2$ and $-1/2$ accordingly correspond to the covariant components $\eta_{\dot1}$ and $\eta_{\dot2}$.

Spinors of higher rank are defined as sets of quantities that transform like products of components of several first-rank spinors. The indices of a higher-rank spinor may include both dotted and undotted ones. For instance, three types of second-rank spinors exist:

$$\xi^{\alpha\beta} \sim \xi^\alpha\Xi^\beta, \qquad \zeta^{\alpha\dot\beta} \sim \xi^\alpha\eta^{\dot\beta}, \qquad \eta^{\dot\alpha\dot\beta} \sim \eta^{\dot\alpha}H^{\dot\beta}.$$

Merely stating the total rank of a spinor is therefore insufficient for an unambiguous definition of the concept; when necessary we shall specify the rank as a pair of numbers (k, l)—the number of undotted and the number of dotted indices.

Since the transformations (17.1) and (17.8) are algebraically independent, there is no need to fix the ordering of dotted and undotted indices (in this sense, for example, the spinors $\zeta^{\alpha\dot{\beta}}$ and $\zeta^{\dot{\beta}\alpha}$ are one and the same).

For an expression involving spinors to have an invariant character, it must carry the same number of undotted and dotted indices on both sides; otherwise the expression will certainly be violated upon passing from one reference frame to another. One should keep in mind here that complex conjugation entails the interchange of dotted and undotted indices. Therefore the relation $\eta^{\dot{\alpha}\dot{\beta}} = \left(\xi^{\alpha\beta}\right)^*$ between two spinors does have an invariant character.

Contraction of spinors or their products can be performed only over pairs of indices of the same kind—two dotted or two undotted. Summation over a pair of indices of different kinds is not an invariant operation. For this reason, from the spinor

$$\zeta^{\alpha_1\alpha_2\ldots\alpha_k\dot{\beta}_1\dot{\beta}_2\ldots\dot{\beta}_l}, \tag{17.10}$$

which is symmetric in all k undotted and in all l dotted indices, it is impossible to form a spinor of lower rank (recall that contracting over a pair of indices in which the spinor is symmetric gives zero). This means that from the quantities (17.10) one cannot assemble any smaller set of their linear combinations that would transform among themselves under every group transformation. Stated differently, symmetric 4-spinors realize the irreducible representations of the proper Lorentz group. Every such representation is labeled by a pair of numbers (k, l).

Since each spinorial index runs over two values, there are $k + 1$ essentially distinct sets of numbers $\alpha_1\alpha_2\ldots\alpha_k$ in (17.10) (containing $0, 1, 2, \ldots, k$ ones and $k, k-1, \ldots, 0$ twos) and $l + 1$ sets of numbers $\dot{\beta}_1\dot{\beta}_2\ldots\dot{\beta}_l$. Altogether, therefore, a symmetric spinor of rank (k, l) has $(k + 1)(l + 1)$ independent components; this is the dimensionality of the irreducible representation it carries.

§ 18. Relation of spinors to 4-vectors

A spinor $\zeta^{\alpha\dot{\beta}}$ carrying one dotted and one undotted index possesses $2 \cdot 2 = 4$ independent components—precisely the number of components belonging to a 4-vector. It is therefore evident that both objects furnish one and the same irreducible representation of the proper Lorentz group, and a definite correspondence must hold between their respective components.

To establish this correspondence we turn first of all to the analogous correspondence in the three-dimensional case, bearing in mind that with respect to purely spatial rotations the behavior of 3- and 4-spinors must be identical.

For a three-dimensional spinor $\psi^{\alpha\beta}$ the correspondence formulas hold (see III, § 57), which we write here in the form

$$
\begin{aligned}
a_x &= \tfrac{1}{2}(\psi^{22} - \psi^{11}) = \tfrac{1}{2}(\psi_1^2 + \psi_2^1),\\
a_y &= -\tfrac{i}{2}(\psi^{22} + \psi^{11}) = \tfrac{i}{2}(\psi_2^1 - \psi_1^2),\\
a_z &= \tfrac{1}{2}(\psi^{12} + \psi^{21}) = \tfrac{1}{2}(\psi_1^1 - \psi_2^2),
\end{aligned}
\tag{III.1}
$$

where a_x, a_y, a_z are the components of a certain three-dimensional vector **a**. In passing to the four-dimensional case one must replace the components ψ^α_β by $\zeta^{\alpha\dot\beta}$, and interpret a_x, a_y, a_z as the contravariant components a^1, a^2, a^3 of a 4-vector. As for the expression for the fourth component, a^0, its form is already clear from the circumstance noted in § 17: the quantity (17.6) must transform as a^0. Hence $a^0 \sim \zeta^{1\dot 1} + \zeta^{2\dot 2}$; the proportionality coefficient is fixed by requiring that the scalar $\zeta_{\alpha\dot\beta}\zeta^{\alpha\dot\beta}$ coincide with $2a_\mu a^\mu \equiv 2a^2$.

We thus arrive at the following correspondence formulas:

$$
\begin{aligned}
a^1 &= \tfrac{1}{2}(\zeta^{1\dot 2} + \zeta^{2\dot 1}), \qquad & a^2 &= \tfrac{i}{2}(\zeta^{1\dot 2} - \zeta^{2\dot 1}),\\
a^3 &= \tfrac{1}{2}(\zeta^{1\dot 1} - \zeta^{2\dot 2}), \qquad & a^0 &= \tfrac{1}{2}(\zeta^{1\dot 1} + \zeta^{2\dot 2}).
\end{aligned}
\tag{18.1}
$$

The inverse formulas are:

$$
\begin{aligned}
\zeta^{1\dot 1} &= \zeta_{2\dot 2} = a^3 + a^0, \qquad & \zeta^{2\dot 2} &= \zeta_{1\dot 1} = a^0 - a^3,\\
\zeta^{1\dot 2} &= -\zeta_{2\dot 1} = a^1 - ia^2, \qquad & \zeta^{2\dot 1} &= -\zeta_{1\dot 2} = a^1 + ia^2.
\end{aligned}
\tag{18.2}
$$

Here

$$\zeta_{\alpha\dot\beta}\zeta^{\alpha\dot\beta} = 2a^2. \tag{18.3}$$

We note also that

$$\zeta_{\alpha\dot\beta}\zeta^{\gamma\dot\beta} = \delta^\gamma_\alpha\, a^2. \tag{18.4}$$

To see why this last relation holds, observe that the quantity $\zeta_{\alpha\dot\beta}\zeta_\gamma^{\dot\beta}$, viewed as a second-rank spinor in the indices α and γ, is antisymmetric and hence must be proportional to the metric spinor.

The link between $\zeta^{\alpha\dot\beta}$ and a 4-vector is actually one instance of a broader principle: any symmetric spinor of rank (k, k) can be placed in one-to-one correspondence with an irreducible symmetric 4-tensor of rank k (irreducibility meaning that contraction over every index pair gives zero).

The relationship linking a spinor to a 4-vector can be written compactly with the aid of the two-by-two Pauli matrices [2]:

$$
\sigma_x = \begin{pmatrix} 0 & 1\\ 1 & 0\end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i\\ i & 0\end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0\\ 0 & -1\end{pmatrix}.
\tag{18.5}
$$

Denoting symbolically by ζ the matrix of the quantities $\zeta^{\alpha\dot\beta}$ with upper indices (the first being undotted), formulas (18.2) take the form

$$\zeta = \mathbf{a}\boldsymbol{\sigma} + a^0 \tag{18.6}$$

[2] For brevity of notation, operators (matrices) that act on spin variables will be denoted by letters without carets.

(in the second term one understands, of course, the product of a^0 with the unit matrix). The inverse relations are:

$$\mathbf{a} = \tfrac{1}{2}\,\mathrm{Sp}(\zeta\boldsymbol{\sigma}), \qquad a^0 = \tfrac{1}{2}\,\mathrm{Sp}\,\zeta. \tag{18.7}$$

With the help of formulas (18.6), (18.7) one can establish the connection between the transformation laws for the 4-vector and the spinor, thereby expressing the spinor transformation law through the parameters of rotations of the 4-coordinate system.

Let us write the transformation of the spinor ξ^α in the form

$$\xi^{\alpha\prime} = (B\xi)^\alpha, \qquad B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \tag{18.8}$$

where B is a two-by-two matrix assembled from the coefficients of the binary transformation. The transformation of the dotted spinor is then:

$$\eta^{\dot\beta\prime} = (B^*\eta)^{\dot\beta} = (\eta B^+)^{\dot\beta}, \tag{18.9}$$

and the transformation of the second-rank spinor $\zeta^{\alpha\dot\beta} \sim \xi^\alpha\eta^{\dot\beta}$ can be written symbolically as $\zeta' = B\zeta B^{+1}$). For an infinitesimally small transformation $B = 1 + \lambda$, where λ is a small matrix, and to first-order accuracy

$$\zeta' = \zeta + (\lambda\zeta + \zeta\lambda^+). \tag{18.10}$$

Consider first the Lorentz transformation to a reference frame moving with an infinitesimally small velocity $\delta\mathbf{V}$ (without altering the orientation of the spatial axes). Under this transformation the 4-vector $a^\mu = (a^0, \mathbf{a})$ changes according to

$$\mathbf{a}' = \mathbf{a} - a^0\delta\mathbf{V}, \qquad a^{0\prime} = a^0 - \mathbf{a}\delta\mathbf{V}. \tag{18.11}$$

Let us now make use of formulas (18.7). The transformation of a^0 can be expressed, on the one hand, as

$$a^{0\prime} = a^0 - \mathbf{a}\delta\mathbf{V} = a^0 - \tfrac{1}{2}\,\mathrm{Sp}(\zeta\boldsymbol{\sigma}\,\delta\mathbf{V}),$$

and on the other hand as

$$a^{0\prime} = \tfrac{1}{2}\,\mathrm{Sp}\,\zeta' = a^0 + \tfrac{1}{2}\,\mathrm{Sp}\,\zeta\,(\lambda + \lambda^+).$$

These expressions must coincide identically (i.e. for arbitrary ζ). From this we obtain the equality:

$$\lambda + \lambda^+ = -\boldsymbol{\sigma}\,\delta\mathbf{V}.$$

By the same method, examining the transformation of $\mathbf{a}$, we find

$$\boldsymbol{\sigma}\lambda + \lambda^+\boldsymbol{\sigma} = -\delta\mathbf{V}.$$

These equalities, regarded as equations for λ, yield the solution:

$$\lambda = \lambda^+ = -\tfrac{1}{2}\boldsymbol{\sigma}\,\delta\mathbf{V}.$$

Thus the infinitesimally small Lorentz transformation of the spinor ξ^α is effected by the matrix

$$B = 1 - \tfrac{1}{2}(\boldsymbol{\sigma}\mathbf{n})\,\delta V, \tag{18.12}$$

where $\mathbf{n}$ is a unit vector along the velocity $\delta\mathbf{V}$. From this it is straightforward to obtain the transformation for a finite velocity $\mathbf{V}$. To do so, recall that the Lorentz transformation amounts (geometrically) to a rotation of the 4-coordinate system in the $t\,\mathbf{n}$ plane through an angle φ related to the velocity $\mathbf{V}$ by $\operatorname{th}\varphi = V$[3]). An infinitesimally small transformation corresponds to the angle $\delta\varphi = \delta V$, while a rotation through a finite angle φ is achieved by repeating the rotation through $\delta\varphi$ a total of $\varphi/\delta\varphi$ times. Raising the operator (18.12) to the power $\varphi/\delta\varphi$ and passing to the limit $\delta\varphi \to 0$, we obtain

$$B = e^{-\frac{\varphi}{2}\,\mathbf{n}\boldsymbol{\sigma}}. \tag{18.13}$$

The mathematical meaning of this operator becomes transparent once one observes that, by the properties of the Pauli matrices, all even powers of $\mathbf{n}\boldsymbol{\sigma}$ equal 1 while all odd powers equal $\mathbf{n}\boldsymbol{\sigma}$. Noting that ch decomposes into even powers and sh into odd powers of its argument, we arrive at the final result

$$B = \operatorname{ch}\frac{\varphi}{2} - \mathbf{n}\boldsymbol{\sigma}\operatorname{sh}\frac{\varphi}{2}, \qquad \operatorname{th}\varphi = V. \tag{18.14}$$

We note that the Lorentz-transformation matrices B turn out to be Hermitian: $B = B^+$.

Consider now an infinitesimally small rotation of the spatial coordinate system. Under such a rotation the three-dimensional vector $\mathbf{a}$ transforms according to

$$\mathbf{a}' = \mathbf{a} - [\delta\theta\,\mathbf{a}], \tag{18.15}$$

where $\delta\boldsymbol{\theta}$ is the vector of the infinitesimally small rotation angle. The corresponding spinor transformation could be found in an analogous manner. There is no need for this, however, since under spatial rotations the behavior of 4-spinors coincides with that of 3-spinors, and for the latter the transformation is known beforehand from the general connection between the spin operator and the operator of an infinitesimally small rotation:

$$B = 1 + \tfrac{i}{2}\boldsymbol{\sigma}\,\delta\boldsymbol{\theta}. \tag{18.16}$$

The passage to a rotation through a finite angle θ is carried out in the same way as the passage from (18.12) to (18.14):

$$B = \exp\!\Big(\frac{i\theta}{2}\,\mathbf{n}\boldsymbol{\sigma}\Big) = \cos\frac{\theta}{2} + i\mathbf{n}\boldsymbol{\sigma}\sin\frac{\theta}{2}, \tag{18.17}$$

where $\mathbf{n}$ is the unit vector along the rotation axis. This matrix is unitary ($B^+ = B^{-1}$), as must be the case for a spatial rotation.

[3]We recall that in planes containing the time axis the metric is pseudo-Euclidean.

§ 19. Inversion of spinors

When developing the three-dimensional spinor theory (in Vol. III) we did not consider how spinors behave with respect to the spatial inversion operation, inasmuch as in the nonrelativistic framework this would not have produced any new physical consequences. It is worth pausing on this matter now, however, for a clearer understanding of the inversion properties of 4-spinors that will be discussed below.

The inversion operation does not alter the sign of an axial vector, and the spin vector is precisely of this type. Consequently the value of its projection s_z remains unchanged as well. It follows that under inversion each component of the spinor ψ^α can transform only into itself, i.e. one must have

$$\psi^\alpha \to P\psi^\alpha, \tag{19.1}$$

where P is a constant coefficient. Applying inversion twice returns us to the original coordinate system. For spinors, however, the return to the initial configuration can be understood in two distinct senses: as a rotation of the system by 0° or by 360°. For spinors these two interpretations are not equivalent, since ψ^α reverses sign under a 360° rotation. Two alternative definitions of inversion therefore arise: in the first,

$$P^2 = 1, \qquad P = \pm 1, \tag{19.2}$$

and in the second,

$$P^2 = -1, \qquad P = \pm i. \tag{19.3}$$

A crucial requirement is that inversion be defined in the same manner for every spinor. It would be inconsistent for distinct spinors to follow different inversion rules (one obeying (19.2), another (19.3)), for then one could not always build a scalar (or pseudoscalar) out of an arbitrary pair of spinors: if ψ^α were governed by (19.2) while φ^α were governed by (19.3), the product $\psi^\alpha\varphi_\alpha$ would pick up a factor $\pm i$ upon inversion instead of staying fixed (or simply flipping sign).

It should also be stressed that (under either definition of inversion) the assignment of one or another parity P to a spinor carries no absolute significance, because spinors change sign under a rotation through 2π, which can always be performed simultaneously with the inversion. What does possess an absolute character, however, is the "relative parity" of two spinors, defined as the parity of the scalar $\psi^\alpha\varphi_\alpha$ formed from them; since a 2π rotation reverses the sign of all spinors simultaneously, the associated ambiguity does not affect the parity of the scalar in question.

Let us turn now to four-dimensional spinors.

We note first that since inversion changes the sign of only three (x, y, z) out of the four (t, x, y, z) coordinates, it commutes with spatial rotations but does not commute with transformations that rotate the t axis. If $\widehat{L}$ denotes the Lorentz transformation to a frame moving with velocity $\mathbf{V}$, then $\widehat{PL} = \widehat{L}'\widehat{P}$, where $\widehat{L}'$ is the transformation to a frame moving with velocity $-\mathbf{V}$.

It follows from this that under inversion the components of a 4-spinor ξ^α cannot transform through themselves alone. If the inversion of the spinor $\xi^{\dot\alpha}$ were still given by the transformation (19.1) (i.e. were represented by a matrix proportional to the identity), it would commute with all Lorentz transformations whatsoever, which certainly must not be the case (since the operations $\widehat{L}$ and $\widehat{L}'$ applied to ξ^α manifestly do not coincide).

Inversion must therefore map the components of the spinor ξ^α into other quantities. These can only be the components of some other spinor $\eta_{\dot\alpha}$ whose transformation properties differ from those of ξ^α. Since inversion leaves the z-projection of the spin unchanged (as already noted above), the components ξ^1 and ξ^2 can pass under inversion only into $\eta_{\dot 1}$ and $\eta_{\dot 2}$, which correspond to the same values $s_z = 1/2$ and $s_z = -1/2$. Taking inversion to be an operation that yields 1 when applied twice, one can specify its action by the formulas

$$\xi^\alpha \to \eta_{\dot\alpha}, \qquad \eta_{\dot\alpha} \to \xi^\alpha. \tag{19.4}$$

For the covariant components ξ_α and the contravariant $\eta^{\dot\alpha}$ these transformations carry the opposite sign:

$$\xi_\alpha \to -\eta^{\dot\alpha}, \qquad \eta^{\dot\alpha} \to -\xi_\alpha, \tag{19.4a}$$

since lowering and raising one and the same index are performed with opposite signs, see (17.5) and (17.9)[4].

A certain distinction between the two definitions of inversion lies in the fact that under the second definition complex-conjugate spinors transform identically: if $\Xi_\alpha = \eta^*_{\dot\alpha}$, $H^{\dot\alpha} = \xi^{\alpha *}$, then from (19.5) one obtains $\Xi_\alpha \to -iH^{\dot\alpha}$, $H^{\dot\alpha} \to -i\Xi_\alpha$, i.e. the same rule as for $\xi_\alpha, \eta^{\dot\alpha}$. Under definition (19.4), by contrast, one would obtain the transformation $\Xi_\alpha \to H^{\dot\alpha}$, $H^{\dot\alpha} \to \Xi_\alpha$, which is the reverse in sign of the transformation of the spinors $\xi_\alpha, \eta^{\dot\alpha}$. We shall return to possible physical aspects of this distinction in § 27.

In what follows we shall everywhere adopt definition (19.5) for definiteness.

Under the rotation subgroup the spinors ξ^α and $\eta_{\dot\alpha}$ transform, as we know, in the same way. Forming from their components the combinations

$$\xi^\alpha \pm \eta_{\dot\alpha}, \tag{19.6}$$

one would obtain quantities transforming under inversion according to (19.1) with $P = \pm i$. These combinations, however, do not behave as spinors with respect to all transformations of the Lorentz group.

Incorporating inversion into the symmetry group therefore demands that one treat a pair of spinors $(\xi^\alpha, \eta_{\dot\alpha})$ jointly; such a pair goes by the name of a first-rank *bispinor*. The four components of a bispinor realize one of the irreducible representations that belong to the extended Lorentz group.

Given two bispinors $(\xi^\alpha, \eta_{\dot\alpha})$ and $(\Xi^\alpha, H_{\dot\alpha})$, their scalar product admits two distinct constructions. The expression

$$\xi^\alpha \Xi_\alpha + \eta_{\dot\alpha} H^{\dot\alpha} \tag{19.7}$$

---

[4] If instead inversion is understood in the sense that $P^2 = -1$, its action is specified by the formulas

$$\xi^\alpha \to i\eta_{\dot\alpha}, \qquad \eta_{\dot\alpha} \to i\xi^\alpha \tag{19.5}$$

or, equivalently,

$$\xi_\alpha \to -i\eta^{\dot\alpha}, \qquad \eta^{\dot\alpha} \to -i\xi_\alpha. \tag{19.5a}$$

is altogether unchanged by inversion, i.e. it constitutes a true scalar. The quantity

$$\xi^{\alpha}\Xi_{\alpha} - \eta_{\dot{\alpha}}H^{\dot{\alpha}} \tag{19.8}$$

is likewise invariant under rotations of the 4-coordinate system but reverses sign under inversion; in other words, it is a pseudoscalar.

A second-rank spinor $\zeta^{\alpha\dot{\beta}}$ can also be defined in two ways. Specifying its transformation law as

$$\zeta^{\alpha\dot{\beta}} \sim \xi^{\alpha}H^{\dot{\beta}} + \Xi^{\alpha}\eta^{\dot{\beta}}, \tag{19.9}$$

one obtains quantities that transform under inversion according to

$$\zeta^{\alpha\dot{\beta}} \to \zeta_{\dot{\alpha}\beta}. \tag{19.10}$$

The 4-vector a^{μ} to which such a spinor is equivalent undergoes, by virtue of (18.1), the mapping $(a^0, \mathbf{a}) \to (a^0, -\mathbf{a})$, so it constitutes a genuine 4-vector (while the spatial part $\mathbf{a}$ is polar).

One may, however, also define $\zeta^{\alpha\dot{\beta}}$ according to

$$\zeta^{\alpha\dot{\beta}} \sim \xi^{\alpha}H^{\dot{\beta}} - \Xi^{\alpha}\eta^{\dot{\beta}}. \tag{19.11}$$

Then [5]

$$\zeta^{\alpha\dot{\beta}} \to -\zeta_{\dot{\alpha}\beta}. \tag{19.12}$$

The 4-vector corresponding to such a spinor undergoes the inversion transformation $(a^0, \mathbf{a}) \to (-a^0, \mathbf{a})$, i.e. it is a 4-pseudovector (while the three-dimensional vector $\mathbf{a}$ is axial).

Symmetric second-rank spinors whose indices are of the same type obey the transformation laws:

$$\xi^{\alpha\beta} \sim \xi^{\alpha}\Xi^{\beta} + \xi^{\beta}\Xi^{\alpha}, \qquad \eta_{\dot{\alpha}\dot{\beta}} \sim \eta_{\dot{\alpha}}H_{\dot{\beta}} + \eta_{\dot{\beta}}H_{\dot{\alpha}}. \tag{19.13}$$

Under inversion they pass into one another:

$$\xi^{\alpha\beta} \to -\eta_{\dot{\alpha}\dot{\beta}}. \tag{19.14}$$

The pair $(\xi^{\alpha\beta}, \eta_{\dot{\alpha}\dot{\beta}})$ constitutes a second-rank bispinor. Its number of independent components equals $3 + 3 = 6$. An antisymmetric second-rank 4-tensor $a^{\mu\nu}$ possesses the same number of independent components. A definite correspondence must therefore exist between them (both furnish equivalent irreducible representations of the extended Lorentz group).

Since the spinors $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ transform independently under the proper Lorentz group, the components of the 4-tensor $a^{\mu\nu}$ can likewise be partitioned into two groups of quantities that mix only among themselves under all rotations of the 4-coordinate system. This decomposition is carried out as follows.

[5] We emphasize that the transformation laws (19.10) and (19.12), which differ by a sign on the right-hand side, are by no means equivalent, since both sides of each involve the components of one and the same spinor (cf. the footnote on p. 92).

We introduce a polar three-dimensional vector **p** and an axial three-dimensional vector **a**, related to the components of the 4-tensor $a^{\mu\nu}$ by

$$a^{\mu\nu} = \begin{pmatrix} 0 & p_x & p_y & p_z \\ -p_x & 0 & -a_z & a_y \\ -p_y & a_z & 0 & -a_x \\ -p_z & -a_y & a_x & 0 \end{pmatrix} \equiv (\mathbf{p}, \mathbf{a}) \tag{19.15}$$

($(\mathbf{p}, \mathbf{a})$ is a shorthand notation that we shall employ for listing the components of such a tensor). Here $a^{\mu\nu} = (-\mathbf{p}, \mathbf{a})$, and of the two quantities

$$\mathbf{a}^2 - \mathbf{p}^2 = \tfrac{1}{2} a_{\mu\nu} a^{\mu\nu}, \qquad \mathbf{a}\mathbf{p} = \tfrac{1}{8} e_{\mu\nu\rho\sigma} a^{\mu\nu} a^{\rho\sigma}$$

the first is a scalar and the second a pseudoscalar; with respect to the proper Lorentz group both are equally invariant. Equally invariant alongside them are the squared moduli of the three-dimensional vectors $\mathbf{f}^{\pm} = \mathbf{p} \pm i\mathbf{a}$. This means that every rotation in 4-space amounts, for the vectors $\mathbf{f}^{\pm}$, to a "rotation" in three-dimensional space—generally speaking through complex angles (the six rotation angles in 4-space correspond to three complex "rotation angles" of the three-dimensional system). The operation of spatial inversion, on the other hand, reverses the sign of **p** (but not of **a**) and thereby interchanges the vectors $\mathbf{f}^+$ and $-\mathbf{f}^-$. The components of these vectors form precisely the two sought-after groups of quantities built from the components of the tensor $a^{\mu\nu}$.

At the same time the correspondence between the components of the 4-tensor $a^{\mu\nu}$ and those of the spinors $\xi^{\alpha\beta}$, $\eta_{\dot\alpha\dot\beta}$ becomes evident. Since spatial rotations enter the Lorentz group as a subgroup, the relations linking the spinor components to the components of the three-dimensional vectors must be the same as for three-dimensional spinors:

$$\begin{aligned} f^+{}_x &= \tfrac{1}{2}(\xi^{22} - \xi^{11}), \qquad & f^+{}_y &= \tfrac{i}{2}(\xi^{22} + \xi^{11}), \qquad & f^+{}_z &= \xi^{12}; \\ f^-{}_x &= \tfrac{1}{2}(\eta_{\dot2\dot2} - \eta_{\dot1\dot1}), \qquad & f^-{}_y &= \tfrac{i}{2}(\eta_{\dot2\dot2} + \eta_{\dot1\dot1}), \qquad & f^-{}_z &= \eta_{\dot1\dot2}. \end{aligned} \tag{19.16}$$

Problem Derive the general correspondence that connects spinors of even rank to 4-tensors.

Solution. Every spinor whose total rank $k + l$ is even realizes a single-valued (though in general reducible) representation of the extended Lorentz group; it is consequently equivalent to a 4-tensor that realizes the same representation [6].

A spinor of rank (k, k) transforms under inversion according to

$$\zeta^{\alpha\beta\ldots\dot\gamma\dot\delta\ldots} \to \pm\zeta_{\dot\alpha\dot\beta\ldots\gamma\delta\ldots}.$$

Such a spinor is equivalent to a symmetric irreducible 4-tensor of rank k—a true tensor or a pseudotensor depending on the sign in (1).

Spinors of ranks (k, l) and (l, k) making up a bispinor transform under inversion according to

$$\zeta^{\overbrace{\alpha\beta\ldots}^{k}\;\overbrace{\dot\gamma\dot\delta\ldots}^{l}} \to (-1)^{\frac{k-l}{2}} \chi_{\dot\alpha\dot\beta\ldots\gamma\delta\ldots}. \tag{2}$$

[6]Spinors whose rank is odd realize, by contrast, double-valued representations: a spatial rotation through a full turn of 360° reverses the sign of these spinors, so that each group element is associated with a pair of matrices differing in sign.

When $l = k+2$ the bispinor is equivalent to an irreducible 4-tensor $a_{[\mu\nu]\rho\sigma\ldots}$ of rank $k+2$, antisymmetric in the indices $[\mu\nu]$ and symmetric in all the rest. The irreducibility of this tensor means that it vanishes upon contraction over any pair of indices and upon dualization with respect to any triple of indices (i.e. $e^{\lambda\mu\nu\sigma} a_{[\mu\nu]\rho\sigma\ldots} = 0$); the latter condition signifies that the tensor vanishes upon antisymmetrization over any index triple.

More broadly, when $l = k+2n$ the bispinor corresponds to an irreducible 4-tensor of rank $k+2n$ that is antisymmetric in n index pairs $[\lambda\mu][\nu\rho]\ldots$ and symmetric in the remaining k indices. Tensors possessing antisymmetry of yet higher degree (in triples, quadruples, and so on of indices) are absent from this classification for a transparent reason: a completely antisymmetric tensor of third rank is dual to a pseudovector, one of fourth rank collapses to a scalar (being proportional to the Levi-Civita pseudotensor $e^{\lambda\mu\nu\rho}$), and antisymmetry in more than four indices cannot arise in a four-dimensional space at all.

§ 20. The Dirac equation in spinor representation

In its own rest frame a particle of spin 1/2 is described by a two-component wave function—a 3-spinor. From the "four-dimensional" standpoint this quantity may originate as either an undotted or a dotted 4-spinor. To describe the particle in an arbitrary frame of reference one needs both kinds of 4-spinor; let us label them ξ^α and $\eta_{\dot\alpha}$ [7].

In the case of a free particle the sole operator entering the wave equation can be (as was noted already in § 10) the 4-momentum $\widehat{p}_\mu = i\partial_\mu$. Expressed through spinor indices, this 4-vector corresponds to an operator spinor $\widehat{p}_{\alpha\dot\beta}$ with components

$$\begin{aligned} \widehat{p}^{1\dot1} &= \widehat{p}_{2\dot2} = \widehat{p}_0 + \widehat{p}_z, & \widehat{p}^{2\dot2} &= \widehat{p}_{1\dot1} = \widehat{p}_0 - \widehat{p}_z, \\ \widehat{p}^{1\dot2} &= -\widehat{p}_{2\dot1} = \widehat{p}_x - i\widehat{p}_y, & \widehat{p}^{2\dot1} &= -\widehat{p}_{1\dot2} = \widehat{p}_x + i\widehat{p}_y. \end{aligned} \tag{20.1}$$

The wave equation is a linear differential relation linking the components of the spinors through the operator $\widehat{p}_{\alpha\dot\beta}$. The demand of relativistic invariance singles out the following system of equations:

$$\widehat{p}^{\alpha\dot\beta}\eta_{\dot\beta} = m\xi^\alpha, \qquad \widehat{p}_{\dot\beta\alpha}\xi^\alpha = m\eta_{\dot\beta}, \tag{20.2}$$

where m is a dimensional constant. Introducing different constants m_1 and m_2 into these two equations (or reversing the sign in front of m) would be pointless, since by a suitable redefinition of ξ^α or $\eta_{\dot\alpha}$ the equations could always be brought back to the form given above.

Eliminating one of the two spinors from equations (20.2) by substituting $\eta_{\dot\beta}$ from the second equation into the first, we obtain:

$$\widehat{p}^{\alpha\dot\beta}\eta_{\dot\beta} = \frac{1}{m}\widehat{p}^{\alpha\dot\beta}\widehat{p}_{\gamma\dot\beta}\xi^\gamma = m\xi^\alpha.$$

[7] A three-dimensional first-rank spinor can also "descend" from 4-spinors of higher odd rank that become antisymmetric in one or several index pairs in the rest frame. Such variants, however, would lead to equations of higher order (cf. the footnote on p. 52).

But by (18.4) $\widehat{p}^{\alpha\dot{\beta}}\widehat{p}_{\gamma\dot{\beta}} = \widehat{p}^2\delta^\alpha_\gamma$, so that we arrive at

$$(\widehat{p}^2 - m^2)\,\xi^\gamma = 0, \tag{20.3}$$

from which it is clear that m is the mass of the particle.

We draw attention to the fact that the necessity of introducing a mass into the wave equation requires the simultaneous treatment of a pair of spinors (ξ^α and $\eta_{\dot{\alpha}}$): from only one of them it is impossible to compose a relativistically invariant equation that would contain any dimensional parameter. By the same token the wave equation turns out to be automatically invariant with respect to spatial inversion, provided one defines the transformation of the wave function as

$$P: \quad \xi^\alpha \to i\eta_{\dot{\alpha}}, \qquad \eta_{\dot{\alpha}} \to i\xi^\alpha. \tag{20.4}$$

One readily checks that this substitution (accompanied by the replacement $\widehat{p}^{\dot{\alpha}\beta} \to \widehat{p}_{\alpha\dot{\beta}}$, obvious from (20.1)) causes the two equations (20.2) to interchange. A pair of spinors that transform into one another under inversion make up a four-component object—a bispinor.

The relativistic wave equation embodied in the system (20.2) bears the name of the *Dirac equation* (it was derived by *Dirac* in 1928). To pursue the further study and application of this equation we shall now examine the different representations in which it can be written.

With the aid of formula (18.6) we rewrite equations (20.2) as

$$(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma})\eta = m\xi, \qquad (\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma})\xi = m\eta. \tag{20.5}$$

Here the symbols ξ and η stand for two-component quantities—spinors

$$\xi = \begin{pmatrix} \xi^1 \\ \xi^2 \end{pmatrix}, \qquad \eta = \begin{pmatrix} \eta_{\dot{1}} \\ \eta_{\dot{2}} \end{pmatrix} \tag{20.6}$$

(the first carrying upper, the second lower indices), and when the matrices σ are multiplied by any two-component quantity f the ordinary rule of matrix multiplication is always understood here and below:

$$(\sigma f)_\alpha = \sigma_{\alpha\beta} f_\beta. \tag{20.7}$$

For ease of subsequent reference we write out the Pauli matrices once more:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{20.8}$$

and recall their principal properties:

$$\sigma_i\sigma_k + \sigma_k\sigma_i = 2\delta_{ik}, \qquad \sigma_i\sigma_k = ie_{ikl}\sigma_l + \delta_{ik} \tag{20.9}$$

(see III, § 55).

Let us also write down the wave equation obeyed by the complex-conjugate wave function, which is built from the spinors

$$\xi^* = \left(\xi^{1*}, \xi^{2*}\right), \qquad \eta^* = \left(\eta^*_{\dot{1}}, \eta^*_{\dot{2}}\right). \tag{20.10}$$

Since every operator $\widehat{p}_\mu$ contains the factor i, one has $\widehat{p}^*_\mu = -\widehat{p}_\mu$. In taking the complex conjugate of both sides of equations (20.5) one must also take into account that, by the Hermiticity of the matrices σ ($\sigma^* = \widetilde{\sigma}$),

$$(\sigma f)^*_\alpha = \sigma^*_{\alpha\beta} f^*_\beta = f^*_\beta \sigma_{\beta\alpha} = (f^*\sigma)_\alpha,$$

and one obtains equations of the form

$$\eta^*(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma}) = -m\xi^*, \qquad \xi^*(\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma}) = -m\eta^*. \tag{20.11}$$

In this way of writing it is tacitly assumed that the operator $\widehat{p}_\mu$ acts on the function placed to its left. The display of ξ^* and η^* as horizontal rows reflects the matrix multiplication convention adopted in these equations: the row f is contracted with the columns of the σ matrices:

$$(f^*\sigma)_\alpha = f^*_\beta \sigma_{\beta\alpha}. \tag{20.12}$$

The inversion transformation for ξ^*, η^* is defined as the complex conjugate of the transformation (20.4):

$$P: \quad \xi^{\alpha *} \to -i\eta^*_{\dot{\alpha}}, \qquad \eta^*_{\dot{\alpha}} \to -i\xi^{\alpha *}. \tag{20.13}$$

§ 21. Symmetric form of the Dirac equation

Among all ways of writing the Dirac equation the spinor form stands out as the one that displays relativistic invariance most transparently. For practical calculations, though, it is often advantageous to pass to a different representation by selecting another quartet of independent components of the wave function.

Let ψ stand for the four-component wave function (its entries being ψ_i, $i = 1, 2, 3, 4$). Written in spinor language this object is a bispinor:

$$\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}. \tag{21.1}$$

With equal right, however, one may choose as the independent components of ψ any linear combinations of the components of the spinors ξ and η [8]. We shall restrict the admissible linear transformations solely by the requirement of unitarity; such transformations leave invariant the bilinear forms composed of ψ and ψ^* (see § 28).

For a completely general selection of the components of ψ one can cast the Dirac equation as

$$\widehat{p}_\mu \gamma^\mu_{ik} \psi_k = m\psi_i,$$

where γ^μ ($\mu = 0, 1, 2, 3$) are certain 4×4 matrices (the *Dirac matrices*). It is customary to write this equation in symbolic form, suppressing the matrix indices:

$$(\gamma\widehat{p} - m)\psi = 0, \tag{21.2}$$

[8] For brevity we shall speak of the four-component quantity ψ as a bispinor also in its non-spinor representations.

where

$$\gamma\widehat{p} \equiv \gamma^\mu \widehat{p}_\mu = \widehat{p}_0\gamma^0 - \widehat{\mathbf{p}}\boldsymbol{\gamma} = i\gamma^0\frac{\partial}{\partial t} + i\boldsymbol{\gamma}\boldsymbol{\nabla}, \quad \boldsymbol{\gamma} = (\gamma^1, \gamma^2, \gamma^3).$$

In the spinor form of the equation with the components ψ from (21.1) the corresponding matrices are[9]

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \tag{21.3}$$

as is readily seen by writing equations (20.5) in the form

$$\begin{pmatrix} 0 & \widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma} \\ \widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma} & 0 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = m \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$

and comparing with (21.2).

In the general case the matrices γ need only satisfy the conditions ensuring that $\widehat{p}^2 = m^2$. To determine these conditions, we multiply equation (21.2) on the left by $\gamma\widehat{p}$. We have

$$(\gamma^\mu \widehat{p}_\mu)(\gamma^\nu \widehat{p}_\nu)\psi = m(\widehat{p}_\mu\gamma^\mu)\psi = m^2\psi.$$

Since $\widehat{p}_\mu\widehat{p}_\nu$ is a symmetric tensor (all operators $\widehat{p}_\mu$ commute), this equation can be rewritten as

$$\tfrac{1}{2}\widehat{p}_\mu\widehat{p}_\nu(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) = m^2\psi,$$

from which it is clear that one must have

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}. \tag{21.4}$$

Thus all pairs of distinct matrices γ^μ anticommute, and the square of each is

$$\left(\gamma^1\right)^2 = \left(\gamma^2\right)^2 = \left(\gamma^3\right)^2 = -1, \qquad \left(\gamma^0\right)^2 = 1. \tag{21.5}$$

Under an arbitrary unitary transformation of the components of ψ ($\psi' = U\psi$, with U a unitary 4×4 matrix) the matrices γ transform according to

$$\gamma' = U\gamma U^{-1} = U\gamma U^+ \tag{21.6}$$

(so that the equation $(\gamma\widehat{p} - m)\psi = 0$ goes over into $(\gamma'\widehat{p} - m)\psi' = 0$). The commutation relations (21.4), of course, remain unchanged.

The matrix γ^0 of (21.3) is Hermitian, and the matrices $\boldsymbol{\gamma}$ are anti-Hermitian. These properties are preserved under every unitary transformation (21.6), so that we shall always have[10]:

$$\boldsymbol{\gamma}^+ = -\boldsymbol{\gamma}, \qquad \gamma^{0+} = \gamma^0. \tag{21.7}$$

[9]Here and below a compact notation is used for 4×4 matrices in terms of 2×2 blocks: each symbol in the expressions (21.3) represents a 2×2 matrix.

[10]These relations can be written together as

$$\gamma^{\lambda +} = \gamma^0\gamma^\lambda\gamma^0. \tag{21.7 a}$$

We now derive the equation satisfied by ψ^*. Complex-conjugating both sides of (21.2) and invoking (21.7) gives

$$(\widehat{p}_0\gamma^0 - \widehat{\mathbf{p}}\widetilde{\boldsymbol{\gamma}} - m)\psi^* = 0.$$

We rearrange ψ^* by $\widetilde{\gamma}^\mu\psi^* = \psi^*\gamma^\mu$ and then multiply the equation on the right by γ^0; noting that $\boldsymbol{\gamma}\gamma^0 = -\gamma^0\boldsymbol{\gamma}$ and introducing the new bispinor

$$\bar{\psi} = \psi^*\gamma^0, \qquad \psi^* = \bar{\psi}\gamma^0, \tag{21.8}$$

we obtain

$$\bar{\psi}(\gamma\widehat{p} + m) = 0. \tag{21.9}$$

Just as in (20.11), $\widehat{p}$ is understood to differentiate the function standing to its left. One calls $\bar{\psi}$ the *Dirac conjugate* (or *relativistic conjugate*) of ψ. The purpose of the γ^0 factor is to swap ξ^* and η^* (in the spinor representation), so that within $\bar{\psi} = (\eta^*, \xi^*)$ the undotted spinor occupies the first position—matching the ordering in ψ—while the dotted spinor occupies the second. This is why $\bar{\psi}$, rather than ψ^*, serves as the natural companion of ψ whenever the two enter bilinear combinations together (see § 28).

Under spatial inversion the wave function transforms according to

$$P: \quad \psi \to i\gamma^0\psi, \qquad \bar{\psi} \to -i\bar{\psi}\gamma^0. \tag{21.10}$$

In the spinor representation of ψ the matrix γ^0 interchanges, as it should under inversion, the components ξ and η. The invariance of the Dirac equation under the transformation (21.10) is in the general case obvious and can be verified directly: replacing $\widehat{\mathbf{p}} \to -\widehat{\mathbf{p}}$ and simultaneously $\psi \to i\gamma^0\psi$ in equation (21.2), we get

$$(\widehat{p}_0\gamma^0 + \widehat{\mathbf{p}}\boldsymbol{\gamma} - m)\gamma^0\psi = 0.$$

Multiplying this equation on the left by γ^0 and using the anticommutativity of γ^0 and $\boldsymbol{\gamma}$, we recover the original equation.

Left-multiplying $(\gamma\widehat{p} - m)\psi = 0$ by $\bar{\psi}$, right-multiplying $\bar{\psi}(\gamma\widehat{p} + m) = 0$ by ψ, and summing the two expressions yields

$$\bar{\psi}\gamma^\mu\left(\widehat{p}_\mu\psi\right) + \left(\widehat{p}_\mu\bar{\psi}\right)\gamma^\mu\psi = \widehat{p}_\mu\left(\bar{\psi}\gamma^\mu\psi\right) = 0,$$

where parentheses mark the function acted upon by $\widehat{p}$. Because this has the structure of a continuity equation $\partial_\mu j^\mu = 0$, the quantity

$$j^\mu = \bar{\psi}\gamma^\mu\psi = \left(\psi^*\psi, \psi^*\gamma^0\boldsymbol{\gamma}\psi\right) \tag{21.11}$$

is the particle current density 4-vector. We note that its time component $j^0 = \psi^*\psi$ is positive definite.

The Dirac equation can be written in a form explicitly solved for the time derivative:

$$i\frac{\partial\psi}{\partial t} = \widehat{H}\psi, \tag{21.12}$$

where $\widehat{H}$ is the particle Hamiltonian [11]. To this end it suffices to multiply equation (21.2) on the left by γ^0. For the Hamiltonian one obtains the expression

$$\widehat{H} = \boldsymbol{\alpha}\widehat{\mathbf{p}} + \beta m, \tag{21.13}$$

where the standard notation has been introduced for the matrices appearing here:

$$\boldsymbol{\alpha} = \gamma^0\boldsymbol{\gamma}, \qquad \beta = \gamma^0. \tag{21.14}$$

We note that

$$\alpha_i\alpha_k + \alpha_k\alpha_i = 2\delta_{ik}, \qquad \beta\boldsymbol{\alpha} + \boldsymbol{\alpha}\beta = 0, \qquad \beta^2 = 1, \tag{21.15}$$

i.e. all the matrices $\boldsymbol{\alpha}$, β anticommute with one another and their squares are unity; all of them are Hermitian. In the spinor representation

$$\boldsymbol{\alpha} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & -\boldsymbol{\sigma} \end{pmatrix}, \qquad \beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{21.16}$$

When velocities are small the particle should reduce, as in non-relativistic quantum mechanics, to a single two-component spinor. And indeed, taking $\mathbf{p} \to 0$, $\varepsilon \to m$ in (20.5) yields $\xi = \eta$: the two spinors inside the bispinor coincide. A drawback of the spinor representation shows up at this point, however—all four entries of ψ stay non-vanishing despite only two being genuinely independent. One would prefer a representation in which two of the four components of ψ go to zero in the non-relativistic limit.

Accordingly we introduce, in place of ξ and η, their linear combinations φ and χ:

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}, \qquad \varphi = \frac{1}{\sqrt{2}}(\xi + \eta), \qquad \chi = \frac{1}{\sqrt{2}}(\xi - \eta). \tag{21.17}$$

Then for a particle at rest $\chi = 0$. This representation of ψ will be called the *standard* representation. Under inversion φ and χ transform into themselves:

$$P: \quad \varphi \to i\varphi, \qquad \chi \to -i\chi. \tag{21.18}$$

The equations for φ and χ are obtained by adding and subtracting equations (20.5):

$$\widehat{p}_0\varphi - \widehat{\mathbf{p}}\boldsymbol{\sigma}\chi = m\varphi, \qquad -\widehat{p}_0\chi + \widehat{\mathbf{p}}\boldsymbol{\sigma}\varphi = m\chi. \tag{21.19}$$

From these one sees that the standard representation corresponds to the matrices

$$\gamma^0 \equiv \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}. \tag{21.20}$$

Since in (21.17) the first and second components of ξ and η are combined separately, in the standard representation—as in the spinor one—the components ψ_1 and ψ_3 correspond to

[11] For a particle of spin 0 the wave equation could not be cast in this form: equation (10.5) for the scalar ψ is of second order in time, while the system (10.4) of first-order equations for the five-component quantity (ψ, ψ_k) contains time derivatives not of all of its components.

the eigenvalue $+1/2$ of the spin projection, while ψ_2 and ψ_4 correspond to the projection $-1/2$. In both representations, therefore, the matrix $\frac{1}{2}\boldsymbol{\Sigma}$ with

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}, \tag{21.21}$$

is the three-dimensional spin operator: the action of $\frac{1}{2}\Sigma_z$ on a bispinor containing only the components ψ_1, ψ_3 or ψ_2, ψ_4 multiplies the bispinor by $+1/2$ or $-1/2$, respectively. In an arbitrary representation this matrix can be written as

$$\boldsymbol{\Sigma} = -\boldsymbol{\alpha}\gamma^5 = -\frac{i}{2}[\mathbf{aa}] \tag{21.22}$$

(for the definition of γ^5 see below, (22.14)).

Problems

1. Find the transformation formulae for the wave function under an infinitesimal Lorentz transformation and an infinitesimal spatial rotation.

   Solution. In the spinor representation of ψ, under an infinitesimal Lorentz transformation

$$\xi' = \left(1 - \frac{1}{2}\boldsymbol{\sigma}\delta\mathbf{V}\right)\xi, \qquad \eta' = \left(1 + \frac{1}{2}\boldsymbol{\sigma}\delta\mathbf{V}\right)\eta$$

   (see (18.8), (18.8 a), (18.12)). Both formulae can be written together as

$$\psi' = \left(1 - \frac{1}{2}\boldsymbol{\alpha}\delta\mathbf{V}\right)\psi. \tag{1}$$

   Similarly, the transformation law under an infinitesimal rotation is

$$\psi' = \left(1 + \frac{i}{2}\boldsymbol{\Sigma}\delta\boldsymbol{\theta}\right)\psi. \tag{2}$$

   In this form the formulae are valid in any representation of ψ, provided one understands $\boldsymbol{\alpha}$ and $\boldsymbol{\Sigma}$ as the matrices in the same representation.

   It is easy to verify that the matrices $\boldsymbol{\alpha}$ and $\boldsymbol{\Sigma}$ constitute the components of the antisymmetric "matrix 4-tensor"

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma})$$

   (the listing of the components follows the rule (19.15)). Let us also introduce the infinitesimal antisymmetric tensor $\delta\varepsilon^{\mu\nu} = (\delta\mathbf{V}, \delta\boldsymbol{\theta})$. Then

$$\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu} = 2i\boldsymbol{\Sigma}\delta\boldsymbol{\theta} - 2\boldsymbol{\gamma}\delta\mathbf{V}$$

   and both formulae (1), (2) can be written in the unified form:

$$\psi' = \left(1 + \frac{1}{4}\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu}\right)\psi. \tag{3}$$

2. Write the Dirac equation in a representation that contains no imaginary coefficients (*E. Majorana*, 1937).

   Solution. In the standard representation, in the equation

$$\left(\frac{\partial}{\partial t} + \alpha_x \frac{\partial}{\partial x} + \alpha_y \frac{\partial}{\partial y} + \alpha_z \frac{\partial}{\partial z} + im\beta\right)\psi = 0$$

   the only imaginary matrices are α_y and $i\beta$. This imaginary character can be eliminated by performing a transformation $\psi' = U\psi$ that interchanges the imaginary matrix α_y with the real matrix β. To this end one sets

$$U = \frac{1}{\sqrt{2}}(\alpha_y + \beta) = U^{-1}.$$

   Then

$$\alpha'_x = U\alpha_x U = -\alpha_x, \quad \alpha'_y = \beta, \quad \alpha'_z = -\alpha_z, \quad \beta' = \alpha_y,$$

   and the Dirac equation takes the form

$$\left(\frac{\partial}{\partial t} - \alpha_x \frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y} - \alpha_z \frac{\partial}{\partial z} + im\alpha_y\right)\psi' = 0,$$

   in which all the coefficients are real.

§ 22. Algebra of Dirac matrices

Calculations involving the Dirac equation make extensive use of the γ matrices without reference to their explicit form in any particular representation. All the algebraic rules governing these matrices follow entirely from the anticommutation relations

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}, \qquad \mu, \nu = 0, 1, 2, 3, \tag{22.1}$$

which encode every general property that the γ matrices possess.

In this section we collect a number of formulae and algebraic identities for the γ matrices that prove useful in various computations.

The "scalar square" of γ with itself is $g_{\mu\nu}\gamma^\mu\gamma^\nu = 4$. For compact notation we introduce, in analogy with the covariant components of a 4-vector, the symbol $\gamma_\mu = g_{\mu\nu}\gamma^\nu$. Then

$$\gamma_\mu \gamma^\mu = 4. \tag{22.2}$$

When γ_μ and γ^μ are separated by one or more γ-factors, one can bring them next to each other by applying the commutation rule (22.1) one or more times, after which the contraction over μ is performed using (22.2). In this way one obtains the identities

$$\begin{gathered}
\gamma_\mu \gamma^\nu \gamma^\mu = -2\gamma^\nu, \\
\gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\mu = 4g^{\lambda\nu}, \\
\gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\mu = -2\gamma^\rho \gamma^\nu \gamma^\lambda, \\
\gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^\mu = 2(\gamma^\sigma \gamma^\lambda \gamma^\nu \gamma^\rho + \gamma^\rho \gamma^\nu \gamma^\lambda \gamma^\sigma).
\end{gathered} \tag{22.3}$$

The γ^μ factors usually appear contracted with various 4-vectors, forming "scalar products"[12]

$$\gamma a \equiv \gamma^\mu a_\mu. \tag{22.4}$$

For products of this kind the relations (22.1) become

$$(a\gamma)(b\gamma) + (b\gamma)(a\gamma) = 2(ab), \qquad (a\gamma)(a\gamma) = a^2 \tag{22.5}$$

and the identities (22.3) take the form

$$\begin{aligned}
\gamma_\mu(a\gamma)\gamma^\mu &= -2(a\gamma),\\
\gamma_\mu(a\gamma)(b\gamma)\gamma^\mu &= 4(ab),\\
\gamma_\mu(a\gamma)(b\gamma)(c\gamma)\gamma^\mu &= -2(c\gamma)(b\gamma)(a\gamma),\\
\gamma_\mu(a\gamma)(b\gamma)(c\gamma)(d\gamma)\gamma^\mu &= 2[(d\gamma)(a\gamma)(b\gamma)(c\gamma) + (c\gamma)(b\gamma)(a\gamma)(d\gamma)].
\end{aligned} \tag{22.6}$$

A widely used operation is the evaluation of the trace of a product of several γ matrices. Consider the quantities

$$T^{\mu_1\mu_2\ldots\mu_n} \equiv \tfrac{1}{4}\,\mathrm{Sp}\big(\gamma^{\mu_1}\gamma^{\mu_2}\ldots\gamma^{\mu_n}\big). \tag{22.7}$$

By the well-known cyclic property of the matrix trace, the tensor T is symmetric under cyclic permutations of the indices $\mu_1\mu_2\ldots\mu_n$.

Because the γ matrices take the same form in every reference frame, the quantities T are likewise frame-independent. They therefore constitute a tensor expressible solely through the metric tensor $g_{\mu\nu}$, which is the only tensor sharing this property.

From the rank-2 tensor $g_{\mu\nu}$ one can build only tensors of even rank. It follows at once that the trace of any product containing an odd number of γ factors vanishes. In particular, the trace of each individual γ is zero[13]:

$$\mathrm{Sp}\,\gamma^\mu = 0. \tag{22.8}$$

The trace of the 4×4 unit matrix (which is implicitly present on the right-hand side of the anticommutation relation (22.1)) equals 4. Taking the trace of both sides of (22.1) therefore gives

$$T^{\mu\nu} = g^{\mu\nu}. \tag{22.9}$$

The trace of a product of four matrices is

$$T^{\lambda\mu\nu\rho} = g^{\lambda\mu}g^{\nu\rho} - g^{\lambda\nu}g^{\mu\rho} + g^{\lambda\rho}g^{\mu\nu}. \tag{22.10}$$

This formula can be derived, for instance, by "dragging" γ^λ to the right through $\mathrm{Sp}(\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho)$ with the help of the anticommutation relation (22.1); each commutation produces one of the terms appearing in (22.10):

$$\begin{aligned}
T^{\lambda\mu\nu\rho} &= 2g^{\lambda\mu}T^{\nu\rho} - T^{\mu\lambda\nu\rho} = 2g^{\lambda\mu}g^{\nu\rho} - T^{\mu\lambda\nu\rho},\\
T^{\mu\lambda\nu\rho} &= 2g^{\lambda\nu}T^{\mu\rho} - T^{\mu\nu\lambda\rho} = 2g^{\lambda\nu}g^{\mu\rho} - T^{\mu\nu\lambda\rho},\\
T^{\mu\nu\lambda\rho} &= 2g^{\lambda\rho}T^{\mu\nu} - T^{\mu\nu\rho\lambda}
\end{aligned} \tag{22.11}$$

[12] In this edition we do not use any special notation for such a product. The literature frequently employs letters with a hat or with a slash (Feynman slash notation).

[13] The trace is invariant under similarity transformations $\gamma = U\gamma U^{-1}$. Hence (22.8) is also obvious from the explicit matrices (21.3).

and so on. After all permutations the right-hand side contains $-T^{\mu\rho\lambda} = -T^{\lambda\mu\nu\rho}$, which is brought to the left. By the same method the trace of a product of six γ's reduces to traces of products of four, etc. Thus,

$$T^{\lambda\mu\nu\rho\sigma\tau} = g^{\lambda\mu}T^{\nu\rho\sigma\tau} - g^{\lambda\nu}T^{\mu\rho\sigma\tau} + g^{\lambda\rho}T^{\mu\nu\sigma\tau} - \\ - g^{\lambda\sigma}T^{\mu\nu\rho\tau} + g^{\lambda\tau}T^{\mu\nu\rho\sigma}. \quad (22.11)$$

We remark that every trace $T^{\lambda\mu\cdots}$ is real, and that it is non-vanishing only when each of the matrices $\gamma^0, \gamma^1, \ldots$

occurs an even number of times in the product; both statements are plain from the formulae just obtained. From this one easily deduces that the trace is unchanged when the order of all factors is reversed:

$$T^{\lambda\mu\ldots\rho\sigma} = T^{\sigma\rho\ldots\mu\lambda}. \quad (22.12)$$

As already noted, the γ factors usually appear as scalar products with various 4-vectors. In that case formulae (22.9) and (22.10), for example, read

$$\begin{gathered} \tfrac{1}{4}\,\mathrm{Sp}(a\gamma)(b\gamma) = ab, \\ \tfrac{1}{4}\,\mathrm{Sp}(a\gamma)(b\gamma)(c\gamma)(d\gamma) = (ab)(cd) - (ac)(bd) + (ad)(bc). \end{gathered} \quad (22.13)$$

A special role is played by the product $\gamma^0\gamma^1\gamma^2\gamma^3$, for which the standard notation is

$$\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (22.14)$$

One readily verifies that

$$\gamma^5\gamma^\mu + \gamma^\mu\gamma^5 = 0, \qquad (\gamma^5)^2 = 1, \quad (22.15)$$

so that γ^5 anticommutes with every γ^μ. With respect to the matrices $\boldsymbol{\alpha}$ and β the rules are

$$\boldsymbol{\alpha}\gamma^5 - \gamma^5\boldsymbol{\alpha} = 0, \qquad \beta\gamma^5 + \gamma^5\beta = 0 \quad (22.16)$$

(the commutativity with $\boldsymbol{\alpha}$ follows from the fact that $\boldsymbol{\alpha} = \gamma^0\boldsymbol{\gamma}$ is a product of two γ^μ matrices).

The matrix γ^5 is Hermitian; indeed,

$$\gamma^{5+} = i\gamma^{3+}\gamma^{2+}\gamma^{1+}\gamma^{0+} = -i\gamma^3\gamma^2\gamma^1\gamma^0,$$

and because the sequence 3210 is obtained from 0123 by an even number of transpositions,

$$\gamma^{5+} = \gamma^5. \quad (22.17)$$

We also give the explicit form of this matrix in the two concrete representations:

$$\begin{aligned} &\text{spinor} \quad \gamma^5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}; \\ &\text{standard} \quad \gamma^5 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}. \end{aligned} \quad (22.18)$$

The trace of γ^5 vanishes:

$$\operatorname{Sp}\gamma^5 = 0 \tag{22.19}$$

(as is also directly apparent from (22.18)). Likewise zero are the traces of $\gamma^5\gamma^\mu\gamma^\nu$. For the product of γ^5 with four γ^μ factors, however, one has

$$\tfrac{1}{4}\operatorname{Sp}\,\gamma^5\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho = ie^{\lambda\mu\nu\rho}. \tag{22.20}$$

We note one further identity:

$$\gamma N = i\gamma^5(\gamma a)(\gamma b)(\gamma c), \qquad N^\lambda = e^{\lambda\mu\nu\rho}a_\mu b_\nu c_\rho, \tag{22.21}$$

valid for mutually orthogonal 4-vectors a, b, c: $ab = ac = bc = 0$.

There are situations (problems with non-relativistic particles) where one must compute traces of products containing γ^0 and the spatial "vector" $\boldsymbol{\gamma}$ as separate entities. A non-vanishing trace requires that γ^0 as well as $\boldsymbol{\gamma}$ each enter an even number of times. Every γ^0 factor then simplifies to the identity, and for products carrying two or four $\boldsymbol{\gamma}$ factors the traces read

$$\begin{gathered}\tfrac{1}{4}\operatorname{Sp}(\mathbf{a}\boldsymbol{\gamma})(\mathbf{b}\boldsymbol{\gamma}) = -\mathbf{ab},\\ \tfrac{1}{4}\operatorname{Sp}(\mathbf{a}\boldsymbol{\gamma})(\mathbf{b}\boldsymbol{\gamma})(\mathbf{c}\boldsymbol{\gamma})(\mathbf{d}\boldsymbol{\gamma}) = (\mathbf{ab})(\mathbf{cd}) - (\mathbf{ac})(\mathbf{bd}) + (\mathbf{ad})(\mathbf{bc}).\end{gathered} \tag{22.22}$$

§ 23. Plane waves

A free particle with definite momentum and energy is described by a plane wave, which we write as

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}}\,u_p\,e^{-ipx}. \tag{23.1}$$

The subscript p labels the 4-momentum; the wave amplitude u_p is a bispinor normalised in a specific way.

When carrying out the second quantisation later on, we shall need, alongside the wave functions (23.1), also the "negative-frequency" solutions that arise in relativistic theory, as explained in § 11, because of the two-valued square root $\pm\sqrt{\mathbf{p}^2 + m^2}$. Following the convention of § 11 we shall everywhere take ε to be the positive quantity $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$, so that the "negative frequency" is $-\varepsilon$; reversing also the sign of $\mathbf{p}$ we arrive at a function that is naturally denoted ψ_{-p}:

$$\psi_{-p} = \frac{1}{\sqrt{2\varepsilon}}\,u_{-p}\,e^{ipx}. \tag{23.2}$$

The physical meaning of these functions will become clear in § 26. Below we shall write formulae for ψ_p and ψ_{-p} side by side.

The components of the bispinor amplitudes u_p and u_{-p} satisfy the algebraic systems

$$(\gamma p - m)u_p = 0, \qquad (\gamma p + m)u_{-p} = 0, \tag{23.3}$$

obtained by substituting (23.1), (23.2) into the Dirac equation (which amounts to replacing the operator $\widehat{p}$ by $\pm p$)[14]. The relation $p^2 = m^2$ serves here as the consistency condition for each system. We shall always normalise the bispinor amplitudes by the invariant conditions

$$\bar{u}_p u_p = 2m, \qquad \bar{u}_{-p} u_{-p} = -2m \tag{23.4}$$

(the bar denotes, as throughout, Dirac conjugation: $\bar{u} = u^* \gamma^0$). Left-multiplying equations (23.3) by $\bar{u}_{\pm p}$ gives $(\bar{u}_{\pm p} \gamma u_{\pm p}) p = 2m^2 = 2p^2$, whence

$$\bar{u}_p \gamma u_p = \bar{u}_{-p} \gamma u_{-p} = 2p. \tag{23.5}$$

We note that the formulae for u_{-p} are obtained from those for u_p by changing the sign of m.

The current-density 4-vector is

$$j = \bar{\psi}_{\pm p} \gamma \psi_{\pm p} = \frac{1}{2\varepsilon} \bar{u}_{\pm p} \gamma u_{\pm p} = \frac{p}{\varepsilon}, \tag{23.6}$$

i.e. $j^\mu = (1, \mathbf{v})$, where $\mathbf{v} = \mathbf{p}/\varepsilon$ is the particle velocity. This shows that the functions ψ_p are normalised to "one particle per unit volume $V = 1$".

Because of equations (23.3) the components of the wave amplitude are related to one another; the explicit form of these relations depends, of course, on the choice of representation for ψ. Let us find them for the standard representation.

From equations (21.19) we have, for a plane wave,

$$(\varepsilon - m)\varphi - \mathbf{p}\boldsymbol{\sigma}\chi = 0, \qquad (\varepsilon + m)\chi - \mathbf{p}\boldsymbol{\sigma}\varphi = 0. \tag{23.7}$$

These relations yield the connection between φ and χ in two equivalent forms:

$$\varphi = \frac{\mathbf{p}\boldsymbol{\sigma}}{\varepsilon - m}\chi, \qquad \chi = \frac{\mathbf{p}\boldsymbol{\sigma}}{\varepsilon + m}\varphi \tag{23.8}$$

(the equivalence is plain: multiplying the first relation on the left by $\mathbf{p}\boldsymbol{\sigma}/(\varepsilon + m)$ and using $(\mathbf{p}\boldsymbol{\sigma})^2 = \mathbf{p}^2$, $\varepsilon^2 - m^2 = \mathbf{p}^2$ reproduces the second). The overall factor in φ and χ is chosen so as to satisfy the normalisation condition (23.4). The result for u_p (and likewise for u_{-p}) is

$$u_p = \begin{pmatrix} \sqrt{\varepsilon + m}\; w \\ \dfrac{\mathbf{p}\boldsymbol{\sigma}}{\sqrt{\varepsilon + m}}\, w \end{pmatrix}, \qquad u_{-p} = \begin{pmatrix} \dfrac{-\mathbf{p}\boldsymbol{\sigma}}{\sqrt{\varepsilon + m}}\, w' \\ \sqrt{\varepsilon + m}\; w' \end{pmatrix}, \tag{23.9}$$

(the second expression is obtained from the first by reversing the sign of m and relabelling $w \to (\mathbf{n}\boldsymbol{\sigma})w'$). Here $\mathbf{n}$ is the unit vector along $\mathbf{p}$, and w is an arbitrary two-component quantity subject only to the normalisation

$$w^* w = 1. \tag{23.10}$$

[14]We note also the analogous systems that follow from the Dirac equation (21.9) for the complex-conjugate function:

$$\bar{u}_p(\gamma p - m) = 0, \qquad \bar{u}_{-p}(\gamma p + m) = 0. \tag{23.3 a}$$

For $\bar{u} = u^*\gamma^0$ (with γ^0 from (21.20)) we have

$$\begin{aligned} \bar{u}_p &= \left(\sqrt{\varepsilon + m}\, w^*,\ -\sqrt{\varepsilon - m}\, w^*(\mathbf{n}\boldsymbol{\sigma})\right), \\ \bar{u}_{-p} &= \left(\sqrt{\varepsilon - m}\, w'^*(\mathbf{n}\boldsymbol{\sigma}),\ -\sqrt{\varepsilon + m}\, w'^*\right), \end{aligned} \tag{23.11}$$

and direct multiplication confirms that $\bar{u}_{\pm p} u_{\pm p} = \pm 2m$.

In the rest frame, i.e. when $\varepsilon = m$, we have

$$u_p = \sqrt{2m}\begin{pmatrix} w \\ 0 \end{pmatrix}, \qquad u_{-p} = \sqrt{2m}\begin{pmatrix} 0 \\ w' \end{pmatrix}, \tag{23.12}$$

i.e. w is precisely the 3-spinor that each wave amplitude becomes in the slow-motion limit. Observe that for the bispinor u_{-p} the components that vanish at rest are the upper two, not the lower two. This behaviour of the negative-frequency Dirac solutions is easy to understand: putting $\mathbf{p} = 0$ and $\varepsilon \to -m$ in (23.7) yields $\varphi = 0$[15].

The plane-wave amplitude contains one arbitrary two-component quantity. In other words, for a given momentum there exist two distinct independent states, corresponding to the two possible values of the spin projection. The projection of the spin on an arbitrary axis z cannot, however, possess a definite value. This is because the Hamiltonian of a particle with definite $\mathbf{p}$ (i.e. the matrix $H = \boldsymbol{\alpha}\mathbf{p} + \beta m$) does not commute with $\Sigma_z = -i\alpha_x\alpha_y$. In agreement with the general statements of § 16, however, the helicity λ—the projection of the spin onto the direction of $\mathbf{p}$—is conserved: the Hamiltonian commutes with $\mathbf{n}\boldsymbol{\Sigma}$.

Helicity eigenstates correspond to plane waves whose 3-spinor $w = w^{(\lambda)}(\mathbf{n})$ is an eigenfunction of the operator $\mathbf{n}\boldsymbol{\sigma}$:

$$\frac{1}{2}(\mathbf{n}\boldsymbol{\sigma})w^{(\lambda)} = \lambda w^{(\lambda)}. \tag{23.13}$$

The explicit form of these spinors is

$$w^{(\lambda=1/2)} = \begin{pmatrix} e^{-i\varphi/2}\cos\frac{\theta}{2} \\ e^{i\varphi/2}\sin\frac{\theta}{2} \end{pmatrix}, \qquad w^{(\lambda=-1/2)} = \begin{pmatrix} -e^{-i\varphi/2}\sin\frac{\theta}{2} \\ e^{i\varphi/2}\cos\frac{\theta}{2} \end{pmatrix}, \tag{23.14}$$

where θ and φ are the polar angle and azimuth of the direction $\mathbf{n}$ with respect to the fixed axes xyz[16].

An alternative choice of two independent states of a free particle with given $\mathbf{p}$ (simpler, though less transparent physically) corresponds to the two values of the z-projection of the spin in the rest frame; we denote it by σ. The associated spinors are

$$w^{(\sigma=1/2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad w^{(\sigma=-1/2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{23.15}$$

As the two linearly independent negative-frequency solutions we choose plane waves with 3-spinors

$$w^{(\sigma)\prime} = -\sigma_y w^{(-\sigma)} = 2\sigma i w^{(\sigma)} \tag{23.16}$$

[15] In the spinor representation one finds $\xi = -\eta$ rather than the relation $\xi = \eta$ that holds at rest for positive-frequency solutions.

[16] The solution of equations (23.13) admits multiplication by an arbitrary phase factor, reflecting the freedom of an arbitrary rotation about $\mathbf{n}$.

(the rationale behind this choice will become clear in § 26).

One can construct a representation of the plane wave in which, in every frame (not only in the rest frame), it has just two non-vanishing components that correspond to definite values of the same physical observable—the spin projection in the rest frame (*L. Foldy, S. A. Wouthuysen*, 1950).

Taking u_p in the standard representation (23.9) as the starting point, we look for a unitary map into such a representation, writing

$$u'_p = Uu_p, \qquad U = e^{W\boldsymbol{\gamma}\mathbf{n}},$$

where W is real; since $\boldsymbol{\gamma}^+ = -\boldsymbol{\gamma}$ the unitarity $U^+ = U^{-1}$ is automatic. Expanding in a series and noting that $(\boldsymbol{\gamma}\mathbf{n})^2 = -1$, we can write U as

$$U = \cos W + \boldsymbol{\gamma}\mathbf{n}\sin W$$

(cf. the passage from (18.13) to (18.14)). Requiring that the last two components of the transformed amplitude u'_p vanish, we find

$$\operatorname{tg} W = \frac{|\mathbf{p}|}{m+\varepsilon},$$

so that

$$U = \frac{m+\varepsilon+(\boldsymbol{\gamma}\mathbf{n})|\mathbf{p}|}{\sqrt{2\varepsilon(\varepsilon+m)}}.$$

In the new representation

$$u'_p = \sqrt{2\varepsilon}\begin{pmatrix} w \\ 0 \end{pmatrix}. \tag{23.17}$$

The particle Hamiltonian in this representation becomes

$$\widehat{H}' = U(\boldsymbol{\alpha}\mathbf{p} + \beta m)U^{-1} = \beta\varepsilon \tag{23.18}$$

(here β, $\boldsymbol{\alpha}$, $\boldsymbol{\gamma}$ are the standard-representation matrices). This Hamiltonian commutes with

$$\boldsymbol{\Sigma} = -\boldsymbol{\alpha}\gamma^5 = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix},$$

which, in the Foldy–Wouthuysen representation, plays the role of the operator for the conserved rest-frame spin.

§ 24. Spherical waves

The wave functions of a free spin-1/2 particle with definite values of the total angular momentum j are spinor spherical waves. We shall determine their form, first recalling the analogous expressions from non-relativistic theory.

The non-relativistic wave function is a 3-spinor $\psi = \binom{\psi^1}{\psi^2}$. For a state with definite energy ε (and hence definite magnitude of the momentum p [17]), orbital angular momentum l, total angular momentum j, and projection m, it takes the form

$$\psi = R_{pl}(r)\Omega_{jlm}(\theta, \varphi). \tag{24.1}$$

[17] Throughout this section p denotes $|\mathbf{p}|$.

The angular factor Ω_{jlm} is a 3-spinor whose components (for the two values $j = l \pm 1/2$ possible at a given l) are

$$\Omega_{l+1/2,\,l,\,m} = \begin{pmatrix} \sqrt{\dfrac{j+m}{2j}}\, Y_{l,\,m-1/2} \\ \sqrt{\dfrac{j-m}{2j}}\, Y_{l,\,m+1/2} \end{pmatrix}, \tag{24.2}$$

$$\Omega_{l-1/2,\,l,\,m} = \begin{pmatrix} -\sqrt{\dfrac{j-m+1}{2j+2}}\, Y_{l,\,m-1/2} \\ \sqrt{\dfrac{j+m+1}{2j+2}}\, Y_{l,\,m+1/2} \end{pmatrix}$$

(see III, § 106, problem). We shall call Ω_{jlm} the *spherical spinors*. Their normalisation is

$$\int \Omega^*_{jlm}\Omega_{j'l'm'}\, do = \delta_{jj'}\delta_{ll'}\delta_{mm'}. \tag{24.3}$$

The radial functions R_{pl} are a common factor in both spinor components of ψ and are given by the formula

(33.10) (see III):

$$R_{pl} = \sqrt{\frac{2\pi p}{r}}\, J_{l+1/2}(pr). \tag{24.4}$$

They are normalised according to

$$\int_0^\infty r^2 R_{p'l} R_{pl}\, dr = 2\pi\delta(p'-p). \tag{24.5}$$

Turning now to the relativistic case, we recall first of all that for a moving particle no separate conservation laws for spin and orbital angular momentum exist: the operators $\widehat{\mathbf{s}}$ and $\widehat{\mathbf{l}}$ individually fail to commute with the Hamiltonian. The parity of the state, however, remains conserved (for a free particle). Consequently the quantum number l ceases to specify a definite value of the orbital angular momentum, but it does determine (see below) the parity of the state.

We shall work with the desired wave function (a bispinor) in the standard representation: $\psi = \binom{\varphi}{\chi}$. Under rotations φ and χ transform as 3-spinors, so their angular dependence is again carried by the spherical spinors Ω_{jlm}. Let $\varphi \propto \Omega_{jlm}$, where l is one (definite) of the two values $j + 1/2$ or $j - 1/2$. Under inversion $\varphi(\mathbf{r}) \to i\varphi(-\mathbf{r})$ (see (21.18)), and since $\Omega_{jlm}(-\mathbf{n}) = (-1)^l\Omega_{jlm}(\mathbf{n})$,

$$\varphi(\mathbf{r}) \to i(-1)^l\varphi(\mathbf{r}).$$

The lower component χ, on the other hand, transforms under inversion as $\chi(\mathbf{r}) \to -i\chi(-\mathbf{r})$. For the state to possess a definite parity (i.e. for inversion to multiply every component by the same phase), the angular dependence of χ must involve $\Omega_{jl'm}$ with the other value of l: because the two allowed values differ by unity, $(-1)^{l'} = -(-1)^l$.

Furthermore, the radial dependence of φ and χ is governed by the same functions R_{pl} and $R_{pl'}$ (with l and l' matching the order of the spherical harmonics entering Ω_{jlm}).

This follows from the fact that each component of ψ satisfies the second-order equation $(\hat{p}^2 - m^2)\psi = 0$, which for a given $|\mathbf{p}|$ reads

$$(\Delta + \mathbf{p}^2)\psi = 0,$$

coinciding in form with the free-particle Schrödinger equation of the non-relativistic theory.

Thus

$$\varphi = AR_{pl}\Omega_{jlm}, \qquad \chi = BR_{pl'}\Omega_{jl'm}, \tag{24.6}$$

and it remains to fix the constant coefficients A and B. To this end we examine the far zone, where the spherical wave may be treated as a plane wave. By the asymptotic formula (33.12) (see III)

$$R_{pl} \approx \frac{1}{ir}\left\{e^{i\left(pr-\frac{\pi l}{2}\right)} - e^{-i\left(pr-\frac{\pi l}{2}\right)}\right\}, \tag{24.7}$$

so that φ is a superposition of two plane waves propagating along $\pm\mathbf{n}$ ($\mathbf{n} = \mathbf{r}/r$). For each of these, by (23.8),

$$\chi = \frac{p}{\varepsilon + m}(\pm\mathbf{n}\boldsymbol{\sigma})\varphi.$$

From what has been said above (formulae (24.6)) it is clear that $(\mathbf{n}\boldsymbol{\sigma})\Omega_{jlm} = a\Omega_{jl'm}$ with a a constant. This constant is easily found by evaluating both sides at $m = 1/2$ with $\mathbf{n}$ along the z-axis. Using (7.2a) one obtains

$$(\mathbf{n}\boldsymbol{\sigma})\Omega_{jlm} = i^{l'-l}\Omega_{jl'm}. \tag{24.8}$$

Collecting the formulae and comparing with (24.6), we find

$$B = -\frac{p}{\varepsilon + m}A.$$

Finally, the coefficient A is fixed by the overall normalisation of ψ. Adopting the condition

$$\int \psi^*_{pjlm}\psi_{p'j'l'm'}\, d^3x = 2\pi\delta_{jj'}\delta_{ll'}\delta_{mm'}\delta(p - p'), \tag{24.9}$$

we arrive at the final result

$$\psi_{pjlm} = \frac{1}{\sqrt{2\varepsilon}}\begin{pmatrix}\sqrt{\varepsilon + m}\,R_{pl}\,\Omega_{jlm}\\ -\sqrt{\varepsilon - m}\,R_{pl'}\,\Omega_{jl'm}\end{pmatrix}, \qquad l' = 2j - l. \tag{24.10}$$

At fixed j and m (and energy ε) there are therefore two states that differ in parity. Which parity a state carries is uniquely determined by l ($= j \pm 1/2$): inversion multiplies the bispinor (24.10) by $i(-1)^l$. The upper and lower components of this bispinor do, however, involve spherical harmonics of both orders l and l'—a direct manifestation of the fact that the orbital angular momentum is not separately well-defined.

As $r \to \infty$ one can, in any small region of space, regard the spherical waves (24.7) as plane waves with momentum $\mathbf{p} = \pm p\mathbf{n}$. It is therefore clear that the momentum-space wave functions differ from (24.10) mainly in the

absence of radial factors, with $\mathbf{n}$ now denoting the direction of the momentum.

To pass directly to the momentum representation one performs the Fourier transform:

$$\psi(\mathbf{p}') = \int \psi(\mathbf{r})\, e^{-i\mathbf{p}'\mathbf{r}}\, d^3x, \tag{24.11}$$

The integral is evaluated with the aid of the plane-wave expansion in spherical harmonics (see III, (34.3)):

$$e^{i\mathbf{pr}} = \frac{2\pi}{p} \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l R_{pl}(r)\, Y^*_{lm}\Big(\frac{\mathbf{p}}{p}\Big) Y_{lm}\Big(\frac{\mathbf{r}}{r}\Big). \tag{24.12}$$

Substituting $e^{-i\mathbf{p}'\mathbf{r}}$ into (24.11) via this expansion and using (24.5), the Fourier components of the function

$$\psi(\mathbf{r}) = R_{pl}(r)\, \Omega_{jlm}\Big(\frac{\mathbf{r}}{r}\Big)$$

are found to be

$$\psi(\mathbf{p}') = \frac{(2\pi)^2}{p}\, \delta(p'-p)\, i^{-l} Y_{lm'}\Big(\frac{\mathbf{p}'}{p'}\Big) \int \Omega_{jlm}\Big(\frac{\mathbf{r}}{r}\Big) Y^*_{lm'}\Big(\frac{\mathbf{r}}{r}\Big)\, do.$$

The integral here equals the coefficients of the spherical harmonics in the definition of the spherical spinors (24.2), and together with the factor $Y_{lm'}(\mathbf{p}'/p')$ it reconstructs the same spherical spinor, now depending on $\mathbf{p}'/p'$:

$$\psi(\mathbf{p}') = \frac{(2\pi)^2}{p}\, \delta(p'-p)\, i^{-l}\, \Omega_{jlm}\Big(\frac{\mathbf{p}'}{p'}\Big).$$

Applying this result to the bispinor wave function (24.10), we obtain its momentum-space form

$$\psi_{pjlm}(\mathbf{p}') = \delta(p'-p)\, \frac{(2\pi)^2}{p\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon+m}\, i^{-l}\, \Omega_{jlm}(\mathbf{p}'/p') \\ -\sqrt{\varepsilon-m}\, i^{-l'}\, \Omega_{jl'm}(\mathbf{p}'/p') \end{pmatrix}. \tag{24.13}$$

The states $|pjlm\rangle$ coincide with the states $|pjm|\lambda||$ considered in § 16 (with $|\lambda| = 1/2$): both sets possess definite values of pjm and parity. The spherical spinors Ω_{jlm} are therefore expressible through the functions $D^{(j)}_m$ (both depending on the argument $\mathbf{p}/p$). In the limit $p \to 0$ the wave functions (24.13) reduce to 3-spinors Ω_{jlm} with parity $P = \eta(-1)^l$ (where $\eta = i$ is the "intrinsic parity" of the spinor). Comparison with the results of § 16 leads to the formula

$$\Omega_{jlm} = i^l \sqrt{\frac{2j+1}{8\pi}} \Big(w^{(-1/2)} D^{(j)}_{-1/2,m} \pm w^{(1/2)} D^{(j)}_{1/2,m} \Big) \tag{24.14}$$

(with $l = j \mp 1/2$, where $w^{(\lambda)}$ are the 3-spinors (23.14)).

§ 25. The connection between spin and statistics

Second quantisation of the field of spin-1/2 particles (the spinor field) proceeds in exactly the same way as was done in § 11 for the scalar field.

Without repeating the entire argument, we write down at once the expressions for the field operators, fully analogous to (11.2):

$$\begin{aligned}\widehat{\psi} &= \sum_{\mathbf{p}\sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\widehat{a}_{\mathbf{p}\sigma} u_{p\sigma} e^{-ipx} + \widehat{b}^{+}_{\mathbf{p}\sigma} u_{-p\,-\sigma} e^{ipx}\right),\\ \widehat{\bar{\psi}} &\equiv \widehat{\psi}^{+}\gamma^{0} = \sum_{\mathbf{p}\sigma} \frac{1}{\sqrt{2\varepsilon}} \left(\widehat{a}^{+}_{\mathbf{p}\sigma} \bar{u}_{p\sigma} e^{ipx} + \widehat{b}_{\mathbf{p}\sigma} \bar{u}_{-p\,-\sigma} e^{-ipx}\right);\end{aligned} \tag{25.1}$$

the summation runs over all values of $\mathbf{p}$ and over $\sigma = \pm 1/2$. The antiparticle annihilation operators $\widehat{b}_{\mathbf{p}\sigma}$ (like the particle annihilation operators $\widehat{a}_{\mathbf{p}\sigma}$) stand as coefficients of functions whose coordinate dependence (e^{ipx}) corresponds to a state with momentum $\mathbf{p}$ [18].

To construct the Hamiltonian of the spinor field there is no need to determine its energy–momentum tensor (as we did for the scalar field), because here the single-particle Hamiltonian exists and can be used to write down the wave equation (the Dirac equation) (21.12). The mean energy of a particle in a state with wave function ψ is the integral

$$\int \psi^{*} \widehat{H} \psi \, d^{3}x = \int \psi^{*} \frac{\partial \psi}{\partial t} \, d^{3}x = i \int \bar{\psi} \gamma^{0} \frac{\partial \psi}{\partial t} \, d^{3}x. \tag{25.2}$$

We draw attention to the fact that the "energy density" (the integrand) is not positive definite here.

Replacing ψ and $\bar{\psi}$ in (25.2) by ψ-operators, taking into account the mutual orthogonality of wave functions with different $\mathbf{p}$ or σ, and also the relation $\bar{u}_{\pm p\sigma}\gamma^{0}u_{\pm p\sigma} = 2\varepsilon$ for the wave amplitudes, we obtain the field Hamiltonian

$$\widehat{H} = \sum_{\mathbf{p}\sigma} \varepsilon \left(\widehat{a}^{+}_{\mathbf{p}\sigma}\widehat{a}_{\mathbf{p}\sigma} - \widehat{b}_{\mathbf{p}\sigma}\widehat{b}^{+}_{\mathbf{p}\sigma}\right). \tag{25.3}$$

It follows that quantisation must here be carried out according to Fermi statistics:

$$\{\widehat{a}_{\mathbf{p}\sigma}, \widehat{a}^{+}_{\mathbf{p}\sigma}\}_{+} = 1, \qquad \{\widehat{b}_{\mathbf{p}\sigma}, \widehat{b}^{+}_{\mathbf{p}\sigma}\}_{+} = 1, \tag{25.4}$$

while all other pairs of operators $\widehat{a}$, $\widehat{a}^{+}$, $\widehat{b}$, $\widehat{b}^{+}$ anticommute (see III, § 65). Indeed, the Hamiltonian (25.3) then takes the form

$$\widehat{H} = \sum_{\mathbf{p}\sigma} \varepsilon \left(\widehat{a}^{+}_{\mathbf{p}\sigma}\widehat{a}_{\mathbf{p}\sigma} + \widehat{b}^{+}_{\mathbf{p}\sigma}\widehat{b}_{\mathbf{p}\sigma} - 1\right),$$

and the energy eigenvalues (after subtracting, as always, the infinite additive constant) are

$$E = \sum_{\mathbf{p}\sigma} \varepsilon (N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma}), \tag{25.5}$$

which are, as required, positive definite. Had we quantised according to Bose statistics, equation (25.3) would have given the meaningless eigenvalues

$$\sum_{\mathbf{p}\sigma} \varepsilon (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma}),$$

[18] Both types of function also correspond to the same value of σ, the rest-frame spin projection: for the functions $\bar{\psi}_{-p,-\sigma}$ this will be demonstrated in § 26, see (26.10).

which are not positive definite.

The analogous expression

$$\mathbf{P} = \sum_{\mathbf{p}\sigma} \mathbf{p}(N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma}) \tag{25.6}$$

is obtained for the momentum of the system—the eigenvalues of the operator $\int \widehat{\psi}^{+}\widehat{\mathbf{p}}\widehat{\psi}\, d^3x$.

The 4-current operator is

$$\widehat{j}^{\mu} = \widehat{\bar{\psi}}\gamma^{\mu}\widehat{\psi}, \tag{25.7}$$

and for the "charge" operator of the field we get

$$\widehat{Q} = \int \widehat{\bar{\psi}}\gamma^0\widehat{\psi}\, d^3x = \sum_{\mathbf{p}\sigma}(\widehat{a}^{+}_{\mathbf{p}\sigma}\widehat{a}_{\mathbf{p}\sigma} + \widehat{b}_{\mathbf{p}\sigma}\widehat{b}^{+}_{\mathbf{p}\sigma}) = \sum_{\mathbf{p}\sigma}(\widehat{a}^{+}_{\mathbf{p}\sigma}\widehat{a}_{\mathbf{p}\sigma} - \widehat{b}^{+}_{\mathbf{p}\sigma}\widehat{b}_{\mathbf{p}\sigma} + 1), \tag{25.8}$$

with eigenvalues

$$Q = \sum_{\mathbf{p}\sigma}(N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma}). \tag{25.9}$$

We thus arrive once more at the picture of particles and antiparticles; everything said about them in § 11 applies here as well.

Whereas spin-0 particles are bosons, however, spin-1/2 particles turn out to be fermions. Tracing this difference to its formal origin, one finds that it stems from a difference in the character of the "energy density" for the scalar and spinor fields. In the scalar case the energy density is positive definite, so that both terms in the Hamiltonian (11.3)

($\widehat{a}^{+}\widehat{a}$ and $\widehat{b}\widehat{b}^{+}$) enter with a plus sign. To guarantee positive energy eigenvalues, the replacement of $\widehat{b}\widehat{b}^{+}$ by $\widehat{b}^{+}\widehat{b}$ must then proceed without a change of sign, i.e. by the Bose commutation rule. For the spinor field, on the other hand, the energy density is not positive definite, so that the $\widehat{b}\widehat{b}^{+}$ term in the Hamiltonian (25.3) carries a minus sign; obtaining positive eigenvalues now requires that the rearrangement $\widehat{b}\widehat{b}^{+} \to \widehat{b}^{+}\widehat{b}$ be accompanied by a sign change, i.e. that it follow the Fermi commutation rule.

On the other hand, the form of the energy density is directly tied to the transformation properties of the wave function and to the demands of relativistic invariance. One can therefore say that the connection between a particle's spin and the statistics it obeys is itself a direct consequence of these requirements.

From the fact that spin-1/2 particles are fermions one can also draw a general conclusion: every particle with half-integer spin is a fermion, and every particle with integer spin is a boson (the latter includes the spin-0 result proved in § 11) [19].

This becomes obvious once one notes that a particle of spin s may be thought of as "assembled" from $2s$ particles of spin $1/2$. When s is half-integer the number $2s$ is odd, while for integer s it is even. Now a composite object built from an even count

[19] The origin of the connection between the spin of a particle and the statistics it obeys was elucidated by *Pauli* (*W. Pauli*, 1940).

of fermions behaves as a boson, whereas one containing an odd count of fermions is a fermion[20].

When a system contains particles of several species, each species must be equipped with its own creation and annihilation operators. Operators belonging to different bosons, or to a boson and a fermion, commute with one another. As for operators belonging to different fermions, in non-relativistic theory they could be taken either commuting or anticommuting (III, § 65). In a relativistic theory, however, which allows mutual transformations of particles, the creation and annihilation operators of distinct fermion species must be regarded as anticommuting—just

like operators referring to different states of the same fermion.

Problem

Find the Lagrangian of the spinor field.

Solution. A real, Lorentz-scalar Lagrangian that yields the Dirac equation is

$$L = \frac{i}{2}\left(\bar{\psi}\gamma^{\mu}\partial_{\mu}\psi - \partial_{\mu}\bar{\psi}\cdot\gamma^{\mu}\psi\right) - m\bar{\psi}\psi. \tag{1}$$

Treating the components of ψ and $\bar{\psi}$ as "generalised coordinates" q, one easily verifies that the corresponding Euler–Lagrange equations (10.10) coincide with the Dirac equations for $\bar{\psi}$ and ψ. The overall sign of the Lagrangian (as well as the overall numerical coefficient) is a matter of convention here. Because L is linear in the derivatives of ψ and $\bar{\psi}$, the action $S = \int L\, d^4x$ can possess neither a minimum nor a maximum. The condition $\delta S = 0$ determines in this case only a stationary point, not an extremum, of the integral.

The field-theoretic Lagrangian of the spinor field is obtained by replacing ψ in (1) with the operator $\widehat{\psi}$. Applying to this Lagrangian the formula (12.12), one recovers the current operator (25.7).

§ 26. Charge conjugation and time reversal of spinors

The factors $\psi_{p\sigma} = u_{p\sigma}e^{-ipx}$ that multiply the operators $\widehat{a}_{\mathbf{p}\sigma}$ in (25.1) are the wave functions of free particles (which we shall call "electrons") carrying momentum $\mathbf{p}$ and polarisation σ:

$$\psi^{(\mathrm{e})}_{p\sigma} = \psi_{p\sigma}.$$

The factors $\bar{\psi}_{-p\,-\sigma}$ accompanying the operators $\widehat{b}_{p\sigma}$, on the other hand, should be regarded as positron wave functions for the same $\mathbf{p}$ and σ. It turns out, however, that the electron and positron functions are expressed in different bispinor representations. This is apparent from the fact that ψ and $\bar{\psi}$ have distinct transformation properties, their components obeying different systems of equations. To eliminate this drawback one must carry out a specific unitary transformation of the components of $\bar{\psi}_{-p\,-\sigma}$—one that

[20]This reasoning tacitly assumes that all particles of the same spin obey the same statistics, regardless of how they are "assembled". That this is indeed so can be seen from an analogous argument. If spin-0 fermions existed, then combining a spin-0 fermion with a spin-1/2 fermion would produce a spin-1/2 particle that is a boson—in contradiction with the general result established above for spin 1/2.

ensures the resulting four-component function satisfies the same equation as $\psi_{p\sigma}$ [21]. It is precisely such a function that we shall call the positron wave function (with momentum **p** and polarisation σ). Denoting the matrix of the required unitary
transformation by U_C, we write

$$\psi^{(\mathrm{p})}_{p\sigma} = U_C \bar{\psi}_{-p\,-\sigma}. \tag{26.1}$$

The operation C by which this function is constructed from $\psi_{-p\,-\sigma}$ is called the *charge conjugation* of the wave function (*H. A. Kramers*, 1937). This notion is by no means limited to plane waves. For any function ψ whatsoever there exists a "charge-conjugate" function

$$\widehat{C}\psi(t, \mathbf{r}) = U_C \bar{\psi}(t, \mathbf{r}), \tag{26.2}$$

which transforms in the same way as ψ and obeys the same equation.

The properties of U_C follow from this definition. If ψ solves the Dirac equation $(\gamma\widehat{p} - m)\psi = 0$, then $\bar{\psi}$ satisfies

$$\bar{\psi}(\gamma\widehat{p} + m) = 0, \qquad \text{i.e.} \quad (\widetilde{\gamma}\widehat{p} + m)\widetilde{\bar{\psi}} = 0.$$

Multiplying this equation on the left by U_C:

$$U_C\widetilde{\gamma}\widehat{p}\,\widetilde{\bar{\psi}} + mU_C\widetilde{\bar{\psi}} = 0,$$

and demanding that $U_C\widetilde{\bar{\psi}}$ obey the same equation as ψ:

$$(\gamma\widehat{p} - m)U_C\widetilde{\bar{\psi}} = 0,$$

we compare the two equations and obtain the following "commutation relation" between U_C and the matrices $\gamma^{\mu\,-1)}$:

$$U_C\widetilde{\gamma}^{\mu} = -\gamma^{\mu}U_C. \tag{26.3}$$

We shall henceforth assume that the wave functions are given in the spinor or standard representation (the case of an arbitrary representation will be discussed only at the end of this section). In these representations

$$\gamma^{0,2} = \widetilde{\gamma}^{0,2}, \qquad \gamma^{1,3} = -\widetilde{\gamma}^{1,3},$$

$$\left(\gamma^{0,1,3}\right)^* = \gamma^{0,1,3}, \qquad \gamma^{2*} = -\gamma^2. \tag{26.4}$$

The conditions (26.3) are then met by the matrix $U_C = \eta_C\gamma^2\gamma^0$ with an arbitrary constant η_C. The requirement $\widehat{C}^2 = 1$ implies $|\eta_C|^2 = 1$, so U_C is fixed up to
a phase factor. We shall set $\eta_C = 1$, giving

$$U_C = \gamma^2\gamma^0 = -\alpha_y. \tag{26.5}$$

[21] For spin-0 particles this question did not arise, since the scalar functions ψ and ψ^* both satisfy the same equation, and ψ^*_{-p} simply coincides with ψ_p.

Noting further that $\bar{\psi} = \psi^*\gamma^0 = \widetilde{\gamma}^0\psi^* = \gamma^0\psi^*$, one can write the action of $\widehat{C}$ as

$$\widehat{C}\psi = \gamma^2\gamma^0\widetilde{\bar{\psi}} = \gamma^2\psi^*. \tag{26.6}$$

Written out explicitly, the transformation (26.6) reads in the spinor representation

$$\begin{aligned} C:\quad & \xi^\alpha \to -i\eta^{\dot{\alpha}*}, \quad \eta_{\dot{\alpha}} \to -i\xi^*_\alpha, \\ C:\quad & \xi_\alpha \to i\xi^{\alpha*}, \qquad \eta^\alpha \to -i\eta^*_{\dot{\alpha}}. \end{aligned} \tag{26.7a, b}$$

The charge-conjugation transformation for the plane waves $\psi_{\pm p\sigma}$ is easily carried out using their explicit forms (23.9) together with the matrix U_C in the standard representation:

$$U_C = \begin{pmatrix} 0 & -\sigma_y \\ -\sigma_y & 0 \end{pmatrix}. \tag{26.8}$$

Noting that

$$\sigma_y\sigma^* = -\boldsymbol{\sigma}\sigma_y,$$

and using the definition of $w^{(\sigma)\prime}$ from (23.16), we obtain

$$U_C\bar{u}_{-p\,-\sigma} = u_{p\sigma}, \qquad U_C u_{-p\,-\sigma} = \bar{u}_{p\sigma}. \tag{26.9}$$

Consequently,

$$\widehat{C}\psi_{-p\,-\sigma} = \psi_{p\sigma}, \tag{26.10}$$

confirming that the functions $\psi_{-p\,-\sigma}$ appearing in the ψ-operators (25.1) alongside $\widehat{b}_{p\sigma}$ do indeed describe states of a particle with momentum $\mathbf{p}$ and polarisation σ. We also see that the electron and positron states are described by the same functions:

$$\psi^{(\mathrm{e})}_{p\sigma} = \psi^{(\mathrm{p})}_{p\sigma} = \psi_{p\sigma}.$$

This is entirely natural, since the functions $\psi_{p\sigma}$ encode only the momentum and polarisation of the particle.

The operation of time reversal can be treated in an analogous manner. Reversing the sign of time must be accompanied by complex conjugation of the wave function. In order for the resulting "time-reversed" wave function ($\widehat{T}\psi$) to belong to the same representation as the original ψ, one must

perform a further unitary transformation on the components of ψ^* (or $\bar{\psi}$). Proceeding in analogy with (26.2), we represent the action of $\widehat{T}$ on ψ as

$$\widehat{T}\psi(t, \mathbf{r}) = U_T\bar{\psi}(-t, \mathbf{r}), \tag{26.11}$$

where U_T is a unitary matrix.

Writing once more the Dirac equation for ψ:

$$\left(i\gamma^0\frac{\partial}{\partial t} + i\boldsymbol{\gamma}\boldsymbol{\nabla} - m\right)\psi(t, \mathbf{r}) = 0,$$

and the corresponding equation for $\bar{\psi}$:

$$\left(i\widetilde{\gamma}^0\frac{\partial}{\partial t} + i\widetilde{\boldsymbol{\gamma}}\boldsymbol{\nabla} + m\right)\widetilde{\bar{\psi}}(t, \mathbf{r}) = 0.$$

Replacing $t \to -t$ in the latter and multiplying on the left by $-U_T$:

$$\left(iU_T\widetilde{\gamma}^0\frac{\partial}{\partial t} - iU_T\widetilde{\boldsymbol{\gamma}}\boldsymbol{\nabla}\right)\widetilde{\overline{\psi}}(-t,\mathbf{r}) - mU_T\widetilde{\overline{\psi}}(-t,\mathbf{r}) = 0.$$

We require $U_T\widetilde{\overline{\psi}}(-t,\mathbf{r})$ to satisfy the same equation as $\psi(t,\mathbf{r})$:

$$\left(i\gamma^0\frac{\partial}{\partial t} + i\boldsymbol{\gamma}\boldsymbol{\nabla}\right)U_T\widetilde{\overline{\psi}}(-t,\mathbf{r}) - mU_T\widetilde{\overline{\psi}}(-t,\mathbf{r}) = 0.$$

Comparison yields the conditions

$$U_T\widetilde{\gamma}^0 = \gamma^0 U_T, \qquad U_T\widetilde{\boldsymbol{\gamma}} = -\boldsymbol{\gamma}U_T. \tag{26.12}$$

In the spinor and standard representations, these conditions are satisfied by the matrix[22]

$$U_T = i\gamma^3\gamma^1\gamma^0. \tag{26.13}$$

The action of $\widehat{T}$ is therefore given by

$$\widehat{T}\psi(t,\mathbf{r}) = i\gamma^3\gamma^1\gamma^0\widetilde{\overline{\psi}}(-t,\mathbf{r}) = i\gamma^3\gamma^1\psi^*(-t,\mathbf{r}). \tag{26.14}$$

Explicitly, in the spinor representation this transformation reads

$$T: \quad \xi^\alpha \to -i\xi^*_\alpha, \qquad \eta_{\dot\alpha} \to i\eta^{\dot\alpha *} \tag{26.15 a}$$

or

$$T: \quad \xi_\alpha \to i\xi^{\alpha *}, \qquad \eta^\alpha \to -i\eta^*_{\dot\alpha}. \tag{26.15 b}$$

In the standard representation

$$U_T = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix}. \tag{26.16}$$

Let us now find the result of applying all three operations P, T, and C to ψ. Writing them out in succession:

$$\widehat{T}\psi(t,\mathbf{r}) = -i\gamma^1\gamma^3\psi^*(-t,\mathbf{r}),$$
$$\widehat{P}\widehat{T}\psi(t,\mathbf{r}) = i\gamma^0(\widehat{T}\psi) = \gamma^0\gamma^1\gamma^3\psi^*(-t,-\mathbf{r}),$$
$$\widehat{C}\widehat{P}\widehat{T}\psi(t,\mathbf{r}) = \gamma^2(\gamma^0\gamma^1\gamma^3\psi^*)^* = \gamma^2\gamma^0\gamma^1\gamma^3\psi(-t,-\mathbf{r}),$$

which gives

$$\widehat{C}\widehat{P}\widehat{T}\psi(t,\mathbf{r}) = i\gamma^5\psi(-t,-\mathbf{r}). \tag{26.17}$$

In the spinor representation

$$CPT: \quad \xi^\alpha \to -i\xi^\alpha, \qquad \eta_{\dot\alpha} \to i\eta_{\dot\alpha}, \tag{26.18}$$

[22] The choice of phase factor in (26.13) is tied to the choice made in (26.5) by considerations discussed below in the footnote on p. 124.

as can also be verified directly from the transformation rules (20.4), (26.7), (26.15)[23].

The expressions for the matrices U_C and U_T written above assumed the spinor or standard representation of ψ. Let us finally clarify which properties of these expressions remain valid for an arbitrary representation.

If ψ undergoes a unitary transformation:

$$\psi' = U\psi, \quad \gamma' = U\gamma U^{-1}, \quad \bar{\psi}' = \psi'^{*}\gamma^{0\prime} = \bar{\psi}U^{+} = \bar{\psi}\widetilde{U}^{-1}, \tag{26.19}$$

then in the new representation

$$(\widehat{C}\psi)' = U(\widehat{C}\psi) = UU_C\widetilde{\bar{\psi}} = UU_C(\widetilde{\bar{\psi}'U}) = UU_C\widetilde{U}\widetilde{\bar{\psi}}'.$$

Comparing with the definition of U'_C in the new representation ($(\widehat{C}\psi)' = U'_C\widetilde{\bar{\psi}}'$), we find

$$U'_C = UU_C\widetilde{U}. \tag{26.20}$$

The transformation (26.20) coincides with the transformation law of the γ matrices only when U is real. For this reason the expression (26.5) holds only in representations obtainable from the spinor or standard representation by a real transformation.

The matrix (26.5) is unitary, and transposition reverses its sign:

$$U_C U_C^{+} = 1, \qquad \widetilde{U}_C = -U_C. \tag{26.21}$$

These properties are invariant under the transformation (26.20) and hence hold in every representation. The matrix (26.5) is also Hermitian ($U_C = U_C^{+}$), but this property is in general not preserved by (26.20).

Everything just said (including (26.21)) applies equally to the properties of the matrix U_T.

Within the second-quantisation formalism, the transformations C, P, T for ψ-operators must be formulated as transformation rules for the creation and annihilation operators of the particles. These rules can be established (similarly to how it was done in § 13 for spin-0 particles) by requiring that the transformed ψ-operators admit a representation of the form

$$\begin{aligned} \widehat{\psi}^{C}(t,\mathbf{r}) &= U_C\widetilde{\widehat{\bar{\psi}}}(t,\mathbf{r}), \\ \widehat{\psi}^{P}(t,\mathbf{r}) &= i\gamma^0\widehat{\psi}(t,-\mathbf{r}), \\ \widehat{\psi}^{T}(t,\mathbf{r}) &= U_T\widetilde{\widehat{\bar{\psi}}}(-t,\mathbf{r}). \end{aligned} \tag{26.22}$$

Problem

Find the charge-conjugation operator in the Majorana representation (see Problem 2, § 21).

Solution. The matrix U'_C in the Majorana representation is obtained from $U_C = -\alpha_y$ in the standard representation via the transformation (26.20) with $U = (\alpha_y + \beta)/\sqrt{2}$,

[23] The notation $\widehat{C}\widehat{P}\widehat{T}$ implies that the operators act from right to left. The overall sign in (26.17), (26.18) depends on this ordering because $\widehat{T}$ does not commute with $\widehat{C}$ and $\widehat{P}$ (in their action on a bispinor).

and equals $U'_C = \alpha_y$ (α_y and β denote the standard-representation matrices). Denoting quantities in the Majorana representation by a prime, we have $\widehat{C}\psi' = U'_C(\psi'^*\beta')$, and since $\beta' = \alpha_y$,

$$\widehat{C}\psi' = \alpha_y(\psi'^*\alpha_y) = \alpha_y\widetilde{\alpha}_y\psi'^* = \psi'^*,$$

i.e. charge conjugation reduces to complex conjugation.

§ 27. Internal symmetry of particles and antiparticles

In its rest frame the wave function of a spin-$1/2$ particle reduces to a single 3-spinor (which we denote Φ^α). The behaviour of this spinor under inversion is linked to the notion of the particle's intrinsic parity. However (as already noted in § 19), although the two possible transformation laws for 3-spinors ($\Phi^\alpha \to \pm i\Phi^\alpha$) are inequivalent, assigning a definite parity to a spinor has no absolute meaning. It is therefore meaningless to speak of the intrinsic parity of a spin-$1/2$ particle by itself. One can, however, speak of the *relative* intrinsic parity of two such particles.

From two (three-dimensional) spinors $\Phi^{(\mathrm{e})}$ and $\Phi^{(\mathrm{p})}$ one can form the scalar $\Phi^{(\mathrm{e})}_\alpha\Phi^{(\mathrm{p})\alpha}$. If this quantity is a true scalar, the particles described by these spinors are said to have the same parity; if it is a pseudoscalar, their intrinsic parities are said to be opposite.

We shall show that the intrinsic parities of a particle and its antiparticle (with spin $1/2$) are opposite (*V. B. Berestetskii*, 1948).

To this end, note that if the operation C (26.7) is applied to both sides of the P-transformation (19.5) (in the spinor representation)

$$P: \quad \xi^\alpha \to i\eta_{\dot\alpha}, \qquad \eta_{\dot\alpha} \to i\xi^\alpha, \tag{27.1}$$

one obtains

$$\eta^{c\alpha*} \to i\xi^{c*}_\alpha, \qquad \xi^{c*}_\alpha \to i\eta^{c\alpha*},$$

where the superscript c labels the components of the bispinor $\psi^C = \binom{\xi^c}{\eta^c}$ that is charge-conjugate to $\psi = \binom{\xi}{\eta}$. Taking the complex conjugate and rearranging the indices, we find

$$P: \quad \eta^c_{\dot\alpha} \to i\xi^{c\alpha}, \qquad \xi^{c\alpha} \to i\eta^c_{\dot\alpha}. \tag{27.2}$$

We see that charge-conjugate bispinors transform under inversion according to the same law.

Let $\psi^{(\mathrm{e})}$ be the wave function of the particle (electron) and $\psi^{(\mathrm{p})}$ that of the antiparticle (positron). The latter is a bispinor obtained by charge-conjugating a certain "negative-frequency" solution of the Dirac equation. In the rest frame each reduces to a 3-spinor:

$$\xi^{(\mathrm{e})\alpha} = \eta^{(\mathrm{e})}_{\dot\alpha} = \Phi^{(\mathrm{e})\alpha}, \qquad \xi^{(\mathrm{p})\alpha} = \eta^{(\mathrm{p})}_{\dot\alpha} = \Phi^{(\mathrm{p})\alpha}.$$

According to (27.1) and (27.2), these spinors transform under inversion as

$$\Phi^\alpha \to i\Phi^\alpha, \tag{27.3}$$

with the same rule for both $\Phi^{(\mathrm{e})}$ and $\Phi^{(\mathrm{p})}$. The product $\Phi^{(\mathrm{e})}\Phi^{(\mathrm{p})}$, however, changes sign—which proves the assertion.

A particle that coincides with its own antiparticle is called truly neutral (see § 12). The ψ-operator of the field of such particles satisfies the condition

$$\widehat{\psi}(t, \mathbf{r}) = \widehat{\psi}^C(t, \mathbf{r}).$$

For spin-$1/2$ particles this yields the conditions (in the spinor representation)[24]

$$\xi^\alpha = -i\bar{\eta}^{\dot{\alpha}+}, \qquad \widehat{\eta}_{\dot{\alpha}} = -i\widehat{\xi}^+_\alpha. \tag{27.4}$$

Like any relations expressing physical properties, these conditions are invariant under the CPT transformation[25]. It is readily verified that they are in fact invariant not only under CPT as a whole, but also under each of the three transformations individually.

In § 19 we agreed to define spinor inversion as the transformation for which $\widehat{P}^2 = -1$, and we have followed this convention throughout. It is easy to see that the result obtained above concerning the relative parity of particle and antiparticle is independent, as it should be, of the particular convention adopted for the definition of inversion.

If inversion is defined by the condition $\widehat{P}^2 = 1$, then in place of (27.1) we have

$$P: \quad \xi^\alpha \to \eta_{\dot{\alpha}}, \qquad \eta_{\dot{\alpha}} \to \xi^\alpha. \tag{27.5}$$

The charge-conjugate function then transforms according to the rule

$$\xi^{c\,\alpha} \to -\eta^c_{\dot{\alpha}}, \qquad \eta^c_{\dot{\alpha}} \to -\xi^{c\,\alpha},$$

which differs from (27.5) by a sign. Accordingly, the three-dimensional spinors Φ transform as

$$\Phi^{(\mathrm{e})\,\alpha} \to \Phi^{(\mathrm{e})\,\alpha}, \qquad \Phi^{(\mathrm{p})\,\alpha} \to -\Phi^{(\mathrm{p})\,\alpha},$$

so that the product $\Phi^{(\mathrm{e})}\Phi^{(\mathrm{p})}$ remains a pseudoscalar, as before.

The only possible physical distinction between the two conventions for inversion is that under the definition (27.5) the true-neutrality condition for the field would not be invariant under this transformation (or under CP): it would reverse the relative sign of the two sides of the equalities (27.4). In practice, truly neutral particles with spin $1/2$ are unknown, and at present one cannot say whether the difference between the two definitions of inversion carries any real physical significance[26].

Problem

Determine the charge parity of positronium (a hydrogen-like system of an electron and a positron).

Solution. The wave function of two fermions must be antisymmetric under the simultaneous interchange of the coordinates, spins, and charge variables of the particles

[24]In the Majorana representation, true neutrality simply means Hermiticity of the operator $\widehat{\psi}'$ (see the problem to § 26).

[25]More precisely, the CPT transformation must be defined in this case so as to leave relations of the type (27.4) invariant. This is achieved by an appropriate choice of the phase factor in the definition of U_T (see the footnote on p. 121).

[26]The incomplete equivalence of the two definitions of inversion was pointed out by *Racah* (*G. Racah*, 1937).

(cf. the problem to § 13). Interchanging the first multiplies the function by $(-1)^l$, the second by $(-1)^{1+S}$ (where $S = 0$ or 1 is the total spin of the system), and the third by the sought quantity C. From the condition $(-1)^l(-1)^{1+S}C = -1$ we find

$$C = (-1)^{l+S}.$$

Since the intrinsic parities of electron and positron are opposite, the spatial parity of the system is $P = (-1)^{l+1}$. The combined parity is $CP = (-1)^{S+1}$.

§ 28. Bilinear forms

Let us examine the transformation properties of the various bilinear forms that can be constructed from the components of ψ and ψ^*. Such forms are of great importance throughout quantum mechanics; among them is the 4-current density (21.11).

Since ψ and ψ^* each have four components, one can construct $4 \cdot 4 = 16$ independent bilinear combinations. Their classification by transformation properties follows immediately from the multiplication rules for two arbitrary bispinors listed in § 19 (which in the present case are ψ and ψ^*). Specifically, one can form a scalar (denoted S), a pseudoscalar (P), a mixed spinor of rank two equivalent to a true 4-vector V^μ (four independent quantities), a mixed rank-two spinor equivalent to a 4-pseudovector A^μ (four quantities), and a rank-two bispinor equivalent to an antisymmetric 4-tensor $T^{\mu\nu}$ (six quantities).

In a representation-independent notation these combinations are written as follows:

$$S = \bar\psi\psi, \qquad P = i\bar\psi\gamma^5\psi, \tag{III.2}$$

$$V^\mu = \bar\psi\gamma^\mu\psi, \qquad A^\mu = \bar\psi\gamma^\mu\gamma^5\psi, \qquad T^{\mu\nu} = i\bar\psi\sigma^{\mu\nu}\psi, \tag{28.1}$$

where

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma}) \tag{28.2}$$

(the listing of components in (28.2) follows (19.15))[27]. All the expressions written above are real.

The scalar and pseudoscalar character of S and P is evident from their spinor-representation forms: $S = \xi^*\eta + \eta^*\xi$, $P = i(\xi^*\eta - \eta^*\xi)$, which are precisely the expressions (19.7) and (19.8). That V^μ is a vector then follows from the Dirac equation: multiplying $\widehat{p}_\mu\gamma^\mu\psi = m\psi$ on the left by $\bar\psi$ gives

$$(\bar\psi\widehat{p}_\mu\gamma^\mu\psi) = m\bar\psi\psi;$$

the right-hand side is a scalar, so the left-hand side must be one as well.

The pattern governing the construction of the quantities (28.1) is straightforward: they are assembled as though the matrices γ^μ formed a 4-vector, γ^5

were a pseudoscalar, and the flanking factors $\bar\psi$ and ψ combined into a scalar[28].

[27] Under a unitary change of representation $\psi \to U\psi$, $\gamma \to U\gamma U^{-1}$, $\bar\psi \to \bar\psi U^{-1}$, so the invariance of the bilinear forms is obvious.

[28] The absence of bilinear forms having the character of a symmetric 4-tensor, obvious from the spinor representation, is clear from this rule as well: the symmetric combination $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$ would reduce the form to a scalar.

Second-quantised bilinear forms are obtained by replacing the ψ-functions in (28.1) with ψ-operators. For greater generality we shall assume that the two ψ-operators refer to the fields of different particles, distinguishing them by subscripts a and b. Let us determine how these operator forms transform under charge conjugation. Noting that[29]

$$\widehat{\psi}^C = U_C \widetilde{\widehat{\bar{\psi}}}, \qquad \widehat{\bar{\psi}}^C = U_C^+ \widehat{\psi}, \tag{28.3}$$

and employing (26.3) and (26.21):

$$\widehat{\bar{\psi}}_a^C \widehat{\psi}_b^C = \widehat{\psi}_a U_C^+ U_C \widetilde{\widehat{\bar{\psi}}}_b = -\widehat{\psi}_a U_C^+ U_C \widehat{\bar{\psi}}_b = -\widehat{\psi}_a \widehat{\bar{\psi}}_b,$$

$$\widehat{\bar{\psi}}_a^C \gamma^\mu \widehat{\psi}_b^C = \widehat{\psi}_a U_C^+ \gamma^\mu U_C \widetilde{\widehat{\bar{\psi}}}_b = -\widehat{\psi}_a U_C^+ \gamma^\mu U_C \widehat{\bar{\psi}}_b = \widehat{\psi}_a \widetilde{\gamma}^\mu \widehat{\bar{\psi}}_b.$$

Restoring the operators to their original order ($\bar{\psi}$ to the left of $\widehat{\psi}$) reverses the sign of the product by the Fermi commutation rules (25.4) (and also generates state-independent terms, which we discard as in the analogous derivation of § 13). We thus obtain

$$\widehat{\bar{\psi}}_a^C \widehat{\psi}_b^C = \widehat{\bar{\psi}}_b \widehat{\psi}_a, \qquad \widehat{\bar{\psi}}_a^C \gamma^\mu \widehat{\psi}_b^C = -\widehat{\bar{\psi}}_b \gamma^\mu \widehat{\psi}_a.$$

Treating the remaining forms in the same way, we find the following charge-conjugation rules[30]:

$$\begin{aligned} C:\quad & \widehat{S}_{ab} \to \widehat{S}_{ba}, \quad \widehat{P}_{ab} \to \widehat{P}_{ba}, \quad (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} \to (\widehat{V}^0, -\widehat{\mathbf{V}})_{ba}, \\ & (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} \to (\widehat{A}^0, -\widehat{\mathbf{A}})_{ba}, \quad \widehat{T}^{\mu\nu}_{ab} \to (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} \to (\widehat{\mathbf{p}}, -\widehat{\mathbf{a}})_{ba} \end{aligned} \tag{28.4}$$

($\widehat{\mathbf{p}}$, $\widehat{\mathbf{a}}$ are the three-dimensional vectors equivalent to the components of $\widehat{T}^{\mu\nu}$ according to (19.15)).

The behaviour of the same forms under time reversal is found analogously. One must bear in mind (see § 13) that this operation entails a reversal of the operator ordering, so that, for instance,

$$\left(\widehat{\bar{\psi}}_a \widehat{\psi}_b\right)^T = \widehat{\psi}_b^T \widehat{\bar{\psi}}_a^T.$$

Substituting

$$\widehat{\psi}^T = U_T \widetilde{\widehat{\bar{\psi}}}, \qquad \widehat{\bar{\psi}}^T = -U_T^+ \widehat{\psi}, \tag{28.5}$$

we get

$$\left(\widehat{\bar{\psi}}_a \widehat{\psi}_b\right)^T = -\widehat{\bar{\psi}}_b U_T U_T^+ \widehat{\psi}_a = \widehat{\bar{\psi}}_b \widehat{\psi}_a.$$

Carrying out the same procedure for all the other forms yields

$$\begin{aligned} T:\quad & \widehat{S}_{ab} \to \widehat{S}_{ba}, \quad \widehat{P}_{ab} \to -\widehat{P}_{ba}, \quad (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} \to (\widehat{V}^0, -\widehat{\mathbf{V}})_{ba}, \\ & (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} \to (\widehat{A}^0, -\widehat{\mathbf{A}})_{ba}, \quad \widehat{T}^{\mu\nu}_{ab} \to (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} \to (\widehat{\mathbf{p}}, -\widehat{\mathbf{a}})_{ba} \end{aligned} \tag{28.6}$$

---

[29]To derive the second relation from the first, write $\bar{\psi}^C = \left[U_C(\widetilde{\widehat{\psi}^* \gamma^{0*}})\right]\gamma^0 = -\widetilde{\gamma}^0 U_C^+ \gamma^0 \widehat{\psi} = \widetilde{\gamma}^0 \gamma^{0*} U_C^+ \widehat{\psi} = U_C^+ \widehat{\psi}$ (using (26.3), (26.21) and the Hermiticity of γ^0).

[30]Note that for bilinear forms built from ψ-functions (rather than ψ-operators), the transformations (28.4) would carry the opposite sign, since restoring the original ordering of the factors $\bar{\psi}$ and ψ would not entail a sign change.

($\widehat{\mathbf{p}}$, $\widehat{\mathbf{a}}$ are the three-dimensional vectors equivalent to the components of $\widehat{T}^{\mu\nu}$ according to (19.15)).

Under spatial inversion, on the other hand, in accordance with the tensor character of the quantities [31],

$$\begin{aligned} P: \quad & \widehat{S}_{ab} \to \widehat{S}_{ab}, \quad \widehat{P}_{ab} \to -\widehat{P}_{ab}, \quad (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} \to (\widehat{V}^0, -\widehat{\mathbf{V}})_{ab}, \\ & (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} \to (-\widehat{A}^0, \widehat{\mathbf{A}})_{ab}, \quad \widehat{T}^{\mu\nu}_{ab} \to (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} \to (-\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab}. \end{aligned} \tag{28.7}$$

Finally, the joint application of all three operations leaves $\widehat{S}_{ab}$, $\widehat{P}_{ab}$, $\widehat{T}^{\mu\nu}_{ab}$ unchanged and reverses the sign of every $\widehat{V}^{\mu}_{ab}$, $\widehat{A}^{\mu}_{ab}$. This is exactly what one expects from the interpretation of this combined transformation as a 4-inversion: since a 4-inversion is equivalent to a rotation of the four-dimensional coordinate system, it draws no distinction between true tensors and pseudotensors of any rank.

Consider now pairwise products of bilinear forms constructed from four distinct functions ψ^a, ψ^b, ψ^c, ψ^d. Different results are obtained depending on which pairs of functions are multiplied together. It turns out, however, that any such product can be reduced to products of bilinear forms with fixed pairings of the factors (*W. Pauli, M. Fierz,* 1936). We shall derive the identity underlying this reduction.

Consider the set of 4×4 matrices

$$1, \quad \gamma^5, \quad \gamma^\mu, \quad i\gamma^\mu\gamma^5, \quad i\sigma^{\mu\nu} \tag{28.8}$$

(1 being the unit matrix). Numbering these 16 (= 1 + 1 + 4 + 4 + 6) matrices in some definite order, we denote them by γ^A ($A = 1, \dots, 16$), and the same matrices with the 4-tensor indices (μ, ν) lowered by γ_A. They possess the following properties:

$$\operatorname{Sp}\gamma^A = 0 \quad (\gamma^A \neq 1), \qquad \gamma^A\gamma_A = 1, \qquad \frac{1}{4}\operatorname{Sp}\gamma^A\gamma_B = \delta^A_B. \tag{28.9}$$

By virtue of the last of these properties, the matrices γ^A are linearly independent. Since their number equals the number of elements (4×4) of a 4×4 matrix, the γ^A form a complete set in terms of which any 4×4 matrix Γ can be expanded:

$$\Gamma = \sum_A c_A\gamma^A, \qquad c_A = \frac{1}{4}\operatorname{Sp}\gamma_A\Gamma, \tag{28.10}$$

or, written out explicitly with matrix indices ($i, k = 1, 2, 3, 4$):

$$\Gamma_{ik} = \frac{1}{4}\sum_A \Gamma_{lm}\gamma^A_{ml}\gamma_{Aik}.$$

Supposing, in particular, that Γ has only one non-vanishing element (Γ_{lm}), we obtain the desired identity (the "completeness relation")

$$\delta_{il}\delta_{km} = \frac{1}{4}\sum_A \gamma_{Aik}\gamma^A_{ml}. \tag{28.11}$$

[31] To avoid misunderstanding, recall that the T and P transformations also require a change in the arguments of the functions on the right-hand sides (the transformed forms in (28.6), (28.7)) to $x^T = (-t, \mathbf{r})$ and $x^P = (t, -\mathbf{r})$, respectively, if the left-hand sides are functions of $x = (t, \mathbf{r})$.

Multiplying both sides by $\bar{\psi}_i^a \psi_k^b \bar{\psi}_m^c \psi_l^d$ gives

$$(\bar{\psi}^a \psi^b)(\bar{\psi}^c \psi^d) = \frac{1}{4} \sum_A (\bar{\psi}^a \gamma_A \psi^b)(\bar{\psi}^c \gamma^A \psi^d). \tag{28.12}$$

This is one identity of the kind described above: it expresses the product of two scalar bilinear forms in terms of products of forms built from different pairings of the factors [32].

Further identities of this type are obtained from (28.12) by substituting

$$\psi^d \to \gamma^B \psi^d, \qquad \psi^b \to \gamma^C \psi^b$$

and using the expansion (see the problem)

$$\gamma^A \gamma^B = \sum_R c_R \gamma^R, \qquad c_R = \frac{1}{4} \operatorname{Sp} \gamma^A \gamma^B \gamma_R.$$

We also record here, for later reference, the analogue of (28.11) for 2×2 matrices. A complete set of linearly independent 2×2 matrices σ^A ($A = 1, \dots, 4$) is formed by

$$1, \quad \sigma_x, \quad \sigma_y, \quad \sigma_z. \tag{28.13}$$

For these

$$\operatorname{Sp} \sigma^A = 0 \quad (\sigma^A \neq 1), \qquad \frac{1}{2} \operatorname{Sp} \sigma^A \sigma^B = \delta_{AB}. \tag{28.14}$$

The completeness relation reads:

$$\delta_{\alpha\gamma} \delta_{\beta\delta} = (1/2) \sum_A \sigma^A_{\alpha\beta} \sigma^A_{\delta\gamma} = (1/2) \delta_{\alpha\beta} \delta_{\delta\gamma} + (1/2) \delta_{\alpha\delta} \delta_{\beta\gamma} \tag{28.15}$$

($\alpha, \beta, \dots = 1, 2$), or equivalently:

$$\sigma_{\alpha\beta} \sigma_{\delta\gamma} = -(1/2) \sigma_{\alpha\gamma} \sigma_{\delta\beta} + (3/2) \delta_{\alpha\gamma} \delta_{\delta\beta}. \tag{28.16}$$

Problem

Derive formulae analogous to (28.12) for the scalar products of two bilinear forms of types P, V, A, T.

Solution. Denote:

$$J_S = (\bar{\psi}^a \psi^b)(\bar{\psi}^c \psi^d), \qquad J_P = (\bar{\psi}^a \gamma^5 \psi^b)(\bar{\psi}^c \gamma^5 \psi^d),$$
$$J_V = (\bar{\psi}^a \gamma^\mu \psi^b)(\bar{\psi}^c \gamma_\mu \psi^d), \qquad J_A = (\bar{\psi}^a i\gamma^\mu \gamma^5 \psi^b)(\bar{\psi}^c i\gamma_\mu \gamma^5 \psi^d),$$
$$J_T = (\bar{\psi}^a i\sigma^{\mu\nu} \psi^b)(\bar{\psi}^c i\sigma_{\mu\nu} \psi^d),$$

[32] We recall, to avoid confusion, that the forms here are built from ψ-functions. For forms built from anticommuting ψ-operators the sign of the rearrangement would be reversed.

and let primed letters denote the same products with ψ^b and ψ^d interchanged. Applying the procedure described in the text yields

$$
\begin{aligned}
4J'_S &= J_S + J_V + J_T + J_A + J_P,\\
4J'_V &= 4J_S - 2J_V + 2J_A - 4J_P,\\
4J'_T &= 6J_S - 2J_T + 6J_P,\\
4J'_A &= 4J_S + 2J_V - 2J_A - 4J_P,\\
4J'_P &= J_S - J_V - J_T - J_A + J_P
\end{aligned}
$$

(the first row is formula (28.12)).

§ 29. Polarisation density matrix

The coordinate dependence of the wave function ψ describing free motion with momentum $\mathbf{p}$ (a plane wave) reduces to the common factor $e^{i\mathbf{pr}}$, while the amplitude u_p plays the role of a spin wave function. In such a (pure) state the particle is fully polarised (see III, § 59). In the non-relativistic theory

this means that the particle's spin has a definite spatial direction (more precisely, there exists a direction along which the spin projection takes the definite value $+1/2$). In a relativistic theory such a characterisation of the state is impossible in an arbitrary reference frame, owing to the non-conservation of the spin vector (already noted in § 23). Purity of the state signifies only that the spin has a definite direction in the particle's rest frame.

In a state of partial polarisation no definite amplitude exists; there is only a *polarisation density matrix* ρ_{ik} ($i, k = 1, 2, 3, 4$ being bispinor indices). We define this matrix so that in a pure state it reduces to the product

$$\rho_{ik} = u_{pi}\bar{u}_{pk}. \tag{29.1}$$

The corresponding normalisation condition for ρ is

$$\operatorname{Sp}\rho = 2m \tag{29.2}$$

(cf. (23.4)).

In a pure state the mean value of the spin is given by

$$\mathbf{s} = \frac{1}{2}\int \psi^*\mathbf{\Sigma}\psi\, d^3x = \frac{1}{4\varepsilon}u_p^+\mathbf{\Sigma}u_p = \frac{1}{4\varepsilon}\bar{u}_p\gamma^0\mathbf{\Sigma}u_p. \tag{29.3}$$

The corresponding expression for a partially polarised state is

$$\mathbf{s} = \frac{1}{4\varepsilon}\operatorname{Sp}(\rho\gamma^0\mathbf{\Sigma}) = \frac{1}{4\varepsilon}\operatorname{Sp}(\rho\gamma^5\boldsymbol{\gamma}). \tag{29.4}$$

The amplitudes u_p and $\bar{u}_p$ satisfy the algebraic equations

$$(\gamma p - m)u_p = 0, \qquad \bar{u}_p(\gamma p - m) = 0.$$

The matrix (29.1) therefore obeys

$$(\gamma p - m)\rho = 0, \qquad \rho(\gamma p - m) = 0. \tag{29.5}$$

The density matrix must satisfy the same linear equations in the general case of a mixed (in spin) state as well (cf. the analogous argument in III, § 14).

When a free particle is considered in its rest frame, the non-relativistic theory applies. In that theory the state of partial polarisation is completely specified by three parameters—the components of the mean spin vector **s** (see III, § 59). It is therefore clear that the same parameters will determine the polarisation state after any Lorentz boost, i.e. for a moving particle.

Denote the doubled mean spin vector in the rest frame by $\boldsymbol{\zeta}$ (in a pure state $|\boldsymbol{\zeta}| = 1$; in a mixed state $|\boldsymbol{\zeta}| < 1$). For a four-dimensional description of the polarisation state it is convenient to introduce a 4-vector a^μ that coincides in the rest frame with the three-vector $\boldsymbol{\zeta}$; since $\boldsymbol{\zeta}$ is an axial vector, a^μ is a 4-pseudovector. This 4-vector is orthogonal to the 4-momentum in the rest frame (where $a^\mu = (0, \boldsymbol{\zeta})$ and $p^\mu = (m, 0)$), and hence in every frame:

$$a_\mu p^\mu = 0. \tag{29.6}$$

In an arbitrary frame one also has

$$a_\mu a^\mu = -\zeta^2. \tag{29.7}$$

The components of a^μ in a frame where the particle moves with velocity $\mathbf{v} = \mathbf{p}/\varepsilon$ are obtained by Lorentz-boosting from the rest frame:

$$a^0 = \frac{|\mathbf{p}|}{m}\zeta_\parallel, \qquad \mathbf{a}_\perp = \zeta_\perp, \quad a_\parallel = \frac{\varepsilon}{m}\zeta_\parallel, \tag{29.8}$$

where the subscripts $\parallel$ and $\perp$ denote the components of $\boldsymbol{\zeta}$ and **a** parallel and perpendicular to **p**[33]. These formulae can be written in vector form:

$$\mathbf{a} = \boldsymbol{\zeta} + \frac{\mathbf{p}(\boldsymbol{\zeta}\cdot\mathbf{p})}{m(\varepsilon + m)}, \qquad a^0 = \frac{\mathbf{a}\cdot\mathbf{p}}{\varepsilon} = \frac{\mathbf{p}\cdot\boldsymbol{\zeta}}{m}, \qquad \mathbf{a}^2 = \boldsymbol{\zeta}^2 + \frac{(\mathbf{p}\cdot\boldsymbol{\zeta})^2}{m^2}. \tag{29.9}$$

Consider first the unpolarised state ($\boldsymbol{\zeta} = 0$). The density matrix can then depend only on the 4-momentum p. The unique matrix of this kind satisfying the equations (29.5) is

$$\rho = \frac{1}{2}(\gamma p + m) \tag{29.10}$$

(*I. E. Tamm*, 1930; *H. B. G. Casimir*, 1933). The numerical coefficient is fixed by the normalisation condition (29.2).

[33]By their transformation properties, the components of the mean spin vector **s** (like those of any angular momentum) are, in relativistic mechanics, the spatial components of an antisymmetric tensor $S^{\lambda\mu}$. The 4-vector a^λ is related to this tensor by

$$S^{\lambda\mu} = \frac{1}{2m}\epsilon^{\lambda\mu\nu\rho}a_\nu p_\rho, \qquad a^\lambda = -\frac{2}{m}\epsilon^{\lambda\mu\nu\rho}S_{\mu\nu}p_\rho.$$

Note that in an arbitrary frame the spatial part of a^λ does not coincide with 2**s**. One easily finds $2\mathbf{s}_\parallel = \frac{1}{m}(a_\parallel\varepsilon - a^0|\mathbf{p}|) = \zeta_\parallel$, $2\mathbf{s}_\perp = \frac{\varepsilon}{m}\mathbf{a}_\perp = \frac{\varepsilon}{m}\boldsymbol{\zeta}_\perp$.

In the general case of partial polarisation ($\zeta \neq 0$) we seek the density matrix in the form

$$\rho = \frac{1}{4m}(\gamma p + m)\rho'(\gamma p + m), \tag{29.11}$$

which automatically satisfies (29.5). When $\zeta \neq 0$ the auxiliary matrix ρ' must reduce to the identity; since

$$(\gamma p + m)^2 = 2m(\gamma p + m),$$

equation (29.11) then coincides with (29.10). Furthermore, ρ' must contain the 4-vector a linearly as a parameter, i.e. it must have the form

$$\rho' = 1 - A\gamma^5(\gamma a); \tag{29.12}$$

the second term involves the scalar product of the pseudovector a with the "matrix 4-pseudovector" $\gamma^5\gamma$. To fix the coefficient A, write the density matrix in the rest frame:

$$\rho = \frac{m}{4}(1+\gamma^0)(1 + A\gamma^5\boldsymbol{\gamma}\cdot\boldsymbol{\zeta})(1+\gamma^0) = \frac{m}{2}(1+\gamma^0)(1 + A\gamma^5\boldsymbol{\gamma}\cdot\boldsymbol{\zeta}),$$

and compute the mean spin via (29.4). Using the trace rules listed in § 22, one readily finds that the only non-vanishing contribution to the required trace is

$$2\mathbf{s} = \frac{1}{2m}\operatorname{Sp}(\rho\gamma^5\gamma) = -\frac{A}{4}\operatorname{Sp}\big((\boldsymbol{\gamma}\cdot\boldsymbol{\zeta})\gamma\big) = A\boldsymbol{\zeta}.$$

Setting this equal to $\boldsymbol{\zeta}$ gives $A = 1$. To obtain the final expression for ρ, substitute (29.12) into (29.11) and commute ρ' past $(\gamma p + m)$; because a and p are orthogonal, γp anticommutes with γa:

$$(\gamma a)(\gamma p) = 2a\cdot p - (\gamma p)(\gamma a) = -(\gamma p)(\gamma a),$$

and therefore commutes with $\gamma^5(\gamma a)$.

The density matrix of a partially polarised electron is thus

$$\rho = \frac{1}{2}(\gamma p + m)[1 - \gamma^5(\gamma a)] \tag{29.13}$$

(*L. Michel, A. S. Wightman*, 1955). If ρ is known, the 4-vector a characterising the state (and with it the vector $\boldsymbol{\zeta}$) can be recovered from

$$a^\mu = \frac{1}{2m}\operatorname{Sp}\big(\rho\gamma^5\gamma^\mu\big). \tag{29.14}$$

The formulae for the positron density matrix are analogous to those for the electron. Had we described a positron (with

4-momentum p) by a positron amplitude $u_p^{(\text{pos})}$ and a correspondingly defined density matrix $\rho^{(\text{pos})}$, there would be no difference from the electron case and $\rho^{(\text{pos})}$ would be given by the same formula (29.13). In actual calculations of scattering cross sections involving positrons, however, one deals (as we shall see later) not with $u_p^{(\text{pos})}$ but with the "negative-frequency" amplitudes u_{-p}. Accordingly, the polarisation density matrix (which we denote $\rho^{(-)}$) should be defined so that for a pure state it reduces to $u_{-p\,i}\bar{u}_{-p\,k}$.

By (26.1) the positron amplitude $u_p^{(\mathrm{pos})} = U_C \bar{u}_{-p}$. Inversely:

$$u_{-p} = U_C u_p^{(\mathrm{pos})}, \qquad \bar{u}_{-p} = U_C^+ u_p^{(\mathrm{pos})} = u_p^{(\mathrm{pos})} U_C^*$$

(cf. (28.3)). If

$$\rho_{ik}^{(-)} = u_{-p\,i}\bar{u}_{-p\,k}, \qquad \rho_{ik}^{(\mathrm{pos})} = u_{pi}^{(\mathrm{pos})}\bar{u}_{pk}^{(\mathrm{pos})},$$

then these relations give

$$\rho^{(-)} = U_C \widetilde{\rho}^{(\mathrm{pos})} U_C^*. \tag{29.15}$$

Substituting (29.13) for $\rho^{(\mathrm{pos})}$ and carrying out straightforward algebra (using (26.3) and (26.21)), we obtain

$$\rho^{(-)} = \frac{1}{2}(\gamma p - m)[1 - \gamma^5(\gamma a)]. \tag{29.16}$$

In particular, for an unpolarised state

$$\rho^{(-)} = \frac{1}{2}(\gamma p - m). \tag{29.17}$$

Henceforth, when speaking of the positron density matrix we shall mean $\rho^{(-)}$ and drop the superscript $(-)$ (the matrix $\rho^{(\mathrm{pos})}$ is in practice never needed).

In various calculations one frequently has to average over spin states expressions of the form $\bar{u}Fu (\equiv \bar{u}_i F_{ik} u_k)$, where F is some (4×4) matrix and u is the bispinor amplitude for a state with definite 4-momentum p. Such averaging is equivalent to replacing the products $u_k \bar{u}_i$ by the density matrix ρ_{ik} of a partially polarised state.

In particular, a complete average over the two independent spin states amounts to passing to the unpolarised state; by (29.10) we then have

$$\frac{1}{2}\sum_{\mathrm{pol}} \bar{u}_p F u_p = \frac{1}{2}\,\mathrm{Sp}(\gamma p + m)F. \tag{29.18}$$

Similarly, for the negative-frequency wave functions

$$\frac{1}{2}\sum_{\mathrm{pol}} \bar{u}_{-p} F u_{-p} = \frac{1}{2}\,\mathrm{Sp}(\gamma p - m)F. \tag{29.19}$$

If one is summing over spin states rather than averaging, the result is twice as large.

Let us trace how the density matrix (29.13) passes to its non-relativistic limit. To this end we go to the electron rest frame. In the standard representation the amplitudes u_p become two-component in this frame, and the density matrix must correspondingly become a 2×2 matrix. Indeed, in the rest frame

$$\rho = \frac{m}{2}(\gamma^0 + 1)(1 + \gamma^5 \boldsymbol{\gamma}\cdot\boldsymbol{\zeta}),$$

and using the explicit forms of the γ matrices (21.20) and (22.18) we find

$$\rho = \begin{pmatrix} \rho_{\mathrm{nr}} & 0 \\ 0 & 0 \end{pmatrix}, \qquad \rho_{\mathrm{nr}} = m(1 + \boldsymbol{\sigma}\cdot\boldsymbol{\zeta}) \tag{29.20}$$

(the zeros denote 2×2 null matrices). Adopting the normalisation $\mathrm{Sp}\,\rho_{\mathrm{nr}} = 1$ conventional in the non-relativistic theory (instead of $2m$), one must divide this expression by $2m$, yielding

$$\frac{1}{2}(1 + \boldsymbol{\sigma} \cdot \boldsymbol{\zeta}),$$

in agreement with formula (59.6) (see III).

The non-relativistic limit of the positron density matrix is obtained in a similar way:

$$\rho = \begin{pmatrix} 0 & 0 \\ 0 & \rho_{\mathrm{nr}} \end{pmatrix}, \qquad \rho_{\mathrm{nr}} = -m(1 + \boldsymbol{\sigma} \cdot \boldsymbol{\zeta}).$$

Finally, let us write down a simplified expression for the density matrix in the ultrarelativistic case. Setting $|\mathbf{p}| \approx \varepsilon$ in (29.8) (thereby neglecting quantities of relative order $(m/\varepsilon)^2$), substituting into (29.13) or (29.16), and choosing the direction of $\mathbf{p}$ as the x-axis, we have

$$\rho = \frac{1}{2}[\varepsilon(\gamma^0 - \gamma^1) \pm m]\left[1 - \gamma^5\left(\frac{\varepsilon}{m}(\gamma^0 - \gamma^1)\zeta_{\parallel} - \zeta_{\perp}\gamma_{\perp}\right)\right],$$

where the upper sign refers to the electron and the lower to the positron. On expanding the product the leading terms cancel, and the next-order terms give

$$\rho = \frac{1}{2}\varepsilon(\gamma^0 - \gamma^1)[1 + \gamma^5(\pm\zeta_{\parallel} + \zeta_{\perp}\gamma_{\perp})]$$

or, writing $\varepsilon(\gamma^0 - \gamma^1)$ as γp:

$$\rho = \frac{1}{2}(\gamma p)[1 + \gamma^5(\pm\zeta_{\parallel} + \boldsymbol{\zeta}_{\perp} \cdot \boldsymbol{\gamma}_{\perp})]. \tag{29.21}$$

This is the desired ultrarelativistic density matrix. Note that all components of the polarisation vector $\boldsymbol{\zeta}$ enter on an equal footing, as terms of the same order. Recall that $\zeta_{\parallel}$ is the component of this vector parallel ($\zeta_{\parallel} > 0$) or antiparallel ($\zeta_{\parallel} < 0$) to the particle's momentum. In particular, for a helicity eigenstate $\zeta_{\parallel} = 2\lambda = \pm 1$; the density matrix then takes the especially simple form

$$\rho = \frac{1}{2}(\gamma p)(1 \pm 2\lambda\gamma^5), \tag{29.22}$$

which coincides, as it should, with the density matrix of a neutrino or antineutrino—a particle with zero mass and definite helicity (see § 30).

§ 30. Two-component fermions

We saw in § 20 that the need to describe a spin-1/2 particle by two spinors (ξ and η) is tied to the particle having a mass. This requirement disappears when the mass vanishes. The wave equation for such a particle can be formulated in terms of a single spinor—say the dotted spinor η:

$$\widehat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} = 0, \tag{30.1}$$

or equivalently,

$$(\widehat{p}_0 + \widehat{\mathbf{p}} \cdot \boldsymbol{\sigma})\eta = 0. \tag{30.2}$$

In § 20 it was also noted that the wave equation containing the mass m is automatically symmetric under inversion (the transformation (20.4)). When the particle is described by a single spinor this symmetry is lost. There is, however, no necessity for it, since symmetry under inversion is not a universal law of nature.

The energy and momentum of a particle with $m = 0$ are related by $\varepsilon = |\mathbf{p}|$. For a plane wave ($\eta_p \propto e^{-ipx}$) the equation (30.2) therefore gives

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\eta_p = -\eta_p, \tag{30.3}$$

where $\mathbf{n}$ is the unit vector along $\mathbf{p}$. The same equation

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\eta_{-p} = -\eta_{-p} \tag{30.4}$$

holds also for the "negative-frequency" waves ($\eta_{-p} \propto e^{ipx}$). The second-quantised ψ-operator is

$$\widehat{\eta} = \sum_{\mathbf{p}}(\eta_p \widehat{a}_{\mathbf{p}} + \eta_{-p}\widehat{b}^+_{\mathbf{p}}), \qquad \widehat{\eta}^+ = \sum_{\mathbf{p}}(\eta^*_p \widehat{a}^+_{\mathbf{p}} + \eta^*_{-p}\widehat{b}_{\mathbf{p}}). \tag{30.5}$$

From this it follows, as usual, that η^*_{-p} are the antiparticle wave functions.

From the definition of the operators $\widehat{p}^{\alpha\dot{\beta}}$ (20.1) it is clear that $\widehat{p}^{\alpha\dot{\beta}*} = -\widehat{p}^{\dot{\alpha}\beta}$. Hence the complex-conjugate spinor η^* satisfies $\widehat{p}^{\dot{\alpha}\beta}\eta^*_{\beta} = 0$, i.e.

$$\widehat{p}_{\dot{\alpha}\beta}\eta^{\beta*} = 0.$$

Denoting $\eta^{\beta*} = \xi^\beta$ (thus expressing the fact that complex conjugation converts a dotted spinor into an undotted one), we see that the antiparticle wave functions satisfy

$$\widehat{p}_{\dot{\alpha}\beta}\xi^\beta = 0, \tag{30.6}$$

or

$$(\widehat{p}_0 - \widehat{\mathbf{p}} \cdot \boldsymbol{\sigma})\xi = 0. \tag{30.7}$$

For a plane wave this gives

$$(\mathbf{n} \cdot \boldsymbol{\sigma})\xi_p = \xi_p. \tag{30.8}$$

Now $\frac{1}{2}(\mathbf{n} \cdot \boldsymbol{\sigma})$ is the operator of the spin projection on the direction of motion. Equations (30.3) and (30.8) therefore mean that states with a definite momentum are automatically helicity eigenstates—the spin projection along the direction of motion takes a definite value. If the particle's spin is antiparallel to its momentum (helicity $-1/2$), then the antiparticle's spin is aligned with its momentum (helicity $+1/2$). Particles with these properties are, perhaps, realised in nature as **neutrinos**. The particle with helicity $-1/2$ is conventionally called the neutrino, while the one with helicity $+1/2$ is called the antineutrino [34].

[34] The existence of the neutrino was predicted theoretically by *Pauli* to explain β-decay properties (1931). Equation (30.1) was first considered by *Weyl* (*H. Weyl*, 1929). A neutrino theory based on these equations was formulated by *L. D. Landau*; *Lee, Yang* and *Salam* (*T. D. Lee, C. N. Yang, A. Salam*) in 1957.

The experimental question of whether the neutrino mass is exactly zero has not been settled definitively up to the present time. We shall henceforth use the term "neutrino" conventionally to designate a particle described by equation (30.3).

In connection with the non-degeneracy of the neutrino states with respect to the spin direction, recall the remark made in § 8 that a massless particle (with spin $1/2$) possesses only an axial symmetry about the direction of its momentum. For a truly neutral particle—the photon—this symmetry includes both rotations about the axis and reflections in planes passing through it. For the neutrino, however, there is no symmetry under reflections, and one is left only with the group of rotations about the axis, which preserves the momentum projection of the spin but does not reverse its sign. Symmetry under reflections exists only on the condition that particle and antiparticle are simultaneously interchanged.

It should also be noted that the obligatory longitudinal polarisation means that, for the neutrino, spin is altogether inseparable from orbital angular momentum (just as, for the photon, it is inseparable from the obligatory transversality of the field; see § 6).

From a single spinor η (or ξ) one can form only four bilinear combinations, which together make up a 4-vector

$$j^\mu = (\eta^*\eta,\ \eta^*\boldsymbol{\sigma}\eta). \tag{30.9}$$

It is easily verified that, by virtue of the equations

$$(\widehat{p}_0 + \widehat{\mathbf{p}}\cdot\boldsymbol{\sigma})\eta = 0, \qquad \eta^*(\widehat{p}_0 - \widehat{\mathbf{p}}\cdot\boldsymbol{\sigma}) = 0,$$

the continuity equation $\partial_\mu j^\mu = 0$ is satisfied, i.e. j^μ serves as the particle current-density 4-vector.

The neutrino plane waves are conveniently normalised in the same fashion as was done in § 23 for massive particles:

$$\eta_p = \frac{1}{\sqrt{2\varepsilon}} u_p e^{-ipx}, \qquad \eta_{-p} = \frac{1}{\sqrt{2\varepsilon}} u_{-p} e^{ipx}, \tag{30.10}$$

with the spinor amplitudes satisfying the invariant normalisation condition

$$u^*_{\pm p}(1, \boldsymbol{\sigma})u_{\pm p} = 2(\varepsilon, \mathbf{p}). \tag{30.11}$$

The current density is then $j^0 = 1$, $\mathbf{j} = \mathbf{p}/\varepsilon = \mathbf{n}$.

Since a free neutrino with a given momentum is always fully polarised, the concept of a mixed spin state does not arise. It may nonetheless be useful to introduce a 2×2 polarisation "density matrix" defined simply as a rank-two spinor:

$$\rho_{\dot\alpha\dot\beta} = u_{\dot\alpha}u^*_{\dot\beta} \tag{30.12}$$

(with $\mathrm{Sp}\,\rho = 2\varepsilon$). The expression for this matrix is obtained by noting that it must satisfy

$$(\varepsilon + \mathbf{p}\cdot\boldsymbol{\sigma})\rho = \rho(\varepsilon + \mathbf{p}\cdot\boldsymbol{\sigma}) = 0.$$

It follows that

$$\rho = \varepsilon - \mathbf{p}\cdot\boldsymbol{\sigma}. \tag{30.13}$$

When considering various interaction processes, the neutrino may appear alongside other particles (with spin $1/2$) that have a mass and are therefore described by four-component wave functions. In such cases it is convenient to maintain a uniform notation by formally introducing a "bispinor" wave function for the neutrino as well, $\psi = \binom{0}{\eta}$, two of whose components are, however, identically zero. But such a form of ψ would, generally speaking, be spoiled by a change to a different (non-spinor) representation. This difficulty is circumvented by observing that in the spinor representation one has the identity

$$\frac{1+\gamma^5}{2}\begin{pmatrix}\xi\\ \eta\end{pmatrix} = \begin{pmatrix}0\\ \eta\end{pmatrix}, \qquad (\eta^* \ \eta^*)\frac{1-\gamma^5}{2} = (\eta^* \ 0),$$

where ξ is an arbitrary "ballast" spinor that drops out thanks to the matrix γ^5 from (22.18). The condition of true "two-componentness" of the neutrino is therefore maintained when describing it by a four-component ψ in any representation, provided one means by ψ a solution of the Dirac equation with $m = 0$:

$$(\gamma p)\psi = 0, \tag{30.14}$$

subject to the additional constraint $\frac{1}{2}(1+\gamma^5)\psi = \psi$, i.e.

$$\gamma^5\psi = \psi. \tag{30.15}$$

This condition can be taken into account, once and for all, in every formula involving ψ and $\bar{\psi}$ by making the substitution

$$\psi \to \frac{1+\gamma^5}{2}\psi, \qquad \bar{\psi} \to \bar{\psi}\frac{1-\gamma^5}{2}. \tag{30.16}$$

The current-density 4-vector then takes the form (cf. the replacement (30.16) in the expression $\bar{\psi}\gamma^\mu\psi$)

$$j^\mu = \frac{1}{4}\bar{\psi}(1-\gamma^5)\gamma^\mu(1+\gamma^5)\psi = \frac{1}{2}\bar{\psi}\gamma^\mu(1+\gamma^5)\psi. \tag{30.17}$$

In accordance with the same rule, the 4×4 density matrix of the neutrino should be written as

$$\rho = \frac{1}{4}(1+\gamma^5)(\gamma p)(1-\gamma^5) = \frac{1}{2}(1+\gamma^5)(\gamma p). \tag{30.18}$$

In the spinor representation it reduces, as it should, to the 2×2 matrix (30.13):

$$\rho = \begin{pmatrix} 0 & 0\\ \varepsilon - \boldsymbol{\sigma}\cdot\mathbf{p} & 0\end{pmatrix}.$$

The corresponding formulae for the antineutrino differ from those written above by a change of sign in front of γ^5.

The neutrino is an electrically neutral particle. A neutrino with the properties described above is not, however, a truly neutral particle. We note in this connection that the "neutrino field" described by a two-component spinor is, by the number of particle states it permits (but not, of course, by any other physical property), equivalent to a truly neutral field described by a four-component bispinor. In place of particles and antiparticles with definite helicities, there would be the same number of states of a single particle with two possible helicity values, and symmetry under inversion would be automatically preserved.

Note, however, that the vanishing of the "four-component" neutrino mass would be, so to speak, "accidental"—it would not be related to any symmetry property of the wave equation (which permits a non-zero mass as well). For this reason the account of various interactions of such a particle would automatically generate a mass that, while small, would not be strictly zero.

§ 31. Wave equation for a spin-3/2 particle

A spin-3/2 particle is described in its rest frame by a symmetric 3-spinor of rank three (with $2s + 1 = 4$ independent components). Accordingly, in an arbitrary frame the description may involve the 4-spinors $\xi^{\alpha\beta\dot{\gamma}}$, $\eta_{\dot{\alpha}\dot{\beta}\gamma}$, $\zeta^{\alpha\beta\gamma}$, and $\chi_{\dot{\alpha}\dot{\beta}\dot{\gamma}}$, each symmetric under interchange of all identical (dotted or undotted) indices; under inversion the spinors in the first pair, and likewise those in the second pair, transform into each other.

For the rest-frame 4-spinors $\xi^{\alpha\beta\dot{\gamma}}$ and $\eta_{\dot{\alpha}\dot{\beta}\gamma}$, symmetric over all three indices, to reduce to 3-spinors they must satisfy the conditions

$$\widehat{p}^{\dot{\alpha}\beta}\eta_{\dot{\alpha}\dot{\beta}\gamma} = 0, \qquad \widehat{p}_{\alpha\dot{\beta}}\xi^{\alpha\beta\dot{\gamma}} = 0. \tag{31.1}$$

Indeed, in the rest frame

$$\widehat{p}^{\dot{\alpha}\beta} \to \widehat{p}_0\delta^\beta_\alpha = m\delta^\beta_\alpha$$

(as is clear from (20.1)). The conditions (31.1) therefore lead to the equalities

$$\delta^\beta_\alpha\eta'^{\alpha}{}_{\beta\gamma} = 0, \qquad \delta^\beta_\alpha\xi'^{\alpha}{}_{\beta\gamma} = 0,$$

where the primed letters denote the corresponding 3-spinors; in other words, these spinors vanish upon contraction over the in-

dices $\alpha\beta$, which means that they are symmetric in those indices, and hence in all three.

The differential relation between the spinors ξ and η is established by the equations

$$\widehat{p}^{\dot{\beta}\gamma}\eta^{\beta}{}_{\dot{\alpha}\dot{\beta}} = m\xi^{\beta\gamma}{}_{\dot{\alpha}}, \qquad \widehat{p}_{\dot{\beta}\gamma}\xi^{\beta\dot{\gamma}}{}_{\alpha} = m\eta_{\alpha\dot{\beta}}{}^{\dot{\gamma}}. \tag{31.2}$$

The symmetry of the left-hand sides of these equations (in the indices β, $\dot{\gamma}$ or α, $\dot{\beta}$) is ensured by the conditions (31.1), by virtue of which they vanish upon contraction over all indices. In the rest frame the 3-spinors ξ' and η' coincide because of equations (31.2). Eliminating η (or ξ) from (31.2), one finds that each component of the spinors ξ and η satisfies the second-order equation

$$(\widehat{p}^2 - m^2)\xi^{\alpha\beta\dot{\gamma}} = 0. \tag{31.3}$$

The system of equations (31.1), (31.2) constitutes the complete set of wave equations for a spin-3/2 particle[35]. Adding the spinors ζ, χ would contribute nothing new. They are constructed according to

$$m\zeta^{\alpha\beta\gamma} = \widehat{p}^{\alpha\dot{\delta}}\eta^{\beta\gamma}{}_{\dot{\delta}}, \qquad m\chi_{\dot{\alpha}\dot{\beta}\dot{\gamma}} = \widehat{p}_{\dot{\alpha}\delta}\xi^{\delta}{}_{\dot{\beta}\dot{\gamma}}.$$

[35] For the Lagrangian formulation of these equations see the article by *Fierz* and *Pauli* cited on p. 76.

The equations for spin-3/2 particles can also be cast in a different form that exploits the vector properties of the spinors (*W. Rarita, J. Schwinger*, 1941; *A. S. Davydov, I. E. Tamm*, 1942). A pair of spinor indices $\alpha\dot{\beta}$ is associated with a single four-dimensional vector index μ. The components of a rank-three spinor $\xi^{\alpha\beta\dot{\gamma}}$ can therefore be brought into correspondence with the components of "mixed" quantities $\psi^{\mu}{}_{l}$ carrying one vector and one spinor index. Likewise, the spinor $\eta^{\dot{\beta}\alpha\gamma}$ corresponds to quantities $\psi^{\mu}{}_{l}$, and the totality of both spinors to a "vector" bispinor ψ_μ (with the bispinor index suppressed). The wave equation is then written as a "Dirac equation" for each of the vector components ψ_μ:

$$(\gamma\widehat{p} - m)\psi_\mu = 0 \tag{31.4}$$

with the supplementary condition

$$\gamma^\mu\psi_\mu = 0. \tag{31.5}$$

Using the expressions for the matrices γ^μ in the spinor representation (18.6), (18.7), it is straightforward to verify that the equations (31.2) are contained in (31.4), and the condition (31.5) is equivalent to the symmetry requirement on the spinors $\xi^{\alpha\beta\dot{\gamma}}$ and $\eta^{\dot{\alpha}\dot{\beta}\gamma}$ in the indices $\beta\dot{\gamma}$ or $\dot{\beta}\gamma$. Multiplying equation (31.4) on γ^μ and using (31.5):

$$\gamma^\mu\gamma^\nu\widehat{p}_\nu\psi_\mu = 0$$

or, applying the commutation rules for the γ matrices,

$$2g^{\mu\nu}\widehat{p}_\nu\psi_\mu - \gamma^\nu\widehat{p}_\nu\gamma^\mu\psi_\mu = 0. \tag{31.6}$$

The second term vanishes by (31.5), and the first gives

$$\widehat{p}^\mu\psi_\mu = 0. \tag{31.7}$$

It is easy to see that this condition, which follows automatically from (31.4) and (31.5), is equivalent to (31.1).

Finally, there is yet another way to formulate the wave equation. It consists in introducing quantities ψ_{ikl} $(i, k, l = 1, 2, 3, 4)$ with three bispinor indices, symmetric under permutation (*V. Bargmann, E. P. Wigner*, 1948). The totality of these quantities is equivalent to the set of all four spinors ξ, η, ζ, χ. The wave equation is written as a system of "Dirac equations"

$$\widehat{p}_\mu\gamma^\mu_{im}\psi_{mkl} = m\psi_{ikl}. \tag{31.8}$$

It is readily seen that these equations already yield the correct number of independent components (four), and no additional constraints are needed. Indeed, in the rest frame (31.8) reduces to

$$\gamma^0_{im}\psi_{mkl} = \psi_{ikl},$$

by virtue of which all components with i, k, $l = 3, 4$ (in the standard representation) vanish, i.e. ψ_{ikl} reduces to the components of a rank-three 3-spinor.

The results presented here generalise in an obvious way to particles of any half-integer spin s. In the description via equations of the type (31.4), (31.5) the wave function is a symmetric 4-tensor of rank $(2s-1)/2$ carrying one bispinor index. In the description via equations of the type (31.8) the wave function has $2s$ bispinor indices over which it is symmetric.

CHAPTER IV

A Particle in an External Field

§ 32. The Dirac equation for an electron in an external field

The wave equations for free particles express, in essence, only those properties that follow from the general requirements of space–time symmetry. The physical processes that occur with particles depend on the nature of their interactions.

A description of the electromagnetic interactions of particles in a relativistic quantum theory turns out to be possible by generalising the method employed for this purpose in both classical and non-relativistic quantum theories.

This method is, however, applicable only to the electromagnetic interactions of particles that are incapable of strong interactions. Electrons (and positrons) belong to this class, and thus the entire vast domain of quantum electrodynamics of electrons becomes accessible to the present theory. Muons are likewise incapable of strong interactions; they too are described by quantum electrodynamics in the regime of phenomena whose time scales are short compared with the muon lifetime (which is governed by weak interactions).

In this chapter we consider the class of problems in quantum electrodynamics that fall within the framework of the single-particle theory, where the number of particles is fixed and the interaction can be described in terms of an external electromagnetic field. Besides the conditions that permit the external field to be treated as given, the range of applicability of such a theory is further restricted by considerations connected with the so-called radiative corrections.

The wave equation of an electron in a given external field is obtained in the same way as in the non-relativistic theory (see III, § 111). Let $A^\mu = (\Phi, \mathbf{A})$ be the 4-potential of the external electromagnetic field ($\mathbf{A}$ is the vector potential, Φ the scalar potential). The desired equation is obtained by replacing in the Dirac equation the 4-momentum operator $\widehat{p}$ with the difference $\widehat{p} - eA$, where e is

the charge of the particle[1]:

$$[\gamma(\widehat{p} - eA) - m]\psi = 0. \tag{32.1}$$

The Hamiltonian corresponding to this equation is obtained from (21.13) by the same substitution:

$$\widehat{H} = \boldsymbol{\alpha}(\widehat{\mathbf{p}} - e\mathbf{A}) + \beta m + e\Phi. \tag{32.2}$$

The invariance of the Dirac equation under a gauge transformation of the electromagnetic potentials is expressed by the fact that its form remains unchanged when the transformation $A \to A + i\widehat{p}\chi$ (where χ is an arbitrary function) is accompanied by the

[1]The charge is understood together with its sign, so that for the electron $e = -|e|$.

transformation of the wave function[2]

$$\psi \to \psi e^{ie\chi} \tag{32.3}$$

(cf. the analogous transformation for the Schrödinger equation in III, § 111).

The current density, expressed in terms of the wave function, is given by the same formula (21.11) $j = \bar{\psi}\gamma\psi$ as in the absence of an external field. It is easily verified that, on substituting (32.1) (together with equation (32.4) below) into the same calculation that was used to derive (21.11), the external field drops out and the continuity equation remains valid in its previous form.

Let us apply to equation (32.1) the operation of charge conjugation. To this end, write the equation as

$$\bar{\psi}[\gamma(\widehat{p} + eA) + m] = 0, \tag{32.4}$$

which is obtained by complex conjugation from (32.1) in the same way as was done previously for equation (19.9) (bearing in mind that the 4-vector A is real). Rewriting this equation as

$$[\widetilde{\gamma}(\widehat{p} + eA) + m]\widetilde{\bar{\psi}} = 0,$$

multiplying on the left by the matrix U_C and using the relations (26.3), we find

$$[\gamma(\widehat{p} + eA) - m](\widehat{C}\psi) = 0. \tag{32.5}$$

Thus the charge-conjugate wave function satisfies an equation differing from the original one only by a

reversal of the sign of the charge. On the other hand, the operation of charge conjugation signifies a transition from particles to antiparticles. We see, therefore, that if the particles carry an electric charge, the charges of electron and positron are automatically opposite.

The first-order equation (32.1) can be converted into a second-order equation by applying the operator $\gamma(\widehat{p} - eA) + m$:

$$[\gamma^\mu\gamma^\nu(\widehat{p}_\mu - eA_\mu)(\widehat{p}_\nu - eA_\nu) - m^2]\psi = 0.$$

Replacing $\gamma^\mu\gamma^\nu$ by

$$\gamma^\mu\gamma^\nu = \tfrac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) + \tfrac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = g^{\mu\nu} + \sigma^{\mu\nu},$$

where $\sigma^{\mu\nu}$ is the antisymmetric "matrix 4-tensor" (28.2). On multiplying by $\sigma^{\mu\nu}$ one may antisymmetrise, i.e.

$$(\widehat{p}_\mu - eA_\mu)(\widehat{p}_\nu - eA_\nu) \to \tfrac{1}{2}\{(\widehat{p}_\mu - eA_\mu), (\widehat{p}_\nu - eA_\nu)\}_- =$$

$$= \tfrac{1}{2}e(-A_\mu\widehat{p}_\nu + \widehat{p}_\nu A_\mu - \widehat{p}_\mu A_\nu + A_\nu\widehat{p}_\mu) =$$

$$= \tfrac{1}{2}ie(\partial_\nu A_\mu - \partial_\mu A_\nu) = -\tfrac{ie}{2}F_{\mu\nu}$$

[2]The transformation (32.3) with a function $\chi(t, \mathbf{r})$ is sometimes called a "local gauge transformation", as distinct from the "global gauge transformation" (12.10) with a constant phase α.

($F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ being the electromagnetic field tensor). The resulting second-order equation is

$$\left[(\widehat{p} - eA)^2 - m^2 - \frac{i}{2}eF_{\mu\nu}\sigma^{\mu\nu}\right]\psi = 0. \tag{32.6}$$

The product $F_{\mu\nu}\sigma^{\mu\nu}$ can be written in three-dimensional form by expressing it in terms of its components:

$$\sigma^{\mu\nu} = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma}), \qquad F^{\mu\nu} = (-\mathbf{E}, \mathbf{H}).$$

We then obtain the second-order equation as

$$[(\widehat{p} - eA)^2 - m^2 + e\boldsymbol{\Sigma}\cdot\mathbf{H} - ie\boldsymbol{\alpha}\cdot\mathbf{E}]\psi = 0, \tag{32.7}$$

or, in ordinary units,

$$\left[\left(\frac{i\hbar}{c}\frac{\partial}{\partial t} - \frac{e}{c}\Phi\right)^2 - \left(i\hbar\nabla + \frac{e}{c}\mathbf{A}\right)^2 - m^2c^2 + \frac{e\hbar}{c}\boldsymbol{\Sigma}\cdot\mathbf{H} - i\frac{e\hbar}{c}\boldsymbol{\alpha}\cdot\mathbf{E}\right]\psi = 0. \tag{32.7a}$$

The appearance in these equations of the terms containing **E** and **H** is connected with the spin of the particle; we shall return to their discussion in the next section.

Among the solutions of the second-order equation there are, of course, "spurious" ones—solutions that do not satisfy the original first-order equation (32.1) (they represent solutions

of equation (32.1) with the opposite sign in front of m). Selecting the correct solutions in specific cases is usually straightforward. A systematic selection procedure is the following: if φ is an arbitrary solution of the second-order equation, the correct solution of the first-order equation is

$$\psi = [\gamma(\widehat{p} - eA) + m]\varphi. \tag{32.8}$$

Indeed, multiplying this relation by $\gamma(\widehat{p} - eA) - m$, one sees that the right-hand side vanishes if φ satisfies equation (32.6).

It should be emphasised that the method of introducing an external field into the relativistic wave equation—by replacing $\widehat{p}$ with $\widehat{p} - eA$—is not entirely self-evident. In carrying it out we essentially relied on an additional principle: the indicated replacement should be performed in the first-order equation. It is precisely for this reason that in equation (32.6) there appeared extra terms that would not have arisen had the substitution been made directly in the second-order equation.

Among the stationary solutions of the Dirac equation in an external field there may exist states belonging to both the continuous and the discrete spectrum. As in the non-relativistic theory, the continuous spectrum corresponds to infinite motion, in which the particle can be found at arbitrarily large distances where it may be regarded as free. Since the eigenvalues of the free-particle Hamiltonian are $\pm\sqrt{\mathbf{p}^2 + m^2}$, it is clear that the continuous spectrum of eigenvalues lies at $\varepsilon > m$ and at $\varepsilon \leq -m$. For $-m < \varepsilon < m$ the particle cannot escape to infinity, so that the motion is finite and the state belongs to the discrete spectrum.

As for free particles, so also for particles in an external field the wave functions with "positive frequency" ($\varepsilon > 0$) and "negative frequency" ($\varepsilon < 0$) enter the scheme of second quantisation in a definite way. For a particle in an external field this scheme is naturally generalised by replacing the plane waves in the formulae (25.1) with the correspondingly normalised eigenfunctions of the Dirac equation $\psi_n^{(+)}$ and $\psi_n^{(-)}$, belonging to the positive ($\varepsilon_n^{(+)}$) and negative ($-\varepsilon_n^{(-)}$) frequencies:

$$\begin{aligned} \widehat{\psi} &= \sum_n \{\widehat{a}_n \psi_n^{(+)} \exp\left(-i\varepsilon_n^{(+)} t\right) + \widehat{b}_n^+ \psi_n^{(-)} \exp\left(i\varepsilon_n^{(-)} t\right)\}, \\ \widehat{\bar{\psi}} &= \sum_n \{\widehat{a}_n^+ \bar{\psi}_n^{(+)} \exp\left(i\varepsilon_n^{(+)} t\right) + \widehat{b}_n \bar{\psi}_n^{(-)} \exp\left(-i\varepsilon_n^{(-)} t\right)\}. \end{aligned} \tag{32.9}$$

One must bear in mind here that, as the potential well deepens, the energy levels may cross the boundary $\varepsilon = 0$, i.e. the energies that were

positive may become negative (or, for a potential of the opposite sign, initially negative energies may become positive). Nonetheless, on grounds of continuity one must continue to regard these levels as electronic (rather than positronic). In other words, an electron includes all states which, in the limit of an infinitely slow removal of the field, tend to the positive boundary of the continuous spectrum ($\varepsilon = m$).

Although the Dirac equation for an electron in an external field does, as stated above, permit the solution of a wide class of quantum-electrodynamic problems, one must at the same time stress that the applicability of the external-field concept in relativistic theory is itself limited. This limitation is connected with the spontaneous creation of electron–positron pairs that arises in sufficiently strong fields (see below, §§ 35, 36).

We shall not consider in this book the introduction of an external field into the wave equations of particles with spin other than 1/2, since this has no direct physical relevance—the real particles with such spins are hadrons, and their electromagnetic interactions cannot be described by wave equations. In this connection it should be noted that for spin-0 particles the wave equation has complex eigenvalues (with imaginary parts of both signs) in a sufficiently deep potential well. The wave equation for a spin-3/2 particle leads to a violation of causality, manifested in the appearance of solutions that propagate faster than light.

Problem

Determine the energy levels of an electron in a constant magnetic field.

Solution. Vector potential: $A_x = A_z = 0$, $A_y = Hx$ (the field H directed along the z-axis). The quantities that are conserved (along with the energy) are the components p_y and p_z of the generalised momentum.

We use the second-order equation for the auxiliary function φ (see (32.8)) and assume that φ is an eigenfunction of the operator Σ_z (with eigenvalue $\sigma = \pm 1$), as well as of the operators $\widehat{p}_y$ and $\widehat{p}_z$. The equation for φ then takes the form

$$\left\{-\frac{d^2}{dx^2} + (eHx - p_y)^2 - eH\sigma\right\}\varphi = (\varepsilon^2 - m^2 - p_z^2)\varphi.$$

This equation is identical in form to the Schrödinger equation for a linear harmonic oscillator. Its eigenvalues are given by the formula

$$\varepsilon^2 - m^2 - p_z^2 = |e|H(2n+1) - eH\sigma, \qquad n = 0, 1, 2, \ldots$$

(cf. III, § 112). Note that the wave function ψ, which should be determined from φ by formula (32.8), is not an eigenfunction of the operator Σ_z, in accordance with the fact that for a moving particle the spin is not a conserved quantity.

§ 33. Expansion in powers of $1/c$

We saw (see § 21) that in the non-relativistic limit ($v \to 0$) the two components (χ) of the bispinor $\psi = \binom{\varphi}{\chi}$ vanish. At low electron velocities, therefore, $\chi \ll \varphi$, and it becomes possible to derive an approximate equation involving only the two-component quantity φ, by formally expanding the wave function in powers of $1/c$ [3].

Starting from the Dirac equation for an electron in an external field written in the form

$$i\hbar\frac{\partial\psi}{\partial t} = \left\{c\boldsymbol{\alpha}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right) + \beta mc^2 + e\Phi\right\}\psi. \tag{33.1}$$

The relativistic energy of the particle includes the rest energy mc^2. To pass to the non-relativistic approximation this must be removed, which we do by introducing in place of ψ a function ψ' defined by

$$\psi = \psi' e^{-imc^2t/\hbar}.$$

Then

$$\left(i\hbar\frac{\partial}{\partial t} + mc^2\right)\psi' = \left\{c\boldsymbol{\alpha}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right) + \beta mc^2 + e\Phi\right\}\psi'.$$

Writing $\psi' = \binom{\varphi'}{\chi'}$ yields the system of equations:

$$\left(i\hbar\frac{\partial}{\partial t} - e\Phi\right)\varphi' = c\boldsymbol{\sigma}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\chi', \tag{33.2}$$

$$\left(i\hbar\frac{\partial}{\partial t} - e\Phi + 2mc^2\right)\chi' = c\boldsymbol{\sigma}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\varphi' \tag{33.3}$$

(hereafter we drop the primes on φ and χ, since this will cause no confusion given that in this section we work exclusively with the transformed function ψ').

In the leading approximation we retain only the term $2mc^2\chi$ on the left of equation (33.3) and obtain

$$\chi = \frac{1}{2mc}\boldsymbol{\sigma}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\varphi \tag{33.4}$$

(note that $\chi \sim \varphi/c$). Substituting this expression into (33.2) gives

$$\left(i\hbar\frac{\partial}{\partial t} - e\Phi\right)\varphi = \frac{1}{2m}\left(\boldsymbol{\sigma}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\right)^2\varphi.$$

[3] In this section we use ordinary units.

For the Pauli matrices one has the identity

$$(\boldsymbol{\sigma}\mathbf{a})(\boldsymbol{\sigma}\mathbf{b}) = \mathbf{a}\cdot\mathbf{b} + i\boldsymbol{\sigma}\cdot[\mathbf{a}\times\mathbf{b}], \tag{33.5}$$

where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors (see (20.9)). Here $\mathbf{a} = \mathbf{b} = \widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}$, but the vector product $[\mathbf{a}\times\mathbf{b}]$ does not vanish because $\widehat{\mathbf{p}}$ and $\mathbf{A}$ do not commute:

$$\left[\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\times\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\right]\varphi = i\frac{e\hbar}{c}\{[\mathbf{A}\cdot\nabla] + [\nabla\cdot\mathbf{A}]\}\varphi = i\frac{e\hbar}{c}\,\mathrm{rot}\,\mathbf{A}\cdot\varphi.$$

Thus

$$\left(\boldsymbol{\sigma}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\right)^2 = \left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 - \frac{e\hbar}{c}\boldsymbol{\sigma}\cdot\mathbf{H} \tag{33.6}$$

(where $\mathbf{H} = \mathrm{rot}\,\mathbf{A}$ is the magnetic field), and for φ we obtain the equation

$$i\hbar\frac{\partial\varphi}{\partial t} = \widehat{H}\varphi = \left[\frac{1}{2m}\left(\widehat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 + e\Phi - \frac{e\hbar}{2mc}\boldsymbol{\sigma}\cdot\mathbf{H}\right]\varphi. \tag{33.7}$$

This is the so-called *Pauli equation*. It differs from the non-relativistic Schrödinger equation by the presence in the Hamiltonian of the last term, which represents the potential energy of a magnetic dipole in an external field (cf. III, § 111). In the leading $(1/c)$ approximation, therefore, the electron behaves as a particle that possesses, in addition to its charge, a magnetic moment[4]:

$$\boldsymbol{\mu} = \frac{e}{mc}\hbar\mathbf{s}. \tag{33.8}$$

The gyromagnetic ratio (e/mc) is twice as large as it would be for a magnetic moment associated with orbital motion.

The density is $\rho = \psi^*\psi = \varphi^*\varphi + \chi^*\chi$. In the leading approximation the second term is to be dropped, so that $\rho = |\varphi|^2$, as it should be for the Schrödinger equation.

The current density is:

$$\mathbf{j} = c\psi^*\boldsymbol{\alpha}\psi = c(\varphi^*\boldsymbol{\sigma}\chi + \chi^*\boldsymbol{\sigma}\varphi).$$

Substituting (33.4):

$$\chi = \frac{1}{2mc}\boldsymbol{\sigma}\left(-i\hbar\nabla - \frac{e}{c}\mathbf{A}\right)\varphi, \qquad \chi^* = \frac{1}{2mc}\left(i\hbar\nabla - \frac{e}{c}\mathbf{A}\right)\varphi^*\boldsymbol{\sigma},$$

and using the products involving two $\boldsymbol{\sigma}$ factors, which are rearranged with the help of (33.5) in the form

$$(\boldsymbol{\sigma}\mathbf{a})\boldsymbol{\sigma} = \mathbf{a} + i[\boldsymbol{\sigma}\times\mathbf{a}], \qquad \boldsymbol{\sigma}(\boldsymbol{\sigma}\mathbf{a}) = \mathbf{a} + i[\mathbf{a}\times\boldsymbol{\sigma}]. \tag{33.9}$$

The result is

$$\mathbf{j} = \frac{i\hbar}{2m}(\varphi\nabla\varphi^* - \varphi^*\nabla\varphi) - \frac{e}{mc}\mathbf{A}\varphi^*\varphi + \frac{\hbar}{2m}\,\mathrm{rot}(\varphi^*\boldsymbol{\sigma}\varphi), \tag{33.10}$$

[4]This remarkable result was obtained by *Dirac* in 1928. The two-component wave function satisfying equation (33.7) had been introduced by *Pauli* (1927) even before Dirac discovered his equation.

in agreement with expression (115.4) (see III) from the non-relativistic theory.

Let us now find the second approximation, continuing the expansion to terms $\sim 1/c^2$ [5]. We shall assume that only an electric field is present ($\mathbf{A} = 0$).

First of all, we note that with the inclusion of terms $\sim 1/c^2$ the density

$$\rho = |\varphi|^2 + |\chi|^2 = |\varphi|^2 + \frac{\hbar^2}{4m^2c^2}|\boldsymbol{\sigma}\cdot\nabla\varphi|^2.$$

This expression differs from the Schrödinger form. To find (in the second approximation) a wave equation analogous to the Schrödinger equation, we must introduce in place of φ a different (two-component) function φ_{Sch}, for which the time-independent integral $\int |\varphi_{\text{Sch}}|^2\, d^3x$ has the same form as in the Schrödinger equation.

To determine the required transformation we write the condition

$$\int \varphi^*_{\text{Sch}}\varphi_{\text{Sch}}\, d^3x = \int \left\{\varphi^*\varphi + \frac{\hbar^2}{4m^2c^2}(\nabla\varphi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla\varphi)\right\} d^3x$$

and integrate by parts:

$$\int (\nabla\varphi^*\cdot\boldsymbol{\sigma})(\boldsymbol{\sigma}\cdot\nabla)\varphi\, d^3x = -\int \varphi^*(\boldsymbol{\sigma}\cdot\nabla)(\boldsymbol{\sigma}\cdot\nabla)\varphi\, d^3x = -\int \varphi^*\Delta\varphi\, d^3x$$

(or the same with φ and φ^* interchanged). Thus

$$\int \varphi^*_{\text{Sch}}\varphi_{\text{Sch}}\, d^3x = \int \left\{\varphi^*\varphi - \frac{\hbar^2}{8m^2c^2}(\varphi^*\Delta\varphi + \varphi\Delta\varphi^*)\right\} d^3x,$$

from which one sees that

$$\varphi_{\text{Sch}} = \left(1 + \frac{\widehat{\mathbf{p}}^2}{8m^2c^2}\right)\varphi, \qquad \varphi = \left(1 - \frac{\widehat{\mathbf{p}}^2}{8m^2c^2}\right)\varphi_{\text{Sch}}. \tag{33.11}$$

For simplicity let us assume a stationary state, i.e. replace the operator $i\hbar\partial/\partial t$ by the energy ε (with the rest energy subtracted). In the next (after (33.4)) approximation we have from (33.3):

$$\chi = \frac{1}{2mc}\left(1 - \frac{\varepsilon - e\Phi}{2mc^2}\right)(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}})\varphi.$$

This expression is to be substituted into (33.2), after which we replace φ by φ_{Sch} according to (33.11), dropping throughout all terms of order higher than $1/c^2$. A straightforward computation yields the

equation $\varepsilon\varphi_{\text{Sch}} = \widehat{H}\varphi_{\text{Sch}}$, with the Hamiltonian

$$\widehat{H} = \frac{\widehat{\mathbf{p}}^2}{2m} + e\Phi - \frac{\widehat{\mathbf{p}}^4}{8m^3c^2} + \frac{e}{4m^2c^2}(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}})\Phi(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}}) - \frac{1}{2}\left(\widehat{\mathbf{p}}^2\Phi + \Phi\widehat{\mathbf{p}}^2\right). \tag{33.12}$$

The expression in braces is rearranged using the formulae

$$(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}})\Phi(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}}) + (\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}}\Phi)(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}}) = \Phi\widehat{\mathbf{p}}^2 + i\hbar(\boldsymbol{\sigma}\cdot\mathbf{E})(\boldsymbol{\sigma}\cdot\widehat{\mathbf{p}}),$$

[5] Here we follow the method of *V. B. Berestetskii* and *L. D. Landau* (1949).

$$\widehat{\mathbf{p}}^2\Phi - \Phi\widehat{\mathbf{p}}^2 = -\hbar^2\Delta\Phi + 2i\hbar\mathbf{E}\cdot\widehat{\mathbf{p}},$$

where $\mathbf{E} = -\nabla\Phi$ is the electric field. The final expression for the Hamiltonian is:

$$\widehat{H} = \frac{\widehat{\mathbf{p}}^2}{2m} + e\Phi - \frac{\widehat{\mathbf{p}}^4}{8m^3c^2} - \frac{e\hbar}{4m^2c^2}\boldsymbol{\sigma}\cdot[\mathbf{E}\times\widehat{\mathbf{p}}] - \frac{e\hbar^2}{8m^2c^2}\operatorname{div}\mathbf{E}. \tag{33.12}$$

The last three terms are the sought-for corrections of order $1/c^2$. The first of them is a consequence of the relativistic dependence of kinetic energy on momentum (the expansion of $\sqrt{\widehat{\mathbf{p}}^2 + m^2c^2} - mc^2$). The second, which may be called the *spin–orbit interaction* energy, represents the interaction of the magnetic moment of a moving charge with the electric field[6]. The last term differs from zero only at those points where the field-producing charges are located; for the Coulomb field of a point charge Ze: $\Delta\Phi = -4\pi Ze\delta(\mathbf{r})$ (*C. G. Darwin*, 1928).

If the electric field is centrally symmetric,

$$\mathbf{E} = -\frac{\mathbf{r}}{r}\frac{d\Phi}{dr}$$

and the spin–orbit interaction operator can be written as

$$\frac{e\hbar}{4m^2c^2r}\boldsymbol{\sigma}\cdot[\mathbf{r}\times\widehat{\mathbf{p}}]\frac{d\Phi}{dr} = \frac{\hbar^2}{2m^2c^2r}\frac{dU}{dr}\widehat{\mathbf{l}}\cdot\widehat{\mathbf{s}}. \tag{33.13}$$

Here $\widehat{\mathbf{l}}$ is the orbital angular momentum operator, $\widehat{\mathbf{s}} = \frac{1}{2}\boldsymbol{\sigma}$ is the electron spin operator, and $U = e\Phi$ is the potential energy of the electron in the field.

§ 34. Fine structure of hydrogen atom levels

Let us determine the relativistic corrections to the energy levels of the hydrogen atom—an electron in the Coulomb field of a stationary nucleus[7]. The velocity of the electron in the hydrogen atom $v/c \sim \alpha \ll 1$. Therefore the desired corrections can be calculated by perturbation theory—as an average over the unperturbed state (i.e. the non-relativistic wave function) of the relativistic terms in the approximate Hamiltonian (33.12). For somewhat greater generality we set the nuclear charge equal to Ze, assuming, however, that $Z\alpha \ll 1$.

The field of the nucleus $\mathbf{E} = Ze\mathbf{r}/r^3$, and its potential satisfies the equation $\Delta\Phi = -4\pi Ze\delta(\mathbf{r})$. Substituting this into (33.12) (the last three terms), and taking into account the negativity of the electron's charge, we obtain the perturbation operator:

$$\widehat{V} = -\frac{\widehat{\mathbf{p}}^4}{8m^3} + \frac{Z\alpha}{2r^3m^2}\widehat{\mathbf{l}}\cdot\widehat{\mathbf{s}} + \frac{Z\alpha\pi}{2m^2}\delta(\mathbf{r}). \tag{34.1}$$

[6] Introducing the magnetic moment (33.8) and the velocity $\mathbf{v} = \mathbf{p}/m$, one can write this energy as $-\frac{1}{2}\boldsymbol{\mu}\cdot[\mathbf{E}\times\mathbf{v}]$. At first sight this result may seem surprising, since on passing to the rest frame of the particle the magnetic field $\mathbf{H} = \frac{1}{c}[\mathbf{E}\times\mathbf{v}]$ arises, relative to which the magnetic moment should carry energy $-\boldsymbol{\mu}\cdot\mathbf{H}$. In reality the factor $1/2$ ("Thomas precession", *L. Thomas*, 1926) is connected with the general requirements of relativistic invariance in conjunction with the specific properties of the electron as a "spinor" particle with its associated value of the gyromagnetic ratio (see § 41).

[7] The effect of nuclear motion on the values of these corrections is of higher-order smallness, which does not interest us here.

Since according to the non-relativistic Schrödinger equation

$$\widehat{\mathbf{p}}^2\psi = 2m\left(\varepsilon_0 + \frac{Z\alpha}{r}\right)\psi$$

($\varepsilon_0 = -mZ^2\alpha^2/2n^2$ is the unperturbed level, n the principal quantum number), the mean value

$$\overline{\mathbf{p}^4} = 4m^2\overline{\left(\varepsilon_0 + \frac{Z\alpha}{r}\right)^2}.$$

This quantity, as well as the mean value of the second term in (34.1), is computed using the formulae (see III, § 36)

$$\overline{r^{-1}} = \frac{m\alpha Z}{n^2}, \qquad \overline{r^{-2}} = \frac{(m\alpha Z)^2}{n^3(l+\frac{1}{2})}, \qquad \overline{r^{-3}} = \frac{(m\alpha Z)^3}{n^3\, l(l+\frac{1}{2})(l+1)} \tag{34.2}$$

(the last formula applies to $l \neq 0$); the eigenvalue

$$\mathbf{l}\cdot\mathbf{s} = \begin{cases} \frac{1}{2}[j(j+1) - l(l+1) - \frac{3}{4}], & l \neq 0, \\ 0, & l = 0. \end{cases}$$

Finally, the averaging of the third term is performed using the formula

$$|\psi(0)|^2 = \begin{cases} \dfrac{1}{\pi}\left(\dfrac{Z\alpha m}{n}\right)^3, & l = 0, \\ 0, & l \neq 0. \end{cases} \tag{34.3}$$

The result of a simple calculation using the formulae written above can be presented in all cases (for all j and l) in the form

$$\Delta\varepsilon = -\frac{m(Z\alpha)^4}{2n^3}\left(\frac{1}{j+\frac{1}{2}} - \frac{3}{4n}\right). \tag{34.4}$$

Formula (34.4) gives the desired relativistic correction to the energy of the hydrogen levels—the fine-structure energy [8]. Recall that in the non-relativistic theory degeneracy occurs both with respect to the direction of spin and the Coulomb degeneracy with respect to l. The fine structure (spin–orbit interaction) removes this degeneracy, but not completely: there remain doubly degenerate levels with the same n, j, but different $l = j \pm \frac{1}{2}$ (non-degenerate turn out to be only the levels with the largest possible $j = j_{\max} = l_{\max} + \frac{1}{2} = n - \frac{1}{2}$ for a given n). Thus the sequence of hydrogen levels with fine structure taken into account is as follows:

$$\begin{array}{lll} 1s_{1/2}; & & \\ 2s_{1/2},\ 2p_{1/2}, & 2p_{3/2}; & \\ 3s_{1/2},\ 3p_{1/2}, & 3p_{3/2},\ 3d_{3/2}, & 3d_{5/2}. \end{array}$$

[8]This formula (and the more exact formula (36.10)) was obtained by *A. Sommerfeld* from the old Bohr theory even before the creation of quantum mechanics.

A level with principal quantum number n splits into n components of fine structure.

Recall that in non-relativistic mechanics the "accidental" degeneracy of energy levels in the Coulomb field is connected with the existence of a conservation law specific to this field: the quantity $\mathbf{A}$ is conserved, whose operator is

$$\widehat{\mathbf{A}} = \frac{\mathbf{r}}{r} + \frac{1}{2m\alpha}\left\{[\widehat{\mathbf{l}}\,\widehat{\mathbf{p}}] - [\widehat{\mathbf{p}}\,\widehat{\mathbf{l}}]\right\}$$

(see III, (36.30)). A specific conservation law is connected with, and persists in, the relativistic case as a twofold degeneracy: the Hamiltonian of the Dirac equation $\widehat{H} = \boldsymbol{\alpha}\widehat{\mathbf{p}} + \beta m - e^2/r$ commutes with the operator

$$\widehat{\Gamma} = \frac{\mathbf{r}}{r}\cdot\boldsymbol{\Sigma} + \frac{i}{m\alpha}\beta(\boldsymbol{\Sigma}\cdot\widehat{\mathbf{l}} + 1)\gamma^5(\widehat{H} - m\beta)$$

(*M. H. Johnson, B. A. Lippmann*, 1950). In the non-relativistic limit this operator $\widehat{\Gamma} \to \boldsymbol{\Sigma}\widehat{\mathbf{A}}$.

We shall see later (§ 123) that this remaining degeneracy is lifted by so-called radiative corrections (the *Lamb shift*), not accounted for by the Dirac equation for the single-electron problem.

Looking ahead, let us note here that the order of magnitude of these corrections is $\sim mZ^4\alpha^5\ln(1/\alpha)$. The second-order correction due to the spin–orbit interaction was $\sim m(Z\alpha)^6$, so that its ratio to the radiative corrections is $\sim Z^2\alpha/\ln(1/\alpha)$. For hydrogen ($Z = 1$) this ratio is quite small, and therefore the problem of an exact solution of the Dirac equation in this case is of no practical interest. This problem may, however, have meaning for the energy levels of an electron in the field of a nucleus with large Z (see § 36).

§ 35. Motion in a centrally symmetric field

Consider the motion of an electron in a centrally symmetric electric field. Since angular momentum and parity are conserved in motion in a central field (relative to the centre of the field, chosen as the coordinate origin), everything said in § 24 about the angular dependence of wave functions of spherical waves of free particles applies to this motion as well. Only the radial functions change. Accordingly we seek the wave function of stationary states (in the standard representation) in the form

$$\psi = \begin{pmatrix}\varphi\\ \chi\end{pmatrix} = \begin{pmatrix} f(r)\,\Omega_{jlm} \\ (-1)^{(1+l-l')/2}\,g(r)\,\Omega_{jl'm}\end{pmatrix}, \tag{35.1}$$

where $l = j \pm \frac{1}{2}$, $l' = 2j - l$, and the factor -1 is introduced to simplify the subsequent formulae.

The Dirac equation in the standard representation gives the following system of equations for φ and χ:

$$(\varepsilon - m - U)\varphi = \boldsymbol{\sigma}\widehat{\mathbf{p}}\,\chi, \qquad (\varepsilon + m - U)\chi = \boldsymbol{\sigma}\widehat{\mathbf{p}}\,\varphi, \tag{35.2}$$

where $U(r) = e\Phi(r)$ is the potential energy of the electron in the field. The calculation of the result of substituting the expressions (35.1) into these equations reduces to computing the right-hand sides.

Expressing the spherical spinor $\Omega_{jl'm}$ through Ω_{jlm} according to

$$\Omega_{jl'm} = i^{l-l'}\left(\frac{\boldsymbol{\sigma}\cdot\mathbf{r}}{r}\right)\Omega_{jlm}$$

(see (24.8)), we write:

$$(\boldsymbol{\sigma}\widehat{\mathbf{p}})\chi = -i(\boldsymbol{\sigma}\nabla)(\boldsymbol{\sigma}\mathbf{r})\,\frac{g}{r}\,\Omega_{jlm}.$$

Transforming now the product $(\boldsymbol{\sigma}\widehat{\mathbf{p}})(\boldsymbol{\sigma}\mathbf{r})$ with the aid of formula (33.5), after expanding the vector operations we find

$$(\boldsymbol{\sigma}\widehat{\mathbf{p}})\chi = -i\{\widehat{p}_r + i\boldsymbol{\sigma}[\widehat{\mathbf{p}}\,\widehat{\mathbf{r}}]\}\,\frac{g}{r}\,\Omega_{jlm} =$$

$$= \left\{-\operatorname{div}\mathbf{r} - (\mathbf{r}\nabla) - \boldsymbol{\sigma}[\mathbf{r}\widehat{\mathbf{r}}]\right\}\frac{g}{r}\,\Omega_{jlm} = -\left\{g' + \frac{2}{r}g + \frac{g}{r}\,\boldsymbol{\sigma}\cdot\widehat{\mathbf{l}}\right\}\Omega_{jlm},$$

where $\widehat{\mathbf{l}} = [\mathbf{r}\widehat{\mathbf{p}}]$ is the orbital angular momentum operator; the prime denotes differentiation with respect to r. The eigenvalues of the product $\boldsymbol{\sigma}\cdot\widehat{\mathbf{l}} = 2\widehat{\mathbf{l}}\cdot\widehat{\mathbf{s}}$ are

$$2\mathbf{l}\cdot\mathbf{s} = \mathbf{j}^2 - \mathbf{l}^2 - \mathbf{s}^2 = j(j+1) - l(l+1) - \tfrac{3}{4} = \begin{cases} j - \frac{1}{2}, & l = j - \frac{1}{2}, \\ -j - \frac{3}{2}, & l = j + \frac{1}{2}. \end{cases}$$

For uniformity of notation in both cases ($l = j \pm \frac{1}{2}$) it is convenient to introduce the notation

$$\varkappa = \begin{cases} -(j+\frac{1}{2}) = -(l+1), & j = l + \frac{1}{2}, \\ +(j+\frac{1}{2}) = l, & j = l - \frac{1}{2}. \end{cases} \tag{35.3}$$

The number $\varkappa$ takes all integer values excluding zero (positive numbers correspond to $j = l - \frac{1}{2}$, and negative ones to $j = l + \frac{1}{2}$). Then $\mathbf{l}\cdot\boldsymbol{\sigma} = -(1+\varkappa)$, so that

$$(\boldsymbol{\sigma}\widehat{\mathbf{p}})\chi = -\left(g' + \frac{1-\varkappa}{r}\,g\right)\Omega_{jlm}.$$

On substituting this expression into the first of equations (35.2), the spherical spinor Ω_{jlm} cancels on both sides of the equation. Proceeding analogously with the second equation, we obtain the following system for the radial functions:

$$\begin{aligned} f' + \frac{1+\varkappa}{r}\,f - (\varepsilon + m - U)\,g = 0, \\ g' + \frac{1-\varkappa}{r}\,g + (\varepsilon - m - U)\,f = 0, \end{aligned} \tag{35.4}$$

or

$$\begin{aligned} (fr)' + \frac{\varkappa}{r}\,(fr) - (\varepsilon + m - U)\,gr = 0, \\ (gr)' - \frac{\varkappa}{r}\,(gr) + (\varepsilon - m - U)\,fr = 0. \end{aligned} \tag{35.5}$$

Let us investigate the behaviour of f and g at small distances, assuming that the field $U(r)$ grows as $r \to 0$ faster than $1/r$. Then in the region of small r equations (35.4) take the form

$$f' + Ug = 0, \qquad g' - Uf = 0.$$

They have real solutions of the form

$$f = \text{const} \cdot \sin\left(\int U\,dr + \delta\right), \qquad g = \text{const} \cdot \cos\left(\int U\,dr + \delta\right), \tag{35.6}$$

where δ is an arbitrary constant. These functions oscillate as $r \to 0$, without tending to any limit. It is easy to see that this situation corresponds in the non-relativistic theory to "falling" of the particle to the centre.

First of all we note that the region of small distances imposes in this case no restrictions on the choice of solution: the condition at $r = 0$ for the oscillating function is absent, and the choice of the constant δ remains arbitrary (the correct behaviour of the wave function at large r can be achieved for any ε by an appropriate choice of δ). This indeterminacy can be removed by considering the singular (at $r = 0$) potential as the limit $r_0 \to 0$ of the potential "cut off" at some small r_0 (i.e. equal to $U(r)$ for $r > r_0$ and $U(r_0)$ for $r < r_0$). With a finite r_0 one obtains, of course, a definite system of energy levels. However, the energy of the ground state tends to $-\infty$ as $r_0 \to 0$.

In the non-relativistic theory this is precisely the "falling" to the centre, since a particle at a deep level is localised in a small region around $r = 0$. In the relativistic theory, however, such a situation is altogether inadmissible, since it signifies the instability of the system with respect to the spontaneous creation of electron–positron pairs. Indeed, if in vacuum the energy needed to create such a pair exceeds $2m$, in the field a smaller energy already suffices. If there exists a bound state of the electron with energy $\varepsilon < m$, pair creation is possible at the cost of only the energy $\varepsilon + m < 2m$, with a free positron created and the electron in the bound state. If the energy of the bound state $\varepsilon < -m$, such a field can create positrons (with energy $-\varepsilon > m$) spontaneously, without any input of energy from an external source. In the field being considered, as $r_0 \to 0$ there is an infinite number of such "anomalous" levels with $\varepsilon < -m$. Therefore a potential $\Phi(r)$ growing as $r \to 0$ faster than $1/r$ cannot at all be considered in the Dirac theory. Let us stress that this applies to potentials of either sign. The "falling" occurs, of course, only in the case of attraction, since the sign of $U = e\Phi$ depends also on the sign of the charge; in one case the electronic, and in the other the positronic levels behave anomalously; in the second case the field creates free electrons.

Let us consider further the behaviour of wave functions at large distances. If the field $U(r)$ falls off sufficiently rapidly at $r \to \infty$, then in determining the asymptotic form of the wave functions at large distances the field can be completely neglected in the equations. For $\varepsilon > m$, i.e. in the continuous-spectrum region, we return then to the free-motion equation, so that the asymptotic form of the wave functions (spherical waves) differs from that for a free particle only by the appearance of additional "phase shifts", whose values are determined by the form of the field at close distances[9]. These shifts depend on the values of j and l, or, equivalently, on the number $\varkappa$ (and also, of course, on the energy ε). Denoting them by $\delta_\varkappa$ and using the expression for the free spherical

[9]Cf. III, § 33. As in the non-relativistic theory, $U(r)$ must fall off faster than $1/r$. The case $U \sim 1/r$ will be considered separately in § 36.

wave (24.7), we can immediately write the desired asymptotic formula

$$\psi \approx \frac{2}{r}\frac{1}{\sqrt{2\varepsilon}}\begin{pmatrix} \sqrt{\varepsilon+m}\,\Omega_{jlm}\sin\left(pr-\dfrac{\pi l}{2}+\delta_\varkappa\right) \\ -\sqrt{\varepsilon-m}\,\Omega_{jl'm}\sin\left(pr-\dfrac{\pi l'}{2}+\delta_\varkappa\right)\end{pmatrix}, \tag{35.7}$$

or, taking into account the definition (35.1):

$$\left.\begin{matrix} f \\ g \end{matrix}\right\} = \frac{\sqrt{2}}{r}\sqrt{\frac{\varepsilon\pm m}{\varepsilon}}\ \sin\left(pr-\frac{\pi l}{2}+\delta_\varkappa\right), \tag{35.8}$$

where $p=\sqrt{\varepsilon^2-m^2}$. The overall coefficient here corresponds to normalisation of the radial functions according to (24.5).

The wave functions of the discrete spectrum ($\varepsilon<m$) decay exponentially at $r\to\infty$ according to the law

$$f=-\sqrt{\frac{m+\varepsilon}{m-\varepsilon}}\,g=\frac{A_0}{r}\exp\left(-r\sqrt{m^2-\varepsilon^2}\right), \tag{35.9}$$

where A_0 is a constant.

As in the non-relativistic theory, the phase shifts $\delta_\varkappa$ (more precisely, the quantities $e^{2i\delta_\varkappa}-1$) determine the scattering amplitude in the given field (this will be discussed in more detail in § 37). We shall not investigate here the analytic properties of these quantities (cf. III, § 128). We note only that $e^{2i\delta_\varkappa}$ as a function of the energy still has poles at the points corresponding to the bound-state energy levels of the particle. The residue of the function $e^{2i\delta_\varkappa}$ at such a pole is related in a definite way to the coefficient in the asymptotic expression of the corresponding wave function of the discrete spectrum. Let us find this relation, which generalises the non-relativistic formula (128.17) (see III). The necessary calculations are entirely analogous to those performed in III, § 128.

Differentiate equations (35.5) with respect to the energy:

$$\left(\frac{\partial rf}{\partial\varepsilon}\right)'+\frac{\varkappa}{r}\frac{\partial rf}{\partial\varepsilon}-(\varepsilon+m-U)\frac{\partial rg}{\partial\varepsilon}=rg,$$

$$\left(\frac{\partial rg}{\partial\varepsilon}\right)'-\frac{\varkappa}{r}\frac{\partial rg}{\partial\varepsilon}+(\varepsilon-m-U)\frac{\partial rf}{\partial\varepsilon}=-rf.$$

Multiply these two equations respectively by rg and $-rf$, and the two equations (35.5) respectively by $-rg$ and rf; then add all four equations term by term. After all cancellations we get

$$\frac{\partial}{\partial r}\left[r^2\left(g\frac{\partial f}{\partial\varepsilon}-f\frac{\partial g}{\partial\varepsilon}\right)\right]=r^2(f^2+g^2).$$

Integrate this relation over r:

$$r^2\left(g\frac{\partial f}{\partial\varepsilon}-f\frac{\partial g}{\partial\varepsilon}\right)=\int_0^r(f^2+g^2)\,r^2\,dr,$$

and then pass to the limit $r \to \infty$. By the normalisation condition the integral on the right-hand side becomes unity. On the left-hand side we take into account that in the asymptotic region the functions f and g are related by

$$rg = \frac{(rf)'}{\varepsilon + m},$$

obtained from (35.5) by neglecting the terms with U and $1/r$. As a result we get

$$\frac{1}{\varepsilon + m}\left[(rf)'\frac{\partial rf}{\partial \varepsilon} - rf\left(\frac{\partial rf}{\partial \varepsilon}\right)'\right] = 1. \tag{35.10}$$

This formula differs from the analogous non-relativistic formula (for the function χ) only by the coefficient ($\varepsilon + m$ instead of $2m$). There is therefore no need to repeat all the subsequent calculations, and we immediately give the final formula, valid near the point $\varepsilon = \varepsilon_0$ (ε_0 being the energy level):

$$e^{2i\delta_\varkappa} = (-1)^l \frac{2A_0^2}{\varepsilon - \varepsilon_0}\sqrt{\frac{m - \varepsilon_0}{m + \varepsilon_0}}, \tag{35.11}$$

where A_0 is the coefficient in the asymptotic expression (35.9).

Problem

Find the limiting form of the wave function at small r in the field $U \sim r^{-s}$, $s < 1$.

Solution. For a free particle we have at small r: $f \sim r^l$, $g \sim r^{l'}$, so that for $l < l'$: $f \gg g$, and for $l > l'$: $f \ll g$. We make the assumption (justified by the result) that such a relation is preserved also in the field being considered. For $l < l'$ (i.e. $l = j - \frac{1}{2}$, $\varkappa = -l - 1$) the g term in the first of equations (35.4) can be dropped, so that $f \sim r^l$ as before. From the second equation we then have $g \sim rfU$, i.e. $g \sim r^{l+1-s}$. The case $l > l'$ is treated analogously. As a result we find:

$$\begin{aligned} &\text{for } l < l': \quad f \sim r^l, \quad\quad g \sim r^{l'-s}; \\ &\text{for } l > l': \quad f \sim r^{l-s}, \quad g \sim r^{l'}. \end{aligned}$$

§ 36. Motion in a Coulomb field

The study of the properties of motion in the most important case of the Coulomb field begins with an investigation of the behaviour of wave functions at small distances. Let us speak for definiteness of the field of attraction: $U = -Z\alpha/r$ [10]. For small r in equations (35.5) one may drop the terms $\varepsilon \pm m$; then

$$(fr)' + \frac{\varkappa}{r}fr - \frac{Z\alpha}{r}gr = 0,$$

$$(gr)' - \frac{\varkappa}{r}gr + \frac{Z\alpha}{r}fr = 0.$$

[10] In ordinary units $U = -Ze^2/r$. In the transition to relativistic units e^2 is replaced by the dimensionless α.

The functions fr and gr enter into each of these equations on an equal footing. We therefore seek both in the form of equal powers of r: $fr = ar^\gamma$, $gr = br^\gamma$. Substitution into the equations gives

$$a(\gamma + \varkappa) - bZ\alpha = 0, \qquad aZ\alpha + b(\gamma - \varkappa) = 0,$$

whence

$$\gamma^2 = \varkappa^2 - (Z\alpha)^2. \tag{36.1}$$

Suppose $(Z\alpha)^2 < \varkappa^2$. Then γ is real, and of the two values the positive one must be chosen: the corresponding solution either does not diverge at $r = 0$, or diverges less rapidly. Such a choice can be justified by considering the potential, cut off at some small r_0, with a subsequent passage to the limit $r_0 \to 0$ (cf. analogous arguments in III, § 35). Thus,

$$f = \frac{Z\alpha}{\gamma + \varkappa}\, g = \text{const} \cdot r^{-1+\gamma}. \tag{36.2}$$

Although the wave function may diverge at $r = 0$ to infinity (if $\gamma < 1$), the integral of $|\psi|^2$ remains, of course, convergent. If $(Z\alpha)^2 > \varkappa^2$, both values of γ from (36.1) are imaginary. The corresponding solutions oscillate as $r \to 0$ (like $r^{-1}\cos(|\gamma|\ln r)$), which, as was already explained above, is an inadmissible situation in the relativistic theory of "falling" to the centre. Since $\varkappa^2 \geqslant 1$, this means that the pure Coulomb field can be considered in the Dirac theory only when $Z\alpha < 1$, i.e. $Z < 137$.

Let us dwell on a qualitative description of the situation that arises for $Z > 137$. Again, to avoid the indeterminacy in the boundary condition at $r = 0$, one should consider the potential cut off at some small r_0 (*I. Ya. Pomeranchuk, Ya. A. Smorodinsky*, 1945). This has not only a formal, but also a direct physical meaning. A charge $Z > 137$ can in fact be concentrated only in a "superheavy" nucleus of some finite radius. Let us therefore consider how the disposition of levels changes with increasing Z for a given r_0.

In the "uncut" Coulomb field the energy ε_1 of the lowest level is unity at $Z\alpha = 1$ and the curve of the dependence $\varepsilon_1(Z)$ terminates at a minimum (see (36.10)). In the "cut" field, at a given $r_0 \neq 0$, the level ε_1 passes through zero only at some value $Z\alpha > 1$. But the value $\varepsilon_1 = 0$ is in no way physically distinguished, and it is not formally distinguished either—the curve of the dependence $\varepsilon_1(Z)$ does not break off here. With further increase of Z the levels continue to descend, and at some "critical" value $Z = Z_c(r_0)$ the energy ε_1 reaches the boundary $(-m)$ of the lower continuum of levels. As was explained in the preceding section, this means that the energy required for the creation of a free positron vanishes. Therefore the critical value Z_c is the maximum charge that a "bare" nucleus can possess at the given r_0.

For $Z > Z_c$ the level $\varepsilon_1 < -m$ and it becomes energetically favourable to create two electron–positron pairs. The positrons escape to infinity, carrying away the kinetic energy $2(|\varepsilon_1| - m)$, and two electrons fill the level ε_1. The result is an "ion" with a filled K-shell and charge $Z_{\text{eff}} = Z - 2$ (*S. S. Gershtein, Ya. B. Zel'dovich*, 1969). This system is stable for $Z > Z_c$, up to values of Z at which the boundary $-m$ is reached by the next level [11].

[11] Thus, if the nuclear charge is distributed uniformly in a sphere of radius $r_0 = 1.2 \cdot 10^{-13}$ cm, the critical

Let us note finally that even in the case of a point charge the behaviour of the potential at small distances is distorted by radiative corrections. Their account leads, however, only to corrections $\sim\alpha$ to the value $Z_c\alpha$.

Let us now turn to the exact solution of the wave equation (*C. G. Darwin*, 1928; *W. Gordon*, 1928).

Discrete spectrum ($\boldsymbol{\varepsilon < m}$). We seek the functions f and g in the form

$$\begin{aligned} f &= \sqrt{m+\varepsilon}\, e^{-\rho/2}\rho^{\gamma-1}(Q_1+Q_2),\\ g &= -\sqrt{m-\varepsilon}\, e^{-\rho/2}\rho^{\gamma-1}(Q_1-Q_2), \end{aligned} \tag{36.3}$$

where we have introduced the notation

$$\rho = 2\lambda r, \qquad \lambda = \sqrt{m^2-\varepsilon^2}, \qquad \gamma = \sqrt{\varkappa^2 - Z^2\alpha^2}. \tag{36.4}$$

This form is of course natural in view of the already known behaviour of the functions at $\rho \to 0$ (36.2) and their exponential decay ($\sim e^{-\rho/2}$) at $\rho\to\infty$. Since at $\rho\to\infty$ the first relation (35.9) should hold and in the case of the Coulomb field one should expect $Q_1 \gg Q_2$.

Substituting (36.3) into (35.4), we obtain the equations

$$\rho(Q_1+Q_2)' + (\gamma+\varkappa)(Q_1+Q_2) - \rho Q_2 + Z\alpha\sqrt{\frac{m-\varepsilon}{m+\varepsilon}}(Q_1-Q_2) = 0,$$

$$\rho(Q_1-Q_2)' + (\gamma-\varkappa)(Q_1-Q_2) + \rho Q_2 - Z\alpha\sqrt{\frac{m+\varepsilon}{m-\varepsilon}}(Q_1+Q_2) = 0$$

(the prime denotes differentiation with respect to ρ). Their sum and difference give

$$\begin{aligned} \rho Q_1' + \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda}\right)Q_1 + \left(\varkappa - \frac{Z\alpha m}{\lambda}\right)Q_2 &= 0,\\ \rho Q_2' + \left(\gamma + \frac{Z\alpha\varepsilon}{\lambda} - \rho\right)Q_2 + \left(\varkappa + \frac{Z\alpha m}{\lambda}\right)Q_1 &= 0, \end{aligned} \tag{36.5}$$

or, after eliminating Q_1 or Q_2,

$$\begin{aligned} \rho Q_1'' + (2\gamma+1-\rho)\,Q_1' - \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda}\right)Q_1 &= 0,\\ \rho Q_2'' + (2\gamma+1-\rho)\,Q_2' - \left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right)Q_2 &= 0 \end{aligned} \tag{36.6}$$

(one should take into account that $\gamma^2 - (Z\alpha\varepsilon/\lambda)^2 = \varkappa^2 - (Z\alpha m/\lambda)^2$). The solutions of these equations, finite at $\rho = 0$:

$$\begin{aligned} Q_1 &= A\,F\left(\gamma - \frac{Z\alpha\varepsilon}{\lambda},\ 2\gamma+1,\ \rho\right),\\ Q_2 &= B\,F\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda},\ 2\gamma+1,\ \rho\right), \end{aligned} \tag{36.6}$$

value $Z_c = 170$, and the next level reaches the boundary $-m$ at $Z = 185$ (*V. S. Popov*, 1970). For a more detailed exposition of the quantitative theory see the review article by *Ya. B. Zel'dovich* and *V. S. Popov* (UFN, 1971, Vol. 105, p. 403).

where $F(\alpha, \beta, z)$ is the confluent hypergeometric function. Setting $\rho = 0$ in either of equations (36.5), we find the relation between A and B:

$$B = -\frac{\gamma - Z\alpha\varepsilon/\lambda}{\varkappa - Z\alpha m/\lambda}\,A. \tag{36.7}$$

Both hypergeometric functions in (36.6) must reduce to polynomials (otherwise they would grow as e^{ρ} at $\rho \to \infty$, and with them the whole wave function would grow as $e^{\rho/2}$). A function $F(\alpha, \beta, z)$ reduces to a polynomial if its first parameter equals a non-positive integer. Denoting

$$\gamma - Z\alpha\varepsilon/\lambda = -n_r. \tag{36.8}$$

If $n_r = 1, 2, \ldots$, both hypergeometric functions reduce to polynomials. If, however, $n_r = 0$, only one of them reduces to a polynomial. But then the equality $n_r = 0$ means that $\gamma = Z\alpha\varepsilon/\lambda$, and then, as is easily verified, $Z\alpha m/\lambda = |\varkappa|$. If $\varkappa < 0$, the coefficient B (36.7) vanishes, so that $Q_2 = 0$, and the required condition is not violated. If, however, $\varkappa > 0$, then $B = -A$, and Q_2 remains a divergent function. Thus the admissible values of the quantum number n_r are:

$$n_r = \begin{cases} 0, 1, 2, \ldots & \text{for } \varkappa < 0; \\ 1, 2, 3, \ldots & \text{for } \varkappa > 0. \end{cases} \tag{36.9}$$

From the definition (36.8) we now find the following expression for the discrete energy levels:

$$\frac{\varepsilon}{m} = \left[1 + \frac{(Z\alpha)^2}{\left(\sqrt{\varkappa^2 - (Z\alpha)^2} + n_r\right)^2}\right]^{-1/2}. \tag{36.10}$$

In particular, the energy of the ground level $1s_{1/2}$ ($|\varkappa| = 1$, $n_r = 0$):

$$\varepsilon_1 = m\sqrt{1 - (Z\alpha)^2}.$$

For $Z\alpha \ll 1$ the first terms of the expansion of formula (36.10) give

$$\frac{\varepsilon}{m} - 1 = -\frac{(Z\alpha)^2}{2(|\varkappa| + n_r)^2} - \frac{(Z\alpha)^2}{(|\varkappa| + n_r)}\left[\frac{1}{|\varkappa|} - \frac{3}{4(|\varkappa| + n_r)}\right]\Bigg\}.$$

Denoting $n_r + |\varkappa| = n$ ($n = 1, 2, \ldots$) and noting that $|\varkappa| = j + \frac{1}{2}$, we recover formula (34.4), obtained earlier by perturbation theory. As was already noted at the end of § 34, the further terms of this expansion are meaningless, since they are inevitably exceeded by the radiative corrections. Formula (36.10), however, makes sense in its exact form when $Z\alpha \sim 1$.

Note that the twofold degeneracy discovered from the approximate formula (34.4) is preserved in the exact formula as well: since only $|\varkappa|$ enters into it, levels with different l at the same n and j still coincide.

It remains to determine the overall normalisation coefficient A in the wave function. As always, the wave function of the discrete spectrum must be normalised by the condition $\int |\psi|^2 \, d^3x = 1$; for the functions f and g this means the condition

$$\int_0^\infty (f^2 + g^2)\, r^2 \, dr = 1.$$

The coefficient A is most easily found from the asymptotic behaviour of the functions at $r \longrightarrow \infty$. Using the asymptotic formula

$$F(-n_r,\, 2\gamma + 1,\, \rho) \approx \frac{\Gamma(2\gamma + 1)}{\Gamma(n_r + 2\gamma + 1)}\, (-\rho)^{n_r}$$

(see III, (d.14)) we find

$$f \approx (-1)^{n_r} A \sqrt{m + \varepsilon}\, \frac{\Gamma(2\gamma + 1)}{\Gamma(n_r + 2\gamma + 1)}\, e^{-\lambda r}\, (2\lambda r)^{\gamma + n_r - 1}.$$

Comparing this formula with the expression (36.22), which will be found below, we determine A. Collecting then the formulae obtained, we write the final expressions for the normalised wave functions:

$$\left.\begin{matrix} f \\ g \end{matrix}\right\} = \frac{\pm (2\lambda)^{\gamma + 1/2}}{\Gamma(2\gamma + 1)} \left[\frac{(m \pm \varepsilon)\, \Gamma(2\gamma + n_r + 1)}{4m\, (Z\alpha m/\lambda)\, (Z\alpha m/\lambda - \varkappa)\, n_r!} \right]^{1/2} (2\lambda r)^{\gamma - 1}\, e^{-\lambda r} \times \tag{36.11}$$

$$\times \left\{ \left(\frac{Z\alpha m}{\lambda} - \varkappa \right) F(-n_r,\, 2\gamma + 1,\, 2\lambda r) \mp n_r\, F(1 - n_r,\, 2\gamma + 1,\, 2\lambda r) \right\}$$

(the upper signs refer to f, the lower to g).

Continuous spectrum ($\varepsilon > m$). There is no need to solve the wave equation anew for the continuous-spectrum states. The wave functions in this case are obtained from those of the discrete spectrum by the substitution [12]

$$\sqrt{m - \varepsilon} \longrightarrow -i\sqrt{\varepsilon - m}, \qquad \lambda \longrightarrow -ip, \qquad -n_r \longrightarrow \gamma - i\,\frac{Z\alpha\varepsilon}{p} \tag{36.12}$$

(regarding the choice of sign in the analytic continuation of the root $\sqrt{m - \varepsilon}$, see III, § 128).

Carrying out the indicated substitution in (36.11), we express the functions f and g in the form

$$\left.\begin{matrix} f \\ g \end{matrix}\right\} = \frac{\sqrt{\varepsilon + m}}{i\sqrt{\varepsilon - m}} \cdot A' e^{ipr}\, (2pr)^{\gamma - 1} \times$$

$$\times \left[e^{i\xi} F(\gamma - i\nu,\, 2\gamma + 1,\, -2ipr) \mp e^{-i\xi} F(\gamma + 1 - i\nu,\, 2\gamma + 1,\, -2ipr) \right],$$

where A' is a new normalisation constant and we have introduced the notation

$$\nu = \frac{Z\alpha\varepsilon}{p}, \qquad e^{-2i\xi} = \frac{\gamma - i\nu}{\varkappa - i\nu m/\varepsilon} \tag{36.13}$$

[12] Hereafter in this section p denotes $|\mathbf{p}| = \sqrt{\varepsilon^2 - m^2}$, and in the indices of the amplitude we write ε and $\mathbf{p}$ separately.

(the quantity ξ is real, since $\gamma^2 + (Z\alpha\varepsilon/p)^2 = \varkappa^2 + (Z\alpha m/p)^2$).

According to the well-known formula

$$F(\alpha, \beta, z) = e^z F(\beta - \alpha, \beta, -z)$$

(see III, (d.10)) we have

$$F(\gamma + 1 - i\nu, 2\gamma + 1, -2ipr) = e^{-2ipr} F(\gamma + i\nu, 2\gamma + 1, 2ipr) =$$

$$= e^{-2ipr} F^*(\gamma - i\nu, 2\gamma + 1, -2ipr),$$

therefore

$$\left.\begin{matrix} f \\ g \end{matrix}\right\} = 2iA'\sqrt{\varepsilon \pm m}\,(2pr)^{\gamma-1} \frac{\mathrm{Im}}{\mathrm{Re}} \left\{e^{i(pr+\xi)} F(\gamma - i\nu, 2\gamma + 1, -2ipr)\right\}. \tag{36.14}$$

The normalisation coefficient A' is determined by comparing the asymptotic expression for this function with the general formula (35.7) for a normalised spherical wave. We write at once the resulting expression for the wave functions of the continuous spectrum (and then verify it) [13]:

$$\left.\begin{matrix} f \\ g \end{matrix}\right\} = 2^{3/2}\sqrt{\frac{m \pm \varepsilon}{\varepsilon}}\, \frac{e^{\pi\nu/2}\,|\Gamma(\gamma + 1 + i\nu)|}{\Gamma(2\gamma + 1)}\, \frac{(2pr)^\gamma}{r} \times \frac{\mathrm{Im}}{\mathrm{Re}} \left\{e^{i(pr+\xi)} F(\gamma - i\nu, 2\gamma + 1, -2ipr)\right\}. \tag{36.15}$$

The asymptotic expression for this function is found using the formula III, (d.14), in which in the present case only the first term is essential (the second falls off with a higher power of $1/r$):

$$\left.\begin{matrix} f \\ g \end{matrix}\right\} = \frac{\sqrt{2}}{r}\sqrt{\frac{\varepsilon \pm m}{\varepsilon}}\, \sin\left(pr - \frac{\pi l}{2} + \delta_\varkappa + \nu \ln 2pr - \frac{\pi l}{2}\right), \tag{36.16}$$

where

$$\delta_\varkappa = \xi - \arg\Gamma(\gamma + 1 + i\nu) - \frac{\pi\gamma}{2} + \frac{\pi l}{2}, \tag{36.17}$$

or

$$e^{2i\delta_\varkappa} = \frac{\varkappa - i\nu m/\varepsilon}{\gamma - i\nu}\, \frac{\Gamma(\gamma + 1 - i\nu)}{\gamma + 1 + i\nu}\, e^{i\pi(l-\gamma)}. \tag{36.18}$$

Let us note for future reference the expression for the phases in the ultra-relativistic case ($\varepsilon \gg m$, $\nu \to Z\alpha$):

$$e^{2i\delta_\varkappa} = \frac{\varkappa}{\gamma - iZ\alpha}\, \frac{\Gamma(\gamma + 1 - iZ\alpha)}{\Gamma(\gamma + 1 + iZ\alpha)}\, e^{i\pi(l-\gamma)}. \tag{36.19}$$

Expression (36.16) differs from (35.8) only by a logarithmic term in the argument of the trigonometric function. As in the case of the Schrödinger equation, the slow fall-off

[13] The wave functions in a repulsive field are obtained from these by changing the sign before $Z\alpha$, i.e. by changing the sign of ν.

of the Coulomb potential leads to a distortion of the phase of the wave, which becomes a slowly varying function of r.

On analytic continuation into the region $\varepsilon < m$ expression (36.18) takes the form

$$e^{2i\delta_\varkappa} = \frac{\varkappa - Z\alpha m/\lambda}{\gamma - Z\alpha\varepsilon/\lambda} \frac{\Gamma(\gamma + 1 - Z\alpha\varepsilon/\lambda)}{\Gamma(\gamma + 1 + Z\alpha\varepsilon/\lambda)} e^{i\pi(l-\gamma)}. \tag{36.20}$$

It has poles at the points where $\gamma + 1 - Z\alpha\varepsilon/\lambda = 1 - n_r$, $n_r = 1, 2, \ldots$ (poles of the Γ-function in the numerator), and also at the point $\gamma - Z\alpha\varepsilon/\lambda = -n_r = 0$ (if $\varkappa < 0$); as was to be expected, these points coincide with the discrete energy levels.

Near one of the poles with $n_r \neq 0$ we have

$$e^{2i\delta_\varkappa} \approx \frac{\left(\dfrac{Z\alpha m}{\lambda} - \varkappa\right) e^{i\pi(l-\gamma)}}{n_r\, \Gamma(2\gamma + 1 + n_r)}\, \Gamma\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right).$$

The form of the Γ-function near its pole is found using the well-known formula $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$:

$$\Gamma\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right) \approx \frac{\pi}{\Gamma(n_r)\sin\pi(\gamma + 1 - Z\alpha\varepsilon/\lambda)},$$

$$\sin\pi\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right) \approx \pi\cos\pi n_r \frac{d}{d\varepsilon}\left(\frac{Z\alpha\varepsilon}{\lambda}\right)(\varepsilon - \varepsilon_0) =$$

$$= (-1)^{n_r} \frac{\pi Z\alpha m^2}{\lambda^3} (\varepsilon - \varepsilon_0)$$

(ε_0 being the energy level). Thus [14],

$$e^{2i\delta_\varkappa} \approx (-1)^{l+n_r} \frac{e^{-i\pi\left(\frac{Z\alpha m}{\lambda} - \varkappa\right)}}{n_r!\,\Gamma(2\gamma + 1 + n_r)} \frac{\lambda^3}{Z\alpha m^2} \frac{1}{\varepsilon - \varepsilon_0}. \tag{36.21}$$

At the end of the preceding section formula (35.11) was obtained, relating the residue of $e^{2i\delta_\varkappa}$ at its pole to the coefficient in the asymptotic expression of the wave function of the corresponding bound state. In the case of the Coulomb field this formula must be somewhat modified in connection with the fact that instead of the constant phase shift $\delta_\varkappa$ (as in (35.7) and (36.16)) one has the sum $\delta_\varkappa + \nu\ln(2pr)$. On the left-hand side of (35.11) one should therefore write not $e^{2i\delta_\varkappa}$, but

$$\exp[2i\delta_\varkappa + 2i\nu\ln(2pr)] \to e^{2i\delta_\varkappa}(2i\lambda r)^{2(n_r+\gamma)}.$$

Using (36.21) and determining from (35.11) the coefficient A_0 (which is now a power function of r), we find the asymptotic normalised wave function of the discrete spectrum:

$$f = \left[\frac{(Z\alpha m/\lambda - \varkappa)(m + \varepsilon)\lambda^2}{2n_r!\,Z\alpha m^2\,\Gamma(2\gamma + 1 + n_r)}\right]^{1/2} \frac{e^{-\lambda r}}{r} (2\lambda r)^{n_r+\gamma}. \tag{36.22}$$

This formula has already been used to determine the coefficient A in (36.11).

[14] It is easy to verify that this formula remains valid for $n_r = 0$ as well.

§ 37. Scattering in a centrally symmetric field

Let us write the asymptotic expression for the wave function of a particle scattered in the field of a stationary force centre, in the form [15]

$$\psi = u_{\varepsilon \mathbf{p}}\, e^{ipz} + u'_{\varepsilon \mathbf{p}'}\, \frac{e^{ipr}}{r}. \tag{37.1}$$

Here $u_{\varepsilon \mathbf{p}}$ is the bispinor amplitude of the incident plane wave. The bispinor $u'_{\varepsilon \mathbf{p}'}$ is a function of the scattering direction $\mathbf{n}'$, and for each given value of $\mathbf{n}'$ coincides in form (but, of course, not in normalisation!) with the bispinor amplitude of a plane wave propagating in the direction $\mathbf{n}'$.

We saw in § 24 that the bispinor amplitude of a plane wave is completely determined by specifying a two-component quantity—a 3-spinor w, representing the non-relativistic wave function in the rest frame of the particle. Through this spinor is expressed the flux density: it is proportional to $w^* w$ (with a coefficient of proportionality depending only on the energy ε) and, consequently, equal for the incident and scattered

particles. The scattering cross section is therefore $d\sigma = (w'^* w'/w^* w)\, do$, or, if (as in § 24) we normalise the incident wave by the condition $w^\dagger w = 1$,

$$d\sigma = w'^* w'\, do.$$

We introduce the scattering operator $\hat{f}$ by the definition

$$w' = \hat{f}\, w. \tag{37.2}$$

By virtue of the two-component character of w, w', the operator defined in this way is analogous to the operator scattering amplitude that appears in the non-relativistic theory of scattering with spin taken into account (see III, § 140). Therefore one can transfer directly the formulae obtained there, expressing the operator through the phase shifts of the wave functions in the scattering field. One need only re-express the notation of these phases, expressing the shifts δ_l^+ and δ_l^- introduced in III, § 140 through the shift $\delta_\varkappa$ that appears in the relativistic formula (35.7). Recall that the phases δ_l^+ and δ_l^- referred to the states with orbital angular momentum l and total angular momentum $j = l + \frac{1}{2}$ and $j = l - \frac{1}{2}$. According to the definition (35.3), $\varkappa = -l - 1$ for $j = l + \frac{1}{2}$ and $\varkappa = l$ for $j = l - \frac{1}{2}$. We must therefore re-label

$$\delta_l^+ \to \delta_{-(l+1)}, \qquad \delta_l^- \to \delta_l$$

(and bear in mind that the index on δ now refers to the value of $\varkappa$!). Thus we obtain the following formulae:

$$\hat{f} = A + B\boldsymbol{\nu}\cdot\boldsymbol{\sigma}, \tag{37.3}$$

$$A = \frac{1}{2ip}\sum_{l=0}^{\infty}\left[(l+1)(e^{2i\delta_{-l-1}} - 1) + l(e^{2i\delta_l} - 1)\right] P_l(\cos\theta), \tag{37.4}$$

$$B = \frac{1}{2p}\sum_{l=1}^{\infty}(e^{2i\delta_{-l-1}} - e^{2i\delta_l})\, P_l^1(\cos\theta), \tag{37.5}$$

[15] In §§ 37, 38 p denotes $|\mathbf{p}|$, and in the indices of the amplitude we write ε and $\mathbf{p}$ separately.

where $\boldsymbol{\nu}$ is the unit vector in the direction $[\mathbf{n}\,\mathbf{n}']$.

Since w is the spinor wave function in the rest frame, the polarisation properties of the scattering are described with the aid of $\widehat{f}$ by the same formulae as in III, § 140.

In the case of the Coulomb field it turns out to be possible to express both functions $A(\theta)$ and $B(\theta)$ through a single one. Let us indicate briefly the course of the corresponding calculations [16].

For the Coulomb field the phases $\delta_\varkappa$ are given by formula (36.18), which we present in the form

$$e^{2i\delta_\varkappa} = -\left(\varkappa - i\,\frac{Ze^2m}{p}\right)\frac{\varkappa}{|\varkappa|}\,C_\varkappa, \tag{37.6}$$

$$C_\varkappa = -\frac{\Gamma(\gamma - i\nu)}{\Gamma(\gamma + 1 + i\nu)}\,e^{i\pi(|\varkappa| - \gamma)}$$

(noting that $e^{i\pi l} = e^{i\pi\varkappa}$ for $\varkappa > 0$ and $e^{i\pi l} = -e^{i\pi\varkappa}$ for $\varkappa < 0$). With the help of the quantities $C_\varkappa$ introduced in this way the series (37.4), (37.5) can be presented in the form

$$\begin{aligned} A(\theta) &= \frac{1}{p}\,G(\theta) - i\,\frac{Ze^2m}{p^2}\,F(\theta), \\ B(\theta) &= -\frac{i}{p}\,\tan\frac{\theta}{2}\,G(\theta) + \frac{Ze^2m}{p^2}\,\cot\frac{\theta}{2}\,F(\theta), \end{aligned} \tag{37.7}$$

where

$$G(\theta) = \frac{i}{2}\sum_{l=1}^{\infty} l^2 C_l (P_l + P_{l-1}), \qquad F(\theta) = \frac{i}{2}\sum_{l=1}^{\infty} l C_l (P_l - P_{l-1}). \tag{37.8}$$

In transforming the series for $B(\theta)$ the following recurrence relations among the Legendre polynomials have been used:

$$P_l^1 + P_{l-1}^1 = \cot\frac{\theta}{2}\cdot l(P_l - P_{l-1}), \tag{37.9}$$

$$P_l^1 - P_{l-1}^1 = \tan\frac{\theta}{2}\cdot l(P_l + P_{l-1}). \tag{37.10}$$

On the other hand, by virtue of the identity

$$(1+\cos\theta)\,\frac{d}{d\cos\theta}\big[P_l(\cos\theta) - P_{l-1}(\cos\theta)\big] == l\big[P_l(\cos\theta) + P_{l-1}(\cos\theta)\big] \tag{37.11}$$

the functions $F(\theta)$ and $G(\theta)$ are related to each other by

$$G = (1-\cos\theta)\,\frac{dF}{d\cos\theta} = -\cot\frac{\theta}{2}\,\frac{dF}{2\,d\theta}. \tag{37.12}$$

Both $A(\theta)$ and $B(\theta)$ are thereby expressed through a single function $F(\theta)$ [17].

[16] *R. L. Gluckstern, S. R. Lin* // J. Math. Phys. 1964, Vol. 5, p. 1594.

[17] The function $F(\theta)$ cannot be expressed in closed form through elementary functions. It can, however, be written in the form of a certain double integral—see the article cited above.

§ 38. Scattering in the ultra-relativistic case

Let us consider in more detail the scattering in the ultra-relativistic case ($\varepsilon \gg m$). In the first approximation we completely neglect in the wave equation the mass m. It is then convenient to use the spinor representation for $\psi = \binom{\xi}{\eta}$, since the equations for ξ and η at $m = 0$ separate:

$$-i\boldsymbol{\sigma}\nabla\xi = (\varepsilon - U)\xi, \qquad -i\boldsymbol{\sigma}\nabla\eta = -(\varepsilon - U)\eta \tag{38.1}$$

(acquiring a "neutrinolike" form, see § 30).

A helical state of the electron, polarised in the direction $\mathbf{p}$, is described by the wave function $\psi = \binom{\xi}{0}$, and one polarised against $\mathbf{p}$: $\psi = \binom{0}{\eta}$. By virtue of the independence of the equations for ξ and η it is clear that this property is preserved under scattering. In other words, in ultra-relativistic scattering of electrons helicity is conserved. From considerations of symmetry (longitudinal polarisation) it is obvious that in the scattering of helical particles there is no azimuthal asymmetry. One can also assert that the scattering cross section for helical electrons does not depend on the sign of the helicity; this follows from the fact that the central field is invariant under inversion, and the sign of helicity reverses under inversion.

In the ultra-relativistic case the formulae (37.3)–(37.5) can be significantly simplified (*D. R. Yennie, D. G. Ravenhall, R. N. Wilson*, 1954).

Let the incident electron be polarised, say, along the direction of motion $\mathbf{n}$. For a plane wave with the helicity value $\mathbf{n}\boldsymbol{\sigma}$ the spinor $\zeta(= (\varphi + \chi)/\sqrt{2})$ is proportional to the same 3-spinor w that appeared in the standard representation of the wave. Therefore the relation between the spinor amplitudes of the incident and scattered waves is given by the same operator $\widehat{f}$.

As a result of scattering the polarisation vector rotates together with the momentum, acquiring the direction $\mathbf{n}'$. The action of the operator $\widehat{f}$ on the spinor wave function of the electron therefore reduces to a rotation of the spin through the angle θ (the angle between $\mathbf{n}$ and $\mathbf{n}'$) about the axis $\boldsymbol{\nu}$. A rotation by $-\theta$ is in turn equivalent to a rotation of the coordinate system about the same axis in the opposite direction, i.e. to the replacement $\theta \to -\theta$. It follows that the operator $\widehat{f}$ must coincide (up to a coefficient) with the operator effecting the transformation of the wave function under the indicated change of coordinates, i.e. with the operator (18.17) with the substitution $\theta \to -\theta$.

Comparing (37.3) with (18.17), we find that it should be

$$\frac{B}{A} = -i\tan\frac{\theta}{2}. \tag{38.2}$$

Thus, in the ultra-relativistic limit,

$$\widehat{f} = A(\theta)\left[1 - i\tan\frac{\theta}{2}\,\boldsymbol{\nu}\cdot\boldsymbol{\sigma}\right]. \tag{38.3}$$

The expression for $A(\theta)$ (37.4) can also be simplified, exploiting the relation between the phases $\delta_\varkappa$ and $\delta_{-\varkappa}$ that arises in the same limit. For its derivation we note that the equations (35.4) for the functions f and g, after striking out the terms containing m, become invariant with respect to the substitution

$$\varkappa \to -\varkappa, \qquad f \to g, \qquad g \to -f,$$

not affecting the parameters of the particle or of the field. It should therefore be that $f_\varkappa / g_\varkappa = -g_{-\varkappa}/f_{-\varkappa}$, and after substitution of the asymptotic expressions we find

$$\tan\left(pr - \frac{l\pi}{2} + \delta_\varkappa\right) = -\cot\left(pr - \frac{l'\pi}{2} + \delta_{-\varkappa}\right),$$

$$\delta_\varkappa = \delta_{-\varkappa} - (l' - l)\,\frac{\pi}{2} + \left(n + \tfrac{1}{2}\right)\pi,$$

whence

$$e^{2i\delta_l} = e^{2i\delta_{-l}}. \tag{38.4}$$

Using this relation (and replacing in the first term of the sum in (37.4) the summation index l by $l - 1$), we obtain

$$A(\theta) = \frac{1}{2ip}\sum_{l=1}^{\infty} l\,(e^{2i\delta_l} - 1)\big[P_l(\cos\theta) + P_{l-1}(\cos\theta)\big]. \tag{38.5}$$

From (38.2) it follows that $\mathrm{Re}(AB^*) = 0$. This means that in the approximation being considered the cross section does not depend on the initial polarisation of the particle, and a non-polarised beam remains non-polarised after scattering (see the formulae III, (140.8)–(140.10)). We note also that as $\theta \to \pi$ the expression $A(\theta)$ (38.5) tends to zero as $(\pi - \theta)^2$ (recall that $P_l(-1) = (-1)^l$). Together with it, B also tends to zero, and so does the cross section

$$\frac{d\sigma}{do} = |A|^2 + |B|^2 = \frac{|A(\theta)|^2}{\cos^2(\theta/2)}. \tag{38.6}$$

The listed properties disappear, of course, in the subsequent approximations in the small quantity m/ε. In particular, the analysis shows that as $\theta \to \pi$ the cross section tends to a limit proportional to $(m/\varepsilon)^2$.

For the Coulomb field in the ultra-relativistic case the phases $\delta_\varkappa$ do not depend on the energy, as is evident from (36.19). Therefore in a pure Coulomb field the scattering cross section for $\varepsilon \gg m$ has the form

$$d\sigma = \frac{\tau(\theta)}{\varepsilon^2}\,do, \tag{38.7}$$

where τ is a function of the angle only.

§ 39. System of wave functions of the continuous spectrum for scattering in a Coulomb field

Later on (see §§ 95, 96) various inelastic processes will be considered that occur in the scattering of ultra-relativistic electrons in the field of a heavy ($Z\alpha \sim 1$) nucleus. For the computation of the corresponding matrix elements we shall need the wave functions, whose asymptotic form (as $r \to \infty$) consists of a plane wave and a spherical wave.

We saw that in the ultra-relativistic case (energy of the electron $\varepsilon \gg m$) the main role in the scattering is played by momentum transfers (from the electron to the nucleus)

$q = |\mathbf{p}' - \mathbf{p}| \sim m$. These values of q correspond to "impact parameters" $\rho \sim 1/q \sim 1/m$, and the electron is deflected through angles [18, 19]

$$\theta \sim \frac{q}{p} \sim \frac{m}{\varepsilon}. \tag{39.1}$$

In terms of the coordinates r (distance from the centre) and $z = r\cos\theta$ this means the region

$$\rho \equiv r\sin\theta \sim 1/m, \qquad p(r-z) = pr(1-\cos\theta) \sim 1. \tag{39.2}$$

Since $r \sim \varepsilon/m^2$, i.e. we are dealing with the region of large distances.

Let us write the Dirac equation in the form

$$(\varepsilon - U - m\beta + i\boldsymbol{\alpha}\nabla)\psi = 0, \qquad U = -Z\alpha/r. \tag{39.3}$$

We convert it into a second-order equation by applying to (39.3) the operator $(\varepsilon - U + m\beta - i\boldsymbol{\alpha}\nabla)$:

$$(\Delta + p^2 - 2\varepsilon U)\psi = (-i\boldsymbol{\alpha}\,\nabla U - U^2)\psi. \tag{39.4}$$

Since in the region under consideration $r \gg Z\alpha/\varepsilon$, we have $U \ll \varepsilon$. In the first approximation one may neglect the right-hand side of (39.4). The remaining equation

$$\left(\Delta + p^2 + \frac{2\varepsilon Z\alpha}{r}\right)\psi = 0 \tag{39.5}$$

coincides in form with the non-relativistic Schrödinger equation in the Coulomb field

$$\left(\frac{1}{2m}\Delta + \frac{p^2}{2m} + \frac{Z\alpha}{r}\right)\psi = 0, \tag{39.5a}$$

differing from it only by an obvious change in the notation of the parameters (the "potential energy" has a superfluous factor ε/m). We can therefore immediately write its solution, which has the required asymptotic form (see III, § 136).

Thus, the wave function containing asymptotically a plane wave ($\propto e^{i\mathbf{pr}}$) and a diverging spherical wave has the form

$$\psi^{(+)}_{\varepsilon\mathbf{p}} = C\,\frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}}\,e^{i\mathbf{pr}}F\left(\frac{iZ\alpha\varepsilon}{p},\ 1,\ i(pr - \mathbf{pr})\right), \tag{39.6}$$

$$C = e^{\pi Z\alpha\varepsilon/(2p)}\,\Gamma\left(1 - i\,\frac{Z\alpha\varepsilon}{p}\right),$$

where F is the confluent hypergeometric function, and $u_{\varepsilon\mathbf{p}}$ is a constant bispinor amplitude of the plane wave, normalised by the condition we have adopted (23.4)

$$\bar{u}_{\varepsilon\mathbf{p}}\,u_{\varepsilon\mathbf{p}} = 2m. \tag{39.7}$$

[18] This is clear directly from equations (38.1), since for the Coulomb field the energy ε can be altogether eliminated from the equations by the substitution $\mathbf{r} \to \mathbf{r}'/\varepsilon$.

[19] In this section p denotes $|\mathbf{p}|$.

The wave function (39.6) is normalised in such a way that the plane wave in its asymptotic expression has the usual form

$$\frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}}\, e^{i\mathbf{pr}},$$

corresponding to one particle in a unit volume. In the ultra-relativistic case $p \approx \varepsilon$, and in (39.6) one can set $Z\alpha\varepsilon/p \approx Z\alpha$:

$$\psi^{(+)}_{\varepsilon\mathbf{p}} = C\,\frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}}\, e^{i\mathbf{pr}} F\big(iZ\alpha,\ 1,\ i(pr - \mathbf{pr})\big), \tag{39.8}$$

$$C = e^{\pi Z\alpha/2}\,\Gamma(1 - iZ\alpha).$$

Let us draw attention to the fact that although the distances we consider are so large that $pr \gg 1$, one cannot replace the hypergeometric function in (39.8) by its asymptotic expression: the argument of the function F is not pr, but $pr(1 - \cos\theta)$, a quantity that is not assumed to be large[20].

In the applications it turns out to be necessary also to include the next approximation in ψ, which has a spinor structure different from (39.8) (reducing to the factor $u_{\varepsilon\mathbf{p}}$). For this computation we write ψ in the form

$$\psi = \frac{C}{\sqrt{2\varepsilon}}\, e^{i\mathbf{pr}}(u_{\varepsilon\mathbf{p}}F + \varphi).$$

On the right-hand side of equation (39.4) we keep the term linear in U, and for the function φ we obtain the equation

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)\varphi = -iu_{\varepsilon\mathbf{p}}(\boldsymbol{\alpha}\nabla U)F. \tag{39.9}$$

Its solution can be found by noting that the function F satisfies the equation

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)F = 0$$

(as can be verified by substituting (39.6) into (39.5)). Applying to this equation the operation ∇, we get

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)\nabla F = 2\varepsilon F\,\nabla U.$$

Comparing with equation (39.9), we find

$$\varphi = -\frac{i}{2\varepsilon}\,(\boldsymbol{\alpha}\nabla)\, u_{\varepsilon\mathbf{p}}\, F.$$

Let us write the final expression for $\psi^{(+)}$ and for a similar function $\psi^{(-)}$, which contains in its asymptotic expression a converging spherical wave:

$$\begin{aligned} \psi^{(+)}_{\varepsilon\mathbf{p}} &= \frac{C}{\sqrt{2\varepsilon}}\, e^{i\mathbf{pr}}\left(1 - \frac{i\boldsymbol{\alpha}}{2\varepsilon}\nabla\right) F\big(iZ\alpha,\ 1,\ i(pr - \mathbf{pr})\big)\, u_{\varepsilon\mathbf{p}}, \\ \psi^{(-)}_{\varepsilon\mathbf{p}} &= \frac{C^*}{\sqrt{2\varepsilon}}\, e^{i\mathbf{pr}}\left(1 - \frac{i\boldsymbol{\alpha}}{2\varepsilon}\nabla\right) F\big(-iZ\alpha,\ 1,\ -i(pr - \mathbf{pr})\big)\, u_{\varepsilon\mathbf{p}}, \end{aligned} \tag{39.10}$$

[20] In III, § 135 we were interested in arbitrarily large r, and therefore such a replacement was possible for all angles θ.

$$C = e^{\pi Z\alpha/2}\,\Gamma(1 - iZ\alpha)$$

(*W. H. Furry*, 1934). Let us write also analogous functions $\psi_{-\varepsilon,-\mathbf{p}}$ with "negative frequency", which will be needed in considering processes involving positrons. They can be obtained from the functions $\psi_{\varepsilon\mathbf{p}}$ by the substitution $\mathbf{p} \to -\mathbf{p}$, $\varepsilon \to -\varepsilon$, where $p = |\mathbf{p}|$ does not change (and by virtue of the last circumstance the parameter $iZ\alpha$ of the hypergeometric function changes sign, as is clear from the original expression (39.6), in which this parameter appears in the form $iZ\alpha\varepsilon/p$). Thus,

$$\psi^{(+)}_{-\varepsilon,-\mathbf{p}} = \frac{C}{\sqrt{2\varepsilon}}\, e^{-i\mathbf{pr}}\left(1 + \frac{i\alpha}{2\varepsilon}\nabla\right) F\big(-iZ\alpha,\ 1,\ i(pr + \mathbf{pr})\big)\, u_{-\varepsilon,-\mathbf{p}},$$

$$\psi^{(-)}_{-\varepsilon,-\mathbf{p}} = \frac{C^*}{\sqrt{2\varepsilon}}\, e^{-i\mathbf{pr}}\left(1 + \frac{i\alpha}{2\varepsilon}\nabla\right) F\big(iZ\alpha,\ 1,\ -i(pr + \mathbf{pr})\big)\, u_{-\varepsilon,-\mathbf{p}}, \tag{39.11}$$

$$C = e^{-\pi Z\alpha/2}\,\Gamma(1 + iZ\alpha).$$

Regarding the calculations performed, one further remark is in order. The asymptotic condition that we imposed is by itself certainly not sufficient to determine the solution of the wave equation uniquely (this is clear already from the fact that one can always add to ψ, without violating this condition, any Coulomb diverging spherical wave). In writing the solution of equation (39.5) in the form (39.6), we thereby tacitly assumed the choice of a solution finite at $r = 0$. Such a requirement was necessary in III, §§ 135, 136, where the exact Schrödinger equation was considered, valid everywhere in space[21]. In the present case, however, equation (39.5) applies only to large distances, and therefore the selection of solutions performed requires additional justification.

This is provided by the fact that for large "impact parameters" $\rho = r\sin\theta$ large orbital angular momenta l and small scattering angles θ are relevant: at $\rho \sim 1/m$ we have

$$l \sim \rho p \sim \rho\varepsilon \sim \varepsilon/m \gg 1,$$

and the angle θ can be estimated quasiclassically:

$$\theta \sim \frac{1}{p}\int \frac{dU}{dr}\,dt \sim \frac{U'(\rho)\rho}{p} \sim \frac{m}{\varepsilon} \ll 1.$$

This means that in the decomposition of ψ into spherical waves the dominant waves (in the region of r and θ being considered) are precisely those with the indicated large values of l. But a spherical wave with large l is necessarily small in approaching the origin of coordinates at the "classically inaccessible" (thanks to the centrifugal barrier) distances $r \ll l/\varepsilon$. Therefore, in "sewing together" the solution of equation (39.5) with the solution of the exact equation (39.4) at small distances at $r \sim r_1$, where $l/\varepsilon \gg r_1 \gg Z\alpha/\varepsilon$, the boundary condition for the solution of equation (39.5) will consist in requiring its smallness, and the selection made is justified.

Problem

[21] As explained in III, § 135, this condition was ensured by the choice of a partial integral of the type (135.1), instead of the sum of integrals over the general contour with different values of β_1, β_2.

For the Coulomb field of attraction with $Z\alpha \ll 1$ find the correction (of relative order $Z\alpha$) to the non-relativistic wave function of the discrete spectrum.

Solution. The velocity of the electron in a bound state $v \sim Z\alpha$, so that for $Z\alpha \ll 1$ the wave function is non-relativistic in the zeroth approximation, i.e.

$$\psi = u\psi_{\mathrm{nr}},$$

where ψ_{nr} is a function satisfying the non-relativistic Schrödinger equation, and u is a bispinor of the form $u = \binom{w}{0}$, where w is a spinor describing the polarisation state of the electron. In the next approximation we write: $\psi = u\psi_{\mathrm{nr}} + \psi^{(1)}$ and, substituting into (39.4), find for $\psi^{(1)}$ the equation

$$\left(\frac{1}{2m}\Delta + |\varepsilon_n| + \frac{Z\alpha}{r}\right)\psi^{(1)} = i\,\frac{Z\alpha}{2m}\left(\nabla\frac{1}{r}\right)(\boldsymbol{\alpha}\,u)\,\psi_{\mathrm{nr}},$$

where ε_n is the non-relativistic discrete energy level. Here the terms of relative order $(Z\alpha)^2$ have been dropped (one should bear in mind that in the non-relativistic case the essential distances are of the order of the Bohr radius: $r \sim 1/mZ\alpha$). The solution of this equation: $\psi^{(1)} = -\frac{i}{2m}\boldsymbol{\alpha}\,u\,\nabla\psi_{\mathrm{nr}}$, so that

$$\psi = \left(1 - \frac{i}{2m}\,\boldsymbol{\alpha}\nabla\right)u\,\psi_{\mathrm{nr}}.$$

§ 40. Electron in the field of a plane electromagnetic wave

The Dirac equation can be solved exactly for an electron moving in the field of a plane electromagnetic wave (*D. M. Volkov*, 1937).

The field of a plane wave with wave 4-vector k ($k^2 = 0$) depends on the 4-coordinates only through the combination $\varphi = kx$, so that the 4-potential

$$A^\mu = A^\mu(\varphi), \tag{40.1}$$

which satisfies the Lorentz gauge condition

$$\partial_\mu A^\mu = k_\mu A'^\mu = 0 \tag{40.2}$$

(the prime denotes differentiation with respect to φ). Since the constant term in A is inessential, in this gauge condition one may drop it and write it in the form

$$kA = 0.$$

We start from the second-order equation (32.6), in which the field tensor

$$F_{\mu\nu} = k_\mu A'_\nu - k_\nu A'_\mu. \tag{40.3}$$

When expanding the square $(i\partial - eA)^2$ one should take into account that by virtue of (40.2) $\partial_\mu(A^\mu\psi) = A^\mu\partial_\mu\psi$. As a result we obtain the equation

$$[-\partial^2 - 2ie(A\partial) + e^2A^2 - m^2 - ie(\gamma k)(\gamma A')]\psi = 0 \tag{40.4}$$

($\partial^2 = \partial_\mu \partial^\mu$).

We seek the solution of this equation in the form

$$\psi = e^{-ipx} F(\varphi), \tag{40.5}$$

where p is a constant 4-vector. Adding to p any vector of the form const $\cdot\, k$ does not change the form of the function ψ (only a corresponding redefinition of the function $F(\varphi)$ is needed). One may therefore impose on p one additional condition without loss of generality. Let

$$p^2 = m^2. \tag{40.6}$$

Then when the field is switched off the quantum numbers p^μ become the components of the 4-momentum of a free particle. The meaning of the components of the 4-vector p in the presence of the field is more transparent in a special reference frame, chosen so that $A_0 = 0$. Let in this frame the vector $\mathbf{A}$ be directed along the x^1 axis, and $\mathbf{k}$ along the x^3 axis (i.e. the electric field of the wave is polarised along x^1, the magnetic along x^2, and the wave propagates along x^3). Then (40.5) takes the form of eigenfunctions of the operators

$$\widehat{p}_1 = i\frac{\partial}{\partial x^1}, \qquad \widehat{p}_2 = i\frac{\partial}{\partial x^2}, \qquad \widehat{p}_0 - \widehat{p}_3 = i\left(\frac{\partial}{\partial x_0} - \frac{\partial}{\partial x^3}\right)$$

with eigenvalues $p_1, p_2, p_0 - p_3$ (the same operators, as is easily verified, are commutative with the Hamiltonian of the Dirac equation). Thus, in this reference frame, p^1, p^2 are the components of the generalised momentum along the axes x^1, x^2, and $p^0 - p^3$ is the difference between the total energy and the component of the generalised momentum along the axis x^3.

On substituting (40.5) into (40.4) we note that

$$\partial^\mu F = k^\mu F', \qquad \partial_\mu \partial^\mu F = k^2 F'' = 0,$$

and find for $F(\varphi)$ the equation

$$2i(kp)F' + [-2e(pA) + e^2A^2 - ie(\gamma k)(\gamma A')]F = 0.$$

The formal solution of this equation is

$$F = \exp\left\{-i\int_0^{kx}\left[\frac{e}{(kp)}\,(pA) - \frac{e^2}{2(kp)}\,A^2\right]d\varphi + \frac{e(\gamma k)(\gamma A)}{2(kp)}\right\}\frac{u}{\sqrt{2p_0}},$$

where $u/\sqrt{2p_0}$ is an arbitrary constant bispinor (its form will be discussed below).

All powers of $(\gamma k)(\gamma A)$ above the first vanish, since

$$(\gamma k)(\gamma A)(\gamma k)(\gamma A) =$$
$$= -(\gamma k)(\gamma k)(\gamma A)(\gamma A) + 2(kA)(\gamma k)(\gamma A) = -k^2A^2 = 0.$$

One can therefore replace

$$\exp\frac{e(\gamma k)(\gamma A)}{2(kp)} = 1 + \frac{e}{2(kp)}\,(\gamma k)(\gamma A),$$

so that ψ takes the form

$$\psi_p = \left[1 + \frac{e}{2(kp)}(\gamma k)(\gamma A)\right] \frac{u}{\sqrt{2p_0}} e^{iS}, \tag{40.7}$$

where [22]

$$S = -px - \int_0^{kx} \left[\frac{e}{(kp)}(pA) - \frac{e^2}{2} A^2\right] d\varphi. \tag{40.8}$$

To establish the conditions imposed on the constant bispinor u, one should consider the wave as infinitely slowly "switched on", starting from $t = -\infty$. Then $A \to 0$ at $kx \to -\infty$ and ψ should go over into the solution of the free Dirac equation. For this $u = u(p)$ must satisfy the equation

$$(\gamma p - m)u = 0. \tag{40.9}$$

This condition discards the "spurious" solutions of the second-order equation. Since it does not depend on time, this condition remains in force at finite kx as well. Thus, $u(p)$ coincides with the bispinor amplitude of a free plane wave; we shall take it normalised by the same condition (23.4): $\bar{u}u = 2m$.

The arguments set forth also immediately allow one to find the normalisation of the wave functions (40.7). An infinitely slowly switched on field does not change the normalisation integral. It follows that the functions (40.7) satisfy the same normalisation condition

$$\int \psi^*_{p'}\psi_p\, d^3x = \int \bar{\psi}_{p'}\gamma^0\psi_p\, d^3x = (2\pi)^3\delta(\mathbf{p}' - \mathbf{p}), \tag{40.10}$$

as free plane waves.

Let us find the current density corresponding to the functions (40.7). Noting that

$$\bar{\psi}_p = \frac{\bar{u}}{\sqrt{2p_0}}\left[1 + \frac{e}{2(kp)}(\gamma A)(\gamma k)\right] e^{-iS},$$

by direct multiplication we get

$$j^\mu = \bar{\psi}_p\gamma^\mu\psi_p = \frac{1}{p_0}\left\{p^\mu - eA^\mu + k^\mu\left(\frac{e(pA)}{(kp)} - \frac{e^2A^2}{2(kp)}\right)\right\}. \tag{40.11}$$

If $A^\mu(\varphi)$ are periodic functions and their average (over time) vanishes, the mean value of the current density is

$$\bar{j}^\mu = \frac{1}{p_0}\left(p^\mu - \frac{e^2\overline{A^2}}{2(kp)} k^\mu\right). \tag{40.12}$$

Let us find also the mean density of kinetic momentum in the state ψ_p. The operator of kinetic momentum is the difference $\widehat{p} - eA = i\partial - eA$. By direct computation we find

$$\psi^*_p(\widehat{p}^\mu - eA^\mu)\psi_p = \bar{\psi}_p\gamma^0(\widehat{p}^\mu - eA^\mu)\psi_p =$$

[22]The expression for S coincides with the classical action for a particle moving in the field of a wave (cf. II, § 47, problem 2).

$$= p^\mu - eA^\mu + k^\mu\left(\frac{e(pA)}{(kp)} - \frac{e^2A^2}{2(kp)}\right) + k^\mu \frac{ie}{8(kp)p_0} F_{\mu\nu}(u^*\sigma^{\lambda\nu}u). \tag{40.13}$$

The time-averaged value of this 4-vector, which we denote by q^μ, is

$$q^\mu = p^\mu - \frac{e^2\overline{A^2}}{2(kp)} k^\mu. \tag{40.14}$$

Its square:

$$q^2 = m_*^2, \qquad m_* = m\sqrt{1 - \frac{e^2}{m^2}\overline{A^2}}; \tag{40.15}$$

m_* plays the role of "effective mass" of the electron in the field. Comparing (40.14) and (40.12), we see that

$$\bar{j}^\mu = q^\mu / p_0. \tag{40.16}$$

Let us note also that the normalisation condition (40.10), expressed with the aid of the vector $\mathbf{q}$, has the form

$$\int \psi^*_{p'}\psi_p\, d^3x = (2\pi)^3 \frac{q_0}{p_0}\,\delta(\mathbf{q}' - \mathbf{q}) \tag{40.17}$$

(the transition from (40.10) to (40.17) is most easily carried out in the special reference frame indicated above).

§ 41. Spin motion in an external field

The passage to the quasiclassical approximation in the Dirac equation is carried out in the same way as in the non-relativistic theory. In the equation of second order (32.7a) we substitute[23] ψ in the form

$$\psi = ue^{iS/\hbar},$$

where S is a scalar, u a slowly varying bispinor. It is assumed that the usual quasiclassical condition is satisfied: the momentum of the particle should change little over distances of the order of the wavelength $\hbar/|\mathbf{p}|$.

In the zeroth approximation in $\hbar$ one obtains the usual classical relativistic Hamilton–Jacobi equation for the action S. All terms containing spin (and proportional to $\hbar$) drop out of the equations of motion. The spin would appear only in the next approximation in $\hbar$. In other words,

the effect of the magnetic moment of the electron on its motion is always of the same order of magnitude as the quantum corrections. This is quite natural in view of the purely quantum nature of the spin angular momentum, which is proportional to $\hbar$.

In this connection the problem acquires physical meaning of the behaviour of the spin of an electron performing a given quasiclassical motion in an external field. The solution of this problem is contained in the next approximation in $\hbar$ of the Dirac equation. We shall, however, apply another, more intuitive method that is not directly connected with the Dirac equation. It has the advantage that it permits one to consider the motion of any

[23] Here we again use ordinary units.

particle, including those possessing an "anomalous" gyromagnetic ratio, not described by the Dirac equation.

Our goal is to establish "equations of motion" for the spin in an arbitrary (given) motion of the particle. Let us begin with the non-relativistic case.

The non-relativistic Hamiltonian of a particle in an external field is

$$\widehat{H} = \widehat{H}' - \mu\boldsymbol{\sigma}\cdot\mathbf{H}, \tag{41.1}$$

where in $\widehat{H}'$ all terms not containing the spin are included (see III, § 111); μ is the magnetic moment of the particle. This form of the Hamiltonian is not tied to a specific sort of particle. For electrons $\mu = e\hbar/2mc$ (the charge of the electron $e = -|e|$), and for nucleons μ contains also an "anomalous part" [24]

$$\mu' = \mu - \frac{e\hbar}{2mc}. \tag{41.2}$$

According to the general rules of quantum mechanics, the operator equation of motion for the spin is obtained from the formula

$$\dot{\widehat{\mathbf{s}}} = \frac{i}{\hbar}(\widehat{H}\widehat{\mathbf{s}} - \widehat{\mathbf{s}}\widehat{H}) = \frac{i}{2\hbar}(\widehat{H}\boldsymbol{\sigma} - \boldsymbol{\sigma}\widehat{H}). \tag{41.3}$$

Substituting here (41.1), we find

$$\dot{\widehat{s}}_i = -\frac{i\mu}{2\hbar}H_k(\sigma_k\sigma_i - \sigma_i\sigma_k) = -\frac{\mu}{\hbar}e_{ikl}H_k\sigma_l,$$

or

$$\dot{\widehat{\mathbf{s}}} = \frac{2\mu}{\hbar}[\widehat{\mathbf{s}}\mathbf{H}]. \tag{41.4}$$

We now average this operator equation over the state of a quasiclassical wave packet moving along a given trajectory. This operation reduces to replacing the operator of spin by its

average value $\bar{\mathbf{s}}$, and the vector $\mathbf{H}$ by the function $\mathbf{H}(t)$ representing the value of the magnetic field at the point of the particle's (wave packet's) location at its given motion along the trajectory. In the non-relativistic approximation, within the framework of the Pauli equation, $\widehat{\mathbf{s}} = \boldsymbol{\sigma}/2$ is the spin operator of the particle in its rest frame, and the average value we denoted in § 29 as $\boldsymbol{\zeta}/2$. Thus, we arrive at the equation [25]

$$\frac{d\boldsymbol{\zeta}}{dt} = \frac{2\mu}{\hbar}[\boldsymbol{\zeta}\,\mathbf{H}(t)]. \tag{41.5}$$

In this form the equation has an essentially classical character. It says that the vector of magnetic moment precesses about the direction of the field $\mathbf{H}$ with the angular velocity $-2\mu\mathbf{H}/\hbar$. In the non-relativistic case, if $\mu' = 0$, then $\mu = e\hbar/2mc$, and this angular

[24] Including radiative corrections, a very small "anomalous part" is present also in the magnetic moment of the electron.

[25] Classically equation (41.5) is obtained directly from the equality $d\mathbf{M}/dt = [\boldsymbol{\mu}\mathbf{H}]$, where $\mathbf{M}$ is the angular momentum of the system, $\boldsymbol{\mu}$ its magnetic moment; $[\boldsymbol{\mu}\mathbf{H}]$ is the torque on the system. Setting $\mathbf{M} = \frac{1}{2}\hbar\boldsymbol{\zeta}$, $\boldsymbol{\mu} = \frac{\mu}{\hbar}\boldsymbol{\zeta} = \mu\boldsymbol{\zeta}$, we obtain (41.5).

velocity coincides with the velocity $-2\mu\mathbf{H}/\hbar$ of rotation of the vector $\boldsymbol{\zeta}$; in other words, the polarisation vector maintains a constant angle with the direction of motion (we shall see below that this result remains valid in the relativistic case as well).

In the same non-relativistic case the velocity $\mathbf{v}$ of the particle changes according to the equation

$$\frac{d\mathbf{v}}{dt} = \frac{e}{mc}\,[\mathbf{v}\,\mathbf{H}],$$

i.e. the vector $\mathbf{v}$ rotates about the direction of $\mathbf{H}$ with the angular velocity $-e\mathbf{H}/mc$. If $\mu' = 0$, this angular velocity coincides with the velocity $-2\mu\mathbf{H}/\hbar$ of rotation of the vector $\boldsymbol{\zeta}$; in other words, the polarisation vector maintains a constant angle with the direction of motion (we shall see below that this result remains valid in the relativistic case as well).

Let us now carry out the relativistic generalisation of equation (41.5). For the covariant description of the polarisation one should use the 4-vector a introduced in § 29, and the equation of spin motion should define the derivative $da^\mu/d\tau$ with respect to the proper time τ [26].

The form of this equation can be established already from considerations of relativistic invariance, if we take into account that its right-hand side must be linear and homogeneous in the tensor of the electromagnetic field $F^{\mu\nu}$ and, besides it, can contain only the 4-velocity $u^\mu = p^\mu/m$. This condition is satisfied only by an equation of the form

$$\frac{da^\mu}{d\tau} = \alpha F^{\mu\nu}a_\nu + \beta u^\mu F^{\nu\lambda}u_\nu a_\lambda, \tag{41.6}$$

where α, β are constants. It is easy to see that by virtue of the condition $a_\mu u^\mu = 0$ and the antisymmetry of the tensor $F^{\mu\nu}$ (so that $F^{\mu\nu}u_\mu u_\nu = 0$) no other expressions of the required form can be constructed.

For $v \to 0$ this equation should coincide with (41.5). Setting $a^\mu = (0, \boldsymbol{\zeta})$, $u^\mu = (1, 0)$, $\tau = t$, we get

$$\frac{d\boldsymbol{\zeta}}{dt} = \alpha[\boldsymbol{\zeta}\,\mathbf{H}].$$

Comparing with (41.5), we find: $\alpha = 2\mu$.

To determine β we note that $a^\mu u_\mu = 0$. Differentiating this relation with respect to τ and using the classical equation of motion of a charge in a field

$$m\,\frac{du^\mu}{d\tau} = eF^{\mu\nu}u_\nu$$

(see II, § 23), we get

$$u_\mu\,\frac{da^\mu}{d\tau} = -a_\mu\,\frac{du^\mu}{d\tau} = -a_\mu\,\frac{e}{m}\,F^{\mu\nu}u_\nu = \frac{e}{m}\,F^{\mu\nu}u_\mu a_\nu.$$

On the other hand, multiplying equation (41.6) on both sides by u_μ, taking into account that $u_\mu u^\mu = 1$ and cancelling the common factor $F^{\mu\nu}u_\mu a_\nu$, we get

$$\beta = -2\Big(\mu - \frac{e}{2m}\Big) = -2\mu'.$$

[26] Hereafter we again set $c = 1$, $\hbar = 1$.

Thus, we find the final relativistic equation of spin motion

$$\frac{da^\mu}{d\tau} = 2\mu F^{\mu\nu} a_\nu - 2\mu' u^\mu F^{\nu\lambda} u_\nu a_\lambda \tag{41.7}$$

(*V. Bargmann, L. Michel, V. Telegdi*, 1959)[27].

Let us go over from the 4-vector a to the quantity $\boldsymbol{\zeta}$, which directly characterises the polarisation of the particle in its "instantaneous" rest frame; the connection between a and $\boldsymbol{\zeta}$ is given by formulae (29.7)–(29.9). We note immediately that from (41.7) it automatically follows that $a_\mu\, da^\mu/d\tau = 0$, i.e. $a_\mu a^\mu = \text{const}$. Since $a_\mu a^\mu = -\zeta^2$, this is a natural result: in the motion of the particle its polarisation remains unchanged in magnitude.

The equation determining the change in the direction of the polarisation is obtained by going over in (41.7) to three-dimensional notation. Expanding the spatial components of this equation, we find

$$\frac{d\mathbf{a}}{dt} = \frac{2\mu m}{\varepsilon}[\mathbf{a}\,\mathbf{H}] + \frac{2\mu m}{\varepsilon}(\mathbf{a}\,\mathbf{v})\mathbf{E} - \frac{2\mu'\varepsilon}{m}\mathbf{v}(\mathbf{a}\,\mathbf{E}) +$$

$$+ \frac{2\mu'\varepsilon}{m}\mathbf{v}(\mathbf{v}[\mathbf{a}\,\mathbf{H}]) + \frac{2\mu'\varepsilon}{m}\mathbf{v}(\mathbf{a}\,\mathbf{v})(\mathbf{v}\,\mathbf{E}).$$

Substituting here (29.9), taking into account on differentiating the equalities $\mathbf{p} = \varepsilon\mathbf{v}$, $\varepsilon^2 = \mathbf{p}^2 + m^2$ and the equations of motion

$$\frac{d\mathbf{p}}{dt} = e\mathbf{E} + e[\mathbf{v}\,\mathbf{H}], \qquad \frac{d\varepsilon}{dt} = e(\mathbf{v}\,\mathbf{E}). \tag{41.8}$$

An elementary, although rather lengthy, computation leads to the following equation[28]:

$$\frac{d\boldsymbol{\zeta}}{dt} = \frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon}[\boldsymbol{\zeta}\,\mathbf{H}] + \frac{2\mu'\varepsilon}{\varepsilon + m}(\mathbf{v}\,\mathbf{H})[\mathbf{v}\,\boldsymbol{\zeta}] + \frac{2\mu m + 2\mu'\varepsilon}{\varepsilon + m}[\boldsymbol{\zeta}\,[\mathbf{E}\,\mathbf{v}]]. \tag{41.9}$$

Of particular interest is not so much the change in the absolute direction of the polarisation in space, as the change of its direction relative to the direction of motion. Representing $\boldsymbol{\zeta}$ in the form

$$\boldsymbol{\zeta} = \mathbf{n}\zeta_{\|} + \boldsymbol{\zeta}_{\perp} \tag{41.10}$$

(where $\mathbf{n} = \mathbf{v}/v$) and writing an equation for the projection $\zeta_{\|}$ of the polarisation on the direction of motion. The computation using (41.8), (41.9) leads to the following result[29]:

$$\frac{\dot{\zeta}_{\|}}{dt} = 2\mu'(\boldsymbol{\zeta}_{\perp}[\mathbf{H}\,\mathbf{n}]) + \frac{2}{v}\left(\frac{\mu m^2}{\varepsilon^2} - \mu'\right)(\boldsymbol{\zeta}_{\perp}\mathbf{E}). \tag{41.11}$$

A number of examples illustrating the application of the obtained equations are considered in the problems to this section. Here we merely note that in the motion in a pure magnetic field the polarisation of a particle without an anomalous magnetic moment

[27]In another form a similar equation was first found by *Ya. I. Frenkel'* (1926).

[28]This equation can be obtained somewhat more briefly by expanding the time component of equation (41.7).

[29]This equation can also be obtained somewhat more briefly by expanding the time component of equation (41.7).

maintains a constant angle with the direction of motion (i.e. $\dot{\zeta}_{\|}$ = const). Thus, this result, indicated above for the non-relativistic case, is in fact of a general character.

Let us refine the conditions of applicability of the obtained equations. The condition mentioned at the outset that the momentum of the particle change slowly amounts to a definite requirement of smallness of the fields $\mathbf{E}$ and $\mathbf{H}$; in particular, the Larmor radius in a magnetic field ($\sim p/eH$) should be large compared with the wavelength of the particle. Besides this, however, a condition must also hold that the fields do not change too rapidly in space, strictly speaking: the field should change little over the dimensions of the quasiclassical wave packet. Thus, the field should change little over distances of the order of the wavelength of the particle ($1/p$), and also of the Compton wavelength, $1/m$ [30].

However, in practical problems of motion in macroscopic fields the condition of slow change of the fields is automatically fulfilled, so that only a sufficient smallness of them is required.

In § 33 the first relativistic corrections to the Hamiltonian of an electron moving in an external field were found. For an electron in an electric field the approximate Hamiltonian has the form (see (33.12))

$$\widehat{H} = \widehat{H}' - \frac{e}{4m}\left(\boldsymbol{\sigma}\left[\mathbf{E}\,\frac{\widehat{\mathbf{p}}}{m}\right]\right), \qquad \widehat{\mathbf{p}} = -i\nabla, \tag{41.12}$$

where $\widehat{H}'$ contains terms not involving the spin. In our case, because the field changes slowly, in $\widehat{H}'$ one should neglect the term with derivatives of $\mathbf{E}$ (i.e. $\operatorname{div}\mathbf{E}$); one may also drop the term with $\widehat{\mathbf{p}}^4$, which has no relation to the effects of the field that interest us here, so that $\widehat{H}'$ (in the absence of a magnetic field) reduces to the non-relativistic Hamiltonian $\widehat{H}' = \widehat{\mathbf{p}}^2/(2m) + e\Phi$.

Formula (41.12) can also be obtained directly from equation (41.9), without invoking the Dirac equation. The generalisation achieved in this way (in the quasiclassical case) applies to particles with an anomalous magnetic moment.

To within terms of first order in the velocity v, the equation for the spin motion in an electric field is obtained from

(41.9) in the form

$$\frac{d\boldsymbol{\zeta}}{dt} = (\mu + \mu')[\boldsymbol{\zeta}\,[\mathbf{E}\,\mathbf{v}]] = \left(\frac{e}{2m} + 2\mu'\right)[\boldsymbol{\zeta}\,[\mathbf{E}\,\mathbf{v}]].$$

If we demand that this equation be obtained quantum-mechanically by commuting the spin operator with the Hamiltonian (in accordance with (41.3)), one must set

$$\widehat{H} = \widehat{H}' = \left(\mu' + \frac{e}{4m}\right)\left(\boldsymbol{\sigma}\left[\mathbf{E}\,\frac{\widehat{\mathbf{p}}}{m}\right]\right). \tag{41.13}$$

This is the desired expression. For $\mu' = 0$ we return to (41.12). Note that the "normal" magnetic moment $e/2m$ enters with an extra factor $\frac{1}{2}$ as compared with the anomalous moment μ' [31].

[30] The last requirement arises from the condition that the spread of velocities in the wave packet be small compared with c; in the contrary case it would be impossible to use the non-relativistic formulae.

[31] This is the "Thomas half", mentioned in the footnote on p. 151. The derivation presented here clearly demonstrates its origin.

Problems

1. Determine the change in the direction of polarisation of a particle in its motion in a plane perpendicular to a uniform magnetic field ($\mathbf{v} \perp \mathbf{H}$).

Solution. On the right-hand side of equation (41.9) only the first term remains, i.e. the vector $\boldsymbol{\zeta}$ precesses about the direction of $\mathbf{H}$ (the z-axis) with the angular velocity

$$-\frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon}\,\mathbf{H} = -\left(\frac{e}{\varepsilon} + 2\mu'\right)\mathbf{H}.$$

The projection of $\boldsymbol{\zeta}$ onto this plane we denote by ζ_1. The vector $\mathbf{v}$ rotates in the same plane with the angular velocity $-e\mathbf{H}/\varepsilon$ (as follows from the equations of motion: $\dot{\mathbf{p}} = \varepsilon\dot{\mathbf{v}} = e[\mathbf{v}\,\mathbf{H}]$). It is clear that ζ_1 rotates relative to the direction of $\mathbf{v}$ with the angular velocity $-2\mu'\mathbf{H}$.

2. The same for motion along the direction of the magnetic field.

Solution. For coinciding directions of $\mathbf{v}$ and $\mathbf{H}$ equation (41.9) reduces to the form

$$\frac{d\boldsymbol{\zeta}}{dt} = \frac{2\mu m}{\varepsilon}\,[\boldsymbol{\zeta}\,\mathbf{H}].$$

i.e. $\boldsymbol{\zeta}$ precesses about the common direction of $\mathbf{v}$ and $\mathbf{H}$ with the angular velocity $-2\mu m\mathbf{H}/\varepsilon$.

3. The same for motion in a uniform electric field.

Solution. Let the field $\mathbf{E}$ be directed along the x-axis, and the motion takes place in the xy-plane (with p_y = const). From (41.9) we find that the vector $\boldsymbol{\zeta}$ precesses about the z-axis with the instantaneous angular velocity

$$-\left(\frac{e}{\varepsilon + m} + 2\mu'\right) E\,\frac{p_y}{\varepsilon}.$$

From (41.11) we find that ζ_1 rotates relative to $\mathbf{v}$ with the instantaneous angular velocity

$$\frac{2v_y}{v^2}\left(\frac{\mu m^2}{\varepsilon^2} - \mu'\right) E = \frac{p_y}{\varepsilon}\left(\frac{em}{p^2} - 2\mu'\right) E.$$

§ 42. Scattering of neutrons in an electric field

In collisions of neutrons with nuclei, scattering at large angles is determined by the principal interaction—nuclear forces. At scattering to small angles, however, the electromagnetic interaction of the magnetic moment of the neutron with the electric field of the nucleus becomes significant (*J. Schwinger*, 1948).

We shall assume the neutron non-relativistic, so that the interaction to be considered is described by the approximate Hamiltonian (41.13). The entire magnetic moment of an electrically neutral particle is "anomalous", and the operator $\widehat{H}'$ reduces in this case to the operator of kinetic energy[32]:

$$\widehat{H} = -\frac{\hbar^2}{2m}\Delta + i\,\frac{\mu\hbar}{mc}\,\boldsymbol{\sigma}[\mathbf{E}\nabla]. \tag{42.1}$$

[32] In this section we use ordinary units, and the letter m denotes the neutron mass.

In view of the smallness of the electromagnetic interaction of the neutron, the scattering amplitude f_{em} caused by it can be computed in the Born approximation:

$$f_{\mathrm{em}} = -\frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{p}'\cdot\mathbf{r}/\hbar} \left(i\,\frac{\mu\hbar}{mc}\,\boldsymbol{\sigma}[\mathbf{E}\nabla] \right) e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\, d^3x \tag{42.2}$$

(see III, § 126), or

$$f_{\mathrm{em}} = \frac{\mu}{qc\hbar^3}\,\boldsymbol{\sigma}[\mathbf{E_q}\,\mathbf{p}], \qquad \mathbf{E_q} = \int \mathbf{E}(\mathbf{r})\, e^{-i\mathbf{q}\cdot\mathbf{r}}\, d^3x$$

($\mathbf{p}$, $\mathbf{p}'$ are the momenta of the neutron before and after the scattering, $\hbar\mathbf{q} = \mathbf{p}' - \mathbf{p}$). In the notation written, the scattering amplitude f_{em} is an operator with respect to the spin variable.

Before proceeding with the further calculation, let us make the following remark. Formula (42.1) was derived in the preceding section for slowly varying fields (which in fact meant the neglect of derivatives of the field in the Hamiltonian, terms containing them). In applications to the Coulomb field of a nucleus this means that the wavelength $\hbar/p$ must be small compared with the distances essential in the integral $\mathbf{E_q}$, i.e. with the distances $r \sim 1/q$. Hence $\hbar q \ll p$, so that the scattering angle $\theta \sim \hbar q/p \ll 1$. Thus, the required condition is indeed satisfied precisely for scattering at small angles.

For the Coulomb field with potential $\Phi = Ze/r$ the Fourier component of the field strength

$$\mathbf{E_q} = -i\mathbf{q}\Phi_q = -\mathbf{iq} \cdot 4\pi Ze/q^2$$

(see II, (51.5)). Substitution into (42.2) gives

$$f_{\mathrm{em}} = i\,\frac{2Ze\mu}{q^2 c\hbar^3}\,(\boldsymbol{\sigma}[\mathbf{p}\,\mathbf{p}']).$$

At small scattering angles $\hbar q \approx p\theta$, $[\mathbf{p}\,\mathbf{p}'] \approx p^2\theta\boldsymbol{\nu}$, where $\boldsymbol{\nu}$ is the unit vector in the direction $[\mathbf{p}\,\mathbf{p}']$. Thus,

$$f_{\mathrm{em}} = i\,\frac{2Ze\mu}{\theta\hbar c}\,\boldsymbol{\sigma\nu}.$$

To this expression one should add the amplitude of nuclear scattering. Owing to the rapid fall-off of the nuclear forces with distance, this amplitude tends at small angles to a finite (depending on the energy) complex value, which we denote by a. Therefore the total scattering amplitude

$$f = a + i\,\frac{b}{\theta}\,\boldsymbol{\sigma\nu}, \qquad b = \frac{2Ze\mu}{\hbar c} = 2Z\alpha\,\frac{\mu}{e}. \tag{42.3}$$

We see that the electromagnetic scattering indeed becomes dominant at sufficiently small angles. The form of expression (42.3) coincides with that considered in III, § 140. We can therefore directly use the formulae derived there. The cross section, summed over all possible final polarisation states:

$$\frac{d\sigma}{do} = |a|^2 + \frac{b^2}{\theta^2} + 2b\,\mathrm{Im}\,a \cdot \boldsymbol{\nu\zeta}, \tag{42.4}$$

where $\boldsymbol{\zeta}$ is the initial polarisation of the neutron beam (P see III, § 140). If the initial state is non-polarised ($\boldsymbol{\zeta} = 0$), the polarisation after scattering is

$$\boldsymbol{\zeta}' = \frac{2b \operatorname{Im} a \cdot \theta}{|a|^2\theta^2 + b^2} \boldsymbol{\nu}. \tag{42.5}$$

This polarisation is maximal at $\theta = b/|a|$, with $\zeta'_{\max} = \operatorname{Im} a/|a|$.

CHAPTER V

Radiation

§ 43. The operator of electromagnetic interaction

The interaction of electrons with the electromagnetic field can, as a rule, be treated by perturbation theory. This circumstance is connected with the comparative weakness of the electromagnetic interaction, expressed in the smallness of the corresponding dimensionless "coupling constant"—the fine-structure constant $\alpha = e^2/\hbar c = 1/137$. This smallness plays a fundamental role in quantum electrodynamics.

In classical electrodynamics (see II, §§ 16, 28) the electromagnetic interaction is described by the term

$$-ej^{\mu}A_{\mu} \tag{43.1}$$

in the Lagrangian density of the system "field + charges" (A is the 4-potential of the field, j the 4-vector of current density of the particles). The current density satisfies the continuity equation

$$\partial_{\mu}j^{\mu} = 0, \tag{43.2}$$

expressing the law of conservation of charge. Recall (see II, § 29) that the gauge invariance of the theory is closely connected with this law. Indeed, under the replacement $A_{\mu} \to A_{\mu} + \partial_{\mu}\chi$ (4.1) the quantity $-ej^{\mu}\partial_{\mu}\chi$ is added to the Lagrangian density (43.1), which by virtue of (43.2) can be written in the form of a 4-divergence

$$-e\partial_{\mu}(\chi j^{\mu})$$

and therefore drops out upon integration over d^4x in the action $S = \int L\, d^4x$.

In quantum electrodynamics the 4-vectors j and A are replaced by the corresponding secondarily quantised operators. The operator of the current is expressed through the ψ-operators according to $\widehat{j} = \widehat{\bar{\psi}}\gamma\widehat{\psi}$. The role of the generalised "coordinates" q in the Lagrangian

$$\int \widehat{L}_{\text{int}}\, d^3x = -e\int (\widehat{j}\widehat{A})\, d^3x$$

is played by the values of $\widehat{\psi}$, $\widehat{\bar{\psi}}$, $\widehat{A}$ at each point of space. Since the Lagrangian density turns out to depend only on the "coordinates" q themselves (and not on their derivatives with respect to x), the transition from the Lagrangian density to the Hamiltonian density according to formula (10.11) reduces merely to a change of sign of the Lagrangian density [1]. Thus, the operator of electromagnetic interaction (the integral over space of the Hamiltonian density of the interaction) has the form

$$\widehat{V} = e\int (\widehat{j}\widehat{A})\, d^3x. \tag{43.3}$$

[1] Independently of these arguments we note that if one is speaking only of a first-order correction, then any small perturbation enters the Hamiltonian with the same sign as the Lagrangian (see I, § 40).

The operator of the free electromagnetic field is a sum

$$\widehat{A} = \sum_n [\widehat{c}_n A_n(x) + \widehat{c}_n^+ A_n^*(x)], \tag{43.4}$$

containing the creation and annihilation operators of photons in various states (labelled by the index n). Each of them has matrix elements only for increasing or decreasing the corresponding occupation number N_n by 1 (with all remaining occupation numbers unchanged). Therefore the operator $\widehat{A}$ also has matrix elements only for transitions with a change in the number of photons by 1. In other words, in the first approximation of perturbation theory only processes of single emission or absorption of a photon arise.

According to (2.15) the matrix elements

$$\langle N_n - 1|\widehat{c}_n|N_n\rangle = \langle N_n|\widehat{c}_n^+|N_n - 1\rangle = \sqrt{N_n}. \tag{43.5}$$

If in the initial state of the field photons (of sort n) are absent, then $\langle 1|\widehat{c}_n^+|0\rangle = 1$. The matrix element of the operator (43.3) for the emission of a photon is

$$V_{fi}(t) = e \int (j_{fi}A_n^*)\, d^3x, \tag{43.6}$$

where $A_n(x)$ is the wave function of the emitted photon, and j_{fi} is the matrix element of the operator $\widehat{j}$ for the transition of the emitter from the initial state i to the final state f [2]. The 4-vector $j^\mu_{fi} = (\rho_{fi}, \mathbf{j}_{fi})$ is called the *transition current*.

Analogously, the matrix element for the absorption of a photon is

$$V_{fi}(t) = e \int (j_{fi}A_n)\, d^3x. \tag{43.7}$$

It differs from (43.6) only in that $A_n^*(x)$ is replaced by $A_n(x)$.

By indicating the argument t in V_{fi} we emphasise that the time-dependent matrix element is meant. Separating in the wave functions the time factors, one can pass in the usual way to the time-independent matrix elements:

$$V_{fi}(t) = V_{fi}\, e^{-i(E_i - E_f \mp \omega)t}, \tag{43.8}$$

(E_i, E_f are the initial and final energies of the emitting system; the signs $\mp$ correspond to emission and absorption of the photon ω).

The wave function of a photon with a definite momentum $\mathbf{k}$ and definite polarisation is

$$A^\mu = \sqrt{4\pi}\sqrt{\frac{1}{2\omega}}\, e^\mu e^{i\mathbf{kr}} \tag{43.9}$$

(see (4.3); the time factor is omitted). Substituting in (43.6), we find the matrix element for the emission of such a photon in the form

$$V_{fi} = e\sqrt{4\pi}\sqrt{\frac{1}{2\omega}}\, e^*_\mu\, j^\mu_{fi}(\mathbf{k}), \tag{43.10}$$

[2] The notation in (43.6) contains a certain inconsistency: the indices of V_{fi} refer to the states of the entire system "emitter + field", while those of j_{fi} refer to the states of the emitter alone.

where $j_{fi}(\mathbf{k})$ is the transition current in the momentum representation, i.e. the Fourier components

$$j_{fi}(\mathbf{k}) = \int j_{fi}(\mathbf{r})\, e^{-i\mathbf{kr}}\, d^3x. \tag{43.11}$$

The analogous formula for the absorption of a photon:

$$V_{fi} = e\sqrt{4\pi}\sqrt{\frac{1}{2\omega}}\, e_\mu\, j^\mu_{fi}(-\mathbf{k}). \tag{43.12}$$

The law of conservation of current in the momentum representation is written as the condition of 4-transversality of the transition currents:

$$k_\mu\, j^\mu_{fi}(\mathbf{k}) = \omega\rho_{fi}(\mathbf{k}) - \mathbf{k}\cdot\mathbf{j}_{fi}(\mathbf{k}) = 0. \tag{43.13}$$

The formulae written in this section, in which the form of the current operator is not specified, have a general character and are valid for electromagnetic processes involving any charged particles. The existing theory provides the means of establishing the form of the current operator (and thereby of computing in principle its matrix elements) only for electrons. In application to strongly interacting particle systems (including nuclei) we shall confine ourselves to a phenomenological theory, in which the transition currents appear as quantities taken from experiment, satisfying only the general requirements of space–time symmetry and the continuity equation.

§ 44. Emission and absorption

The probability of a transition under the influence of a perturbation $\widehat{V}$ in the first approximation is given by known formulae of perturbation theory (III, § 42). Let the initial and final states of the emitting system belong to the discrete spectrum[3]. Then the probability (per unit time) of the transition $i \to f$ with emission of a photon is

$$dw = 2\pi|V_{fi}|^2\delta(E_i - E_f - \omega)\, d\nu, \tag{44.1}$$

where ν conventionally denotes the set of quantities characterising the state of the photon and spanning a continuous range of values (the wave function of the photon being normalised to a δ-function "on the ν scale"). If a photon with a definite value of the angular momentum is emitted, the only continuously varying quantity is the frequency ω. Integration of formula (44.1) over $d\nu \equiv d\omega$ eliminates the δ-function (replacing ω by $E_i - E_f$), and the transition probability becomes

$$w = 2\pi|V_{fi}|^2. \tag{44.2}$$

If instead the emission of a photon with a given momentum $\mathbf{k}$ is considered, then $d\nu = d^3k/(2\pi)^3 = \omega^2\, d\omega\, do/(2\pi)^3$. It is assumed that the wave function of the photon (a plane wave) is normalised to one photon in a volume $V = 1$, as is always adopted

[3]By this we mean, in any event, neglect of the recoil: the emitter as a whole remains stationary.

in this book; $d\nu$ is the number of states in the phase-space volume $V\,d^3k$. Thus, the probability of emission of a photon with a given momentum is

$$dw = 2\pi |V_{fi}|^2\,\delta(E_i - E_f - \omega)\,\frac{d^3k}{(2\pi)^3}, \tag{44.3}$$

or after integrating over $d\omega$:

$$dw = \frac{1}{4\pi^2}\,|V_{fi}|^2\,\omega^2\,do. \tag{44.4}$$

Here the matrix element V_{fi} from (43.10) should be substituted.

In the following sections we shall use these formulae for computing the probability of radiation in various concrete cases. Here we shall consider some general relations among different kinds of radiative processes.

If in the initial state of the field there already existed a non-zero number N_n of the given photons, the matrix element of the transition is multiplied further by

$$\langle N_n + 1|\hat{c}_n^+|N_n\rangle = \sqrt{N_n + 1}, \tag{44.5}$$

i.e. the probability of the transition increases by a factor of $N_n + 1$. The unity in this factor corresponds to *spontaneous* emission, occurring also when $N_n = 0$. The term N_n gives rise to *stimulated* (or *induced*) emission: we see that the presence of photons in the initial state of the field stimulates additional emission of photons of the same kind.

The matrix element V_{if} for the reverse transition of the system ($f \to i$) differs from the element V_{fi} by the replacement of (43.5) by

$$\langle N_n - 1|\hat{c}_n|N_n\rangle = \sqrt{N_n}$$

(and the replacement of all remaining quantities by their complex conjugates). This reverse transition represents the absorption of a photon by the system, going from level E_f to level E_i. Between the probabilities of emission and absorption of a photon (for a given pair of states i, f) the following important relation therefore holds [4]:

$$\frac{w^{(\text{emis})}}{w^{(\text{abs})}} = \frac{N_n + 1}{N_n} \tag{44.6}$$

(first pointed out by *A. Einstein* in 1916).

Let us relate the number of photons to the intensity of the radiation incident on the system from outside. Let

$$I_{\mathbf{ke}}\,d\omega\,do \tag{44.7}$$

be the energy of radiation falling per unit area per unit time, with polarisation $\mathbf{e}$, in the frequency interval $d\omega$ and in the direction of the wave vector within the solid angle do.

Let us note the analogous probabilities for induced emission and absorption. According to (44.6) and (44.8) these probabilities are related to each other by the following equations:

$$dw_{\mathbf{ke}}^{(\text{abs})} = dw_{\mathbf{ke}}^{(\text{ind})} = dw_{\mathbf{ke}}^{(\text{sp})}\,\frac{8\pi^2 c^2}{\hbar\omega^3}\,I_{\mathbf{ke}}. \tag{44.9}$$

[4] By this we mean, in any event, neglect of the recoil: the emitter as a whole remains stationary.

If the incident radiation is isotropic and unpolarised ($I_{\mathbf{ke}}$ does not depend on the directions of **k** and **e**), integrating (44.9) over do and summing over **e** yield analogous relations between total probabilities of radiative transitions (between given states i and f of the system):

$$w^{(\mathrm{abs})} = w^{(\mathrm{ind})} = w^{(\mathrm{sp})} \frac{\pi^2 c^2}{\hbar\omega^3} I, \tag{44.10}$$

where $I = 2 \cdot 4\pi I_{\mathbf{ke}}$ is the total spectral intensity of the incident radiation.

If the states i and f of the emitting (or absorbing) system are degenerate, the total probability of emission (or absorption) of given photons is obtained by summation over all mutually degenerate final states and averaging over all possible initial states. Denoting the multiplicities of degeneracy (statistical weights) of the states i and f by g_i and g_f respectively, for the processes of spontaneous and induced emission the initial states are the states i, and for absorption the states f. Assuming in each case that all g_i or g_f initial states are equally probable, we obtain, obviously, instead of (44.10), the following relations:

$$g_f\, w^{(\mathrm{abs})} = g_i\, w^{(\mathrm{ind})} = g_i\, w^{(\mathrm{sp})} \frac{\pi^2 c^2}{\hbar\omega^3} I. \tag{44.11}$$

In the literature the so-called *Einstein coefficients* are frequently used, defined as

$$A_{if} = w^{(\mathrm{sp})}, \qquad B_{if} = w^{(\mathrm{ind})} \frac{c}{I}, \qquad B_{fi} = w^{(\mathrm{abs})} \frac{c}{I} \tag{44.12}$$

(the quantity I/c is the spatial spectral density of the energy of the radiation). They are related to each other by the equations

$$g_f\, B_{fi} = g_i\, B_{if} = g_i\, A_{if} \frac{\pi^2 c^3}{\hbar\omega^3}. \tag{44.13}$$

§ 45. Dipole radiation

Let us apply the formulae obtained to the emission of a photon by a relativistic electron in a given external field. The transition current in this case is the matrix element of the operator

$$\hat{j} = \hat{\bar{\psi}}\gamma\hat{\psi},$$

in which the ψ-operators are assumed to be expanded over the system of wave functions of stationary states of the electron in the given field (see § 32). The transition of the electron from state i to state f is described by the matrix element $\langle 0_i 1_f | j | 1_i 0_f \rangle$. The change in the occupation numbers is effected by the operator $\hat{a}_f^+ \hat{a}_i$, and for the transition current we obtain

$$j^\mu_{fi} = \bar{\psi}_f \gamma^\mu \psi_i = (\psi_f^* \psi_i,\ \psi_f^* \boldsymbol{\alpha} \psi_i), \tag{45.1}$$

where ψ_i and ψ_f are the wave functions of the initial and final states of the electron.

We choose the wave function of the photon in the three-dimensionally transverse gauge (4-vector of polarisation $e = (0, \mathbf{e})$). Then from (43.10) the product $j_{fi}\, e^* = -\mathbf{j}_{fi} \cdot \mathbf{e}^*$.

Substituting V_{fi} from (44.4), we obtain the following formula for the probability of radiation (per unit time) into the element of solid angle do of a photon with polarisation $\mathbf{e}$:

$$dw_{\mathbf{en}} = e^2 \frac{\omega}{2\pi} |\mathbf{e}^* \cdot \mathbf{j}_{fi}(\mathbf{k})|^2 \, do, \tag{45.2}$$

where

$$\mathbf{j}_{fi}(\mathbf{k}) = \int \psi_f^* \boldsymbol{\alpha} \, \psi_i \, e^{-i\mathbf{kr}} \, d^3x. \tag{45.3}$$

Summation over the polarisations of the photon is carried out by averaging over the directions of $\mathbf{e}$ (in the plane perpendicular to the given direction $\mathbf{n} = \mathbf{k}/\omega$), after which the result is multiplied by 2 to account for the two independent possibilities for transverse polarisation of the photon [5]. Thus we obtain the formula

$$dw_{\mathbf{n}} = e^2 \frac{\omega}{2\pi} |[\mathbf{n}\, \mathbf{j}_{fi}(\mathbf{k})]|^2 \, do. \tag{45.4}$$

A very important case is when the wavelength of the photon λ is large compared with the dimensions of the radiating system a. This situation is usually connected with the smallness of the velocities of particles compared with the speed of light. In the first approximation in a/λ (corresponding to *dipole radiation*—cf. II, § 67) in the transition current (45.3) one can replace the slowly varying factor $e^{-i\mathbf{kr}}$ by unity in the region where ψ_i or ψ_f are appreciably different from zero. This replacement means, in other words, the neglect of the momentum of the photon compared with the momenta of the particles in the system. In the same approximation the integral $\mathbf{j}_{fi}(0)$ may be replaced by its non-relativistic expression, i.e. simply by the matrix element $\mathbf{v}_{fi}$ of the velocity of the electron with respect to the eigenfunction wave functions. In turn this element $\mathbf{v}_{fi} = -i\omega \mathbf{r}_{fi} = \mathbf{d}_{fi}$, where $\mathbf{d}$ is the dipole moment of the electron (in its orbital motion). Thus, we find the following formula for the probability of dipole radiation:

$$dw_{\mathbf{en}} = \frac{\omega^3}{2\pi} |\mathbf{e}^* \cdot \mathbf{d}_{fi}|^2 \, do \tag{45.5}$$

(the direction $\mathbf{n}$ appears here in an implicit form: the vector $\mathbf{e}$ must be perpendicular to $\mathbf{n}$). Summing over polarisations, we get

$$dw_{\mathbf{n}} = \frac{\omega^3}{2\pi} |[\mathbf{n}\, \mathbf{d}_{fi}]|^2 \, do. \tag{45.6}$$

In view of the non-relativistic (with respect to the electron) character of these formulae, their generalisation to arbitrary electronic systems is obvious: under $\mathbf{d}_{fi}$ one should understand the matrix element of the total dipole moment of the system.

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[5] For the averaging one uses the formula

$$\overline{e_i e_k^*} = \tfrac{1}{2}(\delta_{ik} - n_i n_k) \tag{45.4a}$$

or

$$\overline{(\mathbf{ae})(\mathbf{be}^*)} = \tfrac{1}{2}\{\mathbf{ab} - (\mathbf{an})(\mathbf{bn})\} = \tfrac{1}{2}[\mathbf{an}][\mathbf{bn}], \tag{45.4b}$$

where $\mathbf{a}$, $\mathbf{b}$ are constant vectors.

Integrating formula (45.6) over all directions, we find the total probability of radiation:

$$w = \frac{4\omega^3}{3} |\mathbf{d}_{fi}|^2, \tag{45.7}$$

or in ordinary units:

$$w = \frac{4\omega^3}{3\hbar c^3} |\mathbf{d}_{fi}|^2. \tag{45.7a}$$

The intensity I of the radiation is obtained by multiplying the probability by $\hbar\omega$:

$$I = \frac{4\omega^4}{3c^3} |\mathbf{d}_{fi}|^2. \tag{45.8}$$

This formula reveals a direct analogy with the classical formula (see II, (67.11)) for the intensity of dipole radiation by a system of periodically moving particles: the radiation has the frequency $\omega_s = s\omega$ (where ω is the frequency of the motion of the particles, s an integer), and the intensity is

$$I_s = \frac{4\omega_s^4}{3c^3} |\mathbf{d}_s|^2, \tag{45.9}$$

where $\mathbf{d}_s$ are the Fourier components of the dipole moment, i.e. the coefficients of the expansion

$$\mathbf{d}(t) = \sum_{s=-\infty}^{\infty} \mathbf{d}_s \, e^{-is\omega t}. \tag{45.10}$$

The quantum formula (45.8) is obtained from (45.9) by replacing these Fourier components by the matrix elements of the corresponding transitions. This rule (expressing the *correspondence principle* of Bohr) is a special case of the general correspondence between Fourier components of classical quantities and quantum matrix elements of the corresponding transitions (see III, § 48). The radiation is quasiclassical for transitions between states with large quantum numbers; the frequency $\hbar\omega = E_i - E_f$ is then small compared with the energies E_i and E_f. This circumstance, however, does not lead to any changes in the form of formula (45.8), valid for any transitions. By this is explained the (in a certain sense accidental) fact that the correspondence principle for the intensity of radiation turns out to be valid not only in the quasiclassical, but also in the general quantum case.

§ 46. Electric multipole radiation

Instead of considering the emission of a photon in a given direction (i.e. with a given momentum), let us now consider the emission of a photon with definite values of the angular momentum j and its projection m onto some chosen direction z. We saw in § 6 that such photons can be of two types—electric and magnetic; we begin with the emission of photons of the electric type.

It is convenient here to carry out the calculations using the wave functions of the photon in the momentum representation, i.e. representing the 4-vector $A^\mu(\mathbf{r})$ in the form of a Fourier integral. Then the matrix element

$$V_{fi} = e \int d^3x \cdot j^{\mu}_{fi}(\mathbf{r}) \int \frac{d^3k}{(2\pi)^3} A^*(\mathbf{k})\, e^{-i\mathbf{kr}}. \tag{46.1}$$

(to simplify the notation we omit the indices ωjm of the wave functions of the photon).

For an Ej-photon we take the wave function from (7.10), choosing the arbitrary constant C to be

$$C = -\sqrt{\frac{j+1}{j}}.$$

The purpose of this choice is so that in the spatial components of the wave function ($\mathbf{A}$) the terms containing spherical functions of order $j-1$ cancel out (as can be seen from formulae (7.16)). The vector $\mathbf{A}$ will then contain only spherical functions of order $j+1$, as a result of which the corresponding contribution to V_{fi} turns out (as will be clear from the subsequent calculation) to be of higher order of smallness (in a/λ) than the contribution from the component $A^0 \equiv \Phi$, containing spherical functions of lower order j.

Thus, we put

$$A^{\mu} = (\Phi, 0), \qquad \Phi = -\sqrt{\frac{j+1}{j}}\, \frac{4\pi^2}{\omega^{3/2}}\, \delta(|\mathbf{k}| - \omega)\, Y_{jm}(\mathbf{n})$$

($\mathbf{n} = \mathbf{k}/\omega$). Substituting this into (46.1) and integrating over $|\mathbf{k}|$, we get

$$V_{fi} = -e\sqrt{\frac{j+1}{j}}\, \frac{\sqrt{\omega}}{2\pi} \int d^3x \cdot \rho_{fi}(\mathbf{r}) \int do_{\mathbf{n}}\, e^{-i\mathbf{kr}}\, Y^*_{jm}(\mathbf{n}). \tag{46.2}$$

For computing the inner integral we use the expansion (24.12), writing it in the form

$$e^{i\mathbf{pr}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l\, g_l(kr)\, Y^*_{lm}\left(\frac{\mathbf{k}}{k}\right) Y_{lm}\left(\frac{\mathbf{r}}{r}\right), \tag{46.3}$$

where

$$g_l(kr) = \sqrt{\pi/(2kr)}\, J_{l+1/2}(kr) \tag{46.4}$$

(see III, (33.3))[6]. Substituting this expansion into (46.2), we get

$$\int e^{-i\mathbf{kr}}\, Y^*_{jm}(\mathbf{n})\, do_{\mathbf{n}} = 4\pi i^{-j}\, g_j(kr)\, Y^*_{jm}\left(\frac{\mathbf{r}}{r}\right),$$

(the remaining terms vanish by the orthogonality of the spherical functions). In view of the condition $a/\lambda \ll 1$ the integral in (46.2) is restricted to distances r for which $kr \ll 1$.

[6] The normalisation of the functions g_l is such that their asymptotic form at $kr \to \infty$ is:

$$g_l(kr) \approx \frac{\sin(kr - \pi l/2)}{kr}. \tag{46.4a}$$

We can therefore replace the functions $g_j(kr)$ by the first terms of their expansions in powers of kr [7]:

$$g_j(kr) \approx \frac{(kr)^j}{(2j+1)!!}. \tag{46.5}$$

As a result we obtain

$$V_{fi} = (-1)^{m+1} i^j \sqrt{\frac{(2j+1)(j+1)}{\pi j}}\, \frac{\omega^{j+1/2}}{(2j+1)!!}\, e(Q^{(\mathscr{E})}_{j,-m})_{fi}, \tag{46.6}$$

where we have introduced the quantities

$$(Q^{(\mathscr{E})}_{jm})_{fi} = \sqrt{\frac{4\pi}{2j+1}} \int \rho_{fi}(\mathbf{r})\, r^j\, Y_{jm}\left(\frac{\mathbf{r}}{r}\right) d^3x \tag{46.7}$$

(recall that $Y_{j,-m} = (-1)^{j-m} Y^*_{jm}$). The quantities (46.7) are called the 2^j-*pole electric moments of the transition* of the system, by analogy with the corresponding classical quantities (II, § 41) [8].

For an electron in an external field $\rho_{fi} = \psi^*_f \psi_i$, and the quantities (46.7) are then computed as matrix elements of the classical quantity

$$Q^{(\mathscr{E})}_{jm} = \sqrt{\frac{4\pi}{2j+1}}\, r^j\, Y_{jm}.$$

In the non-relativistic (in terms of particle velocity) case, the transition moment can in principle be computed analogously for any system of N interacting particles. The transition charge density is then expressed through the wave functions of the system in the form

$$\rho_{fi}(\mathbf{r}) = \int \psi^*_f(\mathbf{r}_1, \ldots, \mathbf{r}_N)\, \psi_i(\mathbf{r}_1, \ldots, \mathbf{r}_N) \sum_{n=1}^{N} \delta(\mathbf{r} - \mathbf{r}_n) \cdot d^3x_1 \ldots d^3x_N, \tag{46.8}$$

where the integral is taken over the entire configuration space [9].

The wave function of the photon we have used corresponds (in the coordinate representation) to normalisation on a δ-function on the ω-scale, as assumed in formula (44.2). Substituting into it (46.6), we obtain the probability of Ej-radiation [10]

$$w^{(\mathscr{E})}_{jm} = \frac{2(2j+1)(j+1)}{j[(2j+1)!!]^2}\, \omega^{2j+1}\, e^2\, |(Q^{(\mathscr{E})}_{j,-m})_{fi}|^2. \tag{46.9}$$

[7] The power of kr coincides with the order of the spherical function Y_{jm} in the product with which g_j appears, which justifies the neglect of the terms in $\mathbf{A}$ containing spherical functions of higher order.

[8] We define the multipole moments without the factor e, in correspondence with the fact that in this book the current is also defined without the charge factor.

[9] A situation is possible when the transition probability given by approximate selection rules vanishes; in such a case, to obtain a result different from zero one must use wave functions with a relativistic correction that takes into account the spin–orbit interaction.

[10] At first glance it might appear that by virtue of the spatial isotropy the total probability of emission of a photon should not depend on the value of m. This is not so, as is easy to see from the fact that for emission of photons with different values of m the final states of the system must be different (for a given initial state); cf. selection rule (46.16) below.

In particular, for $j = 1$ we have

$$w_{1m}^{(\mathscr{E})} = \frac{4\omega^3}{3} e^2 |(Q_{1,-m}^{(\mathscr{E})})_{fi}|^2. \tag{46.10}$$

The quantities $Q_{1m}^{(\mathscr{E})}$ are related to the components of the vector of the electric dipole moment by the formulae

$$eQ_{10}^{(\mathscr{E})} = id_z, \qquad eQ_{1,\pm 1}^{(\mathscr{E})} = \mp\frac{i}{\sqrt{2}} (d_x \pm id_y). \tag{46.11}$$

Summing (46.10) over the values of m, we return, as expected, to the already known formula (45.7) for the total probability of dipole radiation.

The angular distribution of multipole radiation is determined by formula (7.11). Normalising it to the total probability of emission w_{jm}, we have

$$dw_{jm} = |\mathbf{Y}_{jm}^{(\mathscr{E})}(\mathbf{n})|^2 w_{jm}\, do = \frac{w_{jm}}{j(j+1)} |\nabla_{\mathbf{n}} Y_{jm}|^2\, do. \tag{46.12}$$

In particular, for $j = 1$

$$Y_{10} = i\sqrt{\frac{3}{4\pi}} \cos\theta, \qquad Y_{1,\pm 1} = \mp i\sqrt{\frac{3}{8\pi}} \sin\theta \cdot e^{\pm i\varphi},$$

where θ, φ are the polar angle and the azimuth of the direction $\mathbf{n}$ relative to the z-axis. Computing the gradient, we find that the angular distribution of dipole radiation with definite values of m is given by the expressions

$$dw_{10} = w_{10} \frac{3}{8\pi} \sin^2\theta\, d\theta, \qquad dw_{1,\pm 1} = w_{1,\pm 1} \frac{3}{8\pi} \frac{1 + \cos^2\theta}{2} do. \tag{46.13}$$

These could, of course, also be obtained directly from formula (45.6), setting in it once (for $m = 0$): $d_x = d_y = 0$, $d_z = d$, and another time ($m = \pm 1$): $d_y = \mp id_x = d/\sqrt{2}$, $d_z = 0$.

If the order of magnitude of the dimensions of the system (atom or nucleus) is a, then the order of magnitude of the electric multipole moments is, generally speaking, $Q_{jm}^{\mathscr{E}} \sim a^j$. The probability of multipole radiation is therefore

$$w_{jm}^2 \sim \alpha k (ka)^{2j}. \tag{46.14}$$

Increasing the multipole order by 1 diminishes the probability of radiation by a factor $\sim (ka)^2$.

The laws of conservation of angular momentum and parity lead to definite *selection rules*, restricting the possible changes in the state of the radiating system. If the initial angular momentum of the system equals J_i, then after emission of a photon with angular momentum j the angular momentum

of the system can take only the values J_f determined by the rule of addition of angular momenta ($\mathbf{J}_i - \mathbf{J}_f = \mathbf{j}$):

$$|J_i - J_f| \leqslant j \leqslant J_i + J_f. \tag{46.15}$$

For given values of J_i and J_f the same rule (46.15) determines the possible values of the photon angular momentum j. But since the probability of radiation rapidly diminishes with increasing j, the radiation occurs predominantly with the lowest possible multipole order.

The projections M_i and M_f of the angular momenta $\mathbf{J}_i$ and $\mathbf{J}_f$ together with the projection m of the photon angular momentum satisfy the obvious (from the same law of addition of angular momenta) rule

$$M_i - M_f = m. \tag{46.16}$$

The parities P_i and P_f of the initial and final states of the radiating system must satisfy the condition $P_f P_\Phi = P_i$, where P_Φ is the parity of the emitted photon; since the parities can only take the values ± 1, this condition can also be written in the form

$$P_i P_f = P_\Phi. \tag{46.17}$$

For a photon of the electric type $P_\Phi = (-1)^j$, so that the parity selection rule for electric multipole radiation is:

$$P_i P_f = (-1)^j. \tag{46.18}$$

The selection rules for angular momentum and parity are completely rigorous and must be obeyed in radiation by any systems. Alongside these rules there may exist other, more restrictive ones, connected with specific structural features of particular radiating systems. Such rules inevitably have a more or less approximate character; we shall consider them in subsequent sections of this chapter.

The dependence of the probability of emission on the quantum numbers m, M_i, M_f is determined entirely by the tensor character of the multipole moments. The quantities Q_{jm} with a given j constitute a spherical tensor of rank j. The dependence of their matrix elements on the indicated quantum numbers is given by the formula

$$|\langle n_f J_f M_f | Q_{j,-m} | n_i J_i M_i \rangle|^2 = \begin{pmatrix} J_f & j & J_i \\ M_f & m & -M_i \end{pmatrix}^2 |\langle n_f J_f \| Q_j \| n_i J_i \rangle|^2 \tag{46.19}$$

(see III, (107.6)), where the letter n conventionally denotes the set of remaining quantum numbers, besides J and M, of the state of the system. The reduced matrix elements standing on the right-hand side of (46.19) do not depend on the numbers m, M_i, M_f. Substituting into (46.9) this formula, we determine the desired dependence, which turns out to be proportional to

$$\begin{pmatrix} J_f & j & J_i \\ M_f & m & -M_i \end{pmatrix}^2$$

(it is assumed, of course, that the emitter is not located in an external field; then the transition frequency ω does not depend on the numbers M_i and M_f).

Summing the probability over all values M_f (for given M_i), we obtain the total probability of emission of a photon of given frequency from the initial level $n_i J_i$. By virtue of the spatial isotropy it is obvious that this quantity will not depend also on the initial value of M_i. The summation is carried out using the formula

$$\sum_{M_f} |\langle n_f J_f M_f | Q_{j,-m} | n_i J_i M_i \rangle|^2 = \frac{1}{2J_i + 1} |\langle n_f J_f \| Q_j \| n_i J_i \rangle|^2 \tag{46.20}$$

(see III, (107.11)).

§ 47. Magnetic multipole radiation

The wave function of a photon of the magnetic type $A^\mu = (0, \mathbf{A})$, where $\mathbf{A}$ is given by formula (7.6). Substituting it into (46.1), we obtain for the matrix element of the transition

$$V_{fi} = -e \frac{\sqrt{\omega}}{2\pi} \int d^3x \cdot \mathbf{j}_{fi}(\mathbf{r}) \int do_{\mathbf{n}} \cdot e^{-i\mathbf{kr}} \, \mathbf{Y}^{(M)*}_{jm}(\mathbf{n}). \tag{47.1}$$

The components of the vector $\mathbf{Y}^{(M)}_{jm}$ are expressed according to (7.16) through spherical functions of order j. Substituting the expansion (46.3) into (46.2) and integrating over $do_{\mathbf{n}}$, we get

$$\int e^{-i\mathbf{kr}} \, \mathbf{Y}^{(M)*}_{jm}(\mathbf{n}) \, do_{\mathbf{n}} = 4\pi i^{-j} \, g_j(kr) \, \mathbf{Y}^{(M)*}_{jm}\left(\frac{\mathbf{r}}{r}\right),$$

and after substituting g_j from (46.5) [11]

$$V_{fi} = -e i^{-j} \frac{2\omega^{j+1/2}}{(2j+1)!!} \int \mathbf{j}_{fi}(\mathbf{r}) \, r^j \, \mathbf{Y}^{(M)*}_{jm}\left(\frac{\mathbf{r}}{r}\right) d^3x. \tag{V.1}$$

We transform under the integral sign

$$r^j \, \mathbf{j}_{fi} \cdot [\mathbf{r} \nabla Y^*_{jm}] = -[\mathbf{r} \, \mathbf{j}_{fi}] \nabla (r^j Y^*_{jm})$$

and obtain

$$V_{fi} = (-1)^m i^j \sqrt{\frac{(2j+1)(j+1)}{\pi j}} \, \frac{\omega^{j+1/2}}{(2j+1)!!} \, e(Q^{(M)}_{j,-m})_{fi}, \tag{47.2}$$

where we have introduced the quantities

$$(Q^{(M)}_{jm})_{fi} = \frac{1}{j+1} \sqrt{\frac{4\pi}{2j+1}} \int [\mathbf{r} \, \mathbf{j}_{fi}] \nabla (r^j Y_{jm}) \, d^3x. \tag{47.3}$$

They are called the 2^j*-pole magnetic moments of the transition*. In view of the analogy between expressions (47.2) and (46.6), the formula for the probability of emission is obtained from (46.9) merely by replacing the electric moments by magnetic ones. The angular distribution formula (46.12) also remains in force (as was already noted in connection with (7.11)).

[11] Do not confuse the current $\mathbf{j}$ with the angular momentum j!

Let us examine the structure of expression (47.3) for $j = 1$. In this case the functions

$$\sqrt{\frac{4\pi}{3}}\, rY_{10} = iz, \qquad \sqrt{\frac{4\pi}{3}}\, rY_{1,\pm 1} = \mp\frac{i}{\sqrt{2}}\,(x \pm iy),$$

and their gradients are simply the circular unit vectors $\mathbf{e}^{(0)}$, $\mathbf{e}^{(\pm 1)}$ (7.14). The quantities $e(Q^{(M)}_{1m})_{fi}$ therefore represent the spherical components of the vector

$$\boldsymbol{\mu}_{fi} = \frac{e}{2}\int [\mathbf{r}\,\mathbf{j}_{fi}]\, d^3x, \tag{47.4}$$

which by its structure is analogous to the classical magnetic moment (see II, § 44). The total probability of $M1$-radiation (in ordinary units) is

$$w = \frac{4\omega^3}{3\hbar c^3}\,|\boldsymbol{\mu}_{fi}|^2. \tag{47.5}$$

Let us show how formula (47.4) is connected with the usual non-relativistic expression for the operator of magnetic moment.

The expression for the transition current (see III, § 115):

$$\mathbf{j}_{fi} = -\frac{i}{2m}\,(\psi_f^*\nabla\psi_i - \psi_i\nabla\psi_f^*) + \frac{\mu}{es}\,\mathrm{rot}(\psi_f^*\,\widehat{\mathbf{s}}\,\psi_i), \tag{47.6}$$

where μ is the magnetic moment of the particle, s its spin. Therefore

$$\boldsymbol{\mu}_{fi} = -\frac{ie}{4m}\int \psi_f^*[\mathbf{r}\nabla]\psi_i\, d^3x + \frac{ie}{4m}\int \psi_i[\mathbf{r}\nabla]\psi_f^*\, d^3x + \; + \frac{\mu}{2s}\int [\mathbf{r}\,\mathrm{rot}(\psi_f^*\,\widehat{\mathbf{s}}\,\psi_i)]\, d^3x. \tag{47.7}$$

In the second term we write

$$\int \psi_i[\mathbf{r}\nabla]\psi_f^*\, d^3x = -\int \psi_f^*[\mathbf{r}\nabla]\psi_i\, d^3x + \int \mathrm{rot}(\mathbf{r}\,\psi_f^*\psi_i)\, d^3x.$$

The last integral transforms into an integral over an infinitely remote surface and vanishes. Thus, the first two terms in (47.7) are the same. In the third term we transform the integral as follows (temporarily denoting $\mathbf{F} = \psi_f^*\widehat{\mathbf{s}}\psi_i$):

$$\int [\mathbf{r}[\nabla\mathbf{F}]]\, d^3x = \oint [\mathbf{r}[d\mathbf{f}\cdot\mathbf{F}]] - \int [[\mathbf{F}\nabla]\mathbf{r}]\, d^3x.$$

The integral over the surface vanishes, and in the last integral we have: $[[\mathbf{F}\nabla]\mathbf{r}] = -\mathbf{F}\,\mathrm{div}\,\mathbf{r} + \mathbf{F} = -2\mathbf{F}$. Thus,

$$\int [\mathbf{r}\,\mathrm{rot}\,\mathbf{F}]\, d^3x = 2\int \mathbf{F}\, d^3x.$$

As a result the expression for $\boldsymbol{\mu}_{fi}$ takes the form

$$\boldsymbol{\mu}_{fi} = \int \psi_f^*\Big(\frac{e}{2m}\,\widehat{\mathbf{L}} + \frac{\mu}{s}\,\widehat{\mathbf{s}}\Big)\psi_i\, d^3x, \tag{47.8}$$

where $\widehat{\mathbf{L}} = -i[\mathbf{r}\nabla]$ is the orbital angular momentum operator of the particle. As it should be, $\boldsymbol{\mu}_{fi}$ turns out to be the matrix element of the operator

$$\widehat{\boldsymbol{\mu}} = \frac{e}{2m}\widehat{\mathbf{L}} + \frac{\mu}{s}\widehat{\mathbf{s}}, \tag{47.9}$$

composed of the operators of the orbital and intrinsic magnetic moments of the particle.

The selection rules for magnetic multipole radiation are analogous to those for the electric case: for the total angular momentum the same rules (46.15), (46.16) hold, and for parity the rule is

$$P_i P_f = (-1)^{j+1}, \tag{47.10}$$

obtained by substituting into (46.17) the parity of the Mj-photon: $P_\Phi = (-1)^{j+1}$.

§ 48. Angular distribution and polarisation of radiation

The formulae derived in §§ 46 and 47 referred to the emission of a photon with definite values of the angular momentum j and its projection m. Correspondingly, it was assumed that the radiating system (say, a nucleus) both before and after emission possesses not only definite values of the angular momentum J, but also definite polarisations, i.e. values of M.

Let us now consider the more general case of radiation by a partially polarised nucleus (whose dimensions, as before, are assumed small compared with the wavelength). The emitted photon still possesses a definite angular momentum j, but may be partially polarised. We shall find the probability of emission as a function of the direction $\mathbf{n}$ of the photon. It must be expressed through the density matrices describing the polarisation states of the nucleus and the photon.

To this end let us first write the emission probability as a function of the direction $\mathbf{n}$ and the helicity λ of the photon ($\lambda = \pm 1$) for the case when both the initial and final nuclei have definite values: $J_i M_i$ and $J_f M_f$.

The matrix element of emission of a photon with definite jm is proportional to the matrix element of the 2^j-pole (electric or magnetic) moment of the nucleus:

$$\langle J_f M_f;\, jm|V|J_i M_i\rangle \propto (-1)^m \langle J_f M_f|Q_{j,-m}|J_i M_i\rangle. \tag{48.1}$$

The wave function of the emitted photon (in the momentum representation) is proportional to $\mathbf{Y}^{(\mathscr{E})}_{jm}(\mathbf{n})$ or $\mathbf{Y}^{(M)}_{jm}(\mathbf{n})$. The wave function of a photon with momentum in the direction $\mathbf{n}$ and helicity λ is proportional to the polarisation vector $\mathbf{e}^{(\lambda)}$. The matrix element of emission of a photon $\mathbf{n}\lambda$ is obtained by multiplying (48.1) with the projection of the wave function of the state $|jm\rangle$ onto the wave function of the state $|\mathbf{n}\lambda\rangle$:

$$\langle J_f M_f;\, \mathbf{n}\lambda|V|J_i M_i\rangle \propto (-1)^m \langle J_f M_f|Q_{j,-m}|J_i M_i\rangle(\mathbf{e}^{(\lambda)*}\mathbf{Y}_{jm}).$$

According to (16.23) for photons of both types

$$\mathbf{e}^{(\lambda)*}\mathbf{Y}_{jm}(\mathbf{n}) \propto D^{(j)}_{\lambda m}(\mathbf{n}). \tag{48.2}$$

The matrix element of the multipole moment is expressed in the usual way through the reduced element. As a result we obtain for the transition amplitude the form

$$\langle J_f M_f;\, \mathbf{n}\lambda|V|J_i M_i\rangle \propto (-1)^{J_f-M_f+m}\begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} Q\, D^{(j)}_{\lambda m}(\mathbf{n}), \tag{48.3}$$

where Q denotes $\langle J_f\|Q\|J_i\rangle$.

Now we can proceed to the general case of mixed polarisation states. According to the general rules of quantum mechanics the transition probability will be proportional to the expression [12]

$$\begin{aligned}&\sum_{(m)}(-1)^{2J_i-M_i-M_i'}\, D^{(j)}_{\lambda m}\, D^{(j)*}_{\lambda' m'} \times \\ &\times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix}\begin{pmatrix} J_f & j & J_i \\ -M_f' & -m' & M_i' \end{pmatrix}\langle M_i|\rho^{(i)}|M_i'\rangle \times \\ &\times \langle M_f'|\rho^{(f)}|M_f\rangle\langle\lambda'|\rho^{(\gamma)}|\lambda\rangle,\end{aligned} \tag{48.4}$$

where $\rho^{(i)}$, $\rho^{(f)}$, $\rho^{(\gamma)}$ are the density matrices of the initial nucleus, the final nucleus and the emitted photon; the symbol (m) under the summation sign means that the summation is taken over all doubly repeated m-indices $(M_i M_i' M_f M_f' \lambda\lambda')$. In (48.4) one must substitute (48.3) [13].

Let us denote the probability of emission of a photon into the solid angle do by $w(\mathbf{n})do$. The total probability of emission over all directions and over all polarisations of the photon and the recoil nucleus does not depend, obviously, on the initial polarisation state of the nucleus. It is given by formulae already known to us and does not interest us

[12] If the initial and final states of the system are described by superpositions

$$\psi^{(i)} = \sum_n a_n \psi_n^{(i)}, \qquad \psi^{(f)} = \sum_m b_m \psi_m^{(f)},$$

then the matrix element

$$\langle f|V|i\rangle = \sum_{mn} b_m^* a_n V_{mn},$$

and its square

$$|\langle f|V|i\rangle|^2 = \sum_{nn'mm'} V_{mn} V^*_{m'n'}\, a_n a^*_{n'}\, b^*_m b_{m'}.$$

The transition to mixed states is effected by the replacements

$$a_n a^*_{n'} \to \rho^{(i)}_{nn'}, \qquad b_{m'} b^*_m \to \rho^{(f)}_{m'm},$$

so that

$$|\langle f|V|i\rangle|^2 \to \sum_{nn'mm'} V_{mn} V^*_{m'n'}\, \rho^{(i)}_{nn'} \rho^{(f)}_{m'm}.$$

[13] In the transformations of sign factors one may use the fact that the numbers $2J_i$, $2J_f$, $2M_i$, $2M_f$ are of the same parity. Recall also that the numbers j, m are integers, while $\lambda = \pm 1$.

here. We shall therefore normalise the probability $w(\mathbf{n})$ to 1. For this we obtain [2]

$$w(\mathbf{n}) = \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{2J_i - M_i - M_i'} D^{(j)}_{\lambda m} D^{(j)*}_{\lambda' m'} \times$$

$$\times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f' & -m' & M_i' \end{pmatrix} \langle M_i|\rho^{(i)}|M_i'\rangle \times$$

$$\times \langle M_f'|\rho^{(f)}|M_f\rangle \langle\lambda'|\rho^{(\gamma)}|\lambda\rangle \tag{48.5}$$

(we shall verify the correctness of the normalisation below). We transform this formula by expanding the product of two D-functions into a series (110.2) (see III):

$$D^{(j)}_{\lambda m} D^{(j)*}_{\lambda' m'} = (-1)^{\lambda' + m'} D^{(j)}_{\lambda m} D^{(j)}_{-\lambda', -m'} =$$

$$= (-1)^{\lambda + m} \sum_L (2L+1) \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} D^{(L)}_{\Lambda\mu} \tag{V.2}$$

(indices $\Lambda = \lambda - \lambda'$, $\mu = m - m'$; L are integers, $L \geqslant 2j$). Thus, we obtain finally

$$w(\mathbf{n}) = \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{2J_i - M_i - M_i' + m + 1} (2L+1) \times$$

$$\times \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times$$

$$\times \begin{pmatrix} J_f & j & J_i \\ -M_f' & -m' & M_i' \end{pmatrix} D^{(L)}_{\Lambda\mu}(\mathbf{n}) \langle M_i|\rho^{(i)}|M_i'\rangle \langle M_f'|\rho^{(f)}|M_f\rangle \langle\lambda'|\rho^{(\gamma)}|\lambda\rangle. \tag{48.5}$$

As before, $\sum_{(m)}$ denotes summation over all (doubly repeated) m-indices. In this connection it should be recalled that, unlike the indices λ, λ', the remaining m-indices are summed not over all $2j+1$ possible (for a given j-index) values, but only over two values: $\lambda, \lambda' = \pm 1$, corresponding to the two polarisation states of the photon.

Formula (48.5) contains all the necessary information about the angular distribution of the emitted photons and their polarisation, as well as about the polarisation of the secondary (i.e. the ones that have emitted photons) nuclei. It is assumed that the initial density matrix is given.

Angular distribution. The angular distribution of photons is obtained by summation over all polarisations of the photon and the recoil nucleus. The averaging over polarisations is effected by substituting the density matrices of unpolarised states:

$$\langle\lambda|\rho^{(\gamma)}|\lambda'\rangle = \tfrac{1}{2}\,\delta_{\lambda\lambda'}, \qquad \langle M_f|\rho^{(f)}|M_f'\rangle = \frac{1}{2J_f+1}\,\delta_{M_f M_f'}, \tag{48.6}$$

after which the summation reduces to multiplication by 2 (for the photon) or by $2J_f + 1$ (for the nucleus). In other words, the summation is effected simply by the replacement

$$\langle\lambda|\rho^{(\gamma)}|\lambda'\rangle \to \delta_{\lambda\lambda'}, \qquad \langle M_f|\rho^{(f)}|M_f'\rangle \to \delta_{M_f M_f'}. \tag{48.7}$$

[2] In the transformations of sign factors one may use the fact that the numbers $2J_i$, $2J_f$, $2M_i$, $2M_f$ are of the same parity. Recall also that the numbers j, m are integers, while $\lambda = \pm 1$.

Thus, the angular distribution

$$\overline{w}(\mathbf{n}) = \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{m'+1} (2L+1)\, D^{(L)}_{0\mu}(\mathbf{n}) \times$$
$$\times \begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times$$
$$\times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i|\rho^{(i)}|M'_i\rangle. \tag{V.3}$$

This formula can be substantially simplified by carrying out the summation over the m-indices.

First of all we note that

$$\begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} = (-1)^L \begin{pmatrix} j & j & L \\ -\lambda & \lambda & 0 \end{pmatrix}, \tag{48.8}$$

and therefore

$$\sum_{\lambda=\pm 1} \begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} = \begin{cases} 2\begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} & \text{for even } L, \\ 0 & \text{for odd } L. \end{cases}$$

Thus, in the sum over L only terms with even L remain, i.e. it contains spherical functions ($D^{(L)}_{0\mu}$) of only even orders. This result could have been anticipated: by virtue of parity conservation the probability distribution must be invariant under inversion, i.e. under the replacement $\mathbf{n} \to -\mathbf{n}$.

Thus,

$$\overline{w}(\mathbf{n}) = \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L (2L+1) \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} D^{(L)}_{0\mu}(\mathbf{n}) \times$$
$$\times \sum_{(m)} (-1)^{m'+1} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \times$$
$$\times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i|\rho^{(i)}|M'_i\rangle. \tag{V.4}$$

Let us note that it is easy here to verify the normalisation: by virtue of the formula

$$\int D^{(L)}_{0\mu}(\mathbf{n}) \frac{do}{4\pi} = \delta_{L0}\, \delta_{\mu 0}$$

after integration over directions only the term with $L = \mu = 0$ remains; and with the help of the formulae

$$\begin{pmatrix} j & j & 0 \\ m & -m & 0 \end{pmatrix} = (-1)^{j-m} \frac{1}{\sqrt{2j+1}},$$

$$\sum_{M_f m} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix}^2 = \frac{1}{2J_i + 1}, \qquad \operatorname{Sp} \rho^{(i)} = 1$$

we verify that the integral equals 1.

The further summation over $mm'M_f$ in the inner sum in $\overline{w}(\mathbf{n})$ is carried out with the help of formula (108.4) (see III). As a result we obtain for the angular distribution of photons the following final formula:

$$\overline{w}(\mathbf{n}) = (-1)^{1+J_i+J_f} \frac{(2j+1)\sqrt{2J_i+1}}{4\pi} \sum_{\text{even } L} (-i)^L \sqrt{2L+1} \times$$
$$\times \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \begin{Bmatrix} J_i & J_i & L \\ j & j & J_f \end{Bmatrix} \sum_{\mu} \mathcal{P}^{(i)}_{L\mu} D^{(L)}_{0\mu}(\mathbf{n}), \tag{48.9}$$

where we have introduced the notation

$$\mathcal{P}^{(i)}_{L\mu} = i^L \sqrt{(2L+1)(2J_i+1)} \sum_{M_i M_i'} (-1)^{J_i - M_i'} \begin{pmatrix} J_i & L & J_i \\ -M_i' & \mu & M_i \end{pmatrix} \langle M_i | \rho^{(i)} | M_i' \rangle,$$
$$\mathcal{P}^{(i)*}_{L\mu} = (-1)^{L-\mu} \mathcal{P}^{(i)}_{L,-\mu}. \tag{48.10}$$

The inner sum in (48.9) is taken over all $|\mu| \leqslant L$, and the outer sum over all even values of L satisfying the conditions

$$L \leqslant 2j, \qquad L \leqslant 2J_i \tag{48.11}$$

(these conditions are the triangle rules which the j-indices in the $3j$-symbols appearing in (48.9), (48.10) must satisfy). By virtue of these conditions the number of terms in the sum is usually not large. Thus, for $J_i = 0$ or $\frac{1}{2}$ only the term with $L = 0$ remains, i.e. the radiation is isotropic (it is easy to verify that the term with $L = 0$ equals $1/4$, as it should be by the normalisation condition). For $J_i = 1, \frac{3}{2}$ or for $j = 1$ two terms remain in the sum over L: $L = 0, 2$. Let us also note that if the density matrix $\rho^{(i)}$ is diagonal ($M_i = M_i'$), then $\mu = 0$, and the distribution function (48.9) takes the form of an expansion in Legendre polynomials (according to (16.5) and (58.23) (see III) the functions $D^{(L)}_{00}$ reduce to the polynomials $P_L(\cos\theta)$). Finally, if

$$\langle M_i | \rho^{(i)} | M_i' \rangle = 1/(2J_i + 1)\, \delta_{M_i M_i'},$$

i.e. the initial nucleus is unpolarised, then all the $\mathcal{P}^{(i)}_{L\mu} = 0$ except $\mathcal{P}^{(i)}_{00} = 1$ [14].

[14] Indeed, noting that

$$\begin{pmatrix} J & & J \\ -M' & 0 & M \end{pmatrix} = (-1)^{J-M} \frac{1}{\sqrt{2J+1}} \delta_{MM'},$$

we have

$$\sum_{MM'} (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \delta_{MM'} =$$
$$= \sqrt{2J+1} \sum_{MM'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \begin{pmatrix} J & 0 & J \\ -M' & 0 & M \end{pmatrix} = \sqrt{2J+1}\, \delta_{L0}\, \delta_{\mu 0},$$

from which using the definition (48.10) we find the stated result.

The quantities $\mathcal{P}_{L\mu}$ are convenient characteristics of the polarisation state of the nucleus; we shall call them *polarisation moments*. Formula (48.10) determines these quantities through the density matrix elements $\rho_{MM'}$. By direct verification one can easily establish the validity of the inverse formula, expressing the density matrix through the polarisation moments:

$$\rho_{MM'} = \sum_{L\mu} \sqrt{\frac{2L+1}{2J+1}}\, i^{-L}(-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \mathcal{P}_{L\mu}. \tag{48.12}$$

Let $f_{L\mu}$ be some spherical tensor of rank L depending on the polarisation state of the nucleus. According to the general rules (see III, (14.8)) its mean value in a state with density matrix $\rho_{MM'}$ equals

$$\overline{f}_{L\mu} = \sum_{MM'} \rho_{MM'}\, \langle JM'|f_{L\mu}|JM\rangle. \tag{48.13}$$

Expressing the matrix elements of the quantity $f_{L\mu}$ through the reduced element $\langle J\|f_L\|J\rangle$ according to

$$\langle JM'|f_{L\mu}|JM\rangle = i^L(-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \langle J\|f_L\|J\rangle$$

and introducing the polarisation moments according to the definition (48.10), we obtain

$$\overline{f}_{L\mu} = \frac{\langle J\|f_L\|J\rangle}{\sqrt{(2L+1)(2J+1)}}\, \mathcal{P}_{L\mu}. \tag{48.14}$$

Polarisation of the photon. With given (along with $\rho^{(i)}$) matrices $\rho^{(\gamma)}$ and $\rho^{(f)}$ formula (48.5) determines the probability of a transition in which the emitted photon and the nucleus are found in definite polarisation states. These states are essentially not a characteristic of the emission process as such, but rather of those detectors which register the photon and the nucleus, selecting from the emitted radiation definite polarisations. A more natural formulation of the problem is the other one, in which the final state of the system "nucleus + photon" is not fixed in advance, and it is required to determine the polarisation density matrix of this state for a given direction of emission of the photon.

The answer to this question is given by the same formula (48.5). If we represent it in the form

$$w = \overline{w}(\mathbf{n}) \sum_{(m)} \langle M_f;\, \mathbf{n}\lambda|\rho|M'_f;\, \mathbf{n}\lambda'\rangle\langle\lambda'|\rho^{(\gamma)}|\lambda\rangle\langle M'_f|\rho^{(f)}|M_f\rangle, \tag{48.15}$$

then the expression $\langle M_f;\, \mathbf{n}\lambda|\rho|M'_f;\, \mathbf{n}\lambda'\rangle$ will be the required density matrix, being the "projection" onto the given $\rho^{(\gamma)}\rho^{(f)}$. The factor $\overline{w}(\mathbf{n})$ is separated out in (48.15) in order that this matrix should be normalised by the usual condition

$$\sum_{\lambda M_f} \langle M_f;\, \mathbf{n}\lambda|\rho|M_f;\, \mathbf{n}\lambda\rangle = 1.$$

If we are interested in the polarisation of only the photon, we must sum over $M_f = M'_f$:

$$\langle \mathbf{n}\lambda|\rho|\mathbf{n}\lambda'\rangle = \sum_{M_f} \langle M_f;\, \mathbf{n}\lambda|\rho|M_f;\, \mathbf{n}\lambda'\rangle.$$

Quite analogously to the derivation of formula (48.9) we obtain

$$\langle \mathbf{n}\lambda|\rho|\mathbf{n}\lambda'\rangle = (-1)^{1+J_i+J_f}\,\frac{(2j+1)\sqrt{2J_i+1}}{8\pi\overline{w}(\mathbf{n})}\times$$
$$\times \sum_L (-i)^L\sqrt{2L+1}\begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix}\begin{Bmatrix} J_i & J_i & L \\ j & j & J_f \end{Bmatrix}\sum_\mu \mathcal{P}^{(i)*}_{L\mu}\, D^{(L)}_{\Lambda\mu}(\mathbf{n}) \tag{48.16}$$

($\Lambda = \lambda - \lambda'$), where the summation is taken over all integer values of L satisfying conditions (48.11).

In particular, the circular polarisation is determined by the Stokes parameter

$$\xi_2 = \langle \mathbf{n}1|\rho|\mathbf{n}1\rangle - \langle \mathbf{n},-1|\rho|\mathbf{n},-1\rangle$$

(see the problem to § 8). By virtue of the relations (48.8) in this difference all terms with even L drop out, and for ξ_2 one obtains a formula differing from the expression (48.9) only in that the summation is carried out over odd (instead of even) values of L.

Polarisation of recoil nuclei. Finally, if we are interested only in the final polarisation of the nuclei, we must put $\rho^{(\gamma)} \to \delta$. If we also integrate over the directions of the photon, then the density matrix of the recoil nucleus will be

$$\langle M_f|\rho|M'_f;\, \mathbf{n}\rangle = \int \overline{w}(\mathbf{n})\,\langle M_f;\, \mathbf{n}|\rho|M'_f;\, \mathbf{n}\rangle\, do =$$
$$= (2J_i+1)\sum_{mM_iM'_i}\begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix}\times$$
$$\times\begin{pmatrix} J_f & j & J_i \\ -M'_f & -m & M'_i \end{pmatrix}\langle M_i|\rho^{(i)}|M'_i\rangle. \tag{V.5}$$

The polarisation moments computed from this density matrix are

$$\mathcal{P}^{(f)}_{L\mu} = (-1)^{J_i+J_f+L+j}\sqrt{(2J_i+1)(2J_f+1)}\begin{Bmatrix} J_i & J_i & L \\ J_f & J_f & j \end{Bmatrix}\mathcal{P}^{(i)}_{L\mu}. \tag{48.17}$$

If the initial nucleus is unpolarised, then the final nucleus will also be unpolarised. However, even in this case there will exist a correlation polarisation, i.e. the polarisation of the nucleus after emission of radiation in a given direction. Setting $\rho^{(i)} \to \delta/(2J_i+1)$ (and correspondingly $\overline{w}(\mathbf{n}) = 1/(4\pi)$) and carrying out a calculation analogous to the derivation of (48.9), we obtain for the density matrix describing this polarisation

$$\langle M_f;\, \mathbf{n}|\rho|M'_f;\, \mathbf{n}\rangle = (2j+1)(-1)^{J_i+M'_f+1}\sum_{\text{even } L}(2L+1)\begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix}\begin{pmatrix} J_f & L & J_f \\ -M'_f & \mu & M_f \end{pmatrix}\times$$
$$\times\begin{Bmatrix} J_f & J_f & L \\ j & j & J_i \end{Bmatrix} D^{(L)}_{0\mu}(\mathbf{n}). \tag{48.18}$$

The polarisation moments corresponding to this density matrix are

$$\mathcal{P}^{(f)}_{L\mu} = i^L(-1)^{1+J_i+J_f}(2j+1)\sqrt{(2L+1)(2J_f+1)}\begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix}\begin{Bmatrix} J_f & J_f & L \\ j & j & J_i \end{Bmatrix} D^{(L)}_{0\mu}(\mathbf{n}). \tag{48.19}$$

Only moments of even order arise (which is again a consequence of parity conservation).

If the secondary nucleus in turn radiates, then, being polarised, it gives a non-isotropic distribution of photons. Since the polarisation moments (48.19) depend on the direction $\mathbf{n}$ of the photon emitted in the first decay, there arises a definite angular correlation between the directions of successively emitted photons (with an unpolarised primary nucleus). In an analogous way one can consider other correlation phenomena in cascade emissions (correlation of polarisations, etc.) [15].

Problem

Express the polarisation moments $\mathcal{P}_{1\mu}$ and $\mathcal{P}_{2\mu}$ through the mean values of the angular momentum vector $\mathbf{J}$ and the quadrupole moment tensor Q_{ik}.

Solution. The reduced elements of the vector $\mathbf{J}$ and the tensor Q_{ik} are determined from the equalities (cf. III, (107.10), (107.11)). The operator $\widehat{Q}_{ik}$ is expressed through the angular momentum operators by formula (75.2) (see III):

$$\widehat{Q}_{ik} = \frac{3Q}{2J(2J-1)}\left(\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \tfrac{2}{3}J^2\delta_{ik}\right).$$

From this we find the mean value

$$\overline{Q^2_{ik}} = \frac{3Q^2}{2J^2(2J-1)^2}J^2(4J^2-3) = Q^2\,\frac{3(J+1)(2J+3)}{2J(2J-1)}.$$

The reduced matrix elements:

$$\langle J\|J\|J\rangle = \sqrt{J(J+1)(2J+1)},$$

$$\langle J\|Q\|J\rangle = Q\sqrt{\frac{3(2J+1)(J+1)(2J+3)}{2J(2J-1)}}.$$

From (48.14) we now see that the polarisation moments $\mathcal{P}_{1\mu}$ coincide with the spherical components of the vector

$$\sqrt{\frac{3}{J(J+1)}}\,\overline{\mathbf{J}},$$

and the moments $\mathcal{P}_{2\mu}$ with the spherical components of the tensor

$$\sqrt{\frac{10J(2J-1)}{3(J+1)(2J+3)}}\,\frac{\overline{Q}_{ik}}{Q}.$$

[15] A detailed exposition of these questions can be found in a paper by A. Z. Dolginov in the book "Gamma Rays" (Izd. AN SSSR, 1961).

§ 49. Radiation of atoms. Electric type[16]

The energies of the outer electrons of an atom (participating in optical radiative transitions) are in rough estimate of the order of $E \sim me^4/\hbar^2$, so that the wavelengths radiated are $\lambda \sim \hbar c/E \sim \hbar^2/(\alpha me^2)$. The dimensions of the atom are $a \sim \hbar^2/me^2$. Therefore in optical spectra of atoms, as a rule, the inequality $a/\lambda \sim \alpha \ll 1$ is satisfied. The same order of magnitude is the ratio $v/c \sim \alpha$, where v is the speed of the optical electrons.

Thus, in optical spectra of atoms the condition is fulfilled by virtue of which the probability of electric dipole radiation (if it is allowed by selection rules) substantially exceeds the probability of multipole transitions[17]. In connection with this, in the spectroscopy of atoms the most important role is played by precisely the electric dipole transitions.

As has already been indicated, such transitions are subject to strict selection rules for the total angular momentum J of the atom and for parity P[18]:

$$|J' - J| \leqslant 1 \leqslant J + J', \tag{49.1}$$

$$PP' = -1. \tag{49.2}$$

The inequality $|J' - J| \leqslant 1$ means that the angular momentum J can change only by 0, ±1; by virtue of the inequality $J + J' \geqslant 1$ the transition $0 \to 0$ is additionally forbidden. The parities of the initial and final states must be opposite.

The probability of radiation with the transition $nJM \to n'J'M'$ is determined by the corresponding matrix element of the dipole moment of the atom according to

$$w(nJM \to n'J'M') = \frac{4\omega^3}{3\hbar c^3}\,|\langle n'J'M'|\mathbf{d}_{-m}|nJM\rangle|^2, \tag{49.3}$$

$$\omega = \omega(nJ \to n'J').$$

Summing (49.3) over all values $M' = M - m$ (for a given M), we obtain the total probability of emission of radiation of a given frequency from the atomic level nJ. The summation is carried out with the help of (46.20) and gives

$$w(nJ \to n'J') = \frac{4\omega^3}{3\hbar c^3}\,\frac{1}{2J+1}\,|\langle n'J'\|d\|nJ\rangle|^2. \tag{49.4}$$

The square of the modulus of the reduced matrix element standing here is sometimes called the *line strength of the transition*; this quantity is symmetric with respect to the initial and final states.

The observed intensity of emission is obtained by multiplying w by $\hbar\omega$ and by the number of atoms in the source found on the given excited level ($N_{n,J}$). Thus, in a gas with temperature T the number $N_{n,J} \propto (2J+1)\exp(-E_{nJ}/T)$; the factor $(2J+1)$ is the statistical weight of the level with angular momentum J.

[16] In §§ 49, 51, 53–55 we use ordinary units.

[17] Typical values of the probability of dipole transitions in the optical region of atomic spectra are of the order $10^8\ \mathrm{s}^{-1}$.

[18] The selection rule for parity was established by *Laporte* (*O. Laporte*, 1924).

Further conclusions about the probabilities of transitions in atomic spectra can be made only on specifying the character of the states of the atom. We shall not dwell here on the methods of computing matrix elements, the degree of approximation of which does not have a sharp theoretical character. We shall derive only certain relations for a fairly broad (and especially in light

atoms) category of states, constructed according to the LS-coupling scheme (see III, § 72). Such states are characterised, besides the total angular momentum, also by definite conserved values of the orbital angular momentum L and the spin S in this case.

Since the dipole moment is a purely orbital quantity, its operator commutes with the spin operator, i.e. its matrix is diagonal in the number S. By the number L for the dipole moment the same selection rules apply as for any orbital vector (see III, § 29). Thus, transitions between states constructed according to the LS-scheme are subject to additional (besides (49.1), (49.2)) selection rules:

$$S' - S = 0, \tag{49.5}$$

$$|L' - L| \leqslant 1 \leqslant L + L'. \tag{49.6}$$

Let us emphasise once more that these rules are approximate and break down when the spin–orbit interaction is taken into account. Let us note that the rule (49.5) (the forbiddenness of transitions between terms of different multiplicity) holds not only for dipole but for all transitions of the electric type: the electric multipole moments of all orders are orbital tensors, so that their matrices are diagonal in spin. Thus, for electric quadrupole transitions, besides the general rules

$$|J' - J| \leqslant 2 \leqslant J + J', \qquad PP' = 1, \tag{49.7}$$

in the case of LS-coupling the additional selection rules hold:

$$S' - S = 0, \qquad |L' - L| \leqslant 2 \leqslant L + L'. \tag{49.8}$$

The dependence of the probability of radiation on the numbers S, J, J' can be determined in explicit form. This question is solved directly with the help of general formulae for matrix elements of spherical tensors in the addition of angular momenta. According to formula (109.3) (see III) we have [19]:

$$|\langle n'L'SJ'\|d\|nLSJ\rangle|^2 = (2J+1)(2J'+1)\begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}^2 |\langle n'L'\|d\|nL\rangle|^2. \tag{49.9}$$

Substituting this into (49.4), we get

$$w(nLSJ \to n'L'SJ') = \frac{4\omega^3}{3\hbar c^3}(2J'+1)\begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}^2 |\langle n'L'\|d\|nL\rangle|^2, \tag{49.10}$$

[19] In the formulae of vol. III, § 109 under "angular momenta of subsystems 1 and 2" one should now understand the orbital angular momentum and the spin of the atom, the interaction between which we neglect. The role of the quantity $f_q^{(1)}$ is played by the orbital vector d_q.

where $\omega = \omega(nLS \to n'L'S)$ [20].

For these probabilities one can obtain a definite sum rule. For the squares of the $6j$-symbols the summation formula holds (see III, (108.7)):

$$\sum_{J'} (2J'+1) \begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}^2 = \frac{1}{2L+1}. \tag{49.11}$$

With its help we find from (49.10)

$$\sum_{J'} w(nLSJ \to n'L'SJ') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2L+1} |\langle n'L'\|d\|nL\rangle|^2. \tag{49.12}$$

Let us note that this quantity turns out to be independent of the initial value of J.

If we are dealing with the radiation of a gas with temperature T, much larger than the intervals of the fine structure of the atomic term nSL, then the states with different J are populated uniformly, i.e. all values of J are equally probable. The probability that the atom is on a level with a certain definite value of J in such a case is

$$\frac{2J+1}{(2L+1)(2S+1)}, \tag{49.13}$$

i.e. the ratio of the statistical weight of this level to the full statistical weight of the term nSL. The averaging of the expressions (49.10) or their sums (49.12) over these probabilities reduces to multiplication by the factor (49.13); we shall denote this averaging by a bar over the letter. The total probability of radiation from all lines of the spectral multiplet (formed by all possible transitions between the fine-structure components of the two terms nSL and $n'SL'$) is the sum:

$$\overline{w}(nLS \to n'L'S) = \sum_J \sum_{J'} \overline{w}(nLSJ \to n'L'SJ'). \tag{49.14}$$

Since, of course, $\sum_J (2J+1) = (2S+1)(2L+1)$, for the full probability we obtain an expression coinciding with (49.12).

Therefore for the relative probability (or, which is the same, the relative intensity) of an individual line we obtain

$$\frac{\overline{w}(nLSJ \to n'L'SJ')}{\overline{w}(nLS \to n'L'S)} = \frac{(2J+1)(2J'+1)}{(2S+1)} \begin{Bmatrix} L' & J' & S \\ J & L & 1 \end{Bmatrix}^2. \tag{49.15}$$

Analysis of the numerical values given by this formula reveals that among the lines of the multiplet the most intense are those for which $\Delta J = \Delta L$ (they are called *principal lines*, as distinct from the remaining components of the multiplet, called *satellites*). The intensity of the principal lines is the greater the larger the initial value of J.

Summing (49.15) over J or over J' gives

$$\frac{\sum_{J'} \overline{w}(nLSJ \to n'L'SJ')}{\sum_J \overline{w}(nLSJ \to n'L'SJ')} = \frac{2J+1}{(2L+1)(2S+1)}, \tag{49.16}$$

[20] Neglecting the spin–orbit interaction in the computation of matrix elements, we neglect also the dependence of frequencies on J and J', i.e. the fine structure of the initial and final levels of the atom.

$$\frac{\sum_J \overline{w}(nLSJ \to n'L'SJ')}{\sum_{J'} \overline{w}(nLSJ \to n'L'SJ')} = \frac{2J'+1}{(2L+1)(2S+1)}.$$

Thus, the sum of intensities of all lines of a spectral multiplet having one and the same initial (or final) level is proportional to the statistical weight of the initial (or final) level.

Let us dwell also on the hyperfine structure of spectral lines of the atom. Recall that the hyperfine splitting of atomic levels arises as a result of the interaction of the electrons with the spin of the nucleus, if the latter is different from zero (see III, § 122). The total angular momentum of the atom (including the nucleus) **F** is composed of the total angular momentum of the electrons **J** and the moment of the nucleus **I**. Each component of the hyperfine structure of the level nJ is characterised by its value of the quantum number F.

The strict law of conservation of angular momentum leads now to a strict selection rule for the total angular momentum F; for electric dipole radiation

$$|F' - F| \leqslant 1 \leqslant F + F'. \tag{49.17}$$

But by virtue of the extreme weakness of the interaction of the electrons with the spin of the nucleus it can be entirely neglected in the computation of the matrix elements of the electric (and magnetic) moments of the electron shell of the atom. Therefore the previous selection rules for the electronic angular momentum J and for the electronic parity also remain valid. In particular, by virtue of the latter, electric dipole transitions between components of the hyperfine structure of one and the same term are impossible: all these levels have the same parity, whereas the indicated transitions are possible only between states of different parity.

Since the operator of the dipole moment commutes with the spin of the nucleus, the dependence of the matrix elements on the numbers I and F

can be found in explicit form; the calculations differ from those given above only by an obvious change of notation. The probability of radiation, summed over the final values of the projection of the total angular momentum F:

$$w(nJIF \to n'J'IF') = \frac{4\omega^3}{3\hbar c^3}\,\frac{1}{2F+1}\,|\langle n'J'IF'\|d\|nJIF\rangle|^2, \tag{49.18}$$

$$\omega = \omega(nJ \to n'J'),$$

where the square of the reduced matrix element

$$|\langle n'J'IF'\|d\|nJIF\rangle|^2 = (2F+1)(2F'+1)\begin{Bmatrix} J' & F' & I \\ F & J & 1 \end{Bmatrix}^2 |\langle n'J'\|d\|nJ\rangle|^2. \tag{49.19}$$

Problem

The majority of lines in the spectra of alkali metals can be described as a result of transitions of one outer (optical) electron in the self-consistent field of the atomic core forming a closed configuration; the state of the atom is constructed according to the

LS-coupling scheme. Under these assumptions determine the relative intensities of the fine-structure components of spectral lines.

Solution. The total angular momenta L and $S = \frac{1}{2}$ of the atom coincide with the orbital angular momentum and spin of the optical electron. The parity of the state is $(-1)^l$ (the parity of the closed configuration of the atomic core is positive). The selection rules for parity therefore forbid a dipole transition with $L' = L$, so that only transitions with $L' - L = \pm 1$ are possible. The transitions between components of the doublet levels n, L and $n', L-1$ give in all (by virtue of the selection rules for J) just three lines (Fig. 1).

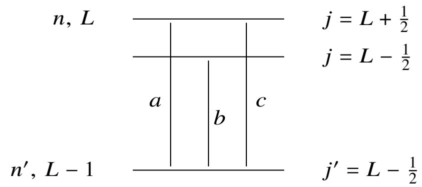

Figure 1

Their relative intensities (denoting them a, b, c) can be determined, without resorting to formula (49.15), from the rule (49.16). Composing the ratios of the summed intensities of lines with one and the same initial (or final) level, we obtain two equalities:

$$\frac{b+c}{a} = \frac{2L}{2L+2}, \qquad \frac{a+b}{c} = \frac{2L}{2L-2},$$

from which

$$a : b : c = [(L+1)(2L-1)] : 1 : [(L-1)(2L+1)].$$

If $L = 1$, the lower level is not split, line c is absent, and $a/b = 2$.

§ 50. Radiation of atoms. Magnetic type

The magnetic moment of an atom is in order of magnitude given by the Bohr magneton: $\mu \sim e\hbar/mc$. This estimate differs by a factor α from the order of magnitude of the electric dipole moment of the atom: $d \sim ea \sim \hbar^2/(me)$ (since $v/c \sim \alpha$, we have $\mu \sim d\omega/c$, as one would expect). It follows therefore that the probability of magnetic dipole ($M1$) radiation by an atom is approximately α^2 times smaller than the probability of electric dipole radiation (of the same frequency). Magnetic radiation therefore plays a role in practice only for transitions forbidden by the selection rules of the electric type.

As regards the electric quadrupole ($E2$) radiation, the ratio of its probability to the probability of $M1$-radiation is in order of magnitude

$$\frac{E2}{M1} \sim \frac{(ea^2)^2\omega^2/c^2}{\mu^2} \sim \frac{a^4 m^2 \omega^2}{\hbar^2} \sim \left(\frac{\Delta E}{E}\right)^2 \tag{50.1}$$

(quadrupole moment $\sim ea^2$, $E \sim \hbar^2/ma^2$ is the energy of the atom, ΔE is the energy change in the transition). We see that for medium atomic frequencies (i.e. for $\Delta E \sim E$)

the probabilities of $E2$- and $M1$-radiation are of the same order (provided, of course, that both are allowed by the selection rules). If $\Delta E \ll E$ (for example, for transitions between fine-structure components of one and the same term), then $M1$-radiation is more probable than $E2$-radiation.

Magnetic dipole transitions are subject to the strict selection rules

$$|J' - J| \leqslant 1 \leqslant J + J', \tag{50.2}$$

$$PP' = 1. \tag{50.3}$$

In the case of LS-coupling additional selection rules arise, even more restrictive than in the electric case. The latter circumstance is connected with the specific property of the magnetic moment of the atom, arising as a result of the equality of all particles in the system (electrons). Namely, the operator of the magnetic moment of the atom is expressed through the operators of its orbital and spin angular momenta:

$$\widehat{\boldsymbol{\mu}} = -\mu_0(\widehat{\mathbf{L}} + 2\widehat{\mathbf{S}}) = -\mu_0(\widehat{\mathbf{J}} + \widehat{\mathbf{S}}), \tag{50.4}$$

where $\mu_0 = |e|\hbar/(2mc)$ is the Bohr magneton (see III, § 113). By virtue of the conservation of the total angular momentum the operator $\widehat{\mathbf{J}}$ does not at all have off-diagonal (in energy) matrix elements; so that for the theory of radiation it suffices to write $\widehat{\boldsymbol{\mu}} = -\mu_0\widehat{\mathbf{S}}$[23].

When the spin–orbit interaction is neglected, the spin operator is diagonal in all the quantum numbers nSL characterising the unsplit term. For any transition at all to take place, the number J must necessarily change. Thus, we have the selection rules:

$$n' = n, \quad S' = S, \quad L' = L, \quad J' - J = \pm 1, \tag{50.5}$$

i.e. transitions are possible only between fine-structure components of one and the same term.

The computation of the probability of radiation in this case can be carried through to the end. Changing the designations correspondingly in formula (49.10), we find

$$w(nLSJ \to nLSJ') = \frac{4\omega^3\mu_0^2}{3\hbar c^3}(2J'+1)\begin{Bmatrix} S & J' & L \\ J & S & 1 \end{Bmatrix}^2 |\langle S\|S\|S\rangle|^2.$$

The reduced matrix element of the spin with respect to the eigenfunction of the spin itself is given by the formula

$$\langle S\|S\|S\rangle = \sqrt{S(S+1)(2S+1)} \tag{50.6}$$

(see III, (29.13)). The $6j$-symbol we need equals

$$\begin{Bmatrix} S & J-1 & L \\ J & S & 1 \end{Bmatrix}^2 = \frac{(L+S+J+1)(L+S-J+1)(L-S+J)(S-L+J)}{S(2S+1)(2S+2)(2J-1)2J(2J+1)} \tag{50.7}$$

[23] An exception is the case when the electronic angular momentum $\mathbf{J}$ of the atom is not conserved: when the hyperfine structure is taken into account, in the presence of an external field, etc. (see the Problems).

(see the table in III, § 108). As a result we obtain

$$w(nLSJ \to nLS,\, J-1) = \frac{2J+1}{2J-1}\, w(nLS,\, J-1 \to nLSJ) =$$
$$= \frac{\omega^3 \mu_0^2}{3\hbar c^3 (2J+1)J}\,(L+S+J+1)(L+S-J+1)(L-S+J)\times$$
$$\times (J+L-S). \tag{50.8}$$

Transitions between components of the hyperfine structure of one and the same level (whose frequencies lie in the radio-wave region) cannot occur as electric-dipole transitions, since all these components possess the same parity. Without change of parity the transitions can occur as $E2$ and $M1$. But in view of the very small magnitude of the intervals of the hyperfine structure the $E2$ emission is highly improbable in comparison with $M1$ (cf. (50.1)), so that the indicated transitions are realised as magnetic-dipole transitions.

Problems

1. Find the probability of the $M1$-transition between components of the hyperfine structure of one and the same level.

Solution. The probability of the transition is given by formulae (49.18), (49.19), in which there will now figure the diagonal reduced matrix element of the magnetic moment: $\langle nJ\|\mu\|nJ\rangle$. Its value can

be written at once if we notice that the full (unreduced) matrix element $\langle nJM|\mu_\lambda|nJM\rangle$ is precisely the splitting of the given level in the Zeeman effect (see III, § 113) and equals $-\mu_0 gM$, where g is the Landé g-factor. The reduced matrix element (see III, (29.7))

$$\langle nJ\|\mu\|nJ\rangle = \frac{1}{M}\sqrt{J(J+1)(2J+1)}\,\langle nJM|\mu_\lambda|nJM\rangle = -\mu_0 g\sqrt{J(J+1)(2J+1)}.$$

As a result we find for the desired probability [24]

$$w(nJIF \to nJI,\, F-1) = \frac{2F+1}{2F-1}\, w(nJI,\, F-1 \to nJIF) =$$
$$= \frac{\omega^3 \mu_0^2 g^2}{3\hbar c^3 (2F+1)F}\,(J+I+F+1)(J+I-F+1)(F+J-I)(F-J+I). \tag{V.6}$$

[24] An interesting example is the transition between the components of the hyperfine structure of the ground level of the hydrogen atom ($1s_{1/2}$), strictly forbidden not only as $E1$ but also as $E2$ (the latter by the rule forbidding the quadrupole transition with $J+J'=1$). This transition has the frequency $\omega = 2\pi\cdot 1.42\cdot 10^9\ \mathrm{s}^{-1}$ (wavelength $\lambda = 21$ cm). Setting $g=2$, $I=\frac{1}{2}$, $J=\frac{1}{2}$, $F=1$, $F'=0$, we obtain

$$w = \frac{4\omega^3\mu_0^2}{3\hbar c^3} = 2.85\cdot 10^{-15}\ \mathrm{s}^{-1}.$$

This expression differs from (50.8) only by an obvious change of notation and an additional factor g^2.

2. Find the probability of the $M1$-transition between Zeeman components of one and the same atomic level.

Solution. We are considering the transition $M \to M-1$ at unchanged nJ; the frequency of the transition (see below, (51.3)): $\hbar\omega = \mu_0 g H$ (g is the Landé factor). The matrix element of the spherical component μ_{-1} of the vector $\boldsymbol{\mu}$:

$$|\langle nJ,\, M-1|\mu_{-1}|nJM\rangle| = \sqrt{\frac{(J-M+1)(J+M)}{2J(J+1)(2J+1)}}\,|\langle nJ\|\mu\|nJ\rangle| =$$

$$= -\mu_0 g\sqrt{\tfrac{1}{2}(J-M+1)(J+M)}$$

(see III, (27.12) and the preceding Problem). The transition probability

$$w = \frac{4\omega^3}{3\hbar c^3}\,|\langle nJ,\, M-1|\mu_{-1}|nJM\rangle|^2 = \frac{2\mu_0^2 H^3}{3\hbar^4 c^3}\,(J-M+1)(J+M).$$

§ 51. Radiation of atoms. Zeeman and Stark effects

In an external magnetic field H (which we assume weak) each atomic level with total angular momentum J splits into $2J+1$ levels

$$E_M = E^{(0)} + \mu_0 g M H, \tag{51.1}$$

where $E^{(0)}$ is the unperturbed level, μ_0 is the Bohr magneton, g is the Landé factor, M is the projection of the angular momentum J onto the direction of the field (see III, § 113). The degeneracy with respect to the directions of the angular momentum is thus completely removed.

Correspondingly, the spectral lines, arising from transitions between two split levels, also split. The number of components of the line is determined by the selection rule for the number M, according to which for dipole radiation it must be

$$m = M - M' = 0,\ \pm 1. \tag{51.2}$$

Additionally the transitions with $M = M' = 0$ are forbidden if at the same time $J' = J$. This is seen directly from the general expressions (29.7) (see III) for the matrix elements of an arbitrary vector.

The components arising from transitions with $m = 0, \pm 1$ are called respectively π- and σ-components. Their frequencies:

$$\hbar\omega_\pi = \hbar\omega^{(0)} + \mu_0 H (g - g')M, \tag{51.3}$$

$$\hbar\omega_\sigma = \hbar\omega^{(0)} + \mu_0 H[gM - g'(M \pm 1)].$$

In the particular case, when $g = g'$, we have

$$\hbar\omega_\pi = \hbar\omega^{(0)}, \qquad \hbar\omega_\sigma = \hbar\omega^{(0)} \mp \mu_0 g H, \tag{51.4}$$

independently of the value of M; in other words, in this case the line splits into a triplet with an undisplaced π-component and two σ-components symmetrically situated on either side of it (the so-called *normal Zeeman effect*).

The total (over all directions) probability of radiation is proportional to the square of the modulus $|\langle n'J'M'|\mathbf{d}_{-m}|nJM\rangle|^2$. Therefore, by virtue of formula (46.19) with $j = 1$, the relative probability of emission of each of the Zeeman components of the spectral line is

$$\begin{pmatrix} J' & 1 & J \\ M' & m & -M \end{pmatrix}^2 . \tag{51.5}$$

In the particular case of the normal Zeeman effect there are in all just three components, each of which arises from transitions with all initial M for a given m. Since

$$\sum_{MM'} \begin{pmatrix} J' & 1 & J \\ M' & m & -M \end{pmatrix}^2 = \frac{1}{3} \tag{51.6}$$

(see III, (106.12)), in this case the radiation of all three components is equiprobable.

Of much greater interest, however, are the relative intensities of the Zeeman components when observed in a definite direction (relative to the direction of the applied magnetic field). According to (45.5) the probability of

radiation (and with it the intensity of the line) in a given direction $\mathbf{n}$ is proportional to $\sum |\mathbf{e}^*\mathbf{d}_{fi}|^2$, where the summation is carried out over the two independent polarisations $\mathbf{e}$ possible for the given $\mathbf{n}$.

In observation along the field (axis z) the sum is

$$|(d_x)_{fi}|^2 + |(d_y)_{fi}|^2.$$

Passing to the spherical components, we obtain

$$|(d_1)_{fi}|^2 + |(d_{-1})_{fi}|^2.$$

This means that in the longitudinal (along the field) direction only the two σ-components ($m = \pm 1$) are observed. Their intensities are proportional to

$$\begin{pmatrix} J' & 1 & J \\ M \mp 1 & \pm 1 & -M \end{pmatrix}^2 . \tag{51.7}$$

Possessing definite values of the angular momentum projection m along the direction of propagation, these lines have right ($m = 1$) and left ($m = -1$) circular polarisations (see § 8).

In observation in the perpendicular-to-field direction (let this be the axis x) the intensity is proportional to the sum

$$|(d_z)_{fi}|^2 + |(d_y)_{fi}|^2 = |(d_0)_{fi}|^2 + \tfrac{1}{2}\{|(d_1)_{fi}|^2 + |(d_{-1})_{fi}|^2\}.$$

Thus, in the transverse direction two σ-components and a π-component are observed, with intensities proportional respectively to

$$\tfrac{1}{2}\begin{pmatrix} J' & 1 & J \\ M \mp 1 & \pm 1 & -M \end{pmatrix}^2 \quad \text{and} \quad \begin{pmatrix} J' & 1 & J \\ M & 0 & -M \end{pmatrix}^2 . \tag{51.8}$$

(the intensities of the σ-components are half those in longitudinal observation). The π-component is polarised linearly along the axis z, and the σ-components are observed in this direction polarised linearly along the axis y.

Let us note that the relative intensities of the Zeeman components are determined entirely by the initial and final values of J and M without any dependence on other characteristics of the levels.

The selection rules forbid electric-dipole transitions between Zeeman components of one and the same level, since all of them have the same parity. For the same reason that was indicated at the end of the preceding paragraph for transitions between components of the hyperfine structure of a level, the indicated transitions are realised as magnetic-dipole ones. By virtue of the selection rule for the number M the transitions occur only between adjacent components ($M' - M = \pm 1$)[25].

The splitting of levels of an atom in a weak electric field (*Stark effect*), unlike the splitting in a magnetic field, does not lead to complete removal of the degeneracy with respect to the directions of the angular momentum. All levels, with the exception of the level $M = 0$, remain doubly degenerate: to each belong two states with projections of the angular momentum M and $-M$.

The computation of the relative intensities of the Stark components of a spectral line is analogous to that presented above for the Zeeman effect. In this connection one must bear in mind that the π-component receives contributions (for $M \neq 0$) from the transitions $M \to M$ and $-M \to -M$, and the σ-component receives contributions from the transitions $M \to M \pm 1$ and $-M \to -(M \pm 1)$. Therefore, for example, upon transverse observation the intensities of the π-components are proportional to

$$2\begin{pmatrix} J' & 1 & J \\ M & 0 & -M \end{pmatrix}^2,$$

and the intensities of the σ-components are proportional to the sums

$$\frac{1}{2}\begin{pmatrix} J' & 1 & J \\ M \pm 1 & \mp 1 & -M \end{pmatrix}^2 + \frac{1}{2}\begin{pmatrix} J' & 1 & J \\ -M \mp 1 & \pm 1 & M \end{pmatrix}^2 = \begin{pmatrix} J' & 1 & J \\ M \pm 1 & \mp 1 & -M \end{pmatrix}^2$$

(recall that on changing the sign of all numbers in the second row the $3j$-symbols can only change sign, while their squares do not change).

In an external field, even a weak one, strictly speaking the total angular momentum $\mathbf{J}$ ceases to be conserved; in a uniform field only the conservation of the projection of the angular momentum M is exactly preserved. Therefore upon radiative transitions in a weak field the conservation of angular momentum is no longer strictly obligatory, and in the spectrum of the atom lines may appear that are forbidden by the usual selection rules.

The computation of the intensities of these lines reduces to the computation of corrections to the matrix of the dipole moment, which in turn requires the determination of corrections to the wave functions of stationary states. In the first approximation of perturbation theory (in the weak external field) the wave function acquires "ad-

[25]These transitions usually have frequencies in the centimetre range and are observed in absorption and stimulated emission (electronic paramagnetic resonance): the absorbing atoms are placed in a strong static magnetic field (producing Zeeman splitting) and a weak radio-frequency field of the resonance frequency.

mixtures" of states connected with the original by non-zero matrix elements of perturbation ($-\mathbf{Ed}$ in an electric field): the admixture of a certain state ψ_2 to the state ψ_1 is

$$\frac{-\mathbf{Ed}_{21}}{E_1 - E_2}\,\psi_2.$$

As a result, in the matrix element of the "forbidden" transition there will appear a term

$$\frac{-(\mathbf{Ed}_{21})\mathbf{d}_{32}}{E_1 - E_2},$$

different from zero if the transitions from the "intermediate" state 2 into both the initial and final states 1 and 3 are allowed.

§ 52. Radiation of atoms. Hydrogen atom

The hydrogen atom represents a unique case in which the computation of the matrix elements of the transition can be carried through to the end in analytical form (*W. Gordon*, 1929).

The parity of the state of the hydrogen atom is $(-1)^l$, i.e. it is unambiguously determined by the orbital angular momentum of the electron (recall that the number l as the quantity determining the parity of the state retains its meaning also for the exact relativistic wave functions, i.e. when the spin–orbit interaction is taken into account). Therefore the selection rule for parity strictly forbids electric-dipole transitions without change of l; only transitions with $l \to l \pm 1$ are possible. Changes of the principal quantum number n, however, are not restricted.

The dipole moment of the hydrogen atom reduces to the radius vector of the electron: $\mathbf{d} = e\mathbf{r}$. Since the wave function of the electron in the hydrogen atom is a product of the angular part and the radial function R_{nl}, the reduced matrix elements of the radius vector are also represented as a product

$$\langle n', l-1\|r\|nl\rangle = \langle l-1\|\nu\|l\rangle \int_0^\infty R_{n',l-1}\, r\, R_{nl}\, r^2\, dr, \qquad (52.1)$$

where $\langle l-1\|\nu\|l\rangle$ are the reduced matrix elements of the unit vector $\boldsymbol{\nu}$ in the direction of $\mathbf{r}$. The latter are

$$\langle l-1\|\nu\|l\rangle = \langle l\|\nu\|l-1\rangle^* = i\sqrt{l}$$

(see III, (29.14)). Thus,

$$\langle n', l-1\|r\|nl\rangle = -\langle nl\|r\|n', l-1\rangle = i\sqrt{l}\int_0^\infty R_{n',l-1}\, R_{nl}\, r^3\, dr. \qquad (52.1)$$

The non-relativistic radial functions of the discrete spectrum of the hydrogen atom are given by formula (36.13) (see III)[26]:

$$R_{nl} = \frac{2}{n^{l+2}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}\,(2r)^l\, e^{-r/n} \times F(-n+l+1,\, 2l+2,\, 2r/n). \quad (52.2)$$

[26]In this paragraph we use atomic units. In ordinary units the expressions given below for matrix elements of coordinates should be multiplied by $\hbar^2/(me^2)$ (if a hydrogen-like ion with atomic number Z is under consideration, then by $\hbar^2/(mZe^2)$).

The integral (52.1) with the product of two hypergeometric functions is evaluated with the help of formulae in III, § f[27]. The computation leads to the result

$$\langle n',\, l-1\|r\|nl\rangle =$$

$$= i\sqrt{l}\,\frac{(-1)^{n'-l}}{4(2l-1)!}\sqrt{\frac{(n+l)!(n'+l-1)!}{(n-l-1)!(n'-l)!}}\;\frac{(4nn')^{l+1}(n-n')^{n+n'-2l-2}}{(n+n')^{n+n'}}\times$$

$$\times\Big\{F\Big(-n+l+1,\,-n'+l,\,2l,\,-\frac{4nn'}{(n-n')^2}\Big)-$$

$$-\Big(\frac{n-n'}{n+n'}\Big)^2F\Big(-n+l-1,\,-n'+l,\,2l,\,-\frac{4nn'}{(n-n')^2}\Big)\Big\},\tag{52.3}$$

where $F(\alpha,\,\beta,\,\gamma,\,z)$ is the hypergeometric function. Since the parameters α,β in this case are negative integers (or zero), the functions reduce to polynomials[28].

For reference we give the expressions obtained from (52.3) in some particular cases (the value of l is indicated by the spectroscopic symbol $s,\,p,\,d,\,\ldots$):

$$\begin{aligned}
|\langle 1s\|r\|np\rangle|^2 &= \frac{2^8n^7(n-1)^{2n-5}}{(n+1)^{2n+5}},\\
|\langle 2s\|r\|np\rangle|^2 &= \frac{2^{17}n^7(n^2-1)(n-2)^{2n-6}}{(n+2)^{2n+6}},\\
|\langle 2p\|r\|nd\rangle|^2 &= \frac{2^{19}n^9(n^2-1)(n-2)^{2n-7}}{3(n+2)^{2n+7}},\\
|\langle 2p\|r\|ns\rangle|^2 &= \frac{2^{15}n^5(n-2)^{2n-6}}{3(n+2)^{2n+6}}.
\end{aligned}\tag{52.4}$$

Formula (52.3) is inapplicable for transitions without change of the principal quantum number n (transitions between fine-structure components of a level). In this case ($n=n'$) for carrying out the integration we start from the representation of the radial functions through the generalised Laguerre polynomials:

$$R_{nl} = -\frac{2}{n^2}\sqrt{\frac{(n-l-1)!}{[(n+l)!]^3}}\;e^{-r/n}\Big(\frac{2r}{n}\Big)^l L_{n+l}^{2l+1}\Big(\frac{2r}{n}\Big).\tag{52.5}$$

In the integral

$$\int_0^\infty R_{n,l-1}\,R_{nl}\,r^3\,dr \propto \int_0^\infty e^{-\rho}\rho^{2l+2}\,L_{n+l}^{2l+1}(\rho)\,L_{n+l-1}^{2l+1}(\rho)\,d\rho$$

[27]The computation involves evaluating the integral $J_{l_2+2}^{l_2}(-n+l+1,\,-n'+l)$. It is carried out with the help of the formulae (f. 12) (f. 16).

[28]Numerical tables of matrix elements and transition probabilities for hydrogen can be found in the book: *H. Bethe, E. Salpeter*, Quantum Mechanics of One- and Two-Electron Atoms. M.: IL, 1960.

we replace one of the polynomials by its expression through the generating function (see III, § d):

$$L_{n+l}^{2l+1}(\rho) = -\frac{(n+l)!}{(n-l-1)!}\, e^{\rho}\rho^{-2l-1}\Big(\frac{d}{d\rho}\Big)^{n-l-1} e^{-\rho}\rho^{n+l}.$$

After $(n-l-1)$-fold integration by parts we obtain an integral of the form

$$\int_0^\infty e^{-\rho}\rho^{n+l}\Big(\frac{d}{d\rho}\Big)^{n-l-1} \rho L_{n+l-1}^{2l+1}(\rho)\, d\rho,$$

in which we replace the Laguerre polynomial by its explicit expression according to the formula

$$L_n^m(\rho) = (-1)^m n! \sum_{k=0}^{n-m} \binom{n}{m+k} \frac{(-\rho)^k}{k!}\,.$$

After carrying out the differentiation only three terms remain in the sum, after which the integration is elementary. The computation leads to the simple result:

$$\langle n,\, l-1\|r\|nl\rangle = i\sqrt{l}\cdot \tfrac{3}{2}n\sqrt{n^2-l^2}. \tag{52.6}$$

The integral

$$\int_0^\infty R_{n',l-1}\, R_{nl}\, r^3\, dr = \int_0^\infty \chi_{n',l-1}(r\chi_{nl})\, dr$$

(where $\chi_{nl} = rR_{nl}$) is the coefficient of the expansion of the function $r\chi_{nl}$ in the system of orthogonal functions $\chi_{n',l-1}$ $(n' = 1,$

$2, \ldots)$. The sum of squares of the moduli of these coefficients equals the integral of the square of the function being expanded[29]. Therefore

$$\sum_{n'} |\langle n',\, l-1\|r\|nl\rangle|^2 = l\int_0^\infty r^2\chi_{nl}^2\, dr. \tag{52.7}$$

Making use of the known expression for the mean value of r^2 in the state nl (see III, (36.16)), we find the following sum rule:

$$\sum_{n'} |\langle n',\, l-1\|r\|nl\rangle|^2 = l\,\frac{n^2}{2}\,[5n^2+1-3l(l+1)]. \tag{52.8}$$

For given values of n, l and large values of n' the matrix element of the transition $nl \to n'l'$ decreases according to the law

$$|\langle n'l'\|r\|nl\rangle|^2 \propto \frac{3}{n'^3}, \tag{52.9}$$

as one can verify both from the particular expressions (52.4) and from the general formula (52.3). This result is entirely natural: the Coulomb levels $E' = -1/2n'^2$ for

[29]The summation is over both discrete and continuous spectrum states.

large n' are spaced quasi-continuously, and the probability of a transition to any level in the interval dE' is proportional to the density of levels, which itself is $\propto n'^{-3}$.

The Stark effect in hydrogen has, as is well known, a specific character (see III, § 77)—the splitting is proportional to the first power of the electric field. At the same time it is assumed that the field is not strong (condition of applicability of perturbation theory), but at the same time such that the splitting of levels is large compared with the fine structure (i.e. large compared to the Lamb shift). Under these conditions the angular momentum is not conserved at all and the levels must be classified by the parabolic quantum numbers n_1, n_2, m. The last of these—the magnetic quantum number m—as before determines the projection of the orbital angular momentum onto the axis z (direction of the field), which under the given conditions (neglect of the spin–orbit interaction) is conserved. Therefore the usual selection rule applies:

$$m' - m = 0, \ \pm 1. \tag{52.10}$$

For the change of the numbers n_1, n_2 there are no restrictions.

The matrix elements of the dipole moment in parabolic coordinates can also be computed analytically. The

resulting formulae are, however, very cumbersome, and we shall not present them here [30].

Problems

1. Find the Stark splitting of the levels of hydrogen in the case when the splitting is small compared with the fine-structure intervals (but large compared with the Lamb shift).

Solution. Under the stated conditions the doubly degenerate levels with $l = j \pm \frac{1}{2}$ remain unperturbed, and the Stark splitting Δ is determined from the secular equation

$$\begin{vmatrix} -\Delta & -E(d_z)_{12} \\ -E(d_z)_{21} & -\Delta \end{vmatrix} = 0, \qquad \Delta = \pm E|(d_z)_{12}|$$

(indices 1, 2 refer to the states with $l = j \pm \frac{1}{2}$ and a given magnetic quantum number m; the perturbation $V = -Ed_z$ is diagonal in m and does not have diagonal (in l) matrix elements). The matrix element of the orbital quantity d_z is computed with the help of formulae (29.7) and (109.3) (see III), according to which

$$\langle j, l-1, m|d_z|jlm\rangle = \frac{m}{\sqrt{j(j+1)(2j+1)}}\langle j, l-1\|d\|jl\rangle,$$

$$\langle j, l-1\|d\|jl\rangle = -(2j+1)\begin{Bmatrix} l-1 & j & \frac{1}{2} \\ j & l & 1 \end{Bmatrix}\langle l-1\|d\|l\rangle,$$

where one should put $l = j + \frac{1}{2}$; the quantity $\langle l-1\|d\|l\rangle$ is taken from (52.6). As a result we obtain $\Delta = \pm\frac{3}{2}\sqrt{n^2 - (j+\frac{1}{2})^2}\ \frac{m}{j(j+1)}\ E$.

[30] These formulae and corresponding numerical tables can be found in the book cited above by *H. Bethe* and *E. Salpeter*.

2. Determine the probability of emission of a photon in the transition between the states $2\,s_{1/2} \to 1\,s_{1/2}$ of the hydrogen atom (*G. Breit, E. Teller*, 1940).

Solution. The process under consideration is strictly forbidden for $E1$-radiation by parity, and for $E2$-radiation by the rule (46.15). We should therefore compute the probability of $M1$-radiation, given by formula (47.5). In the given case ($l = 0$), however, the magnetic moment is a purely spin quantity, and its matrix element in the neglect of the spin–orbit interaction vanishes by virtue of the mutual orthogonality of the orbital wave functions with different principal quantum numbers, so that one must start from the full Dirac equation.

In the standard representation of the wave functions the transition current (see § 35)

$$j_{fi} = \psi_f^\dagger \boldsymbol{\alpha} \psi_i = \varphi_f^\dagger \boldsymbol{\sigma} \chi_i + \chi_f^\dagger \boldsymbol{\sigma} \varphi_i.$$

According to (35.1), (24.2), (24.8), for states with $l = 0$, $j = \frac{1}{2}$, the wave functions have the form

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \frac{1}{4\pi} \begin{pmatrix} f(r)\, w(m) \\ -ig(r)(\boldsymbol{\sigma}\mathbf{n})\, w(m) \end{pmatrix},$$

where $\mathbf{n} = \mathbf{r}/r$, and $w(m)$ is a real unit 3-spinor corresponding to the spin projection m. Thus,

$$j_{fi} = \frac{1}{4\pi}\left\{ f_f g_i\, w_f \boldsymbol{\sigma}(\boldsymbol{\sigma}\mathbf{n})\, w_i - g_f f_i\, w_f (\boldsymbol{\sigma}\mathbf{n})\boldsymbol{\sigma}\, w_i \right\}.$$

Substituting this expression into (47.4) and integrating over directions $\mathbf{n}$, we get

$$\mu_{fi} = -\frac{e}{6i}\, w_f [\boldsymbol{\sigma}\boldsymbol{\sigma}] w_i\, I = -\frac{e}{3}\, w_f \boldsymbol{\sigma}\, w_i\, I$$

(by virtue of the commutation relations of the Pauli matrices $[\boldsymbol{\sigma}\boldsymbol{\sigma}] = 2i\boldsymbol{\sigma}$); here

$$I = \int_0^\infty (f_f g_i + f_i g_f)\, r^3\, dr. \tag{1}$$

The probability of emission of a photon (47.5), summed over the values of m_f, is

$$w = \frac{4e^2\omega^3}{27}\, w_i \boldsymbol{\sigma}^2 w_i\, I^2 = \frac{4e^2\omega^3}{9}\, I^2. \tag{2}$$

From (35.4) we have (taking $\varkappa = -1$)

$$g = \frac{f'}{\varepsilon + m + \alpha/r} \approx \frac{f'}{2m} - \left(\varepsilon - m + \frac{\alpha}{r} - \frac{R'}{4m^2}\right),$$

where in the second term the exact function f is replaced by the non-relativistic radial function R. If we restrict ourselves to the approximation $g = R'/2m$, the integral

$$I = \frac{1}{2m}\int_0^\infty (R_f R_i' + R_f' R_i)\, r^3\, dr = -\frac{3}{2m}\int_0^\infty R_f R_i\, r^2\, dr = 0 \tag{3}$$

by the orthogonality of the functions R_f and R_i. In the next approximation, taking into account (3),

$$I = \frac{1}{2m}\int_0^\infty (f_f f_i')'\, r^3\, dr + \frac{1}{4m^2}\int_0^\infty \left\{ R_f' R_i (\varepsilon_i - \varepsilon_f) - \frac{\alpha}{r}\,(R_f R_i)' \right\} r^3\, dr. \tag{4}$$

Noting that by the orthogonality of the exact functions ψ_i and ψ_f we have (for $\varkappa_i = \varkappa_f$)

$$\int_0^\infty (f_i f_f + g_i g_f)\, r^2\, dr = 0,$$

the first term in (4) can be rewritten after integration by parts as

$$-\frac{3}{2m}\int_0^\infty f_f f_i\, r^2\, dr \approx \frac{3}{8m^3}\int_0^\infty R'_f R'_i\, r^2\, dr,$$

with account of (3).

The computation of the integral with the functions

$$R_f = 2(m\alpha)^3\, e^{-m\alpha r}, \qquad R_i = \frac{(m\alpha)^3}{\sqrt{2}}\left(1 - \frac{m\alpha r}{2}\right)e^{-m\alpha r/2}$$

(see III, § 36) and the difference of energies

$$\omega = \varepsilon_i - \varepsilon_f = \frac{m\alpha^2}{2}\left(1 - \frac{1}{2^2}\right) = \frac{3}{8}\, m\alpha^2$$

gives $I = 2^{3/2}\alpha^2/(9m)$. Whence the probability of the transition (ordinary units)

$$w = \frac{2^5\alpha^7\hbar^2\omega^3}{3^5 m^2 c^4} = \frac{mc^2\alpha^{13}}{2^4\cdot 3^5\hbar} = 5.6\cdot 10^{-6}\,\mathrm{s}^{-1}.$$

The corresponding lifetime of the $2\,s_{1/2}$ state is very large, and in practice the decay occurs much more probably through the simultaneous emission of two photons (see the footnote on p. 263).

CHAPTER VI

Scattering of Light

CHAPTER VII

The Scattering Matrix

CHAPTER VIII

Invariant Perturbation Theory

CHAPTER IX

Interaction of Electrons

CHAPTER X

Interaction of Electrons with Photons

CHAPTER XI

Exact Propagators and Vertex Parts

CHAPTER XII

Radiative Corrections

CHAPTER XIII

Asymptotic Formulae of Quantum Electrodynamics

CHAPTER XIV

Electrodynamics of Hadrons